Power Systems

Ming Zhou
Zhaoyuan Wu
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Power System Flexibility

Modeling, Optimization and Mechanism Design

Science Press Beijing

Springer

Power Systems

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Power Systems is indexed in Scopus

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ISSN 1612-1287 ISSN 1860-4676 (electronic)
Power Systems
ISBN 978-981-19-9074-8 ISBN 978-981-19-9075-5 (eBook)
https://doi.org/10.1007/978-981-19-9075-5

Jointly published with Science Press
The print edition is not for sale in China mainland. Customers from China mainland please order the print book from: Science Press.

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Preface

To combat the worldwide ambitious carbon-neutral target, the share of renewable energy in global annual electricity generation will need to rise from around 30% today to nearly 90% in 2050, and more than 60% will come from variable renewable energy (VRE), such as wind and solar power. The increasing integration of VRE poses specific challenges across different time scales, that is, maintaining the supply and demand balance, which triggers higher requirements for the flexibility of the power system. Power system flexibility here is defined as the ability of a power system to cope with various random factors and uncertainties at reasonable economic cost at different time scales, including planning and operation, under physical and security constraints. In summary, power system flexibility is the ability of the power system to respond quickly to large fluctuations in power supply and demand. In this context, it is necessary to analyze the flexibility of power system, which includes an important number of stochastic renewable production facilities.

The book provides an appropriate blend of theoretical background and practical applications of power system flexibility and provides a detailed description of flexibility in a high share of renewables integrated power systems, including power system flexibility modeling, flexibility-based economic dispatch, demand-side flexibility management, large-scale distributed flexible resources aggregation, and market design for enhancing the flexibility of power system etc.

The above feature strengthens the usefulness of the book for both students and practitioners. Students will gain an insightful understanding of the flexibility problem of the power system with high share of renewables integration including: (1) the formulation of flexibility modeling and flexibility-based economic dispatch models, (2) the familiarization with efficient solution algorithms for such models, (3) insights into these problems through the detailed analysis of numerous illustrative examples, and (4) market design approach for enhancing the flexibility of the power system. Hopefully, this book can greatly benefit readers in the fields of energy economics and engineering.

This book consists of the following eight chapters:

Chapter 1 motivates the subject matter of this book by introducing the power system flexibility and providing an overview of the main problems addressed in the remaining chapters.

Chapter 2 introduces different power system flexibility sources as well as modeling approaches from the perspective of power supply chain, i.e., supply side, grid side, and demand side.

Chapter 3 provides tools for flexibility-based economic dispatch considering the flexibility potential from the supply side.

Chapter 4 introduces distributed dispatch approach in AC/DC hybrid systems with high flexibility requirement.

Chapter 5 provides the flexible operation mechanism of the power grid and proposes the voltage source converter-based high voltage direct current transmission (VSC-HVDC) and the transmission switch (TS) approaches.

Chapter 6 focuses on the operational flexibility of the power system considering the flexibility provision from demand-side resources, and the integrated energy system demand response is also considered.

Chapter 7 provides the aggregation approach of large-scale distributed flexibility resources, and the equivalent aggregation theorem is introduced.

Chapter 8 introduces the market design for enhancing the flexibility of power system, and the interaction between the balancing market settlement rules and the flexibility provision is taken into account.

The power system flexibility here is defined as the ability of a power system to cope with various random factors and uncertainties at reasonable economic cost at different time scales, including planning and operation, under physical and security constraints.

This book opens the door to develop operational tools from the perspective of power system flexibility, especially in the future VRE sources-dominated system. These tools will evolve in an extraordinary way as the share of VRE in the future energy system continues to increase to achieve worldwide sustainability goals. This is a fascinating path that involves important intellectual and practical challenges, and is certainly a way to contribute to the flexible operation of energy system.

Finally, we would be remiss if we were to conclude this preface without expressing our gratitude to a few people and institutions. We would like to take this opportunity to express our sincere appreciation to everyone who directly or indirectly helped in making this book a reality. Our special thanks go to my talented philosophy doctors: Junyi Zhai, Zun Guo, Yan Li, Hanyan Huang, Shiyi Zhang, Zhi Zhang, and Junjie Rong for assisting us in this work. We are also very grateful to our graduates: Mingyang Zhang, Conghao Zhao, and Hongji Yang for their help in revising the language of specific chapters. Special thanks also go to the Institutions for the financial support from the Smart Grid Joint Foundation Program of National Natural

Science Foundation of China and State Grid Corporation of China (U1866204), and the National Key Research and Development Program of China (2016YFB0900100).

Beijing, China

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Chapter 1
Introduction



This introductory chapter serves to motivate the subject matter of this book by providing a brief overview of the power system flexibility challenges caused by the increasing penetration of renewable energy as well as the corresponding technologies and solutions to combat the ambitious carbon neutral targets. Specifically, we first introduce the history and development of the power system flexibility and propose a taxonomy for power system flexibility resources in terms of supply-side flexibility, grid flexibility, demand-side flexibility and flexibility from energy storage. Then, we provide a short overview of the framework of power system flexibility analysis from the perspective of power supply chain. In the end, we briefly introduce how to solve the problem of power system flexibility by effective market mechanism design.

1.1 Why Is Flexibility Necessary for the Power System

To cope with a series of social and environmental problems such as environmental pollution and greenhouse effect caused by increasing fossil energy consumption, the world is currently undergoing a major energy transformation, which has also brought profound changes to the production, structure and consumption mode of energy system [1]. As the core of energy transformation, the construction of low-carbon sustainable power system with renewable energy as the main body has led the way. In recent years, to achieve the key sustainable development and climate goals, especially with the reduction of cost and the progress of technology, renewable energy represented by wind and solar is developing at an unprecedented speed.

Although the rapid development of variable renewable energy has effectively alleviated the current severe problems such as energy shortage, environmental pollution and climate change, a high proportion of variable renewable energy access has had a series of impacts on the operation and planning of power system [2]. For a long time, China's power grid has been required to fully accommodate variable renewable energy such as wind and solar power. However, due to the uncertainty, volatility and

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M. Zhou et al., *Power System Flexibility*, Power Systems,
https://doi.org/10.1007/978-981-19-9075-5_1

reverse peak shaving characteristics of wind and solar power generation output, the further increase of variable renewable energy penetration in the system will bring great challenges to the power and energy balance of the system [3]. In order to eliminate or weaken the impact of uncertainty and volatility of variable renewable energy on system security and economic operation, power system needs to have sufficient response and regulation capability, that is, sufficient flexibility. If the system is not flexible enough and lacks sufficient regulation capacity, it is difficult to fully deal with frequent events such as uncertainty of variable renewable energy output and large fluctuation of net load. In order to ensure the safety and stability of the system operation, emergency measures such as wind and solar energy abandonment and load shedding sometimes must be taken, resulting in great economic impact and social losses. Sufficient flexibility resources has become a necessary condition for the safe and economic operation of power system [4].

In this context, this book provides a detailed description of flexibility in high share of renewables integrated power systems, including power system flexibility modeling, flexibility-based economic dispatch, demand side flexibility management, large-scale distributed flexible resources aggregation, and market mechanism design for enhancing the flexibility of power system etc. The book provides an appropriate blend of theoretical background and practical applications of the power system flexibility, which are developed as working algorithms, coded in MATLAB and GAMS environments. This feature strengthens the usefulness of the book for both students and practitioners. Students will gain an insightful understanding of the flexibility problem of the power system with high share of renewables integration, including: (1) the formulation of flexibility modeling and flexibility-based economic dispatch models, (2) the familiarization with efficient solution algorithms for such models, and (3) insights into these problems through the detailed analysis of numerous illustrative examples. (4) market mechanism design approach on enhancing the flexibility of the power system. Hopefully, this book can greatly benefit readers in the fields of energy economics and engineering.

1.2 Overview of Power System Flexibility

1.2.1 History and Development

In recent years, with the increasing penetration of renewable energy, the operational risk of power system is becoming more and more complex, and flexibility has gradually become a research hotspot. Although the research on flexibility is still in its infancy, power system flexibility is not a new concept. As early as the 1990s, in the field of power system planning, relevant scholars proposed the concept of flexibility for the uncertain factors in planning problems: the ability of power system development planning to adapt to the changes of planning conditions under certain cost constraints [5]. Some scholars also discussed how to use power electronics

and other equipment to improve the flexibility of transmission system, and finally achieve the purpose of enhancing the operation reliability of power system [6]. At that time, renewable energy was not connected to the grid on a large scale, and the regulation capacity of the power system dominated by traditional conventional units was relatively sufficient. Therefore, the relevant research has not attracted extensive attention.

After large-scale renewable energy grid connected operation, the uncertainty and volatility faced by the power system are significantly enhanced. How to give consideration to the economy and security of power system operation under the dual impact of renewable energy and load has become the focus of attention. In this context, power system flexibility has been paid more and more attention. At present, the definition of flexibility proposed by the International Energy Agency and the North American power reliability Commission is more recognized by the academic community. In 2008, the International Energy Agency (IEA) defined power system flexibility as the ability to quickly respond to large fluctuations on the generation side or load side caused by foreseeable and unforeseen events under system boundary constraints [7]. When the power system has enough flexibility, it can not only absorb a large amount of intermittent renewable energy output economically and efficiently, but also provide sufficient power generation capacity when the renewable energy output is low. The North American Electric Reliability Council (NERC) defines power system flexibility as the ability of the system scheduling internal resources to respond to net load changes [8]. NERC also points out that the system flexibility is different under different time scales, and the corresponding adjustment means of the system are also different. From the perspective of operation, the operation flexibility of power system includes second level stability control, minute level automatic generation control or load frequency control, load tracking or economic dispatching from minutes to hours and unit combination from hours to days.

Some scholars have given different definitions of flexibility according to the needs of their own research problems. Literature [9] holds that flexibility refers to the ability of power system to respond to load fluctuations at different time scales in order to maintain the reliability level. Aiming at the problem of insufficient hydropower output in dry season in Brazil, document [10] defines flexibility as power regulation capacity and economic benefits brought by regulation. Literature [11] defines flexibility as the maximum uncertainty range that the power system can cope with. At present, the research on the definition and connotation of power system flexibility has been relatively mature. In general, the research on power system flexibility focuses more on the dynamic response and margin of generation side and demand side.

In this book, we refer to the definition from International Renewable Energy Agency (IRENA) and define the power system flexibility as the capability of a power system to cope with the variability and uncertainty that solar and wind energy introduce at different time scales, from the very short to the long term, avoiding curtailment of power from these variable renewable energy (VRE) sources and reliably supplying all customer energy demand.

1.2.2 Taxonomy-Power System Flexibility Sources

Power system flexibility resources include all physical elements and regulation means that has the ability to flexibly adjust and deal with uncertainty and fluctuation. In this section, we propose a taxonomy for power system flexibility resources in terms of supply-side flexibility, grid flexibility, demand-side flexibility and flexibility from energy storage, and Chap. 2 of this book will elaborate on methods and models to describe the flexibility potential all links in power supply chain.

(1) Supply-side flexibility

In the traditional power system, the generation from the supply side is the only source of the system flexibility. The existing conventional thermal power plants can improve their operation flexibility by reducing the minimum technical output, shortening the start-up time and improving the ramping speed. In addition, in the high proportion of renewable energy power system, the uncertainty and volatility of renewable energy output is one of the main flexibility demands. If the prediction accuracy of renewable energy output can be improved, the scarcity of the system flexibility will be greatly alleviated, and the system flexibility will be improved from the supply side in a disguised manner. That is why most existing studies aim to improve the prediction accuracy of variable renewable energy output by establishing statistical models or based on external environmental factors such as meteorological data.

(2) Grid Flexibility

Grid flexibility refers to the capability of a power grid to maintain balance between generation and load during uncertainty, resulting in increased grid efficiency, resiliency, and the integration of variable renewables into the grid [12]. Power system flexibility has spatial attributes, and flexible interconnected power grid is the basis for realizing flexible supply–demand balance at different spatial scales. Different from the flexibility resources on the supply side or demand side, the ability to provide flexibility to the system mainly depends on their respective regulation capabilities, control and parameters. The power grid flexibility not only depends on the specific topology, control and parameters of the power grid, but also closely related to the flexibility resources on the supply side and demand side, that means the grid flexibility is the backbone of the whole system flexibility.

(3) Demand Side Flexibility

With the development of distributed energy and the support of the Internet of things (IoT), various flexible resources on the demand side can effectively respond to dispatch signals by aids of their fast response capability. Various flexible resources can be aggregated and behave as a controllable battery storage system for various grid supports, e.g., the system balancing, ancillary services, renewable energy accommodation and coordinated control, which can improve the flexibility of the system.

(4) Flexibility from energy storage

Power system flexibility has variable time attributes. Energy storage can connect the whole system to realize the translation of power supply and demand in time scale, so as to provide flexibility. For the supply side, energy storage can store redundant variable renewable energy power generation, so as to improve the flexibility of the power generation side; For the demand side, the electrification of intelligent terminals such as electric vehicles expands the flexibility of the demand side; For the power grid side, energy storage breaks the time boundary of the original supply–demand balance, reduces peak load, alleviates line congestion and improves the flexibility of power grid operation. In addition, with the development of various energy conversion technologies, different energy storage forms, such as electricity, heat and gas, further expand the supply dimension of the system flexibility.

1.2.3 Power System Flexibility Analysis

In this section, we will provide a short overview of the framework of power system flexibility analysis from the perspective of flexibility supply chain, which are subjects of subsequent discussion in Chaps. 3–6 of this book. Specifically, Chaps. 3–6 discusses in detail the flexible operation of power system from the supply-side, grid side and demand side, respectively.

The traditional power system operation pays more attention to the balance of power and electric energy in each independent operation period. In the future, the high proportion of variable renewable energy power system is faced with the demand of power system flexibility balance, that is, on each time and space scale, the flexibility supply capacity of the system needs to exceed the corresponding flexibility demand [13]. That is to say, a flexible power system means that the power system can deal with various random factors and uncertainties at a reasonable economic cost, and maintain reliable power supply under different time scales and physical and security constraints. Therefore, power system flexibility analysis includes not only the modeling and optimization of the operation characteristics of flexible resources and the flexible operation mode of power system at the technical level, but also the value evaluation of flexible resources and the design of incentive mechanism at the economic level, and the fundamental framework of power system flexibility analysis can be summarized as a three-level structure, as shown in Fig. 1.1.

As shown in Fig. 1.1, the lower layer is the analysis and modeling of flexible resource operation characteristics, that is, the power system flexibility potential from all links of the power supply chain, and the middle layer is the flexible operation strategy of variable renewable energy power system, that is, the system flexibility optimization scheduling method under the cooperation of “resource-network-load-storage”, which can be understood as the flexible operation mode of power system, The first two layers of the basic framework are the “technical flexibility” mentioned above. The top layer is designed to improve the market mechanism of the system

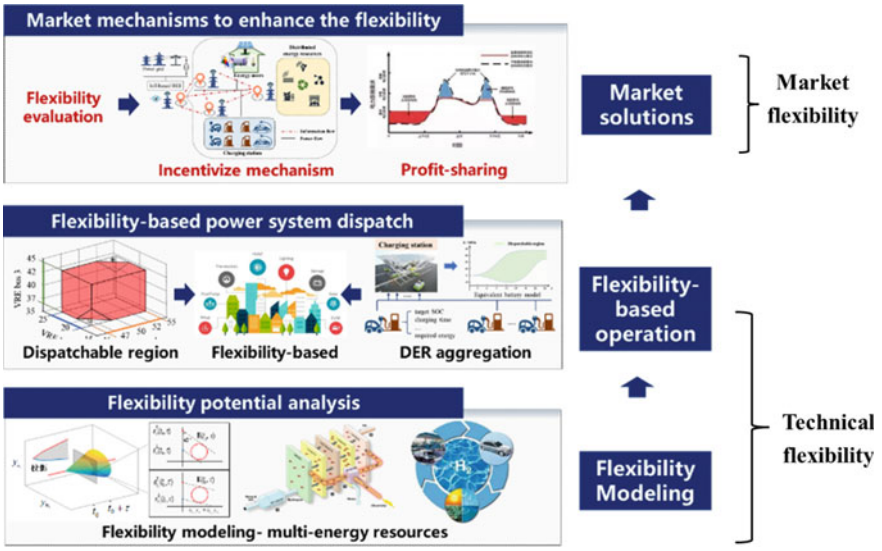


Fig. 1.1 Fundamental framework of power system flexibility analysis

flexibility, namely "market flexibility". The improvement of power system flexibility depends not only on the technical flexibility itself, but also on the effectiveness of the market mechanism. Only through a reasonable market mechanism can we effectively activate the potential of technical flexibility and realize the improvement of the flexibility of variable renewable energy power system.

1.3 Market Solutions

The effectiveness of technology needs to be supported by effective market mechanism. On the one hand, the flexibility of power system depends on the technical characteristics of the above "resource-grid-load-storage" resources; on the other hand, it also depends on the corresponding economic constraints and the design of incentive mechanism. In this section, we briefly introduce how to solve the problem of power system flexibility by means of market, especially considering that the power system will be dominated by renewable energy in the future.

As mentioned above, power system flexibility can be divided into two levels: technology and market. Market flexibility emphasizes that power system should pay attention to responding to the demand for flexibility efficiently and economically, so as to avoid blindly improving technical flexibility and ignoring economic constraints. From the perspective of market solutions, the power system flexibility enhancement can be divided into wholesale and retail markets. Wholesale markets involve a sequence of markets, including year-ahead, month-ahead, day-ahead and

real-time balancing market, while the retail markets related to power system flexibility mainly focus on the activation of flexibility potential from distributed energy resources.

Theoretically, in the short term, the power market needs to transfer the value of flexibility through the time-varying spot market prices, encourages the market members to actively provide flexibility and tap the existing flexibility potential of the system; In the long run, market participants need to be guided to invest in flexibility resources that match the flexibility needs of power system operation, so as to ensure the adequacy of the system flexibility resources. With the energy transformation under the dual carbon goal, the overall structure and function of the power system are changing. In the face of such a large-scale access to variable renewable energy, it is necessary to re-examine whether the current market structure can provide a reasonable return on the value of flexibility and whether it can realize the cross time and space optimal allocation of flexible resources.

In terms of retail markets, with the development of intelligent control and communication technology, these distributed resources also bring greater potential for the flexible operation of power system. However, different from the centralized wholesale market, market dispatching collects massive distributed energy, microgrid and demand side response quotations, which will produce huge transaction costs; In addition, when using distributed energy, users pursue a simple and easy transaction mode, which may lack the willingness to respond to power grid signals in real time. Therefore, for the distributed market, the core of its market mechanism is to design an incentive compatible income distribution mechanism to ensure the willingness of distributed resources to participate and maximize the flexibility potential.

In general, the key to improve the flexibility of power system is to activate the flexibility potential of the whole power supply chain and enhance the flexibility provision capacity of the system. The other is to design efficient market mechanisms to guide market behavior with reasonable and effective price signals and improve the efficiency of allocating flexible resources across time and space. We will discuss this topic in detail in Chaps. 7 and 8, especially in the future RES-dominated power system.

1.4 Summary

Faced with the challenges of environmental pollution and fossil fuel shortage, it has become a global consensus to develop renewable energy technology industry. Consequently, large-scale integration of electronically-coupled renewable generation is displacing conventional fossil-fired units, which dramatically increases the need for system flexibility.

This introductory chapter serves to motivate the subject matter of this book by providing a brief overview of the power system flexibility challenges caused by the increasing penetration of renewable energy as well as the corresponding technologies and solutions to combat the ambitious carbon neutral targets.

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Chapter 2
Power System Flexibility Modelling



The power system flexibility refers to the adaptability of the system to internal and external uncertain factors, that is, the response capability of the system when internal or external variables change. Flexibile resources come from all regulation means that can deal with the system volatility and uncertainty. Flexible resources in power systems can be defined as a set of resources in which the "power source-network-load" link can provide a certain regulation capability to adapt to the random changes of power systems (such as renewable energy output uncertainty, load fluctuation, and DC lock caused by grid failure) under a given time scale. The role of flexible resources is to act as a "flexible power regulation" to provide sufficient margin to meet the flexibility requirements of the system. This chapter divides flexible resources according to the three dimensions of "power source-network-load", and introduces the technical and economic characteristics of flexible resources in the multiple links of "power source-network-load".

2.1 Introduction

Power system operation continues to face the challenge of maintaining power supply and demand balance. This equilibrium is disturbed by three different time range events: fast random fluctuations, slow periodic fluctuations, and rare mutations [1]. The combination of variable production of renewable energies and demand-side fluctuation leads to high fluctuation of the net load curve (load demand minus variable renewable generation). Flexibility is defined in [2–4] as the ability of the system to adapt to the net load change by adjusting the input or output power of the whole power grid over time, in which the main sources of change and uncertainty are variable renewable generation, demand and equipment failure.

Flexible resources are an important part of power system flexibility research. A comprehensive study of the scope, characteristics, and modelling of flexible resources is the premise and basis for optimizing the allocation of flexible resources

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M. Zhou et al., *Power System Flexibility*, Power Systems,
https://doi.org/10.1007/978-981-19-9075-5_2

and improving the flexibility of the power system. Reference [5] divides flexibility resources into transmission system flexibility (including traditional energy, renewable energy with strong controllability and grid interconnection), distribution system flexibility (including electric vehicles, load management, and demand response), and transmission and distribution system dual terminal flexibility (including energy storage and Microgrid), the technical characteristics and economic cost of various flexible resources are preliminarily analyzed. Reference [6] holds that the common features of all kinds of resources in power systems are that they have the attributes of energy, power, and the ramping rate at the same time. Reference [7] analyzes the flexible regulation capacity of conventional thermal power units from the perspective of the power system flexibility balance mechanism. The research shows that the flexible regulation capability of thermal power units is related to their output level, time scale, and regulated direction. Distribution system flexibility resources include load management and load response. In a broad sense, load management and load response can be considered as virtual reserve generation resources and transmission resources. This mechanism increases the reserve capacity of the system and improves the flexibility of the power system [8, 9].

According to the definition of flexibility in the report of International Energy Agency (IEA) of 2016, power system flexibile resources are divided into three categories according to whether "power source-network-load" resources have the flexible regulation ability as the classification standard: (1) rapid response capability of generator units. Traditional power generation resources include thermal power units, hydropower, and nuclear power, which are the power supply with strong controllability and reliability in power systems. They are the important part of flexible resources. (2) The interconnection capacity of adjacent transmission networks that ensures to implement the output when the system flexibility is surplus, and input when the system flexibility is insufficient. Interconnected transmission between multiple power systems can significantly improve power system flexibility and enhance the accommodation capability of uncertain power sources such as wind power. It is also a potential flexibility resource. (3) Demand-side management and response: electricity consumers change the behavior of electricity consumption when the system supply is surplus/shortage. Load management and demand response is regarded as a virtual backup power resource and capacity resource, which provides a certain quantity and quality flexibility for power systems. It is becoming a more and more important flexibility resource. However, for new demand-side loads such as electric vehicles, data centers, 5G base stations, comprehensive energy parks, demand side distributed photovoltaic and their polymers, the response potential of these demand-side resources has not been brought into full play, many flexible resources are sleeping, and lack of linkage means with the power grid, which improves the difficulty of power grid operation and dispatching.

The flexible resources of power systems are widely distributed at the three aspects of "source-network-load". The existing research much focuses on the flexibility resources of the power supply side, less on load side, barely on the flexibility potential of the power grid side. In this chapter, the flexible resources of the power supply

side, power grid side, and demand side are modeled, and the technical and economic characteristics of various flexible resources are analyzed.

2.2 Power System Flexibility Resource Classification

2.2.1 Demand Side Flexibility Resources

Demand response, the main mode of interaction between power grid and customers under power market development, has been applied in China in recent years. In aspect of the system operation, demand response (DR) can improve the load profile and flexibility of the system operation by reducing peak load and peak-valley difference, thus decreasing the operation cost of the system, and alleviating the pressure of the grid investment for load increase. On the other hand, in aspect of electricity consumers, DR reduces cost of customers' electricity consumption without affecting their satisfaction.

As the development of smart grid technology, controllable load and distributed generation (DG) have been gradually integrated into demand side. Smart meters, in addition, have been widely applied to demand side resources. Therefore, a novel DR participant arises from demand side. Figure 2.1 shows the components of demand side flexible resources including smart community load and DG, which is generally divided into interruptible load, controllable load, roof-top solar panel, and storage battery.

China has built numerous demonstration projects of smart community during past several years aiming to reduce peak load, peak-valley difference, and increase energy utilization efficiency in city, and these projects are currently in high-speed progress. The community participates in DR through advanced metering infrastructure (AMI) coordinated by load aggregator (LA), whose structure is shown in Fig. 2.2. In comparison to conventional loads, smart community has greater DR potential leading to that it can smooth the load curve more dramatically. In addition, smart community DR is also crucial for ancillary service entities to get involved in power market.

2.2.2 Power Supply Side Flexibility Resources

The largest source of flexibility in power systems today is dispatchable power plants. A plant is dispatchable if it can respond to commands from a system operator—at any time, within certain availability parameters—to increase or decrease output over a defined period. The power supply resources with large flexible regulation capability are mainly thermal power units (including coal, oil, and natural gas turbines), hydropower units, cogeneration units, etc. These power supplies are flexible resources in the most conventional sense.

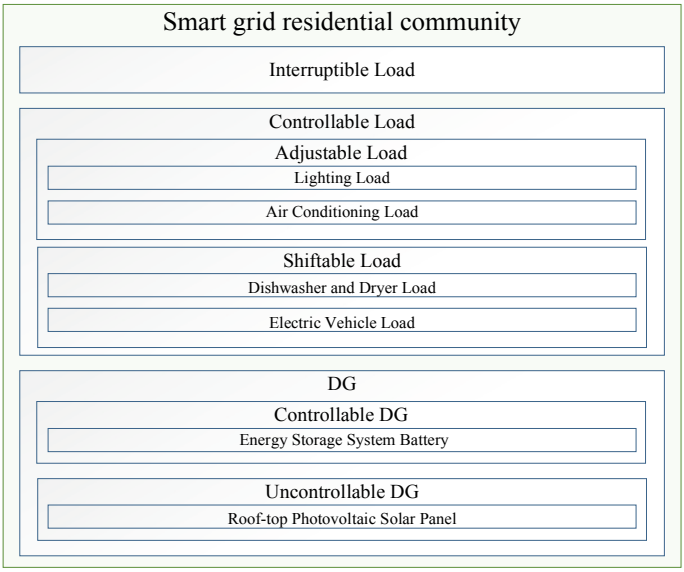


Fig. 2.1 Components of smart residential community load and distributed generation

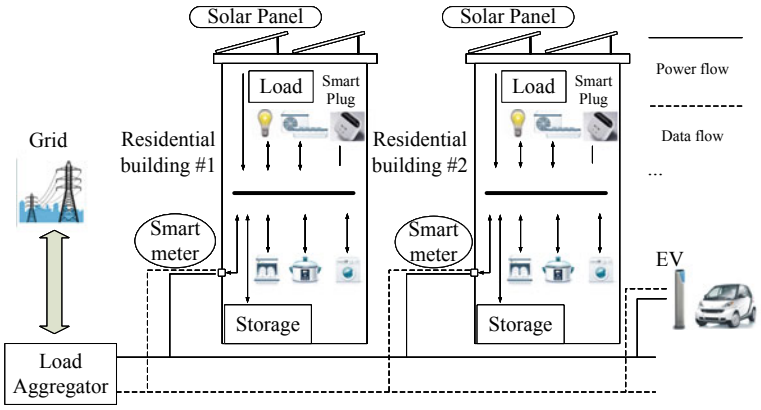


Fig. 2.2 Smart residential community DR structure

They can be divided into three categories: base-load, mid-merit and peaking plant. Peaking plants such as simple cycle gas or diesel turbines and reservoir hydropower can respond immediately. Mid-merit plants typically have the ability to ramp down to a minimum operating level. Among them, thermal power units can be divided into conventional peak shaving thermal power units and thermal power units with deep peak shaving capacity. Baseload plants have slower response times, and will generally have been designed to operate more or less around the clock. With them, the flexibility requirements of minute level and hour level can be met.

2.2.3 Grid Side Flexibility Resources

In traditional AC power grids, power flow follows the natural distribution, with the limited regulation capability. The power regulation of the system depends on the flexible regulation technologies on the generation side and demand side. With the rapid development of power electronics technology and other emerging control technology, the increasing penetration of renewable energy brings more challenges to the flexible operation of the system, meanwhile, the voltage source converter based high voltage direct current transmission (VSC-HVDC) and quick AC transmission switch provide new power regulation means from the side of the power grid.

For the VSC control, it can realize bidirectional control of power flow and decoupling control of active power and reactive power by the power regulation mode and voltage modulation technology, which is more flexible in control and operation than the line commutated converter (LCC) [10]. For the VSC power regulation, it traditionally applies the master–slave operation strategy [11] or voltage margin strategy [12] as the system-level strategy to arrange the operation of multiple VSC stations. Recent years, power droop control is widely focused, which can realize the autonomous decentralized coordination for multiple VSC stations and no communication among the stations required [13]. In a word, the VSC regulation and VSC-based multiple terminal topology provide a more flexible regulation means.

With renewables energy integration rising, more flexibility is required not only from the generation side but also from the grid side [13]. Traditionally, AC transmission lines are operated as uncontrollable static elements in real-time operation. However, the system operators can change the topology of the transmission network to improve voltage profiles [14] or increase transfer capability [15]. Transmission switching (TS) is proven to be an effective control method for line overloading [16] and cost reduction. With large-scale renewables integration, the flexibility by TS should be fully utilized.

2.3 Flexible Power Supply Resources: Analysis and Modelling

The flexible resources on the power supply side mainly include thermal power units, hydropower units, cogeneration units, etc. This section describes the technical and economic characteristics of various power-side flexibility resources.

2.3.1 Technical Characteristics of Flexible Power Supply Resources

2.3.1.1 Thermal Power Unit

Coal, oil, and natural gas are used as fuels in thermal power plants. The chemical energy during fuel combustion is converted into heat energy, and then is converted into mechanical energy with the help of thermal machinery such as steam turbines, and the steam turbine drives the generator to convert mechanical energy into electrical energy. As a flexible resource, thermal power plant mainly reduces or increase generator output according to the command of power grid dispatching to meet the demand of system flexibility.

The operation characteristics of thermal power units can be summarized as follows: the unit startup and shutdown speed are relative slow, the time from startup to the rated output generally needs several hours, and the startup cost is high for cold start-up, so it is not suitable to start up frequently. The load lifting speed of most units are relative slow, and the output must be kept between the minimum technical output and the maximum rated output. When the output is too low, the fuel combustion of the unit is unstable, which would increase the fuel consumption and reduce the unit efficiency.

Based on the operation characteristics of thermal power units, it is not allowed to have too much capacity to participate in flexibility regulation. Considering the proportion of thermal motor assembly capacity in the power grid, in many cases, the regulation capacity of other available flexibility resources can not meet the needs of power grid flexibility. At this time, the participation of thermal power units is required to ensure the reliable operation of the power grid. Therefore, thermal power units should be reasonably arranged to participate in flexibility regulation in a timely and appropriate manner. Among different thermal power units, units with low unit coal consumption should be given priority to bear the load. Considering the economy of power grid operation, priority should be given to units with low efficiency and high coal consumption to participate in flexible regulation.

Take thermal power units as example, they have large regulation capability, as shown in Fig. 2.3. The figure shows the relationship between the flexibility supply capacity of the thermal power unit and the regulation duration at a certain output level. The curve shows that with the increase of regulation duration, the up/down flexibility supply increases, but is limited by the regulation capacity, and finally stabilizes at a certain fixed level, that is, the up/down flexibility shows a saturation trend with the increase of regulation duration. In addition, the units can also provide additional down-regulation flexibility through the shutdown. In Fig. 2.3, the flexibility of the upper adjustment on the left half axis is taken as an example to show the function relationship between flexibility supply capacity and output of the unit at a certain time scale. The technical output range of the unit is $\left[P_{\mathrm{TG}}^{\min}, P_{\mathrm{TG}}^{\max}\right]$. Within the range of low output level, its flexibility regulation capability is constrained by the ramping rate, which is a constant; with the increase of output level, when the increased space

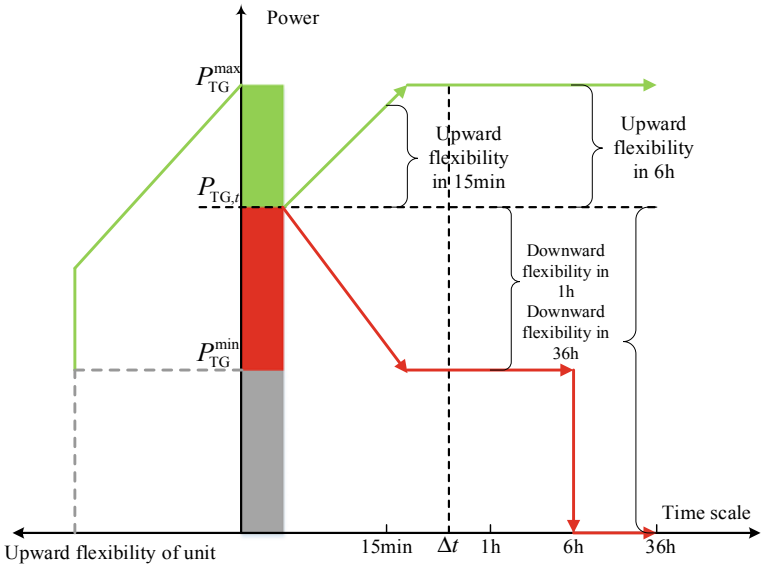


Fig. 2.3 Flexibility regulation range of thermal power unit

is lower than the ramping rate, its flexibility supply capacity is limited by the regulation capability, and the relationship between flexibility and the unit output shows a decreasing trend.

The flexibility duration would affect the active output of thermal power units. Therefore, the technical flexible adjustment range of the active output of thermal power units at time t can be expressed in the following form by integrating the upper and lower output limits of the unit and the constraint of flexible duration and unit ramping rates:

$$\max\left\{P_{TG}^{min}, P_{TG,t} - r_{TG}^{d}\Delta t\right\} \le P_{TG,t+1} \le \min\left\{P_{TG}^{max}, P_{TG,t} + r_{TG}^{u}\Delta t\right\} \quad (2.1)$$

where $P_{TG,t}$, P_{TG}^{max} and P_{TG}^{min} is the output of the unit at time t and corresponding the upper/lower limit; r_{TG}^{u} and r_{TG}^{d} is the upward and downward ramping rate of the unit, respectively.

Deep peaking regulation is an effective way to fully tap the flexibility potential of coal-fired thermal units. The flexible regulation range of the units is expanded by reducing the output of each power plant and exceeding the basic peak shaving range. After the grid-connected power generation with a high proportion of renewable energy, the peak-valley difference of the net load of the system is greatly increased, which puts forward large demand for thermal unit flexible operation. Lowering output level of thermal units can fully tap the reserve space of existing thermal power units, which will be one of the most effective ways to accommodate the fluctuate renewable generation. The output of deep peaking regulation is below the minimum stable combustion load of the boiler in the power plant. Generally, the deep peaking

regulation is up to 60–70% of the boiler maximum continuous rating (BMCR). Deep peak regulation needs to lower the lower limit of the technical output of thermal units. Accordingly, the flexibility adjustment range is increasing.

2.3.1.2 Conventional Hydropower Unit

The hydropower unit operates very flexibly, it can rapidly start/stop and have strong load-following capability. It takes only a few minutes from start-up to the rated output. According to the requirements of power grid dispatch, the loading and unloading of hydropower units can be completed in an instant, the load can be adjusted in a wide range, and the dynamic performance is very ideal. Secondly, in terms of operation cost, there is no fuel consumption for hydropower units. The whole process of a 200 MW hydropower unit from static start-up, speed-up, and grid connection to full load takes only about 2–5 min; For coal-fired thermal power units under the same conditions, it takes about 5–8 h from cold start to full load, and more than 2 h even for the warm startup.

Hydropower can assist the nuclear power units and thermal units in power grid to operate stably at their highly efficient outputs, without frequent startup and shutdown, to save fuel, reduce power generation cost and bring great comprehensive economic benefits to the power system. In addition, hydropower can also undertake the tasks of system frequency regulation, phase modulation, and emergency standby according to the needs of the power system. In the wet season, to avoid the waste of water resources, some hydropower stations with large capacity will bear the base load of the system.

Therefore, in terms of dynamic performance and operation cost, it is a very reasonable choice for hydropower units to participate in flexibility regulation. Similar to thermal power units, the flexible regulation characteristics of hydropower units are also limited by their technical output range and ramping rate.

$$\max\left\{P_{\mathrm{HG}}^{\min},\, P_{\mathrm{HG},t} - r_{\mathrm{HG}}^{\mathrm{d}}\Delta t\right\} \le P_{\mathrm{HG},t+1} \le \min\left\{P_{\mathrm{HG}}^{\max},\, P_{\mathrm{HG},t} + r_{\mathrm{HG}}^{\mathrm{u}}\Delta t\right\} \tag{2.2}$$

where P_{HG} is the output of the hydropower unit; $P_{\mathrm{HG},t}$, $P_{\mathrm{HG}}^{\max}$ and $P_{\mathrm{HG}}^{\min}$ are the output of the hydropower unit at time t and the upper and lower limits of its output; $r_{\mathrm{HG}}^{\mathrm{u}}$ and $r_{\mathrm{HG}}^{\mathrm{d}}$ is the upward and downward ramping rate of the hydropower unit, respectively.

In addition, the hydropower unit is also limited by its water flow volume, that is, comprehensively considering the conversion relationship between hydropower electricity output and generation water consumption, and the restrictions of outbound water flow and waste water, the output range of hydropower unit under the constraint of generation water consumption can be obtained. The output of the hydropower unit can be expressed by power generation and water consumption as follows:

$$P_{\mathrm{HG}} = g\ \eta_{\mathrm{HG}} Q_{\mathrm{HG}} H_{\mathrm{HG}} \tag{2.3}$$

where g is the gravitational acceleration constant, taken as 0.0098 km/s^2; η_{HG} is the conversion efficiency from water energy to the electric energy of the hydropower unit; Q_{HG} represents the generation water consumption of the hydropower unit in Δt period; H_{HG} represents the head height of the hydropower unit in Δt period.

The water consumption of hydropower units for power generation in Δt period is the difference between outgoing water flow Q_{out} and wastewater flow Q_{sp}, i.e. $Q_{\mathrm{HG}} = Q_{\mathrm{out}} - Q_{\mathrm{sp}}$.

Outbound water flow constraint:

$$Q_{\mathrm{out}}^{\min} \le Q_{\mathrm{out}} \le Q_{\mathrm{out}}^{\max} \tag{2.4}$$

where $Q_{\mathrm{out}}^{\min}$ and $Q_{\mathrm{out}}^{\max}$ are the minimum and maximum discharge of the reservoir respectively.

Waste water flow constraint:

$$0 \le Q_{\mathrm{sp}} \le Q_{\mathrm{sp}}^{\max} \tag{2.5}$$

where, $Q_{\mathrm{sp}}^{\max}$ is the maximum waste water flow of the reservoir.

The water consumption constraint for power generation is:

$$Q_{\mathrm{out}}^{\min} - Q_{\mathrm{sp}}^{\max} \le Q_{\mathrm{HG}} \le Q_{\mathrm{out}}^{\max} \tag{2.6}$$

After considering the constraints of generation water consumption, the output range of hydropower unit is as follows:

$$g\eta_{\mathrm{HG}}\left(Q_{\mathrm{out}}^{\min} - Q_{\mathrm{sp}}^{\max}\right)H_{\mathrm{HG}} \le P_{\mathrm{HG}} \le g\eta_{\mathrm{HG}}Q_{\mathrm{out}}^{\max}H_{\mathrm{HG}} \tag{2.7}$$

To sum up, the flexible technical adjustment range of the active power of hydropower unit is:

$$\begin{aligned}&\max\left\{P_{\mathrm{HG}}^{\min},\, P_{\mathrm{HG},t} - r_{\mathrm{HG}}^{\mathrm{d}}\Delta t,\, g\eta_{\mathrm{HG}}\left(Q_{\mathrm{out}}^{\min} - Q_{\mathrm{sp}}^{\max}\right)H_{\mathrm{HG}}\right\} \le P_{\mathrm{HG},t+1}\\ &\le \min\left\{P_{\mathrm{HG}}^{\max},\, P_{\mathrm{HG},t} + r_{\mathrm{HG}}^{\mathrm{u}}\Delta t,\, g\eta_{\mathrm{HG}}Q_{\mathrm{out}}^{\max}H_{\mathrm{HG}}\right\}\end{aligned} \tag{2.8}$$

2.3.1.3 Cogeneration Unit

Cogeneration refers to the combined production of electricity and heat. According to different fuel forms used in cogeneration, and the principal structure of the unit, it can be divided into four forms of cogeneration: internal combustion engine cogeneration, gas turbine cogeneration, steam turbine cogeneration, and nuclear power cogeneration. Among them, steam turbine cogeneration is the main form of cogeneration central heating system at home and abroad. Therefore, this chapter mainly analyzes

steam turbine cogeneration. The correlation and coupling relationship between the generated electric power and heating power of combined heat and power (CHP) units are generally called "electrothermal characteristics", which well reflects the external characteristics of CHP units. Therefore, it is an effective way to analyze the flexible operation capability of CHP units. CHP units are divided into back pressure type and extraction type.

The power output of the backpressure CHP unit depends on the amount of steam passing through the steam turbine, and the amount of steam passing through the backpressure CHP unit is determined by the thermal demand. Therefore, the backpressure CHP unit has the operation characteristics of "electricity determined by heat", and its power generation is strictly restricted by the thermal demand and cannot be adjusted independently. Figure 2.4 shows the thermal-electric operation characteristics of the backpressure unit, and the ratio of thermal-electricity output power is constant, that is, when the thermal output power is constant, its power output power cannot be adjusted.

$$P_{chp} = c_m \cdot h_{chp} \tag{2.9}$$

where, P_{chp} and h_{chp} are the power output and thermal output of the CHP unit respectively; c_m is the ratio of the thermal electric output power of the backpressure CHP unit.

Compared with the backpressure CHP unit, the extraction CHP unit has more flexible thermal electric operation characteristics, and its power output can be flexibly adjusted within a certain range. Figure 2.5 shows the electrothermal characteristics of the steam extraction unit. The thermal electric output power can be flexibly adjusted in the area surrounded by ABCD. The ratio of the thermal electric output power of steam extraction CHP unit is not a constant value. The power output power can be adjusted within a certain range and has a certain flexible adjustment ability.

By analogy with the concept of thermal- electricity ratio of backpressure unit, c_v is defined as the reduction of generating power under multi extraction of unit heating heat when the steam inlet is constant, that is, the slope of the curve, which can be

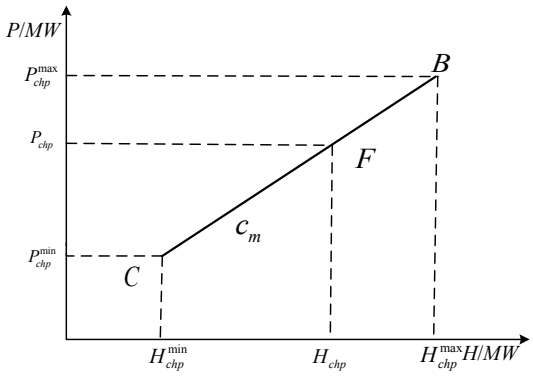


Fig. 2.4 Thermal electric operation characteristics of back pressure unit

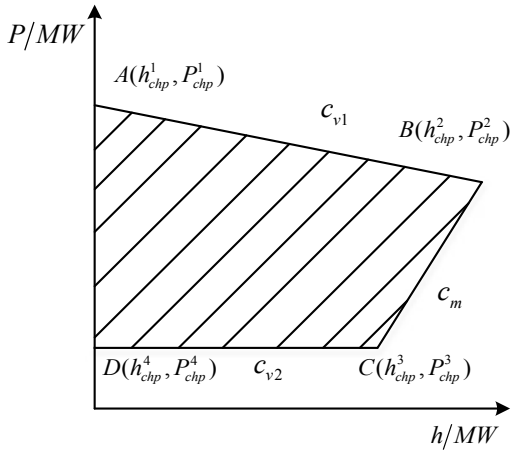


Fig. 2.5 Thermal electric operation characteristics of steam extraction unit

approximately regarded as a constant.

$$c_v = \Delta P_{chp} / \Delta h_{chp} \tag{2.10}$$

In Fig. 2.5, c_{v1} and c_{v2} are the corresponding c_v values under the maximum and minimum electric output. c_m is the value of c_v of electric power and thermal power during backpressure operation.

The electric thermal characteristics of the extraction steam unit can be described as follows:

$$\begin{cases} \max\{P_{\min} - c_{v2}h, c_m(h - h_0)\} \le P \le P_{\max} - c_{v1}h \\ 0 \le h \le h_{\mathrm{T,max}} \end{cases} \tag{2.11}$$

The flexible regulation range of CHP units is not only limited by its "electrothermal characteristics", but also by its own electric output, ramping rate and thermal output.

$$\max\left\{P_{chp}^{\min},\ P_{chp,t} - r_{chp}^{\mathrm{d}}\Delta t\right\} \le P_{chp,t+1} \le \min\left\{P_{chp}^{\max},\ P_{chp,t} + r_{chp}^{\mathrm{u}}\Delta t\right\} \tag{2.12}$$

$$h_{\min} \le h_t \le h_{\max} \tag{2.13}$$

where P_{chp} is the electric output of the CHP unit. $P_{chp,t}$, $P_{chp}^{\max}$ and $P_{chp}^{\min}$ are the current output of the CHP unit and the upper and lower limits of its electrical output respectively. r_{chp}^{u} and r_{chp}^{d} are the upward and downward ramping rate of electric output of CHP units respectively. h_t, $h_{\max}$ and $h_{\min}$ are the current thermal output of the CHP unit and the upper and lower limits of its thermal output respectively.

Considering the above two points, it can be concluded that the technical flexible adjustment range of CHP unit is:

Table 2.1 Comparison of technical characteristics of conventional power supply

Power supply	Ramping type (%/min)	Power type (%)
Conventional thermal power	1–1.5	40–60
Deep peak shaving thermal power	1–1.5	60–80
Conventional hydropower	50	100
CHP	0.5–1	15–20

$$\begin{cases} \max\left\{P_{chp}^{\min} - c_{v2}h_t, c_m(h_t - h_0), P_{chp,t} - r_{chp}^{\mathrm{d}}\Delta t\right\} \le P_{chp,t+1} \\ \quad \le \min\left\{P_{chp}^{\max} - c_{v1}h_t, P_{chp,t} + r_{chp}^{\mathrm{u}}\Delta t\right\} \\ h_{\min} \le h_t \le h_{\max} \end{cases} \tag{2.14}$$

The technical characteristics of conventional power supply can be measured by ramping index and power index. The former refers to the ratio of ramping power per minute to rated power, and the latter refers to the ratio of power-adjustable range to rated power. It can be seen from Table 2.1 that among conventional power units, hydropower units have the fastest ramping speed, followed by thermal power units, and CHP units are the slowest; Hydropower units still account for the largest proportion of adjustable capacity range in rated power. Thermal power units with deep peaking capability expand their regulation range.

2.3.2 Economic Characteristics of Power Supply Resources Flexibility

2.3.2.1 Thermal Power Unit

For traditional thermal power units, the total transfer cost in response to the flexibility demand is mainly the operation cost when the thermal power unit changes its output. The relationship between the consumed fuel and the generated power is the consumption characteristic of the generator unit, which is usually expressed by the secondary curve of active output, that is, the flexibility demand transfer cost $C_{\mathrm{TG},t}$ can be expressed as,

$$C_{\mathrm{TG},t} = aP_{\mathrm{TG},t}^2 + bP_{\mathrm{TG},t} + c \tag{2.15}$$

where a, b and c are fuel consumption coefficients of thermal power units.

2.3.2.2 Conventional Hydropower Unit

Under the flexibility demand of hydropower units, the variation of operating cost in response to the flexibility demand with the output power can be approximately regarded as a first-order function. Therefore, the call cost of the hydro motor unit under the demand of flexibility can be expressed as,

$$C_{\mathrm{HG},t} = a_{\mathrm{n}} P_{\mathrm{HG},t} + b_{\mathrm{n}} \tag{2.16}$$

where a_{n} and b_{n} are the cost coefficients of hydropower units.

2.3.2.3 Combined Heat and Power Unit

According to the operation principle of extraction type cogeneration unit, when the heating power is $P_{chp,t}$ and the generating power is h_t, the relationship between the electric power P converted into pure condensation condition is shown in (2.17).

$$P_{\mathrm{ZS}} = P_{chp,t} + c_{v0} h_t \tag{2.17}$$

where c_{v0} is the reciprocal of thermal power ratio of CHP units.

The operating cost of CHP unit can be obtained by bringing the electric power under pure condensation condition into the quadratic curve of fuel cost, that is, the call cost of cogeneration unit under flexibility demand is expressed as,

$$C_{chp,t} = a_{\mathrm{g}} P_{chp,t}^2 + b_{\mathrm{g}} P_{chp,t} + c_{\mathrm{g}} P_{chp,t} h_t + d_{\mathrm{g}} h_t^2 + e_{\mathrm{g}} h_t + f_{\mathrm{g}} \tag{2.18}$$

where a_{g}, b_{g}, c_{g}, d_{g}, e_{g} and f_{g} are fuel cost coefficients of CHP units.

2.4 Demand Side Flexibility Model

Demand side flexibility resources can be divided into three categories based on different demand response (DR) strategies, which are interruptible load, adjustable load, and shiftable load. Interruptible load participates in DR through the IL program. When the load responds, the total load power would step changes, which means the load is completely curtailed. Unlike interruptible load, adjustable load is not curtailed during response period, but decreases by small proportion of power instead, hence the load curve changes smoothly. Shiftable load differs from those two, it does not reduce the total electricity consumption in DR program, but shifts the load operation time optimally.

2.4.1 Interruptible Load

This category of loads does not include controllable load appliance, which is not responsive to price-based DR program. In other words, these loads do not decrease, they are not affected by electricity price, or shifted to low price hours. However, they can join the IL program through smart plug by signing the contracts. Customers can determine which of their appliances to join IL program voluntarily based on their own electricity consumption habit. During peak hours, when the grid power supply is insufficient for load demand or electricity price is extremely high, customers would curtail partial load according to the IL contract agreement. The interruptible DR is modeled as follows.

$$0 \le P_{j,t}^{l} \le I_{j,t}^{l} P_{j}^{l,\max} \tag{2.19}$$

$$\sum_{t} I_{j,t}^{l} \le X_{j}^{l,\max} \tag{2.20}$$

$$P_{j,t}^{l} \le P_{j,t}^{L} \tag{2.21}$$

where $P_{j}^{l,max}$ is maximum curtailable power of interruptible load j at each hour, $X_{j}^{l,\max}$ is maximum daily curtailable hours of load j, $P_{j,t}^{L}$ forecasted power of load j at hour t, $I_{j,t}^{l}$ is binary status indices of interruptible load j at hour t.

2.4.2 Adjustable Load

Adjustable loads, such as lighting, air conditioning, etc., can adjust their power demand to participate in DR program. When realtime price is high, these loads would partially reduce power to a lower consumption level. Residents can set a price threshold value and power adjustment level of their adjustable loads based on their acceptable comfort. When the hourly real-time electricity price is higher than the price threshold, the power consumption level of load would decrease to a lower volume set by users. For instance, lighting would dim brightness to the preset degree, and air conditioning would adjust temperature set point to reduce power.

Lighting load DR problem constraint is modeled as follows:

$$P_{j,t}^{a} = P_{j,t}^{a,\text{base}}(1 - I_{j,t}^{a}) + P_{j,t}^{a,\min} I_{j,t}^{a} \tag{2.22}$$

$$\varepsilon(\rho_t - \rho_{j,t}^{a,\min}) < I_{j,t}^{a} \le \varepsilon(\rho_t - \rho_{j,t}^{a,\min}) + 1 \tag{2.23}$$

where $P_{j,t}^{a}$ is power of lighting load j at hour t, $P_{j,t}^{a,base}$ is original power of lighting load j at hour t, $\rho_{j,t}^{a,\min}$ is price threshold of lighting load j at hour t, $P_{j,t}^{a,\min}$ is adjusted

power of lighting load j at hour t, ρ_t is power price of grid at hour t, ε is small positive constant, $I^a_{j,t}$ is binary status indices of adjustment of lighting load j at hour t.

When the real-time price ρ_t is higher than the price threshold $\rho^{a,\min}_{j,t}$, lighting load power decreases from $P^{a,base}_{j,t}$ to $P^{a,\min}_{j,t}$.

Air conditioning temperature state evolution is modeled with discrete time difference equation, which is commonly used in literature as follows:

$$T_{k,t+1} = \alpha_k T_{k,t} + (1-\alpha_k)(T^{\alpha}_{k,t} - T^{g}_{k,\mathrm{t}}) \tag{2.24}$$

$$\alpha_k = e^{-\Delta t / C_k R_k} \tag{2.25}$$

$$T^g_{k,t} = \begin{cases} R_k \eta_k P^a_{k,t} & \text{cooling mode} \\ -R_k \eta_k P^a_{k,t} & \text{heating mode} \end{cases} \tag{2.26}$$

$$P^{\min}_k \le P^a_{k,t} \le P^{\max}_k \tag{2.27}$$

$$T_{k,t} = T^s_{k,t} \tag{2.28}$$

where $T_{k,t}$ is air conditioner temperature of air conditioner k at hour t, $T^g_{k,t}$ is temperature adjustment of air conditioner k at hour t when it is turned on, $P^a_{k,t}$ is load power of air conditioner k at hour t, $T^s_{k,t}$ is temperature set point of air conditioner k at hour t, $T^{\alpha}_{k,t}$ is ambient temperature of air conditioner k at hour t, α_k is the system inertia of air conditioner k, Δt is control interval, C_k is thermal capacitance of air conditioner k, R_k is thermal resistance of air conditioner k, η_k is working efficiency factor of air conditioner k, $P^{\min}_k$, $P^{\max}_k$ is minimum and maximum power of air conditioner k at each hour.

We employ the adjustment control for air conditioning in cooling mode based on the above model as follows, which can easily be modified for heating mode as well:

$$T^s_{k,t} = T^{s,\mathrm{base}}_{k,t}(1 - I^a_{k,t}) + T^{s,\max}_{k,t} I^a_{k,t} \tag{2.29}$$

$$\varepsilon(\rho_t - \rho^{a,\min}_{k,t}) < I^a_{k,t} \le \varepsilon(\rho_t - \rho^{a,\min}_{k,t}) + 1 \tag{2.30}$$

where $I^a_{k,t}$ is binary status indices of adjustment of air conditioner k at hour t, $T^{s,\mathrm{base}}_{k,t}$ is original temperature set point of air conditioner k at hour t, $T^{s,\max}_{k,t}$ is adjusted temperature set point of air conditioner k at hour t in cooling mode, $\rho^{a,\min}_{k,t}$ is price threshold of air conditioner k at hour t.

When the real-time price ρ_t is higher than the price threshold $\rho^{a,\min}_{k,t}$, temperature set point rises from $T^{s,\mathrm{base}}_{k,t}$ to $T^{s,\max}_{k,t}$.

2.4.3 Shiftable Load

Operation time of these controllable loads can be shifted to low price period in accordance with the price signal. We divide shiftable load into two categories. One of them includes rice cooker, dryer, washing machine, and other similar loads. The other one is storage-like load, e.g. EV load.

(1) Rice cooker and washing machine load

The feature of this type of load is that it consumes a fixed total amount of electric energy in a fixed period (i.e. a cycle duration). Once the load is turned on, it will remain on for the duration of its cycles until it is operated by one cycle. Although the load may consume different power volume at different time during the cycle (e.g. washing and drying stages of washing machine), the total quantity of energy consumption is fixed. Therefore, we propose a fixed average power to represent the power consumption amount at each hour of duration (2.31). The total energy usage can be represented as a product of average power multiplied by cycle duration time. Customers can set an operating time window (i.e. preferred start and end hours) for these loads. The load can be turned on at any time during the operating time window, and should be operated for one cycle (2.32). The load DR problem constraint is modeled as follows:

$$P_{j,t}^{b} = P_{j}^{b,\text{average}} I_{j,t}^{b} \tag{2.31}$$

$$[X_{j,(t-1)}^{b,\text{on}} - U_{j}^{b}][I_{j,(t-1)}^{b} - I_{j,t}^{b}] \geq 0 \tag{2.32}$$

$$X_{j,t}^{b,\text{on}} = \sum_{t=\tau}^{t} I_{j,t}^{b} \tag{2.33}$$

$$\sum_{t=\tau}^{T} I_{j,t}^{b} = U_{j}^{b} \tag{2.34}$$

where $P_{j,t}^{b}$ is load power of shiftable load j at hour t, $I_{j,t}^{b}$ is binary status indices of shiftable load j at hour t, $P_{j}^{b,\text{average}}$ is average power of shiftable load j, $X_{j,t}^{b,\text{on}}$ is total operated hours of shiftable load j at hour t, ${U_j}^{b}$ is cycle duration of shiftable load j.

(2) EV load

EV charge time window is from the home arrival time of the final trip in the day (i.e. EV plug-in time) to the time of departure of the next day (i.e. end of charge time). Since EV actual charge duration is shorter than the time window, the decision of charge or not and the amount of charge power at each hour can be made (2.35). EV should be fully charged in advance of customer's departure time (2.37).

$$P_{k}^{b,\min} I_{k,t}^{b} \leq P_{k,t}^{b} \leq P_{k}^{b,\max} I_{k,t}^{b} (T_a \leq t \leq T_b) \tag{2.35}$$

$$E^b_{k,t} = E^b_{k,t-1} + P^b_{k,t}\Delta t / E^{\max}_k \tag{2.36}$$

$$E^b_{k,T_b} = 1 \tag{2.37}$$

$$E^b_{k,t} \le 1 \tag{2.38}$$

where $P^b_{k,t}$ is charge power of EV k at hour t, $I^b_{k,t}$ is binary charge status indices of EV k at hour t, $E^b_{k,t}$ is SOC of EV k at hour t, $P^{b,\min}_k$, $P^{b,\max}_k$ is minimum and maximum charge power of EV k at each hour, T_a is user's EV home arrival time, T_b is user's EV charge end time, $E^{\max}_k$ is battery capacity of EV k.

In circumstance of multiple EVs in the community, the stochastic profile of the EV initial state of charge (SOC) E^b_{k,T_a} and home arrival time of the final trip T_a should be concerned. Residents' EV daily mileage is close to a logarithmic normal distribution. Besides, EV initial SOC is approximately linear related to its daily mileage. Thus, the probability density function of EV initial SOC is obtained as follows [8]:

$$f(E^b_{k,T_a}) = \frac{1}{\sqrt{2\pi}\,D(1-E^b_{k,T_a})\sigma_d} \cdot \exp\left\{-\frac{[\ln(1-E^b_{k,T_a}) + \ln D - \mu_d]^2}{2\sigma_d^2}\right\} \tag{2.39}$$

where D is maximum mileage of EV.

Home arrival time of the final trip T_a is close to a normal distribution curve. We uses a normal distribution function to describe the home arrival time of the community EV.

2.5 Power Grid Flexible Regulation Technologies

In this section, two representative flexible regulation technologies on the grid side are introduced: the voltage source converter (VSC) based multiple-terminal DC transmission, and the AC power transmission topology control.

2.5.1 Voltage Source Converter (VSC) Based Multiple-Terminal DC Transmission

Conventionally AC power grid operates at power flow obeying natural distribution, in this regard, there is limited regulation capability. With the rapid development of power electronics technology and other emerging control technology, the voltage source converter based high voltage direct current transmission (VSC-HVDC) provides new power regulation technique from the power grid. The VSC can realize bidirectional

control of power flow and decoupling control of active power and reactive power by the power regulation control mode and voltage modulation technology, which is more flexible in control and operation than line commutated converter (LCC). For the power regulation mode of VSC station, it traditionally applies the master–slave operation strategy or voltage margin strategy as the system-level strategy to arrange the operation of multiple VSC stations. It is difficult for above strategies to meet any power fluctuations or sudden grid failures, the flexible regulation capability of VSC stations cannot be exerted either. For this reason, the scholars focus on the power-voltage droop operation mode, an autonomous decentralized coordination strategy for multiple VSC stations, and no communication among the stations required. What is more, the DC grid can realize the coordinated operation of multiple VSC stations by applying the system-level power regulation strategy and implementing multiple terminal flexible operation.

The detailed VSC power transmission model is shown in Fig. 2.6, without considering the transient process of VSC power devices. The renewable output is sent out through AC line i and connected to dc node i through VSC station i. The subscript s represents the electrical parameters in the grid-connection point. The subscript c and d respectively represent the ac and dc electrical parameters of the VSC station. The equivalent losses caused by converter transformer, converter reactor, conductor and other components are expressed in R_i, and its reactance value is expressed in $\mathrm{j}X_i$.

Let:

$$\delta_i = \theta_{\mathrm{s},i} - \theta_{\mathrm{c},i} \tag{2.40}$$

By derivation, we can get the following relationship:

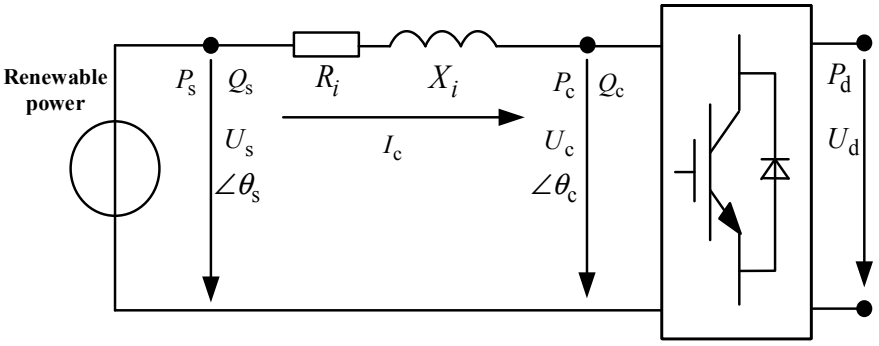


Fig. 2.6 X. Steady-state power transmission model of VSC station

$$\begin{cases} P_{\mathrm{s},i} = P_{\mathrm{c},i} = \dfrac{U_{\mathrm{s},i} \cdot U_{\mathrm{c},i}}{X_i} \sin \delta_i \\ Q_{\mathrm{s},i} = \dfrac{\left(U_{\mathrm{s},i}\right)^2}{X_i} - \dfrac{U_{\mathrm{s},i} \cdot U_{\mathrm{c},i}}{X_i} \cos \delta_i \\ Q_{\mathrm{c},i} = -\dfrac{\left(U_{\mathrm{c},i}\right)^2}{X_i} + \dfrac{U_{\mathrm{s},i} \cdot U_{\mathrm{c},i}}{X_i} \cos \delta_i \end{cases} \tag{2.41}$$

$$P_{\mathrm{d},i} = P_{\mathrm{c},i} - P_{\mathrm{loss},i}^{\mathrm{ac}} \tag{2.42}$$

$$P_{\mathrm{loss},i}^{\mathrm{ac}} = \mathrm{a}_i + \mathrm{b}_i \cdot I_{\mathrm{c},i} + \mathrm{c}_i \cdot \left(I_{\mathrm{c},i}\right)^2 + R_i \cdot \left(I_{\mathrm{c},i}\right)^2 \tag{2.43}$$

$$I_{\mathrm{c},i} = \sqrt{\left(P_{\mathrm{c},i}\right)^2 + \left(Q_{\mathrm{c},i}\right)^2} / U_{\mathrm{c},i} \tag{2.44}$$

where $P_{\mathrm{loss},i}^{\mathrm{ac}}$ is the total ac losses; X_i is the equivalent reactance of the AC line connected to the VSC station i. R_i represents the transmission losses of the collection line. a_i, b_i and c_i are respectively the coefficients of the constant term, first-order term, and second-order term of the operation loss formula of VSC station i.

This model describes the power transmission process and the power losses of the VSC power transmission and commutation. In addition, the power regulation model should also be carried out for exerting the VSC flexible operation, which is currently in the research stage, the relevant research achievements will be introduced in detail in Sect. 5.2.

2.5.2 AC Grid Flexible Topology Control

In AC grid, natural power flow is distributed according to impedance. However, the natural power flow distribution may not meet the requirements of secure, high-quality and economic power supply, which calls for power flow control.

Traditionally, there are three ways to control power flow: series capacitor, series reactance and series voltage injection. With the development of power electronics technology, new methods of power flow control appear, including the improvement of series capacitor (such as thyristor-controlled series capacitors (TCSC)) and series voltage injection and unified power flow controller (UPFC).

Among all above power flow control methods, AC transmission lines are operated as uncontrollable static elements. However, the system operators can change the topology of the transmission network by transmission switching (TS) to improve voltage profiles or increase transfer capability and adds flexibility to the AC grid. With transmission switching, congestion, which is another reason for lowing RES shares in electricity generation, can be prevented by changing the status of transmission lines. Thus, by applying TS operations (switching some lines out of service), RES penetration can be easily increased. Reducing congestion on transmission lines may

also improve the efficiency of other components (e.g. generation units) or other control mechanisms, such as energy storage and demand management.

The difference in power flow model between with TS and without TS mainly lies in the different modeling of network constraints. The network constraints without TS are expressed as follows:

$$p_{lnmt} = \frac{1}{X_{nm}}(\theta_{nt} - \theta_{mt}) \tag{2.45}$$

$$-F_{nm} \leq p_{lnmt} \leq F_{nm} \tag{2.46}$$

where p_{lnmt} is the power flow of line (n, m) at time t, X_{nm} is the reactance of line (n, m), θ_{nt} is the voltage angle of bus n at time t, F_{nm} is the transmission capacity of line (n, m).

Introducing a decision variable q_{nmt} (which is a transmission switching binary variable representing the state of line (n, m) at time t (0 closed, 1 switched off)) and adopting big-M approach, the network constraints with TS are expressed as follows:

$$-M_{nm}(1 - q_{nmt}) \leq p_{lnmt} - \frac{1}{X_{nm}}(\theta_{nt} - \theta_{mt}) \leq M_{nm}(1 - q_{nmt}) \tag{2.47}$$

$$-q_{nmt}F_{nm} \leq p_{lnmt} \leq q_{nmt}F_{nm} \tag{2.48}$$

where the parameter M_{nm} has to be greater than or equal to $|1/X_{nm}|max(|\theta_{nt} - \theta_{mt}|)$. To provide tighter constraints, it is better for M_{nm} to be as small as possible, so M_{nm} is taken as $\left|(1/X_{nm})(\theta^{max} - \theta^{min})\right|$.

In this way, the network constraints work no matter the transmission line is closed or switched off.

2.6 Conclusions

This chapter introduces the types of flexible resources of power systems and models the flexible resources of the power supply side, power grid side, and demand-side respectively. For the flexible resources on the power supply side, this chapter analyzes the flexible regulation capability of thermal power units, hydropower units, and cogeneration units in a short time scale. These power supplies have certain flexible adjustment ability to meet the flexibility requirements of minute level to hour level. On the grid side, two representative flexible regulation technologies are introduced, the voltage source converter (VSC) based multiple-terminal DC transmission, and the AC network transmission switching. The VSC can realize bidirectional control of power flow and decoupling control of active power and reactive power by the power regulation control mode and voltage modulation technology, which is more flexible in control and operation than line commutated converter (LCC). In addition,

the system operators can change the topology of the AC transmission network to improve voltage profiles or increase transfer capacity. In an aspect of the demand side, DR can improve the load profile and flexibility of the system operation by reducing peak load and peak-valley difference. On the other hand, with the development of smart metering technology in China, controllable loads and distributed generation (DG) have been gradually integrated into the demand side.

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Chapter 3
Flexibility-Based Economic Dispatch



Reasonable scheduling flexible resources play an important role in the reliable operation of high proportion renewables energy power systems. However, how to determine the system operation decision respecting the system security constraints and flexibility demand is still an open problem. In this chapter, the concept of wind power accommodation domain (ADWP) is proposed in Sect. 3.2 as a metric for quantifying flexibility demand and an operational guideline, and a flexible look-ahead unit commitment (LAUC) model is established. In Sect. 3.3, a two-level decision model of day-ahead generation-reserve considering flexibility demand is proposed, and the quantitative relationship between operation reserve and flexibility is established through the general generation function method. In Sect. 3.4, an improved endogenous reserve determination method is proposed. A variety of uncertainties are integrated into the security and network constrained economic scheduling model and solved iteratively with the help of Benders Decomposition. In Sect. 3.5, IEEE-30 bus, and IEEE-118 bus systems are taken as case studies to verify the effectiveness of proposed models.

3.1 Introduction

In the last decades, large-scale renewable energy sources (RES) have been integrated into power systems, aiming to alleviate the deterioration of environment and the shortage of fossil energy. The uncertainty and variability of variable renewable energy (VRE) are imposed on the load fluctuation from the generation side [1], contributing to the advent of new challenges in the system operation, especially in the real-time (RT) operation. The high share of VRE escalates the challenges in tackling the ramping demand in both upward and downward directions [2]. The lack of flexible ramping capacities (FRC) in a system could lead to curtailment of renewables or load shedding [3]. Therefore, the power system needs sufficient flexibility resources to

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M. Zhou et al., *Power System Flexibility*, Power Systems,
https://doi.org/10.1007/978-981-19-9075-5_3

deal with the fluctuation of VRE, that is, rapid response and standby power generation capacity.

Flexible look-ahead unit commitment aims to maintain sufficient FRC at a reasonable cost, where the flexible regulation capacities are properly reserved in a specific time slot for future use [4]. Many researches have been reported on this topic. Thatte et al. [5] introduce a scheduling scheme simultaneously meeting lack of ramp probability (LORP) and ramp deliverability in a multi-zone system. Inter-system flexibility set is utilized in [6] to guarantee the adequacy of flexibility and finds a feasible operating point. Simultaneously, many technical solutions have been proposed to alleviate the lack of FRC in power systems with high VRE penetration, including energy storage, demand response [7], and fast ramping units [8]. Increasing penetration of VRE will necessitate steeper ramping requirements from generators, which will result in a high cost on unit operations [9]. In order to further excavate the flexibility potential of the system, RES is no longer treated as non-dispatchable resource, studies on wind dispatchability have already been performed [10]. Typically, the concept of flexible dispatch margin is proposed in [11] and which is realized by underscheduling wind generators in the hour-ahead energy market [12].

Above studies are helpful for handling the optimization problems with uncertainty data, but they are unable to tell operators how much flexibility a system has. Although some exogenous approaches can intuitively reflect the relationship between flexible demand and supply, such as flexibility envelope [13] and probability box [14], they have large calculation burden and are not suitable for real time operation. The operational flexibility metric LORP proposed in [15] can be used in flexible LAUC, but it is a probabilistic metric, lacks of intuitiveness. In fact, decision makers are more inclined to adopt deterministic metrics. Therefore, there is an urgent need to establish an intuitive and indicative metric for flexibility assessment and VRE dispatch for good application in the RT operation of the power systems with high share of VRE. Therefore, in this section:

(1) The system flexibility is reflected by a multi-dimensional polyhedron, which is the maximum domain of wind power output that the system can accommodate, so we call it the accommodated domain of wind power (ADWP). We introduce FRC requirement constraint, which is based on the variability between two consecutive time steps and uncertainty at both time steps, in the proposed ADWP to avoid ramping shortages in the RT operation. Then, two flexible look-ahead unit commitment methods based on ADWP are proposed.
(2) A day ahead generation-reserve bilevel decision model is proposed. Based on historical data and forecast information, the proposed model establishes the quantitative relationship between reserve and flexibility, and ensures the system flexibility to meet the requirements while realizing conventional unit output scheduling. In the upper level, the unit commitment (UC) model is solved on the basis of predicted load and wind power output, and what is different is the relationship between reserve and flexibility established by the lower level is taken as a constraint. In the lower level, the ramping state of load and wind power is formulated based on historical data and forecast information. Then,

the UC scheme is verified by the flexibility requirement, and if it could not provide enough reserve capacity, the UC model of upper level is corrected and re-solved.

(3) An improved endogenous reserve determination method is proposed. Multiple uncertainties, including wind power outputs, load fluctuations, and generator failures, especially power flow uncertainties caused by wind power fluctuations and generator faults, are comprehensively integrated into a security and network-constrained economic dispatch (S&NCED) model to jointly schedule the generation and reserve. Affinely adjustable robust optimization (AARO) is adopted to address wind power and load uncertainties. Then the model is transformed into a two-stage (normal state optimization and fault verification) S&NCED model for iterative solution with the aid of Benders decomposition, and auxiliary constraints are designed to accelerate the convergence.

3.2 Quantifying Accommodated Domain of Wind Power for Flexible Look-Ahead Unit Commitment

This section reflects the flexibility of power systems by a multi-dimensional polyhedron, we call it ADWP. Then, two flexible look-ahead unit commitment methods based on ADWP are proposed.

3.2.1 Formulation of ADWP

3.2.1.1 Definition of ADWP

Similar to Do-Not-Exceed (DNE) limit, accommodated domain of wind power (ADWP) is defined as the M-dimensional domain composed of all the feasible injection power of M VRE buses that the system can accommodate without causing security violation in current interval, either flexible ramping capacities (FRC) shortage in adjacent interval, denoted by $\mathbb{D}$. But different from DNE, ADWP not only guarantees the system reliability when the outputs of wind farms deviate from the predictive ones, but also takes FRC requirement into consideration, which is coupled with next time interval. ADWP can not only be used as a metric of flexibility assessment, but also facilitate the RT operation. And $\mathbb{D}_t \subseteq \mathbb{U}_t$, $\mathbb{U}_t$ is the M-dimensional domain that contains all possible injection powers of M VRE buses in interval t, that is, the uncertainty set of VRE buses.

When the scheduling plan is determined, FRC that the system can provide is also determined under the constraints of transmission flows, unit outputs and ramping constraints. Usually, the FRC reserved at current time should be able to meet the maximum fluctuation range of net load at the end time of an interval. However, due to the uncertainty, the actual output of wind farm may deviate from the predicted one

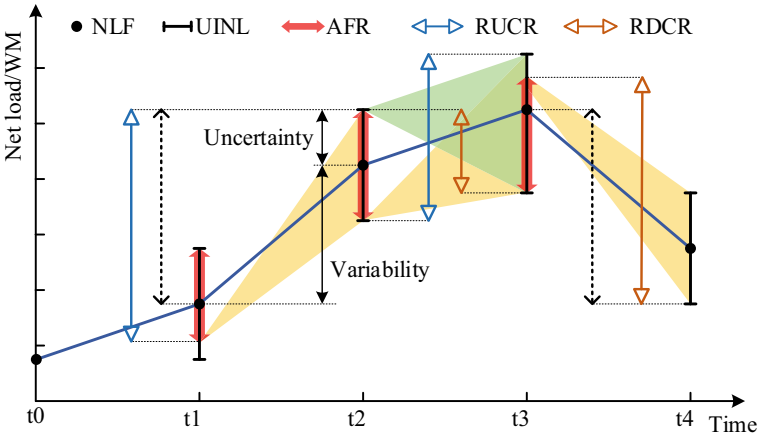


Fig. 3.1 Acceptable fluctuation range of net load and FRC requirement

in the start time of interval, and units will adjust their scheduled outputs to adapt this unexpected deviation. Therefore, the FRC that the system can supply and the FRC requirement will also change. When the FRC requirement is changing greater than the possible FRC supply, there may be a lack of flexibility in the end time of interval. Hence, we employ the design of FRC requirement based on the variability between two consecutive time steps and uncertainty at both time steps, which is showed in Fig. 3.1.

In Fig. 3.1, black dots represent net load forecast (NLF) and the vertical line segments through them indicate the uncertainty interval of net load (UINL). The red arrow is acceptable fluctuation range of net load (AFR), which is the maximum net load range that the power system can accommodate, corresponding to net load range when wind power outputs locate within $\mathbb{D}$. Blue arrow and orange arrow indicate ramping up capacity requirement (RUCR) and ramping down capacity requirement (RDCR), respectively, which together constitute FRC requirements, determined by the variability and uncertainty between two consecutive time steps. As a comparison, dashed arrow represents the reserved FRC, which may be insufficient under new FRC requirement. Shaded area means that the FRC of the system should envelope UINL of next time step when the net load changes within AFR.

In some existing markets, system operators are facing the challenge of temporary price spike in the RT operation, which is an indication of the lack of FRC in the system. It is mainly because the curtailment action is implemented in an *ex-post* fashion (flexible capacities are not reserved in advance). Under the proposed framework, ADWP serves as an operational guideline, which will guide the dispatch of the VRE buses. By knowing ADWP, operators can execute curtailment in an *ex-ante* fashion to enforce wind power output stay within the ADWP, thus avoiding security violation and the risk of price spikes. Moreover, it is helpful for power-to-gas (P2G) operating to improve wind energy utilization.

3.2.1.2 Mathematical Formulation

As mentioned in above section, when the injection power of VRE buses locate within ADWP, the system should have enough ramping capacity to satisfy FRC requirement brought by variability between two consecutive time steps, and the uncertainty of both time steps superimposed on it. $\mathbb{D}_t$ can be described as follows:

$$\mathbb{D}_t = \{\boldsymbol{w} \in \mathbb{U}^M : \exists \boldsymbol{p} \in \mathbb{R}^G, s.t.(\mathbf{2}) - (\mathbf{8})\} \tag{3.1}$$

$$\text{s.t.} \quad \sum_{g=\mathbf{1}}^{N_\text{g}} p_{gt} + \sum_{j=\mathbf{1}}^{N_\text{W}} w_{jt}^f = \sum_{m=\mathbf{1}}^{N_\text{d}} P_{mt}^f \tag{3.2}$$

$$\sum_{g=\mathbf{1}}^{N_\text{G}} p_{gt} + \sum_{i=\mathbf{1}}^{N_a} \Delta p_{it} + \sum_{j=\mathbf{1}}^{N_\text{W}} w_{jt} = \sum_{m=\mathbf{1}}^{N_\text{d}} P_{mt}^f \tag{3.3}$$

$$\left| \sum_n DF_{nl} \times (p_{g(n)} + \Delta p_{i(n)} - P_{m(n)}^f) + \sum_j DF_{jl} \times w_j \right| \le F_l \tag{3.4}$$

$$\Delta p_{it} = \alpha_{it} \sum_{j=\mathbf{1}}^{M} (w_{jt}^f - w_{jt}), \forall \alpha_{it} \ge \mathbf{0}, \ \sum\nolimits_{i=1}^{N_a} \alpha_{it} = \mathbf{1} \tag{3.5}$$

$$\max\{-RD_i \Delta T, \ p_i^{\min} - p_{it}\} \le \Delta p_{it} \le \min\{RU_i \Delta T, \ p_i^{\max} - p_{it}\} \tag{3.6}$$

$$\begin{cases} \delta p_{it+\mathbf{1}}^u = \alpha_{it+\mathbf{1}} \sum\limits_{j=\mathbf{1}}^{N_\text{W}} \lambda (w_{jt+\mathbf{1}}^f - w_{jt+\mathbf{1}}^l) \\ \delta p_{it+\mathbf{1}}^d = \alpha_{it+\mathbf{1}} \sum\limits_{j=\mathbf{1}}^{N_\text{W}} \lambda (w_{jt+\mathbf{1}}^u - w_{jt+\mathbf{1}}^f) \end{cases} \tag{3.7}$$

$$\begin{cases} RU_i \Delta T \ge p_{it+\mathbf{1}} + \delta \ p_{it+\mathbf{1}}^u - p_{it} - \Delta p_{it} \\ RD_i \Delta T \ge p_{it} + \Delta p_{it} - p_{it+\mathbf{1}} + \delta \ p_{it+\mathbf{1}}^d \end{cases} \tag{3.8}$$

where, P_{it} and α_{it} are the base output and participation factor of unit i in period t. Δp_{it}^u and Δp_{it}^d are the scheduled upward/downward flexible regulation capacity for unit i in period t. δp_{it}^u and δp_{it}^d are the upward/downward output deviation of unit i in period t. w_{jt}^f is the predicted wind power fed into the VRE bus m in period t. P_{mt}^f is the predicted value of load j in period t. λ is the adjustment coefficient for conservativeness. N_G, N_a is the number of the online units and AGC units, respectively. N_d and N_W are the number of loads and VRE buses. $i(n)$ is the index of unit i located at bus n. DF_{nl} is the power transfer distribution factor of bus n to line l. P_i^{min} and P_i^{max} are the minimum and maximum power output of unit i. F_l

is the transmission capacity of line l. RU_i and RD_i are the maximum upward and downward ramping rates of unit i. ΔT is the length of one time period.

In the above model, (3.1) shows that the injection power of VRE buses included in $\mathbb{D}_t$ should belong to the M-dimensional uncertainty set of VRE buses $\mathbb{U}^M$, and there is a solution for the outputs of units that satisfy constraints (3.2)–(3.8). (3.2) is power balance constraint for scheduled output of units. (3.3) shows that system fulfills the load by scheduling AGC units when the injection power of VRE buses deviates from its predictive one. (3.4) represents the transmission line limits. (3.5) and (3.6) show that the output change of each AGC unit should meet the ramping rate and its capacity constraints. (3.7) and (3.8) indicate that unit ramping capability should be able to meet FRC requirement brought by the output deviation between two consecutive time steps. It is worth to note that only wind power is considered in VRE buses in this model for simplicity, but the proposed model and methods are highly compatible and easily expanded to photovoltaic (PV) energy.

Since $\mathbb{D}_t$ is related to the fluctuation range of net load at the end time of interval, which influences the FRC requirement. In order to avoid over conservativeness, the uncertainty interval at the end time can be selected as a certain proportion of the maximum uncertainty interval (e.g. 90%). Therefore, we introduce adjustment coefficient λ in (3.7) to regulate the level of conservativeness, which reflect the operator's attitude to risk. In extreme case, when λ is equal to zero, $\delta p^u_{it+\mathbf{1}}$ and $\delta p^d_{it+\mathbf{1}}$ are also equal to zero, which means the uncertainty at the end time of the interval is excluded from the FRC requirement, and ADWP obtained from this case is denoted as $\mathrm{ADWP_V}$. Correspondingly, when λ is equal to 1, ADWP described in (3.1) is denoted as $\mathrm{ADWP_{V/U}}$.

3.2.1.3 Diagrammatic Description

In order to clearly illustrate the concept of the domain in ADWP, Fig. 3.2 is made to present the possible ADWP that may appear in a system with single, two and three VRE buses in a certain time interval, respectively.

In Fig. 3.2a, the horizontal lines of the bars indicate the possible wind power scenarios, which are all included in the uncertainty set of VRE buses. The ADWP of the system can be obtained based on a scheduling scheme. For those wind power scenarios outside that, wind curtailment or load shedding would occur.

The rectangle and cube in Fig. 3.2b, c contain all possible injection powers of VRE buses, that is, the M-dimensional wind power uncertainty set $\mathbb{U}^M$ formed by output ranges of M VRE buses; The shaded area is ADWP, the M-dimensional domain $\mathbb{D}$, which does not consider transmission and voltage constraints in this figure. In reality, the polygon of Fig. 3.2b and polyhedron of Fig. 3.2c may be more complicated, and there may be some infeasible points at edges.

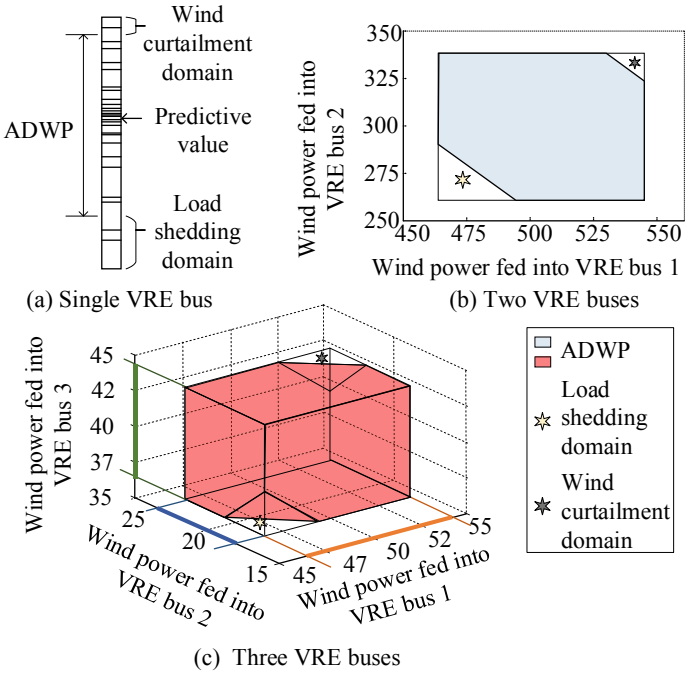


Fig. 3.2 Schematic diagram of ADWP

3.2.2 Flexible Look-Ahead Unit Commitment Models

As shown in (3.1)–(3.8), ADWP is determined by the base points and the flexible regulation capacities of the online units. In other words, different scheduling schemes correspond to different ADWP. The largest ADWP reflects the maximum flexibility potential of the system, but its corresponding scheduling scheme is not necessarily the most economical. Therefore, two approaches for flexible LAUC are proposed for different purposes in this chapter. One is the three-stage interval optimization approach, which aims to determine the largest ADWP and minimize the operating cost of the corresponding scheduling scheme. The other one is a scenario-based stochastic optimization approach, which aims to minimize the overall cost by reaching an equilibrium between operating cost and penalty cost. Due to the immature trading mechanism and its inherent stochastic characteristic, wind power producers are not considered providing regulation reserve in the proposed model.

3.2.2.1 Three-Stage Interval Optimization Approach

(1) Maximizing the ADWP

The aim of the first stage is to maximize the ADWP. As defined in the above section, $\mathbb{D}$ has the property of domain, and its size is difficult to compute directly. Therefore, slack variables sl^u and sl^d are introduced to substitute the size of $\mathbb{D}$: the physical meaning of sl^d is the sum of the wind curtailment of each VRE bus when the injection power of all VRE buses reaches its upper boundary, and the physical meaning of sl^u is the sum of the load shedding of each load when the injection power of all VRE buses reaches its lower boundary. In this way, the problem of maximizing $\mathbb{D}$ becomes the problem of minimizing the sum of these two variables. The objective function of the first stage is then expressed as follows:

$$z_{\mathbf{1}} = \min \sum_{t=\mathbf{1}}^{T} (sl_t^u + sl_t^d) \tag{3.9}$$

where, sl_i^u and sl_i^d *are the* slack variables. T is the number of time periods in optimization horizon.

In this stage of optimization, we do not consider penalty costs associated with wind curtailment and load shedding, so it is worth mentioning that the unit of the expression (3.9) is energy.

(2) Minimizing the operating costs

The second stage is for minimizing the operating costs, which include fuel cost, unit commitment cost, and cost for scheduling regulation resources. In order to simplify the calculation, it is assumed that only AGC units is scheduled in the LAUC. The objective function of the second stage is as follows:

$$z_{\mathbf{2}} = \min \sum_{t=\mathbf{1}}^{T} [\sum_{i=\mathbf{1}}^{N_a} (C_i(p_{it}) + c_i^u \Delta p_{it}^u + c_i^d \Delta p_{it}^d + SU_{it} + SD_{it})] \tag{3.10}$$

where the startup and shutdown cost can be written as:

$$SU_{it} \geq c_i^{su}(u_{it} - u_{it-\mathbf{1}}),\ SU_{it} \geq \mathbf{0} \tag{3.11}$$

$$SD_{it} \geq c_i^{sd}(u_{it-\mathbf{1}} - u_{it}),\ SD_{it} \geq \mathbf{0} \tag{3.12}$$

where $C_i(.)$ is the fuel cost function of unit i. SU_{it} and $\mathrm{SD_{it}}$ are the startup and shutdown cost of unit i in period t. c_i^{su} and c_i^{sd} are the start-up and shut-down cost of unit i. u_{it} is a binary variable, equal to 1 if unit i is online in period t.

As shown in (3.10), in the proposed model, the units would bid for both energy and regulation capacity. c_i^u, c_i^d inherently reflect a lost opportunity cost estimated by the market participant. The operating constraints are as follows:

(1) Power balance constraint: The scheduled output of the AGC units plus the predictive output of wind power should be equal to the load excluding the part P_{non} supplied by non-AGC units.

$$\sum_{i=1}^{N_a} p_{it} + \sum_{j=1}^{N_W} w_{jt}^f = \sum_{m=1}^{N_d} P_{mt}^f - P_{non},\ \forall t \tag{3.13}$$

(2) Capacity constraints of AGC units:

$$\begin{cases} \mathbf{0} \le \Delta p_{it}^u \le \min\{RU_i \Delta T,\ p_i^{\max} - p_{it}\} \\ \mathbf{0} \le \Delta p_{it}^d \le \min\{RD_i \Delta T,\ p_{it} - p_i^{\min}\} \end{cases} \tag{3.14}$$

(3) Regulation capacity requirement constraints:

$$\begin{cases} \sum\limits_{i=1}^{N_a} \Delta p_{it}^u + sl_t^u = \sum\limits_{j=1}^{N_W} (w_{jt}^f - w_{jt}^l) \\ \sum\limits_{i=1}^{N_a} \Delta p_{it}^d + sl_t^d = \sum\limits_{j=1}^{N_W} (w_{jt}^u - w_{jt}^f) \end{cases} \tag{3.15}$$

(4) Power output constraints of generators:

$$p_i^{\min} u_{it} \le p_{it} \le p_i^{\max} u_{it} \tag{3.16}$$

(5) FRC requirement constraints: When the injection power of VRE buses locate within $\mathbb{D}_t$, the system should be able to meet the FRC requirement. The FRC requirement constraints are expressed as:

$$\begin{cases} RU_i \Delta T \ge p_{it+1} + \min(\Delta p_{it+1}^u, \delta p_{it+1}^u) - p_{it} + \Delta p_{it}^d \\ RD_i \Delta T \ge p_{it} + \Delta p_{it}^u - p_{it+1} + \min(\Delta p_{it+1}^d, \delta p_{it+1}^d) \end{cases} \tag{3.17}$$

(6) Minimum up and down time constraints:

$$\begin{cases} \sum\limits_{h=0}^{T_i^{on}-1} u_{it+h} \ge (u_{i,t} - u_{it-1}) \cdot \min(T_i^{on}, T - t + 1) \\ \sum\limits_{h=0}^{T_i^{off}-1} (1 - u_{it+h}) \ge (u_{it-1} - u_{it}) \cdot \min(T_i^{off}, T - t + 1) \end{cases} \tag{3.18}$$

where, T_i^{on} and T_i^{off} are the minimum up time and minimum down time of the unit i.

(7) Transmission line power constraints: Adopting the distribution factor of each bus, the transmission power constraints are the same as (3.4).

(3) Penalty cost evaluation

The penalty cost evaluation stage can embody the performance of the unit commitment results in specific scenarios. Given the schedule for energy and regulation capacity, this phase aims to minimize the realized penalty cost associated with each typical scenario, including wind curtailment and load shedding cost. The objective

function of the third stage is:

$$z_{\mathbf{3}} = \min \sum_{s=\mathbf{1}}^{S} \rho_s \sum_{t=\mathbf{1}}^{T} (\sum_{j=\mathbf{1}}^{N_{\mathrm{W}}} c_j^{cur} CUR_{s,j,t} + \sum_{m=\mathbf{1}}^{N_{\mathrm{d}}} c_m^{ls} LS_{s,m,t}) \tag{3.19}$$

Since the penalty cost for different VRE buses and loads may be different, the decision variables are the amount of wind curtailment of each VRE bus $CUR_{s,j,t}$ and the amount of load shedding of each load $LS_{s,m,t}$, which should meet the following scenario constraints:

$$(w_{s,\mathbf{1},t}, w_{s,\mathbf{2},t}, ..., w_{s,j,t}) - (CUR_{s,\mathbf{1},t}, CUR_{s,\mathbf{2},t}, ..., CUR_{s,j,t}) \in \mathbb{D}_t \tag{3.20}$$

$$(w_{s,\mathbf{1},t}, w_{s,\mathbf{2},t}, ..., w_{s,j,t}) + (ls_{s,\mathbf{1},t}, ls_{s,\mathbf{2},t}, ..., ls_{s,j,t}) \in \mathbb{D}_t \tag{3.21}$$

$$\sum_{j=\mathbf{1}}^{N_{\mathrm{W}}} ls_{s,j,t} = \sum_{m=\mathbf{1}}^{N_{\mathrm{d}}} LS_{s,m,t}, \quad CUR_{s,j,t}, \quad ls_{s,j,t} \geq \mathbf{0} \tag{3.22}$$

where, $ls_{s,j,t}$ is the intermediate variable introduced from the mathematical point of view to calculate $LS_{s,j,t}$. ρ_s is the probability of scenario s. CUR_{smt} is the wind curtailment of VRE bus m in period t under scenario s.

The sum of each VRE bus's $ls_{s,m,t}$ should be equal to the sum of each load's $LS_{s,j,t}$, as shown in (3.22). (3.20) and (3.21) indicate that for wind power realization outside $\mathbb{D}$, after wind curtailment or load shedding, it should belong to $\mathbb{D}$. At the same time, transmission line power flow limits should be satisfied for each scenario, as shown in (3.23).

$$\left| \sum_{n} DF_{n,l} \times (p_{i(n)} + \Delta p_{si(n)} - (d_{m(n)} - LS_{sm(n)})) + \sum_{j} DF_{jl} \times (w_{sj} - CUR_{sj}) \right| \leq F_l, \forall t, l, s \tag{3.23}$$

In conclusion, the three-stage interval optimization model can be formulated as below.

First stage: Maximizing the ADWP.

$$Obj: \quad z_{\mathbf{1}} = \min_{\substack{p_{it}, \alpha_{it}, \Delta\widehat{p}_{it}, \Delta\breve{p}_{it} \\ u_{it}, sl_t^u, sl_t^d}} \sum_{t=\mathbf{1}}^{T} (sl_t^u + sl_t^d) \tag{3.24}$$

s.t. Constraints (3.4), (3.7) and (3.11)–(3.18).

Second stage: Minimizing the operating costs.

$$Obj:\ z_2 = \min_{\substack{p_{it},\alpha_{it},\Delta\widehat{p}_{it},\Delta\breve{p}_{it} \\ u_{it},sl_t^u,sl_t^d}} \sum_{t=1}^{T}[\sum_{i=1}^{N_a}(C_i(p_{it}) + c_i^u\Delta\widehat{p}_{it} + c_i^d\Delta\breve{p}_{it} + SU_{it} + SD_{it})] \tag{3.25}$$

s.t. $\sum_{t=1}^{T}(sl_t^u + sl_t^d) \geq z_1$, constraints (3.4), (3.7) and (3.11)–(3.18).

After solving the above two stages, the scheduling scheme that could obtain the maximum ADWP with the lowest operating cost is determined. Then, $\mathbb{D}$ can be expressed as (for simplicity, the subscript t is omitted):

$$\begin{aligned} &\mathbb{D} := \{\boldsymbol{w} \in \mathbb{U}^M : sum(\underline{\boldsymbol{w}}) + sl^u \leq sum(\boldsymbol{w}) \leq sum(\overline{\boldsymbol{w}}) - sl^d, \\ &A\boldsymbol{w} + B(\boldsymbol{p} + \Delta\boldsymbol{p}) \leq c\} \end{aligned} \tag{3.26}$$

where, $A\boldsymbol{w} + B(\boldsymbol{p} + \Delta\boldsymbol{p}) \leq c$ represents the power flow constraint.

Third stage: Penalty cost evaluation.

$$Obj:\ z_3 = \min_{CUR_{sjt},LS_{sjt}} \sum_{s=1}^{S}\rho_s\sum_{t=1}^{T}(\sum_{j=1}^{N_W}c_j^{cur}CUR_{s,j,t} + \sum_{m=1}^{N_d}c_m^{ls}LS_{s,m,t}) \tag{3.27}$$

s.t. Constraints (3.20)–(3.23).

3.2.2.2 Scenario-Based Stochastic Optimization Approach

When the ADWP reaches maximal domain, all flexible resources have been utilized and the operating cost is the highest accordingly, but the overall cost is not necessarily the lowest. This is because in some realizations, a certain part of the ADWP may be redundant, which increases operating cost but does not reduce the penalty cost. In addition, the system needs to call more expensive flexibility resources in order to meet some scenarios with a low probability of occurrence, which hinders the economic operation of the system. In order to minimize the overall cost in all typical scenarios, a scenario-based stochastic optimization model is established, which make a tradeoff between the operating cost and the penalty cost. The most economical ADWP is also obtained simultaneously from that. Then, the objective function is:

$$\begin{aligned} z_4 = \min &\sum_{t=1}^{T}[\sum_{i=1}^{N_a}(C_i(p_{it}) + c_i^u\Delta p_{it}^u + c_i^d\Delta p_{it}^d + SU_{it} + SD_{i,t})] + \\ &\sum_{s=1}^{S}\rho_s\sum_{t=1}^{T}(\sum_{j=1}^{N_W}c_j^{cur}CUR_{s,j,t} + \sum_{m=1}^{N_d}c_m^{ls}LS_{s,m,t}) \end{aligned} \tag{3.28}$$

The first part of (3.28) corresponds to operating cost, which could determine the ADWP. The second part corresponds to penalty cost, which embodies the adaptation of the scheduling plan in specific scenarios. According to (3.26), $\mathbb{D}$ can be expressed by sl_t^u and sl_t^d, then the scenario constraints (3.20) and (3.21) can be represented as:

$$\sum_{j=1}^{N_W} CUR_{s,j,t} \geq sl_t^d + \sum_{j=1}^{N_W} (w_{s,j,t} - w_{j,t}^u) \tag{3.29}$$

$$\sum_{j=1}^{N_d} LS_{s,j,t} \geq sl_t^u + \sum_{m=1}^{M} (w_{m,t}^l - w_{s,m,t}) \tag{3.30}$$

The scenario-based stochastic optimization model can be explicitly formulated as follows:

$$Obj: \ z_4 \tag{3.31}$$

s.t. Constraints (3.7) and (3.11)–(3.18) and (3.29)–(3.30).

(3.1)–(3.31) propose a flexible look-ahead unit commitment model. The model is a mixed-integer linear programming model, which is solved by CPLEX in MATLAB R2016b. See Sect. 3.5.1 for the corresponding case studies in this section.

3.3 Flexibility Based Day-Ahead Generation–Reserve Bilevel Decision Model

This section proposes a flexibility-based day-ahead generation-reserve bilevel decision model. In the upper level, the day-ahead unit commitment model is constrained by flexibility reserve, which is calculated in the lower level. In the lower level, the ramping probability distribution of an equivalent system is obtained by the universal generating function method, then the quantified relationship between operating reserve and flexibility is established ultimately.

3.3.1 Day-Ahead Unit Commitment Model Considering Flexibility Constraint

Given that operating reserve is closely related with conventional unit operation state, it is essential for a day-ahead unit commitment model to provide necessary boundary conditions if the system flexibility is to be rational measured to assess the reserve capacity required. In the upper level, the day-ahead unit commitment model schedules

the conventional unit start-up and output plan of 96 time periods based on the short-term forecast of wind power and load. Meanwhile, the constraints of the UC model also include the flexibility-based reserve obtained by the lower level. The specific model is as follows.

3.3.1.1 Objective Function

The objective function of day-ahead UC features the system operation cost minimization, and the operation cost includes generation cost and unit start-up cost, expressed as:

$$\min \sum_{t}^{T} (C_1 + C_2) \tag{3.32}$$

(1) Generation cost C_1

The generation cost of conventional unit is normally expressed as quadratic function, namely:

$$C_1 = \sum_{i \in N_G} u_{i,t} C(P_{i,t}) = \sum_{i \in N_G} u_{i,t} \cdot c_i + b_i P_{it} + a_i (P_{it})^2 \tag{3.33}$$

where T is the total number of time period, N_G is the conventional unit set; $u_{i,t}$ is the state of unit i at the time t, 1 for operation and 0 for outage, $P^G{}_{i,t}$ is the planned output of unit i at the time t, a_i, b_i and c_i is the generation cost coefficient of unit i.

(2) Start-up cost C_2

$$C_2 = \sum_{t \in T} \sum_{i \in N_G} u_{it}(1 - u_{it-1}) c_{i,t}^{su} \tag{3.34}$$

where the unit start cost $c^{su}{}_{i,t}$ can be expressed as:

$$c_{i,t}^{su} = \begin{cases} C_i^{hot}, T_i^{off} \le t_{i,t}^{off} \le T_i^{off} + T_i^{cold} \\ C_i^{cold}, t_{i,t}^{off} \ge T_i^{off} + T_i^{cold} \end{cases} \tag{3.35}$$

where, C_i^{hot} and C_i^{cold} represent hot start-up cost and cold start-up cost of unit i, respectively, T_i^{off} and T_i^{cold} respectively represent minimum off time and cold start time, $t_{i,t}^{off}$ is the actual continuous offline time. As the stop cost is small, it is normally expressed as constant, which is not considered here.

3.3.1.2 Constraints

(1) Power balance

$$\sum_{i \in N_{\mathrm{G}}} P_{it} + \sum_{j \in N_{\mathrm{W}}} w_{jt} - \sum_{m \in N_{\mathrm{d}}} P_{mt}^{f} = 0 \tag{3.36}$$

where, N_{W} and N_{d} respectively represent wind farm set and load set, w_{jt} is the predicted output of wind farm j at time t, and P_{mt}^{f} is the predicted load m at time t.

(2) The generation constraints for the power unit are:

$$u_{it} P_i^{\min} \leq P_{it} \leq u_{it} P_i^{\max} \tag{3.37}$$

where, P_i^{max} and P_i^{min} refer respectively to the upper and lower limits of unit i.

(3) The ramping constraints of the power unit are:

$$-\Delta T R D_i \leq P_{it} - P_{it-1} \leq \Delta T R U_i \tag{3.38}$$

$$R_{it}^{\mathrm{U}} \leq T_r R U_i \tag{3.39}$$

$$R_{it}^{\mathrm{D}} \leq T_r R D_i \tag{3.40}$$

where, ΔT is the length of time interval, T_{r} is the spinning reserve response time, R_{it}^{U} and R_{it}^{D} refers to the up and down reserve of unit i at time t.

(4) Minimum up and down time

$$t_{i,on}^{\min} \geq T_i^{on} \tag{3.41}$$

$$t_{i,off}^{\min} \geq T_i^{off} \tag{3.42}$$

where, $t_{i,on}^{min}$ and $t_{i,off}^{min}$ refer to the minimum up and down duration of unit i, $T_i{}^{on}$ and $T_i{}^{off}$ are the allowable minimum up and down time of unit i.

(5) Flexibility constraint

The UC model must arrange enough flexibility resources to maintain the safe and reliable operation of the power system. Loss of flexibility probability (LOFP) is designed as a flexibility index. The UC model needs to keep the LOFP lower than the requirement set by operator to ensure the flexibility of the system operation. Since the calculation of the LOFP is complicated, it is difficult to consider the LOFP in the optimization process. The quantitative relationship between the LOFP and reserve

capacity is established by the reserve decision model in the lower level. In this way, the flexibility constraint is converted into the reserve constraint. In other words, in order to ensure that the flexibility of system meets the given requirement, the total up and down reserve capacity R_t^U and R_t^D provided by conventional unit must meet the reserve requirements calculated by the reserve decision model in the lower level.

$$\left(R_t^{\mathrm{U}}, R_t^{\mathrm{D}}\right) = f_{\mathrm{R}}(\alpha,\ t) \tag{3.43}$$

$$\sum_{i \in N_{\mathrm{G}}} R_{it}^{\mathrm{U}} = R_t^{\mathrm{U}} \tag{3.44}$$

$$\sum_{i \in N_{\mathrm{G}}} R_{it}^{\mathrm{D}} = R_t^{\mathrm{D}} \tag{3.45}$$

$$R_{it}^{\mathrm{U}} \le u_{it}\left(P_i^{\max} - P_{it}\right) \tag{3.46}$$

$$R_{it}^{\mathrm{D}} \le u_{it}\left(P_{it} - P_i^{\min}\right) \tag{3.47}$$

where, $f_R(\alpha, t)$ is the function of reserve established by the reserve decision model in the lower level at time t, and it will be given in detailed in Sect. 3.3.2, α refers to the given tolerance of the LOFP. The initial values of $R_{i,t}^U$ and $R_{i,t}^D$ are set as a certain proportion of the total load.

The solution of the UC model in the upper level, including the start-up plan and output scheme of conventional unit, provides boundary conditions for the reserve decision model in the lower level.

3.3.2 Flexibility Based Reserve Decision Method

The universal generating function (UGF) is an effective discrete random variable expression and calculation tool proposed by Professor Ushakov in the 1980s. It has been widely used in the reliability analysis of multi-state systems. The UGF theory establishes a one-to-one correspondence between the discrete probability distribution and a u-function. Then, the discrete probability distribution of a random function can be calculated by the corresponding UGF operator in a recursive way. Other stochastic approaches in optimization, e.g., interval optimization, only are limited scenarios considered, while probability information is lost. The adoption of the UGF theory makes it possible to evaluate the flexibility of the power system at every time period effectively. Thus, the UGF based flexibility assessment method in this section provides the basis for the flexibility-based UC. In the lower level, the proposed reserve decision model considers various factors of uncertainty and fluctuation like stochastic wind power output and conventional unit forced outage comprehensively,

establishes the quantitative relationship between operating reserve and the flexibility index.

Flexibility refers to the ability of the system to cope with power fluctuations. For the issue of day-ahead scheduling, the system responds to the deviation of the dispatch scheme by the operating reserve. Therefore, the operating reserve can be quantified based on the flexibility assessment. The system ramping is defined as the power change between two adjacent intervals. The key of the reserve decision based on the flexibility is how to evaluate the system ramping reasonably. The power system is considered as a multi-state model, and the UGF models of ramping state for each component are built; then UGF combination operators are used to obtain the ramping state distribution of the equivalent system. The relationship between the flexibility index LOFP and reserve capacity is established by the ramping state probability distribution of the equivalent system, thereby the corresponding reserve demand is calculated by the given LOFP.

3.3.2.1 Universal Generating Function Model for Load

The load forecasting error of the power system generally obeys a normal distribution, so the actual load at time t, $P^{\mathrm{D}}(t)$ conforms to normal distribution ($N(P_{\mathrm{d}}(t), \sigma_{\mathrm{d}}(t))$, where $P_{\mathrm{d}}(t)$ is the predicted load value at time t and $\sigma_{\mathrm{d}}(t)$ is the load forecasting standard deviation. $\sigma_{\mathrm{d}}(t)$ can be obtained through statistical analysis to large amount of historical data, or comes from empirical value.

Step P_{intv} is set for discretization of probability load model. According to probability theory, the study is limited in the range of $[P_{\mathrm{d}}(t) - 3\sigma_{\mathrm{d}}(t), P_{\mathrm{d}}(t) + 3\sigma_{\mathrm{d}}(t)]$ for efficiency. The load is divided into N_{d} states. For a certain load state k, suppose that the load value is $P_k^d(t)$ and the error is $\Delta P_k^d(t)$, then the corresponding probability $p^d\ k(t)$ is:

$$\begin{aligned} p_m^{\mathrm{d}}(t) &= \Pr\left\{ P_m^{\mathrm{d}}(t) - \frac{P_{\mathrm{intv}}}{2} < P^{\mathrm{D}}(t) < P_m^{\mathrm{d}}(t) + \frac{P_{\mathrm{intv}}}{2} \right\} \\ &= \Phi\left(\frac{2\Delta P_m^{\mathrm{d}}(t) + P_{\mathrm{intv}}}{2\sigma_{\mathrm{d}}(t)} \right) - \Phi\left(\frac{2\Delta P_m^{\mathrm{d}}(t) - P_{\mathrm{intv}}}{2\sigma_{\mathrm{d}}(t)} \right) \end{aligned} \tag{3.48}$$

where, $\Pr\{\cdot\}$ is the probability of event $\{\cdot\}$, $\Delta P_m^d(t)$ is the error of load state k, and $\Phi(\cdot)$ is the standard normal distribution function. Then the UGF model of load can be obtained:

$$u_{\mathrm{d}}(z, t) = \sum_{m=1}^{N_{\mathrm{d}}} p_m^f(t) \cdot z^{P_m \mathrm{d}(t)} \tag{3.49}$$

Similar to the ramping of generator, the load change at time t is $R^{\mathrm{d}}(t) = P^{\mathrm{D}}(t+1) - P^{\mathrm{D}}(t)$, the prediction errors of load at adjacent time are independent, so the UGF model of load change is:

$$u_{\mathrm{D}}(z,t) = u_d(z,t+1) \otimes_{-} u_d(z,t) = \sum_{m=1}^{S_{\mathrm{D}}} p_m^f(t) \cdot z^{R_m \mathrm{d}(t)} \tag{3.50}$$

where, $\otimes_{-}$ is the difference operator of UGF method, S_{D} is the total number of load change states, $R^d m(t)$ refers to the load change of state k, and $p^f{}_m(t)$ is the corresponding probability.

3.3.2.2 Universal Generating Function Model for Wind Power

The wind power prediction error is formulated as a normal distribution model, and the modeling method is also applicable to other distribution forms. The output of wind farm $P^{\mathrm{w}}(t)$ at the time t obeys normal distribution $N(^{-}P^{\mathrm{w}}(t), \sigma_{\mathrm{w}}(t))$, where $^{-}P^{\mathrm{w}}(t)$ is the wind farm output predict at time t, $\sigma_{\mathrm{w}}(t)$ is the standard deviation. Same as load, the study is limited in the range of $[P^{\mathrm{w}}(t) - 3\sigma_{\mathrm{w}}(t), {}^{-}P^{\mathrm{w}}(t) + 3\sigma_{\mathrm{w}}(t)]$ for efficiency. The output of wind farm is divided into N_{w} states, the power value of state i is $P_i^w(t)$, the prediction error is $\Delta P_i^w(t)$, then the corresponding probability p $i^w(t)$:

$$\begin{aligned} p_i^{\mathrm{w}}(t) &= \Pr\left\{ P_i^{\mathrm{w}}(t) - \frac{P_{\mathrm{intv}}}{2} < P_{\mathrm{w}}(t) < P_i^{\mathrm{w}}(t) + \frac{P_{\mathrm{intv}}}{2} \right\} \\ &= \Phi\left(\frac{2\Delta P_i^{\mathrm{w}}(t) + P_{\mathrm{intv}}}{2\sigma_{\mathrm{w}}(t)} \right) - \Phi\left(\frac{2\Delta P_i^{\mathrm{w}}(t) - P_{\mathrm{intv}}}{2\sigma_{\mathrm{w}}(t)} \right) \end{aligned} \tag{3.51}$$

The transition probability $P_{ij}^{w1}(t)$ from the output state i at time t to the output state j at time $t+1$ is obtained by statistically analyzing the historical data of wind farm W1. The difference in output from the next moment to the current moment is the wind power ramping state of current moments. From this, the conditional probability of wind power ramping on the current output is obtained. While the output of W1 is in state i, the conditional probability of the ramping state k can be described as $p_{k|i}^{w1}(t)$:

$$\begin{aligned} p_{k|i}^{\mathrm{w1}}(t) &= \Pr\left\{ R_k^{\mathrm{w1}}(t) \middle| P_i^{\mathrm{w1}}(t) \right\} \\ &= p_{ij}^{\mathrm{wl}}(t) | R_k^{\mathrm{w1}}(t) = P_j^{\mathrm{w1}}(t+1) - P_i^{\mathrm{w1}}(t) \end{aligned} \tag{3.52}$$

According to conditional probability formula, the probability $p_{k|i}^{w1}(t)$ of ramping state k is:

$$p_k^{\mathrm{w1}}(t) = \sum_{i=1}^{N_{\mathrm{w1}}} \left(p_{k|i}^{\mathrm{w1}}(t) p_{i,t}^{\mathrm{w1}}(t) \right) = \sum_{i,j \in \Pi_k} \left(p_{ij}^{\mathrm{w1}}(t) p_{i,t}^{\mathrm{w1}}(t) \right) \tag{3.53}$$

where, N_{w1} is the total number of output states of wind farm W1; Π_k is the state set that satisfies $R_k^{wl}(t) = P_j^{w1}(t+1) - P_i^{w1}(t)$. In (3.54), the ramping probability

distribution of wind farm W1 is established on the basis of predicted output and ramping state transition distribution, then the ramping UGF model for wind farm W1 at time t is as follows:

$$u_{\mathrm{w1}}(z,t) = \sum_{k=1}^{S_{\mathrm{w1}}} p_k^{\mathrm{w1}}(t) z^{R_k \mathrm{w1}(t)} \tag{3.54}$$

where S_{w1} is the total number of ramping states. Repeat the process listed above until the ramping probability distributions of all wind farms are obtained.

If wind farms are independent, then the total ramping probability distribution of wind farms is:

$$u_{\mathrm{W}}(z,t) = u_{\mathrm{w1}}(z,t) \otimes_{+} u_{\mathrm{w2}}(z,t) \otimes_{+} \ldots u_{\mathrm{wN^w}}(z,t) = \sum_{j=1}^{S_{\mathrm{W}}} p_j^{\mathrm{W}} \cdot z^{R_j \mathrm{W}(t)} \tag{3.55}$$

where, $\otimes_{+}$ is the summation operator of the UGF method, N^{w} refers to the total number of wind farms, S_{w} is the is the total number of wind farm total ramping states, $R_j{}^{w}(t)$ and $p_j{}^{w}(t)$ are the ramping value and probability of the jth ramping state of total wind farms at time t.

3.3.2.3 Universal Generating Function Model for Conventional Unit

In the flexibility analysis, it is necessary to consider the forced outage of the conventional unit for the accuracy. On the basis of the operating scheme obtained in the upper level, combined with the unit's forced outage rate, the conventional units are modeled as follows.

There are only two states for the ramping of conventional unit caused by forced by outage in flexibility analysis: (1) If conventional generation unit fault at the time $t + 1$ and causes outage of the unit, then the ramping of conventional unit at time t is the negative value of the planned output at time $t + 1$, and the corresponding probability is the forced outage rate. (2) If the conventional unit operates normally, then there is no ramping. Therefore, the UGF model of conventional unit g at time t is:

$$u_g(z,t) = \left(1 - p_g\right) z^0 + p_g z^{-P_{it+1}} \tag{3.56}$$

where, p_g is the forced outage rate of unit g, P_{it+1} is the planned output of unit g at time $t + 1$.

The forced outage events of conventional units are independent so the equivalent multi-state UGF model for conventional units is:

$$u_{\mathrm{G}}(z,t) = \otimes_{+}(u_1(z,t), u_2(z,t), \cdots, u_N(z,t)) = \sum_{i=1}^{S_{\mathrm{G}}} p_i(t) \cdot z^{R_i^{\mathrm{G}}(t)} \tag{3.57}$$

where, S_{G} is the total number of ramping states of equivalent conventional units, N refers to the total number of conventional units, $R_i^G(t)$ and $P_i^G(t)$ are the ramping value and probability of the ith ramping state of equivalent conventional units at time t.

3.3.2.4 Flexibility Based Reserve Capacity Calculation

The system ramping, which refers to the power fluctuation between the adjacent intervals, is the algebraic sum of load change, conventional unit ramping caused by forced outage, and wind power ramping, so the UGF model for the system ramping is:

$$\begin{aligned} u_{\mathrm{R}}(z,t) &= u_{\mathrm{D}}(z,t) \otimes_{-} u_{\mathrm{G}}(z,t) \otimes_{-} u_{\mathrm{W}}(z,t) \\ &= \sum_{k=1}^{S_{\mathrm{D}}} p_k^{\mathrm{dr}} \cdot z^{R_k \mathrm{d}} \otimes_{-} \sum_{j=1}^{S_{\mathrm{W}}} p_j^{\mathrm{W}} \cdot z^{R_j \mathrm{W}} \otimes_{-} \sum_{i=1}^{S_{\mathrm{G}}} p_i^{\mathrm{G}} \cdot z^{R_i \mathrm{G}} \\ &= \sum_{k=1}^{S_{\mathrm{eq}}} p_k^{\mathrm{eq}}(t) \cdot z^{R_k^{eq}(t)} \end{aligned} \tag{3.58}$$

where, S_{eq} is the total number of ramping states of the equivalent system, $R_k^{eq}(t)$ is the system ramping power at state k, and $P_k^{eq}(t)$ is the corresponding probability.

In this way, the probability distribution function $F_t^{eq}(\cdot)$ of the system ramping at time t is obtained.

Loss of flexibility probability means the probability that the system ramping is larger than the operating reserve. It is an intuitive index of the system flexibility. The up and down operating reserves are $R^{\mathrm{U}}(t)$ and $R^{\mathrm{D}}(t)$, respectively. For ramping state k, if $R\,k^{eq}(t) > R^{\mathrm{U}}(t)$, the system cannot provide sufficient generation power in transition from time t to $t+1$, causing loss of load; if $R\,k^{eq}(t) < -R^{\mathrm{D}}(t)$, it cannot lower output in transition from time t to $t+1$, resulting in wind power curtailment. Therefore, the system LOFP can be defined as:

$$LOFP(t) = \sum_{R_i^{eq}(t) > R^{\mathrm{U}}(t)} p_i^{eq}(t) + \sum_{R_j^{eq}(t) < -R^{\mathrm{D}}(t)} p_j^{eq}(t) \tag{3.59}$$

The relationship between LOFP and operating reserve is shown in Fig. 3.3.

The quantified relationship between the system operating reserve and LOFP is established in (3.59), but this definition cannot be calculated directly. The explicit

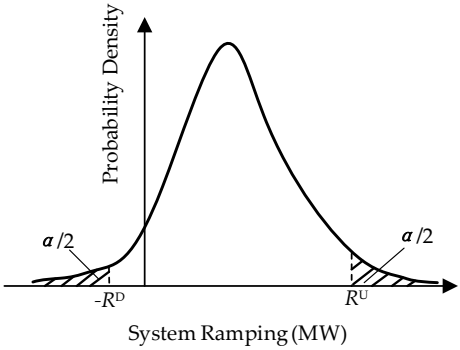


Fig. 3.3 The relationship between the loss of flexibility probability (LOFP) and operating reserve

form of $f_R(\alpha, t)$ still needs further derivation. As the probability distribution function $F_t^{eq}(\cdot)$ of the system ramping at time t is obtained by the u-function (3.58), $R^{\mathrm{U}}(t)$ and $R^{\mathrm{D}}(t)$ can be calculated by the inverse function of $F_t^{eq}(\cdot)$. When the LOFP is set as α, the $f_R(\alpha, t)$ can be expressed as:

$$R^{\mathrm{U}}(t) = G_t^{\mathrm{eq}}\left(1 - \alpha/2\right) \tag{3.60}$$

$$R^{\mathrm{D}}(t) = G_t^{\mathrm{eq}}\left(\alpha/2\right) \tag{3.61}$$

where, $G_t^{eq}(\cdot)$ refers to the inverse function of $F_t^{eq}(\cdot)$.

Just like reliability, the power system needs to keep the LOFP lower than a certain value. According to this flexibility requirement, the total system operating reserve can be calculated through the (3.60) and (3.61). If the reserve obtained in the lower level is greater than the maximal reserve capacity that can be provided by the UC scheme, then the solution of the upper level is not feasible. Therefore, the feedback mechanism between the upper and lower levels is necessary. If the obtained reserve is greater than the reserve capacity that can be provided by the UC scheme, then the flexibility constraint (3.43) in the UC model of upper level is to be modified, then re-solve the whole model; or otherwise, output the results.

Meanwhile, it has to be noted that there is no coupling relationship among the time periods in the reserve decision models, so this model can conduct parallel calculation over a different time period. Moreover, the calculation of the universal generating function features recursiveness, so it significantly improves the efficiency and is extremely applicable to large-scale computer analysis.

The flowchart of the proposed model is as shown in Fig. 3.4.

(3.32)–(3.61) are the flexibility-based day-ahead generation-reserve bilevel decision model. The model is solved by the double-layer iterative method, as shown in Fig. 3.4. See Sect. 3.5.2 for the corresponding case studies.

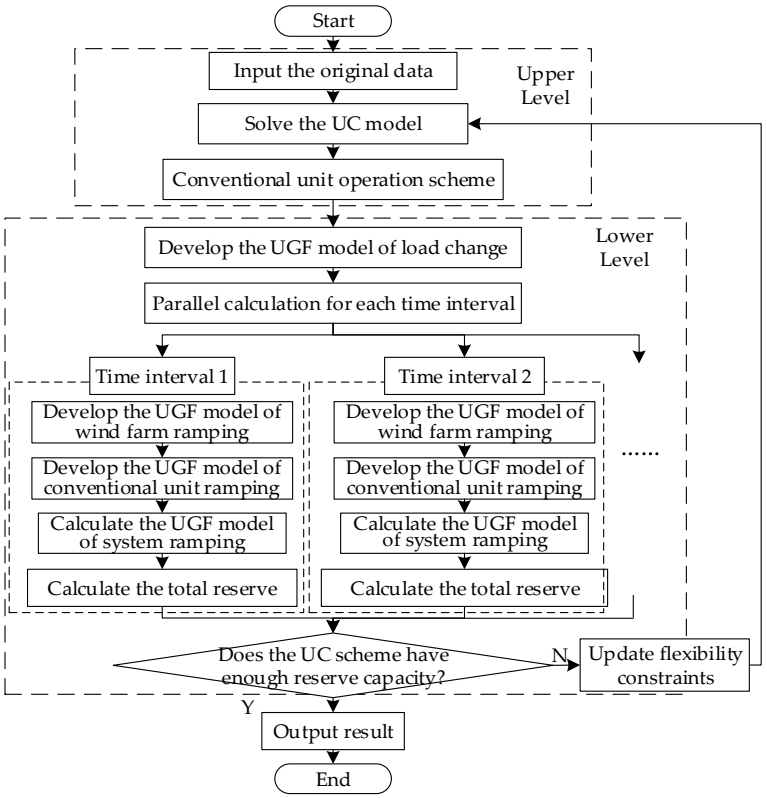


Fig. 3.4 Flowchart of flexibility-based day-ahead generation-reserve bilevel decision model

3.4 An Endogenous Approach to Quantifying the Wind Power Reserve

Due to inherent uncertainty and volatility, large-scale wind power integration requires additional reserve for power balancing. This section addresses this problem with an improved endogenous reserve determination method. Affinely adjustable robust optimization (AARO) is adopted to address wind power and load uncertainties. Then the model is transformed into a two-stage (normal state optimization and fault verification) S&NCED model for iterative solving with the aid of Benders decomposition, and auxiliary constraints are designed to accelerate the convergence.

3.4.1 Dynamic S&NCED Model with AARO

Studies have shown that the determination of the reserve for wind power output and load fluctuations can be separated from that of the contingency reserve for generator "N-1" faults. When establishing an S&NCED model that implements endogenous reserve determination, for different characteristics of multiple uncertainties, different methods are adopted to determine the reserve dealing with the uncertainty of the wind power output (referred to as the "wind power reserve") and the reserve coping with the discrete uncertainty such as generator N − 1 faults (referred to as the "contingency reserve"). The uncertainties of wind power output and load are described by intervals and addressed by the AARO method, the generator participation factors are introduced to realize the corresponding reserve sizing and allocation among generators, and the allocation results can be directly used by automatic generation control (AGC). The contingency reserve for generator failure is determined by establishing fault scenarios and introducing operational constraints after fault.

3.4.1.1 Uncertainty Modeling

In this paper, AARO is used to deal with continuous uncertainties, such as wind power output and load fluctuations. Unlike conventional methods based on the assumption of wind power and load prediction error probability distribution, the AARO method adopts interval representation, and can directly use the confidence interval provided by the prediction method:

$$w_{jt} = w_{jt}^{f} + [-\Delta\varepsilon_{jt}, \Delta\varepsilon_{jt}] \tag{3.62}$$

$$P_{mt} = P_{mt}^{f} + [-\Delta\zeta_{mt}, \Delta\zeta_{mt}] \tag{3.63}$$

where, w_{jt}^{f} is the predicted output of wind farm j at time t. $\Delta\varepsilon_{jt}$ is the maximum power output deviation of wind farm j from w_{jt}^{f} at time t. P_{mt}^{f} is the predicted value of load m at time t. $\Delta\zeta_{mt}$ is the maximum deviation of load m from P_{mt}^{f} at time t.

In order to consider the spatial smoothing effect of wind power fluctuations and control the conservativeness of the AARO method, an uncertainty budget λ is employed, and the uncertain wind power output can be modeled as:

$$w_{jt} = w_{jt}^{f} + \varepsilon_{jt} \tag{3.64}$$

$$\varepsilon_{jt} = \Delta\varepsilon_{jt} z_{wjt}, \quad \left|z_{wjt}\right| \le 1, \quad \sum_{j=1}^{N_{\mathrm{W}}} \left|z_{wjt}\right| \le \lambda \tag{3.65}$$

Similarly, the uncertain load can be modeled as:

$$P_{mt} = P_{mt}^{f} + \zeta_{mt} \tag{3.66}$$

$$\zeta_{mt} = \Delta\zeta_{mt} z_{dmt}, \quad |z_{dmt}| \leq 1 \tag{3.67}$$

Then, these uncertainties are shared by setting the "generator participation factor". During actual operation, when the wind power output and load fluctuate within the uncertain set, the generators can adjust the outputs according to their participation factors to ensure the real-time power balance.

For the generator fault uncertainty, the N − 1 criterion is adopted and the operational constraints after the generator fault are added to the model.

The output of the generator is different under normal conditions and different fault conditions, and to address the wind power and load uncertainties, the generator output is also described as variables with uncertainties:

$$p_{it}=p_{b\,it} - \eta_{it}^{0} \sum_{j=1}^{N_{\mathrm{W}}} \varepsilon_{jt} + \eta_{it}^{0} \sum_{m=1}^{N_{\mathrm{d}}} \zeta_{mt} \tag{3.68}$$

$$p_{it}^{k}=p_{b\,it}^{k} - \eta_{it}^{k} \sum_{j=1}^{N_{\mathrm{W}}} \varepsilon_{jt} + \eta_{it}^{k} \sum_{m=1}^{N_{\mathrm{d}}} \zeta_{mt} \tag{3.69}$$

where, p_{it} and ${p_{it}}^{k}$ are the base outputs of generator i under normal condition and fault condition k at time t. ${\eta_{it}}^{0}$ and ${\eta_{it}}^{k}$ are the participation factors of generator i under normal condition and fault condition k at time t.

3.4.1.2 Dynamic S&NCED Model

We focuse on synchronized reserves, specifically up-spinning and down-spinning. The optimization goal of the proposed dynamic S&NCED model is to minimize the overall operating costs, including the generation costs and reserve costs:

$$\min \sum_{t=1}^{T} \{ \sum_{i=1}^{N_{\mathrm{G}}} (a_i (p_{b\,it})^2 + b_i p_{b\,it} + c_i) + \sum_{i=1}^{N_{\mathrm{G}}} [c_i^{u} (r_{ait}^{U} + r_{mit}^{U}) + c_i^{d} (r_{ait}^{D} + r_{mit}^{D})] \} \tag{3.70}$$

Constraints of the dynamic S&NCED model are:

$$\sum_{i=1}^{N_{\mathrm{G}}} p_{b\,it} + \sum_{j=1}^{N_{\mathrm{W}}} w_{jt}^{f} = \sum_{m=1}^{N_{\mathrm{d}}} P_{mt}^{f} \quad \forall t \tag{3.71}$$

$$P_i^{\min} \leq p_{it} \leq P_i^{\max} \quad \forall i, \forall t \tag{3.72}$$

$$-F_l \le \sum_n DF_{nl}(\sum_i I_{ni}^1 p_{bit} + \sum_j I_{nj}^2 w_{jt} - \sum_m I_{nm}^3 P_{mt}) \le F_l \quad \forall l, \forall t \tag{3.73}$$

$$-RD_i \Delta T \le p_{bi(t+1)} - p_{bit} \le RU_i \Delta T \quad \forall i, t = 1, 2, \cdots, T-1 \tag{3.74}$$

$$r_{ait}^U + r_{mit}^U \le RU_i T_r \quad \forall i, \forall t \tag{3.75}$$

$$r_{ait}^D + r_{mit}^D \le RD_i T_r \quad \forall i, \forall t \tag{3.76}$$

$$p_{gbit}^0 + r_{ait}^U + r_{mit}^U \le P_{Gi}^{Max} \quad \forall i, \forall t \tag{3.77}$$

$$P_{Gi}^{min} \le p_{gbit}^0 - (r_{ait}^D + r_{mit}^D) \quad \forall i, \forall t \tag{3.78}$$

$$r_{ait}^U \ge \eta_{it}^0 \varepsilon_{\lambda t}^{\max} + \eta_{it}^0 \sum_{m=1}^{ND} \Delta\zeta_{mt} \quad \forall i, \forall t \tag{3.79}$$

$$r_{ait}^D \ge \eta_{it}^0 \varepsilon_{\lambda t}^{\max} + \eta_{it}^0 \sum_{m=1}^{ND} \Delta\zeta_{mt} \quad \forall i, \forall t \tag{3.80}$$

$$\sum_{i=1}^{N_G} p_{bit}^k + \sum_{j=1}^{N_W} w_{jt}^f = \sum_{m=1}^{N_d} P_{mt} \quad \forall k, \forall t \tag{3.81}$$

$$P_i^{\min} \le p_{git}^k \le P_i^{\max} \quad \forall i, \forall k, \forall t \tag{3.82}$$

$$-F_l \le \sum_n DF_{nl}(\sum_i I_{ni}^1 p_{git}^k + \sum_j I_{nj}^2 w_{jt} - \sum_m I_{nm}^3 P_{mt}) \le F_l \quad \forall i, \forall k, \forall t \tag{3.83}$$

$$r_{ait}^U \ge \eta_{it}^k \varepsilon_{\lambda t}^{\max} + \eta_{it}^k \sum_{m=1}^{ND} \Delta\zeta_{mt} \quad \forall i, \forall k, \forall t \tag{3.84}$$

$$r_{mit}^U \ge p_{bit}^k - p_{bit} \quad \forall i, \forall k, \forall t \tag{3.85}$$

$$r_{ait}^D \ge \eta_{it}^k \varepsilon_{\lambda t}^{\max} + \eta_{it}^k \sum_{m=1}^{ND} \Delta\zeta_{mt} \quad \forall i, \forall k, \forall t \tag{3.86}$$

$$r_{mit}^D \ge p_{bit} - p_{bit}^k \quad \forall i, \forall k, \forall t \tag{3.87}$$

where, π_{ln} is the power transfer distribution factor of bus n to line l. $I_{ni}{}^1$ is the relevance parameter of generator i to bus n. $I_{nj}{}^2$ is the relevance parameter of wind

farm j to bus n. $I_{nm}{}^{3}$ is the relevance parameter of load m to bus n. $r_{ait}{}^{U}$ and $r_{ait}{}^{D}$ are the upward and downward wind power and load reserves provided by generator i at time t. $r_{mit}{}^{U}$ and $r_{mit}{}^{D}$ are the upward and downward contingency reserves provided by generator i at time t.

(3.71) and (3.81) are the power balance constraints under normal conditions and fault conditions, respectively. (3.72) and (3.82) are the generator output constraints under normal conditions and fault conditions, respectively, and (3.74) is the ramping rate constraint. (3.75) and (3.76) indicate that the reserve capacity cannot exceed the ramping rate of the generator, and (3.77) and (3.78) are generation capacity constraints considering the reserve. (3.85) and (3.87) are the upward and downward contingency reserve constraints, respectively. (3.73) and (3.83) are network constraints under normal conditions and fault conditions, respectively, where the variables contain uncertain intervals, so they will be satisfied under all the actual probable uncertain conditions of wind power and load. (3.79), (3.80), (3.84) and (3.86) are the upward and downward wind power reserve and load reserve demand constraints under normal conditions and fault conditions, respectively, where $\varepsilon_{jt}{}^{\max}$ can be calculated by the following optimization problem:

$$\varepsilon_{\lambda t}^{\max} = \max \sum_{j=1}^{N_{\mathrm{W}}} \left| \Delta\varepsilon_{jt} z_{wjt} \right| \tag{3.88}$$

$$\left| z_{wjt} \right| \le 1, \sum_{j=1}^{NW} \left| z_{wjt} \right| \le \lambda \tag{3.89}$$

The above S&NCED model adopts AARO to address load and wind power output fluctuations, reducing the number of scenarios that the stochastic optimization requires to establish in the face of multiple uncertainties.

3.4.2 Two-Stage Solution Method Based on the Benders Decomposition

3.4.2.1 The Benders Decomposition Method

If the unified solution method is adopted, that is, the proposed model is solved with all constraints of the normal conditions and the fault conditions, then it is a heavy computation burden. Different from the unified solution method, the iterative solution method generates constraints that affect the results in the fault verification process and achieves iterative solving. Identifying and generating the effective constraints that impact the results is the key to the iterative solution method. In this work, the Benders decomposition method is used to realize the extraction and feedback of key constraints. The model established in above section is split into the master problem of

the normal condition and the subproblems of the fault conditions. The optimization values of the first stage variables obtained by solving the master problem are substituted into the subproblems, and the Benders feasibility cuts established in solving the subproblems are added to the master problem constraints. The master problem and subproblems interact, and the model is iteratively optimized. The variables of the master problem are $\boldsymbol{X}^f = \{p_{bit}, \eta_{it}{}^0, r_{ait}{}^U, r_{mit}{}^U, r_{ait}{}^D, r_{mit}{}^D\}$. The constraints of the master problem consist of the Benders feasibility cuts returned by the subproblems and the operational constraints of the normal condition, i.e., the master problem MP1 is as follows:

(MP1)

$$\min_{\boldsymbol{X}^f} \sum_{t=1}^{T} \{\sum_{i=1}^{N_G} (a_i (p_{b\,it})^2 + b_i p_{b\,it} + c_i) + \sum_{i=1}^{N_G} [c_i^u (r_{ait}^U + r_{mit}^U) + c_i^d (r_{ait}^D + r_{mit}^D)]\} \tag{3.90}$$

$$\begin{gathered} (3.71) \text{ to } (3.80) \\ \text{Benders feasibility cuts} \end{gathered} \tag{3.91}$$

After solving the master problem, the optimization values of the first stage variables $\boldsymbol{X}^{f*} = \{p_{bit}{}^*, \eta_{it}{}^{0*}, r_{ait}{}^{U*}, r_{mit}{}^{U*}, r_{ait}{}^{D*}, r_{mit}{}^{D*}\}$ are substituted into the subproblems to verify whether the first-stage variable optimization values obtained by solving the master problem satisfy the generator output and network constraints after the fault. $p_{bit}{}^k$ and $\eta_{it}{}^k$ are the variables of the subproblems. For coupling constraints (that is, the subproblem constraints containing variables of the master problem), relaxation variables $v_{1it}{}^k$, $v_{2it}{}^k$, $v_{3it}{}^k$, and $v_{4it}{}^k$ are introduced in (3.84)–(3.87) to verify the feasibility of the dispatch scheme under fault conditions. The variables of the subproblems are $\boldsymbol{X}^s = \{p_{bit}{}^k, \eta_{it}{}^k, v_{1it}{}^k, v_{2it}{}^k, v_{3it}{}^k, v_{4it}{}^k\}$, and the subproblems are as follows:

(SP)

$$f_t^k = \min_{\boldsymbol{X}^s} \sum_{i=1}^{N_G} (v_{1it}^k + v_{2it}^k + v_{3it}^k + v_{4it}^k) \tag{3.92}$$

$$\eta_{it}^k \varepsilon_{\lambda t}^{\max} + \eta_{it}^k \sum_{m=1}^{ND} \Delta\zeta_{mt} - r_{ait}^{U*} - v_{1it}^k \le 0 \tag{3.93}$$

$$p_{bit}^k - p_{bit}^* - r_{mit}^{U*} - v_{2it}^k \le 0 \tag{3.94}$$

$$\eta_{it}^k \varepsilon_{\lambda t}^{\max} + \eta_{it}^k \sum_{m=1}^{N_d} \Delta\zeta_{mt} - r_{ait}^{D*} - v_{3it}^k \le 0 \tag{3.95}$$

$$p_{b\,it}^* - p_{b\,it}^k - r_{mit}^{D*} - v_{4it}^k \le 0 \tag{3.96}$$

$$v_{1it}^{k}, v_{2it}^{k}, v_{3it}^{k}, v_{4it}^{k} \geq 0 \tag{3.97}$$

If ${f_t}^k > 0$, it means that the $\boldsymbol{X}^{f*}$ obtained by the master problem does not have a feasible dispatch scheme under fault condition k at time t. Then the Benders feasibility cut needs to be generated and added into the master problem to adjust the variables of the master problem to be within the feasible domain; then the Benders feasibility cut is generated as follows:

$$\begin{aligned} &f_t^k + \sum_{i=1}^{N_G} \lambda_{it}^k (p_{b\,it} - p_{b\,it}^*) + \sum_{i=1}^{N_G} \mu_{ait}^{Uk} (r_{ait}^U - r_{ait}^{U*}) + \\ &\sum_{i=1}^{N_G} \mu_{mit}^{Uk} (r_{mit}^U - r_{mit}^{U*}) + \sum_{i=1}^{N_G} \mu_{ait}^{Dk} (r_{ait}^D - r_{ait}^{D*}) + \sum_{i=1}^{N_G} \mu_{mit}^{Dk} (r_{mit}^D - r_{mit}^{D*}) \leq 0 \end{aligned} \tag{3.98}$$

where, ${\lambda_{it}}^k$, ${\mu_{ait}}^{Uk}$, ${\mu_{mit}}^{Uk}$, ${\mu_{ait}}^{Dk}$, and ${\mu_{mit}}^{Dk}$ are the corresponding multipliers of ${p_{gbit}}^0$, ${r_{ait}}^U$, ${r_{mit}}^U$, ${r_{ait}}^D$ and ${r_{mit}}^D$ in (3.93)–(3.96), respectively.

The S&NCED model established in above section is now transformed into a two-stage S&NCED model (master problem MP1 and subproblem SP) that can be solved iteratively. To accelerate the convergence, auxiliary constraints are designed in the master problem to constrain the total reserve capacity considering the generator N − 1 faults. These constraints are implicit in the model when the unified solution is adopted. Adding these constraints during the iterative solution can improve the success rate of fault verification and reduce the number of iterations:

$$\sum_{i=1, i\neq u}^{N_G} r_{mit}^U \geq p_{b\,ut} \ \forall u, \forall t \tag{3.99}$$

$$\sum_{i=1, i\neq u}^{N_G} r_{ait}^U \geq \varepsilon_{\lambda t}^{\max} + \sum_{m=1}^{N_d} \Delta\zeta_{mt} \ \ \forall u, \forall t \tag{3.100}$$

$$\sum_{i=1, i\neq u}^{N_G} r_{ait}^D \geq \varepsilon_{\lambda t}^{\max} + \sum_{m=1}^{N_d} \Delta\zeta_{mt} \ \ \forall u, \forall t \tag{3.101}$$

Then the master problem MP with auxiliary constraints is:

(MP)

$$\min_{\boldsymbol{X}^f} \sum_{t=1}^{T} \{ \sum_{i=1}^{N_G} (a_i (p_{b\,it})^2 + b_i p_{b\,it} + c_i) + \sum_{i=1}^{N_G} [c_i^u (r_{ait}^U + r_{mit}^U) + c_i^d (r_{ait}^D + r_{mit}^D)] \} \tag{3.102}$$

(3.71) to (3.80)

(3.99) to (3.101)

Benders feasibility cuts (3.103)

3.4.2.2 Solution Strategy

The iterative solution process of the two-stage S&NCED is:

(1) Solve the master problem MP and obtain the optimization values of the first stage variables $\boldsymbol{X}^{f*}$.
(2) Verify each fault condition k at each time t. Substitute $\boldsymbol{X}^{f*}$ into the subproblem SP for a solution; if ${f_t}^k > 0$, generate a Benders feasibility cut.
(3) If all subproblems yield ${f_t}^k = 0$, then $\boldsymbol{X}^{f*}$ constitute the S&NCED optimal solution; then terminate the program. Otherwise, go to the next step.
(4) Solve the master problem MP by adding all the Benders feasibility cuts generated in step (2), obtain the optimization values of the first stage variables $\boldsymbol{X}^{f*}$, and return to step (2).

The scale of the master problem and subproblems maintains small for the iterative method, which reduces the complexity of the solution, and there is no coupling between the subproblems. Therefore, distributed computation can be used to solve the subproblems in parallel, which can greatly shorten the calculation time.

Whether using a unified or an iterative solution method, the S&NCED model contains uncertain constraints. To facilitate the solution, transformation is required. The uncertain quantities ε_{jt} and ζ_{mt} are directly or indirectly included in (3.72), (3.73), (3.82) and (3.83), and deterministic transformation is required. For simplicity, the superscript k and the subscript t are omitted, and the variables in (3.64) and (3.66) are converted into matrix form and substituted into (3.72), (3.73), (3.82) and (3.83):

$$\boldsymbol{P}_{\boldsymbol{G}}^{\min} \le \boldsymbol{P}_{\boldsymbol{GB}} - diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{1}}\boldsymbol{\varepsilon} + diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{2}}\boldsymbol{\zeta} \le \boldsymbol{P}_{\boldsymbol{G}}^{\max} \tag{3.104}$$

$$\begin{aligned} -\boldsymbol{F} \le \boldsymbol{\pi}(\boldsymbol{I}^{\mathbf{2}}(\boldsymbol{W}_{\boldsymbol{F}} + \boldsymbol{\varepsilon}) - \boldsymbol{I}^{\mathbf{3}}(\boldsymbol{D}_{\boldsymbol{F}} + \boldsymbol{\zeta}) + \\ \boldsymbol{I}^{\mathbf{1}}(\boldsymbol{P}_{\boldsymbol{GB}} - diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{1}}\boldsymbol{\varepsilon} + diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{2}}\boldsymbol{\zeta})) \le \boldsymbol{F} \end{aligned} \tag{3.105}$$

where $\mathbf{1}^{\mathbf{1}}$ is an $NG \times NW$ matrix with all elements equal to 1 and $\mathbf{1}^{\mathbf{2}}$ is an $NG \times ND$ matrix with all elements equal to 1.

Set:

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{I} \\ -\boldsymbol{I} \\ \boldsymbol{\pi}\boldsymbol{I}^{\mathbf{1}} \\ -\boldsymbol{\pi}\boldsymbol{I}^{\mathbf{1}} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{1}} \\ diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{1}} \\ -\boldsymbol{\pi}\boldsymbol{I}^{\mathbf{1}}diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{1}} + \boldsymbol{\pi}\boldsymbol{I}^{\mathbf{2}} \\ \boldsymbol{\pi}\boldsymbol{I}^{\mathbf{1}}diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{1}} - \boldsymbol{\pi}\boldsymbol{I}^{\mathbf{2}} \end{bmatrix} \tag{3.106}$$

$$\boldsymbol{C}=\begin{bmatrix} diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{2}} \\ -diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{2}} \\ \boldsymbol{\pi}\boldsymbol{I}^{\mathbf{1}}diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{2}}-\boldsymbol{\pi}\boldsymbol{I}^{\mathbf{3}} \\ -\boldsymbol{\pi}\boldsymbol{I}^{\mathbf{1}}diag(\boldsymbol{\eta})\mathbf{1}^{\mathbf{2}}+\boldsymbol{\pi}\boldsymbol{I}^{\mathbf{3}} \end{bmatrix},\ \boldsymbol{e}=\begin{bmatrix} \boldsymbol{P}_{\boldsymbol{G}}^{\boldsymbol{max}} \\ -\boldsymbol{P}_{\boldsymbol{G}}^{\boldsymbol{min}} \\ \boldsymbol{F}-\boldsymbol{\pi}\boldsymbol{I}^{\mathbf{2}}\boldsymbol{W}_{\boldsymbol{F}}+\boldsymbol{\pi}\boldsymbol{I}^{\mathbf{3}}\boldsymbol{D}_{\boldsymbol{F}} \\ \boldsymbol{F}+\boldsymbol{\pi}\boldsymbol{I}^{\mathbf{2}}\boldsymbol{W}_{\boldsymbol{F}}-\boldsymbol{\pi}\boldsymbol{I}^{\mathbf{3}}\boldsymbol{D}_{\boldsymbol{F}} \end{bmatrix} \tag{3.107}$$

where $\boldsymbol{I}$ is an NG-order unit matrix.

Therefore, (3.104) and (3.105) can be expressed as:

$$\boldsymbol{A}\boldsymbol{P}_{\boldsymbol{GB}}+\boldsymbol{B}\boldsymbol{\varepsilon}+\boldsymbol{C}\boldsymbol{\zeta}\leq\boldsymbol{e} \tag{3.108}$$

For the qth row constraints, we find:

$$\boldsymbol{A}_{\boldsymbol{q}}\boldsymbol{P}_{\boldsymbol{GB}}-e_q+\max\boldsymbol{B}_{\boldsymbol{q}}\boldsymbol{\varepsilon}+\max\boldsymbol{C}_{\boldsymbol{q}}\boldsymbol{\zeta}\leq 0 \tag{3.109}$$

Here max $\boldsymbol{B}_{\boldsymbol{q}}\boldsymbol{\varepsilon}$ can be expressed as:

$$\max\sum_{j=1}^{N_{\mathrm{W}}}B_{qj}\Delta\varepsilon_j(z_{wj}^{+}-z_{wj}^{-}) \tag{3.110}$$

$$0\leq z_{wj}^{+}\leq 1,0\leq z_{wj}^{-}\leq 1,z_{wj}^{+}+z_{wj}^{-}\leq 1\ \ \forall j \tag{3.111}$$

$$\sum_{j=1}^{NW}z_{wj}^{+}+z_{wj}^{-}\leq\lambda \tag{3.112}$$

Similarly, max $\boldsymbol{C}_{\boldsymbol{q}}\boldsymbol{\zeta}$ can be expressed as:

$$\max\sum_{m=1}^{ND}C_{qm}\Delta\zeta_m(z_{dm}^{+}-z_{dm}^{-}) \tag{3.113}$$

$$0\leq z_{dm}^{+}\leq 1,0\leq z_{dm}^{-}\leq 1,z_{dm}^{+}+z_{dm}^{-}\leq 1\ \ \forall m \tag{3.114}$$

Taking the dual of (3.110)–(3.112) and (3.113)–(3.114), (3.109) is transformed into:

$$\begin{aligned}&\sum_{i=1}^{NG}A_{qi}p_{gbi}-e_q+\sum_{j=1}^{NW}(\theta_{wqj}^{+}+\theta_{wqj}^{-}+\theta_{wqj}^{1})+\lambda\theta_{wq}^{\lambda}\\&+\sum_{m=1}^{ND}(\theta_{dqm}^{+}+\theta_{dqm}^{-}+\theta_{dqm}^{1})\leq 0\quad\forall q\end{aligned} \tag{3.115}$$

$$\theta_{wqj}^{+}+\theta_{wqj}^{1}+\theta_{wq}^{\lambda}\geq B_{qj}\Delta\varepsilon_j\ \ \forall q,\forall j \tag{3.116}$$

$$\theta_{wqj}^{-} + \theta_{wqj}^{1} + \theta_{wq}^{\lambda} \geq -B_{qj}\Delta\varepsilon_j \quad \forall q, \forall j \tag{3.117}$$

$$\theta_{dqm}^{+} + \theta_{dqm}^{1} \geq C_{qm}\Delta\zeta_m \quad \forall q, \forall m \tag{3.118}$$

$$\theta_{dqm}^{-} + \theta_{dqm}^{1} \geq -C_{qm}\Delta\zeta_m \quad \forall q, \forall m \tag{3.119}$$

$$\theta_{wqj}^{+} \geq 0, \theta_{wqj}^{-} \geq 0, \theta_{wqj}^{1} \geq 0 \quad \forall q, \forall j \quad \theta_{wq}^{\lambda} \geq 0 \quad \forall q \tag{3.120}$$

$$\theta_{dqm}^{+} \geq 0, \theta_{dqm}^{-} \geq 0, \theta_{dqm}^{1} \geq 0 \quad \forall q, \forall m \tag{3.121}$$

After the above transformation, the uncertainties in the constraints are effectively processed, the uncertain constraints are transformed into the deterministic ones, and the S&NCED model is transformed into a deterministic model that can be solved by a mature solver.

Finally, it should be noted that the adjustment mode of adjustable variables is different in AARO and RO. AARO assumes that adjustable variables are affinely related to the uncertainties, while RO is not limited to affine adjustment, adopts fully adjustable mode, in this aspect, AARO is relatively more restrictive than RO. However, it was suggested that the increased conservativeness of AARO is much less pronounced for larger networks and the use of AARO for computing the optimal participation factors is directly compatible with AGC systems.

The proposed model is solved by the Benders Decomposition method introduced above. See Sect. 3.5.3 for the corresponding case studies in this section.

3.5 Case Studies

3.5.1 Case Studies of the Flexible Look-Ahead Unit Commitment

3.5.1.1 Data Description

The optimization horizon and time resolution of flexible LAUC are set to 4 h and 15 min, respectively. Before running the flexible LAUC, a deterministic day ahead UC is executed using the day ahead wind power prediction and load prediction to decide the commitment schedule for the 24 h horizon. The constraints, such as the generator output and ramp rate limit, are included in this stage. During the stage of LAUC, the ON/OFF status and scheduled outputs of non-AGC units are decided. However, the outputs of the AGC units allow to be different from the day ahead schedule, and the status of fast-start units can be changed, which are assumed to

complete their startup and shutdown within 15 min. If higher accuracy is required, the piecewise linear approximation method can also be used. Modified IEEE 30-bus system contains twenty loads and seven units, including two wind farms with installed capacity of 60 and 70 MW linked to Bus 13 and Bus 15. The installed VRE capacity accounts for 30.23% of the total installed generation capacity.

The load profile and wind power data used in this study are obtained from actual data of Gansu Province, China, and scaled down to suit this test system. This study assumes that there is no deviation in the load forecast, while the maximum forecast error of wind power is set to 10% of the forecast. For penalty cost evaluation stage, we assume that the probability density function of wind power on a connection point follows a Gaussian distribution, and 200 wind power scenarios are generated by Latin hypercube sampling, which are reduced to 10 scenarios using the improved K-means clustering technique for decision making.

3.5.1.2 Results of the Flexible Look-Ahead Unit Commitment

(1) Unit commitment results comparison

In this study, it is assumed that the wind curtailment penalty for all VRE buses is \$200/MWh, and the load shedding penalty for all loads is \$5000/MWh. Actually, these prices are decided based on the historical data or the long-term electricity contract, which reflect the risk attitude of the system operator. Table 3.1 shows the economy comparison among LAUC by DNE, the ordinary LAUC model considering FRC requirement (denoted by FRC-LAUC) and the two proposed approaches in Sect. 3.2.2.1 (denoted by IO-LAUC) and Sect. 3.2.2.2 (denoted by SO-LAUC).

It is worth to mention that above costs do not include the operating cost of non-AGC unit because it is the same in above four scheduling approaches. Among these four approaches, the scheduling results of units G1–G4 are the same, and they are always online. The status of the fast-start unit G5 are different. Table 3.2 shows the unit commitment results of different scheduling approaches.

In DNE-LAUC and FRC-LAUC, the fast-start unit G5 is always offline, and only the spinning reserve of the online units are called upon to meet the FRC requirement, so the ADWP is small. Due to the large upward fluctuations of the net load in the 5th and 9th intervals, part of VRE buses uncertainty set that cannot be accommodated by these two approaches is relatively large, resulting in more penalty cost.

Table 3.1 Economy performance comparison of scheduling approaches

	Overall cost (\$)	Operating cost (\$)	Penalty cost (\$)
DNE-LAUC	9326.5	9182.1	144.4
FRC-LAUC	9318.1	9182.7	135.4
IO-LAUC	9491.5	9476.6	14.9
SO-LAUC	9258.6	9243.7	14.9

Table 3.2 Unit commitment results comparison

Interval	Unit commitment (DNE-LAUC/FRC-LAUC/IO-LAUC/SO-LAUC)				
	G1	G2	G3	G4	G5
1st–4th	1/1/1/1	1/1/1/1	1/1/1/1	1/1/1/1	0/0/0/0
5th	1/1/1/1	1/1/1/1	1/1/1/1	1/1/1/1	0/0/1/0
6th	1/1/1/1	1/1/1/1	1/1/1/1	1/1/1/1	0/0/1/1
7th	1/1/1/1	1/1/1/1	1/1/1/1	1/1/1/1	0/0/1/1
8th–9th	1/1/1/1	1/1/1/1	1/1/1/1	1/1/1/1	0/0/0/0
10th	1/1/1/1	1/1/1/1	1/1/1/1	1/1/1/1	0/0/1/1
11th	1/1/1/1	1/1/1/1	1/1/1/1	1/1/1/1	0/0/1/1
12th–16th	1/1/1/1	1/1/1/1	1/1/1/1	1/1/1/1	0/0/0/0

In order to obtain a sufficiently large ADWP, for IO-LAUC, the unit G5 starts up and shuts down twice. During the 5th–7th intervals, due to the existence of G5, the system has a steeper ramping rate, which reduces the wind curtailment in the 5th interval. In the 9th interval, G5 does not directly provide flexibility regulation, but share the load of other units to alleviate the ramping capacity requirements of them through operating in the 10th interval, thereby leaving G2-G4 more downward ramping capacities in the 9th interval mediately, so that the ADWP of IO-LAUC in the 9th interval is expanded. In this case, IO-LAUC significantly reduces the penalty cost compared with FRC-LAUC, but the increased operating cost results in a higher overall cost.

Compared with IO-LAUC, SO-LAUC does not arrange the online operation of the G5 in the 5th interval. Figure 3.5 shows the contribution of each unit to overall FRC that the system can provide in IO-LAUC and SO-LAUC approaches, and we can infer that SO-LAUC approach reduce the reserved upward/downward FRC in most interval, thereby decreasing the operating cost. However, the risk cost has not increased. This is because the SO-LAUC is based on scenarios. Although the ADWP has shrunk, it can still completely envelop most wind power scenarios, the rest of those occurring curtailment have lower probability of occurrences. Therefore, the overall cost of SO-LAUC is the lowest.

(2) ADWP profile

Since the FRC requirement is not considered in DNE-LAUC, for simplicity, we just compare three scheduling approaches FRC-LAUC, IO-LAUC and SO-LAUC, which is enough for illustration. Figure 3.6 shows the $\mathrm{ADWP}_{\mathrm{V/U}}$ and $\mathrm{ADWP}_{\mathrm{V}}$ of three scheduling approaches in the 5th and 9th intervals, because these two intervals are the most representative.

In Fig. 3.6, the largest dashed box indicates the uncertainty set of VRE buses $\mathbb{U}^M$, and the gray area in Fig. 3.6a, b indicate the injection power of VRE buses that cannot be accommodated due to the limitation of the transmission. By comparing Fig. 3.6a–d, the following observations can be obtained.

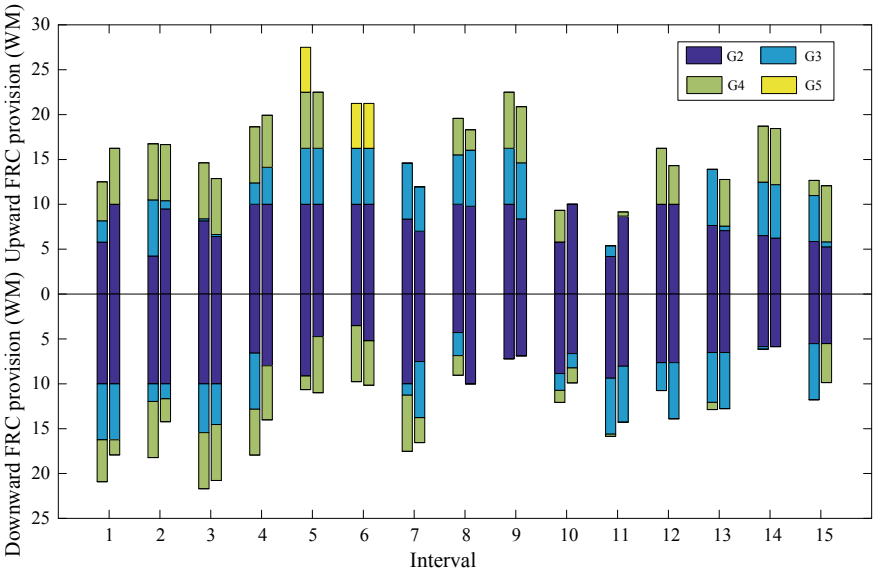


Fig. 3.5 Maximum FRC provision in IO-LAUC and SO-LAUC. Left: IO-LAUC. Right: SO-LAUC

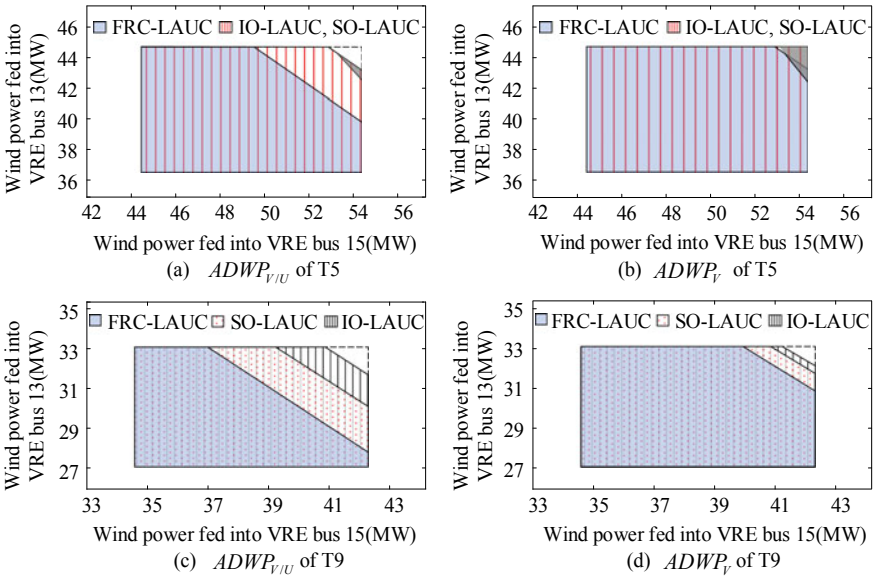


Fig. 3.6 ADWP profiles of different LAUC approaches

(1) In the 5th interval, the $ADWP_V$ and $ADWP_{V/U}$ in IO-LAUC and SO-LAUC are the same. The coverage rate of the $ADWP_{V/U}$ to $\mathbb{U}^M$ is 98.38% and the coverage rate of $ADWP_V$ to $\mathbb{U}^M$ is 98.39%, which indicates that under these two scheduling approaches, wind curtailment will hardly occur in the 5th interval. Considering that the economy of SO-LAUC is better than IO-LAUC, the system operator will prefer to choose SO-LAUC.
(2) In the 5th interval, the coverage rate of $ADWP_{V/U}$ to $\mathbb{U}^M$ in FRC-LAUC is 86.30%, and the coverage rate of $ADWP_V$ to $\mathbb{U}^M$ is 98.38%, which indicates that when the operator does not require the system to be able to fully cope with the uncertainty of the next period, the FRC-LAUC can still meet most wind power scenarios.

(3) Because the 5th interval is the peak load period, when the injection power of VRE buses is too high, there is a congestion on the branch connected to VRE bus 15, which results in wind curtailment. However, there is still a part of flexibility potential in the generation side not be utilized. Therefore, operators can fully utilize the system flexible resources by increasing the capacity of this transmission line.
(4) In the 9th interval, both $ADWP_{V/U}$ and $ADWP_V$ in IO-LAUC are larger than that in SO-LAUC, and the advantage of IO-LAUC is more obvious when the considering ADWP is more conservative (i.e. $ADWP_{V/U}$). The coverage rate of $ADWP_{V/U}$ to $\mathbb{U}^M$ in IO-LAUC is 7.71% higher than that of SO-LAUC.
(5) In the 9th interval, the $ADWP_{V/U}$ of the three scheduling approaches have large difference, and the coverage rate of them to $\mathbb{U}^M$ are 70.52, 90.19, and 97.90%, respectively. Therefore, in the actual operation of FRC-LAUC, if excessive wind power is scheduled in the 9th interval, the probability of insufficient ramping up capacity in the 10th interval will be higher.

(3) Impact of the system parameters

(1) *Impact of ramp rates of AGC units*: Considering that ADWP is greatly affected by the ramp rates of the units, the change of ADWP and the overall cost under different ramp rates of unit G2 and unit G3 are analyzed. In order to better capture the change relationship between them, the power flow limitation is ignored in the calculation. As illustrated in Fig. 3.7, when the ramp rate of G3 increases from 24 to 32 MW/h, ADWP increases by 1.42%, and the growth rate gradually decreases until a stable value, while the overall cost decreases by 1.11%. That is because the improvement in AGC unit's ramping capability avoids calling more expensive flexible resources (i.e. G5). Due to the power flow limitation is ignored, when the same ramp rate increment is added to G2, the change in ADWP is the same, but the overall cost is reduced by 1.25%, which is more economical than changing G3 because c^u/c^d of G2 is lower. Thus, operators can improve the system flexibility by enhancing the ramping capability of the AGC unit, which can reduce operating costs to a certain extent at the same time, and the lower the c^u/c^d of that unit, the more the overall cost will be reduced.

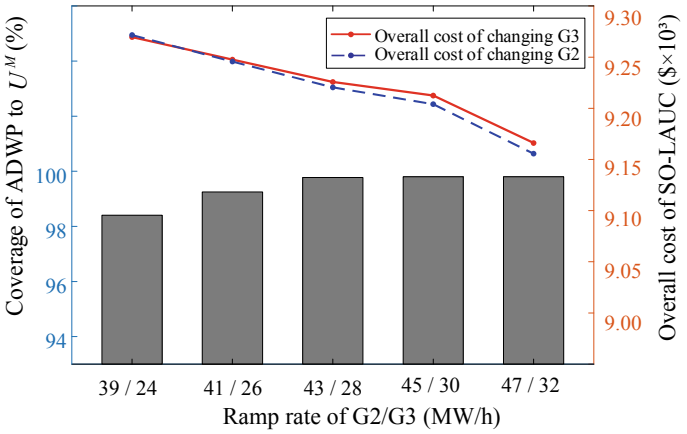


Fig. 3.7 Results under different ramp rates of G2 and G3

(2) *Sensitivity analysis of the adjustment coefficient*: In order to illustrate the effect of introducing the adjustment coefficient λ on controlling the conservativeness of the proposed approaches, the relationship between the overall cost or penalty cost and λ in IO-LAUC are studied, as shown in Fig. 3.8.

It can be seen from Fig. 3.8 that λ has an obvious impact on the penalty cost. With the increase of λ, the FRC requirement and the amount of wind curtailment will also increase, but the overall cost will not change much, that is because the reduced cost of flexible regulation capacity and the increased penalty cost offset each other. However, when λ is greater than 0.88, the overall cost will increase significantly because the FRC requirement calls for the fast unit running for an extra time period. Through

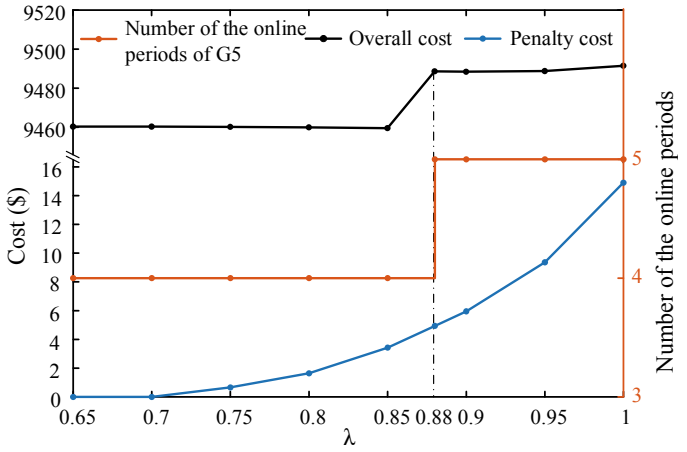


Fig. 3.8 The relationship between the overall cost or penalty cost and λ

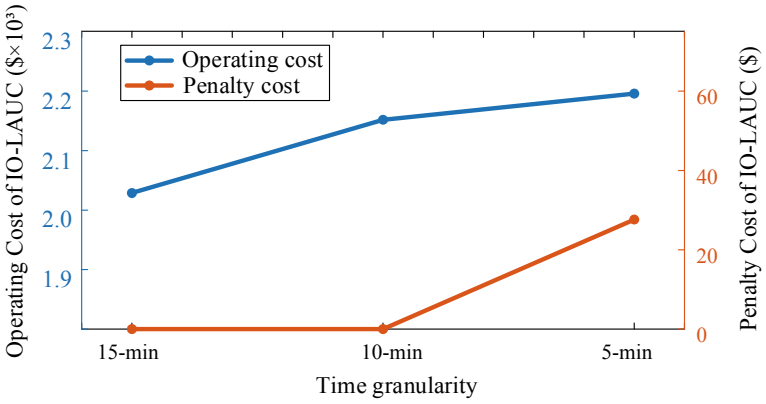


Fig. 3.9 Scheduling results in different time granularity

curve fitting, we determine the quantitative relationship between λ and penalty cost (i.e. z_3): $z_3 = 180.6\lambda^2 - 259.6\lambda + 93.5(\lambda > 0.7)$. Reducing λ can reduce the amount of wind curtailment, but it will increase the risk of load shedding at the next time due to insufficient upward FRC. Therefore, the choice of λ could be used to reflect the risk attitude of the system operator. How to strike a balance between the risk and overall cost is an interesting future research direction.

(4) Impact of time granularity

In different time resolution, which refers to the granularity, the fluctuation characteristics of wind power are diverse, and at the same time, the response abilities of flexible resources to the fluctuation are also different. Thus, the assessment of flexibility is dependent on the granularity. In 1 h time horizon, the scheduling results of IO-LAUC under 5 min-interval, 10 min-interval, 15 min-interval are presented in Fig. 3.7.

As Fig. 3.9 shows, when the granularity varies from 15 to 10 min, the operating cost increases by 6.07%; when the granularity varies from 15 to 5 min, the operating cost raises by 8.24% and the penalty cost raises to $27.63. Therefore, when the time granularity become more refined, the system has to call more flexible resources to meet the steeper ramping requirements, and the probability of wind curtailment/load shedding will also increase.

3.5.2 Case Studies of the Day-Ahead Generation-Reserve Bilevel Decision Model

The proposed unit commitment model in the upper level is converted into a mixed integer linear planning problem, and it is solved by the CPLEX solver on Matlab using the Yalmip toolbox. The parallel computing programming of the reserve decision in

the lower level is conducted in the Matlab. The test is carried out on the computer with CPU i7 3.5 GHz and 16G RAM.

A modified IEEE-118 bus test system is used in case studies, which contained 54 conventional units, and the 24 h load curve is as shown in Fig. 3.10. In addition, two wind farms are added to the IEEE-118 system with the capacities of 200 and 150 MW. The wind farms are modeled by the historical data of two wind farms in northwest China, and the historical wind output data of 24 h as showed in Fig. 3.11 is used as the predicted value.

(2) Flexibility based Reserve Decision Method

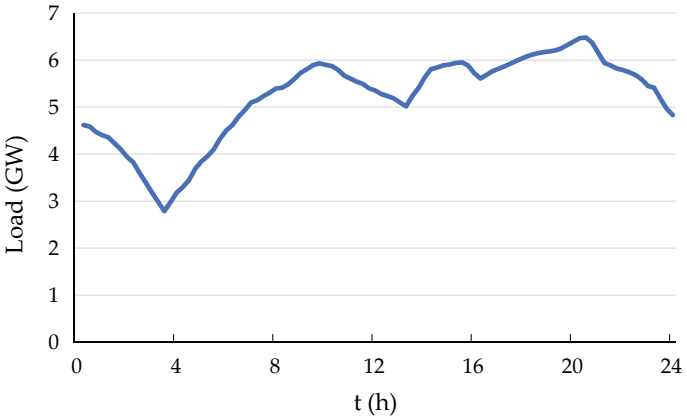


Fig. 3.10 Total load curve of the IEEE-118 bus system

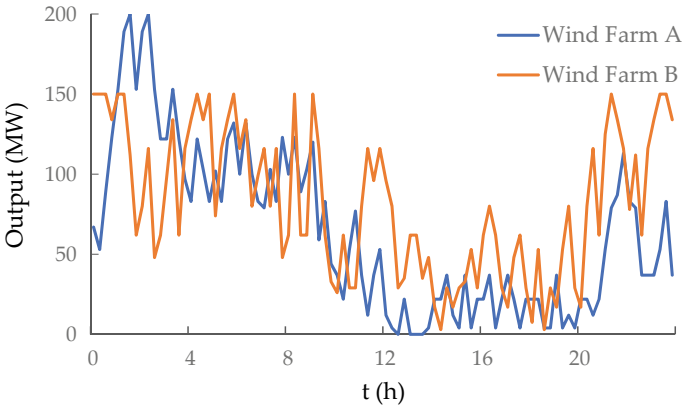


Fig. 3.11 Output data of two wind farms in northwest China

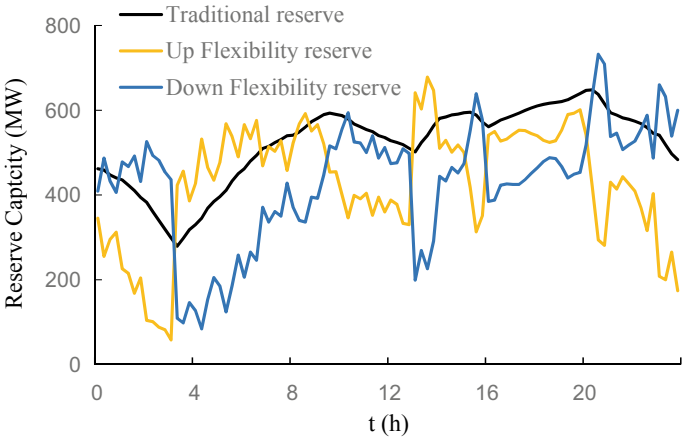


Fig. 3.12 Comparison of reserve between the proposed method and traditional one

Traditional reserve decision model only considers uncertainty in the system, but the flexibility-based reserve model proposed in this study has comprehensively considered the uncertainty and fluctuation of net load as well as the forced outage of conventional unit. Under the joint influence of uncertainty and fluctuation, the system needs reserve in both up and down directions. In order to demonstrate the effectiveness of the proposed method in a more intuitive way, the flexibility-based reserve only considered the uncertainty and fluctuation of load in this case. The LOFP is set as 5%, and the load prediction errors SD is 3% of forecasted load. Figure 3.12 shows the comparison of reserve between the proposed method and the traditional method, which values the reserve as 10% of forecasted load.

As showed in Fig. 3.12, the reserve obtained by the proposed method is less than the traditional method most of the time. However, when the load occurs a rapid rising, such as around six o'clock, the up-flexibility reserve is greater than the traditional one; and when the load sees a rapid dropping such as around 22 o'clock, the down flexibility reserve is greater than the traditional one. This indicates that the traditional reserve model may not be able to satisfy the ramping requirement of the system when load changes rapidly. For traditional reserve model, the operator will increase the reserve amount if the reserve fails frequently, but the insufficient of reserve may have caused losses already before the operator takes action. The risk of loss flexibility is uncontrollable. What is more, increasing the reserve amount for all time intervals will lead to a waste of system resources, which is uneconomical.

Meanwhile, at the load rising stage, e.g., 5–10 a.m., the up-flexibility reserve is greater than the down flexibility reserve; at the load dropping stage, e.g., 0–3 a.m., the up flexibility reserve is smaller than the down flexibility reserve. The flexibility-based reserve is a measure to the system ramping demand. It captures the characteristics of the system power changes between adjacent intervals. It shows a better performance of indicating the potential changes of the system power, where the traditional one

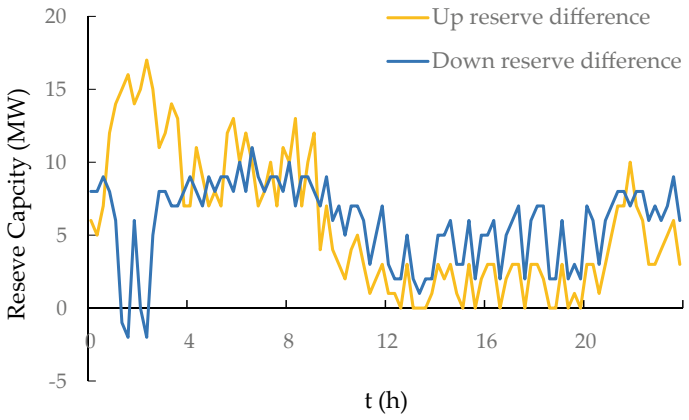


Fig. 3.13 The R_{DW} and R_{D} curve

fails to measure. This shows that the proposed method can more accurately describe the net load characteristics. It can be seen that the proposed flexibility-based reserve decision method is more applicable to the reliable operation of the power system containing fluctuating RE.

In order to analyze the influence of RES upon the flexibility-based reserve, flexibility-based reserve (R_{DW}) comprehensively considering load and wind farm A, and flexibility-based reserve (R_D) only considering the load fluctuation, were compared, and the difference between R_{DW} and R_D could directly reflect the influence of wind farm fluctuation upon flexibility reserve. The curve is as shown in Fig. 3.13.

Due to the intensive uncertainty and fluctuation of the wind farm, the introduction of the wind farm increases the up and down flexibility reserves of the system in general. According to the statistical results of historical data, when the wind farm output is high, the downward ramping probability is larger than upward ramping; when the wind power output is low, it is the opposite. With wind farm A taken into consideration, when the wind farm output is high, for example at 3 a.m., the downward ramping probability of the wind power is greater, so the increment of the up reserve is larger than that of the down reserve. Particularly, when the wind farm output approximates to the rated value, its downward fluctuation probability is far larger than the upward fluctuation. At this moment, the up reserve significantly increases while the down reserve may drop. When the wind farm output is low, for example at 8 p.m., the upward ramping probability is larger than the downward ramping, so the increment of down flexibility is larger than that of the up flexibility.

Meanwhile, to verify the effect of parallel computing, serial computing, 2 threads parallelism, 4 threads parallelism, 6 threads parallelism and 8 threads parallelism are adopted. The computing time for different computing means is as shown in Table 3.3. The comparison of different computing means indicates that parallel computing could reduce the solution time effectively.

Table 3.3 Computing time of different computing means

Computing means	Serial computing	2 threads parallelism	4 threads parallelism	6 threads parallelism	8 threads parallelism
Calculation time (s)	1207	924	642	563	526

Based on the above analyses, the proposed flexibility-based reserve decision model can accurately reflect the uncertainty and fluctuation of load and RES. When compared with the traditional reserve decision method, the proposed flexibility-based reserve can make the dispatch scheme stronger and safer.

(3) Flexibility based Generation-Reserve Bilevel Decision Model

In order to verify the accuracy of the proposed model, four scenarios were studied in the first case of Sect. 3.3.2.3. The parameters of each scenario are listed in Table 3.4.

The obtained unit commitment schemes of the four scenarios are the same. As the predicted value of the load and wind power remained consistent, the objective function values of the four scenarios are the same, which is \$1.96 × 106. As all solutions are obtained through one iteration, the solution time is about 52 min. Figure 3.14 shows the UC scheme of partial conventional units. In Fig. 3.14, the gray bar represents unit on operation while the white represents unit shutdown. The comparison of reserve in different scenarios is shown in Fig. 3.15.

The comparison of scenario 1 and 2 shows the influence of different flexibility requirements upon reserve. When the flexibility requirement rises, the system reserve significantly increases. The comparison of scenario 1 and 3 displays the impact of load prediction error upon reserve. Compared to scenario 1, the load prediction error in scenario 3 is larger, then the corresponding reserve significantly increases. The reserve of scenario 4 is almost the same with scenario 1. This is because the total capacity of wind power is relatively small, so the increase in wind power output fluctuation has less impact on the system flexibility.

The above simulation results show that the reserve increases as flexibility requirement rises. When the uncertainty of the system diminishes, such as the lowered load prediction SD and wind speed prediction SD, the operating reserve will decrease. Those cases listed above demonstrate that the reserve capacity obtained by the

Table 3.4 Parameters of each scenario

Scenarios	LOFP (%)	Load prediction errors SD in percentage of forecasted load (%)	Wind power prediction errors SD in percentage of forecasted wind power (%)
Scenario 1	5	3	5
Scenario 2	2	3	5
Scenario 3	5	5	5
Scenario 4	5	3	3

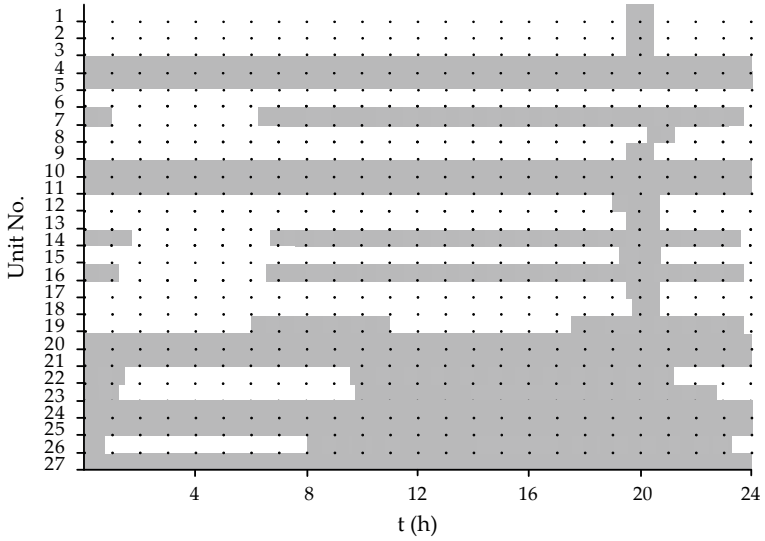


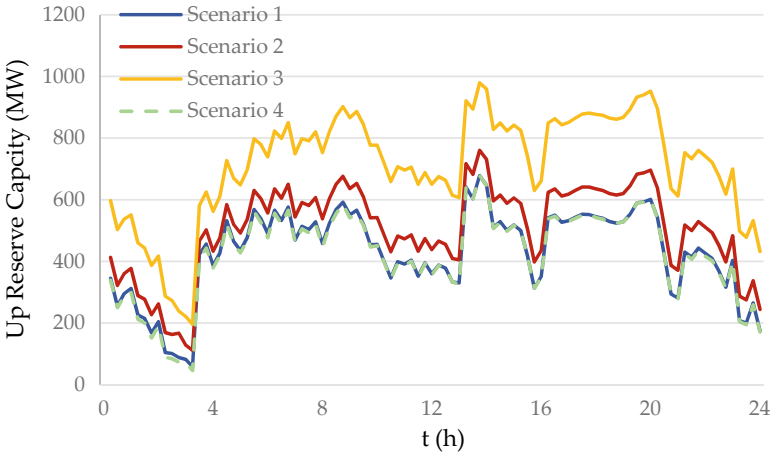
Fig. 3.14 Unit commitment of partial units

proposed model could accurately reflect the system uncertainty and flexibility, hence verifying the validity of the proposed model and algorithm.

In order to further analyze the influence of wind power upon the operating reserve, the system reserve is calculated under different wind power penetration rates. In scenario 1, the total wind power capacity is 350 MW, with a penetration rate of 5%. The scenarios of no wind power and integrated wind power capacity of 800, 1250 and 1800 MW are calculated separately, corresponding to wind power penetration rates of 0, 10, 15 and 20%, respectively. The solution information is listed in Table 3.5.

When the wind power penetration rate is relatively low, the UC scheme of initial solution by the upper level model could provide sufficient reserve, however, the uncertainty in the system increased with the rising of the wind power penetration rate. Although wind power generation cost is neglected, the unit generation cost increases as wind power penetration rises. When the penetration rate reaches 10% or more, the UC scheme of the first iteration fails to meet the reserve requirement, and the reserve model in the lower level adjusts according to the flexibility constraint and then re-solves the entire model. To facilitate analysis, Fig. 3.16 only displays the reserve at 2 a.m. under different wind power penetration rates.

Figure 3.16 shows that wind power has a significant influence on the system flexibility. The flexibility reserve is positively correlated to the wind power penetration rate. The growth rate of the flexibility reserve also shows a positively correlation to the wind power penetration rate.



(a) Comparison of the up reserve



(b) comparison of the down reserve

Fig. 3.15 Comparison of reserve in different scenarios

Table 3.5 Solution information under different wind power penetration rates

Wind power penetration rates (%)	Solution time	Number of iterations	Objective function value
0	51′31″	1	$\$2.00 \times 10^6$
5	52′08″	1	$\$1.96 \times 10^6$
10	93′32″	2	$\$1.92 \times 10^6$
15	91′17″	2	$\$1.92 \times 10^6$
20	94′43″	2	$\$1.95 \times 10^6$

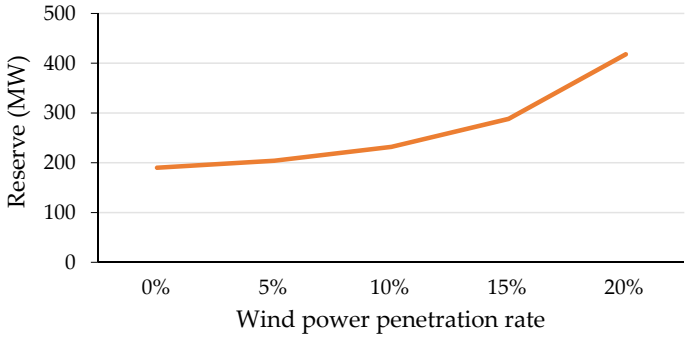


Fig. 3.16 Comparison of reserve under different wind power penetration rates

3.5.3 Case Studies of the Endogenous Approach to Quantifying the Wind Power Reserve

In this section, based on the IEEE118-bus standard system, wind farms are connected to buses 59, 90 and 116. To verify the superiority and robustness of the proposed approach, comparisons are carried out among the proposed approach (denoted by ENDO-RO), the exogenous methods based on LOLP (denoted by EXO-LOLP) and the endogenous methods with stochastic optimization (denoted by ENDO-SO). The comparison of ENDO-RO, EXO-LOLP and ENDO-SO are shown in Table 3.6.

From the table, it can be seen that the proposed approach guarantees a 100% success. Like case 1, EXO-LOLP fails to guarantee a 100% success with an even higher operating cost. Unlike case 1, compared with EXO-LOLP, ENDO-SO shows a smaller success rate as well as a lower operating cost, that is because case 2 has more wind farms than case 1, so there exist more uncertain wind farm power outputs, when the number of scenarios is the same, it provides less coverage of the actual situation. As in case 1, a long solution time is still its largest drawback.

In order to illustrate the effect of introducing the uncertainty budget λ on controlling the conservativeness of robust optimization, four values for λ are considered: 100, 75, 50, and 25% of the number of wind farms (100% is equivalent to the case without an uncertainty budget). The results of the comparison of the operating cost of different uncertainty budgets are listed in Table 3.7.

Table 3.6 Comparison of different approaches (ENDO-RO, EXO-LOLP and ENDO-SO)

	ENDO-RO	EXO-LOLP		ENDO-SO
		LOLP = 0.01	LOLP = 0.05	
Success rate (%)	100.00	99.98	93.20	85.07
Operating cost ($)	461724.66	467407.61	447217.61	438469.64
Solution time (s)	457.12	4.61	4.65	9074.16

Table 3.7 Comparison of operating costs of different uncertainty budgets

λ (%)	100	75	50	25
Operating cost ($)	461724.66	456220.96	450721.36	445234.52

It can be seen that introducing the uncertainty budget has an obvious effect on reducing the operating cost, and thus contributes to reducing the conservativeness of the robust optimization.

The comparisons of the calculation results of different methods are shown in Fig. 3.17 and Table 3.8.

Compared with case 1, the superiority of the iterative solution method in case 2 is more obvious because the number of generators in case 2 is large and more fault conditions need to be verified; if the unified solution method is adopted, the storage space of variables and the computational complexity greatly increase. The master problem and subproblems of the iterative solution method always maintain a small scale (roughly equal to the scale of a normal state model that does not require fault verification), which greatly reduces the calculating amount. Additionally, the effectiveness of adding auxiliary constraints is also more obvious, effectively improving the solution efficiency.

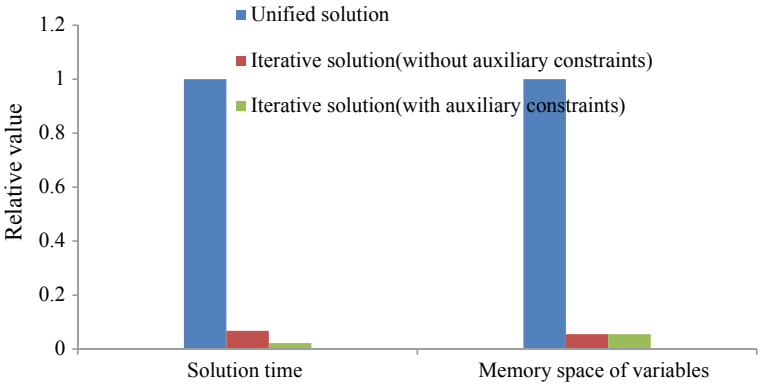


Fig. 3.17 Comparison of the unified solution and iterative solution

Table 3.8 Comparison of different iterative solution methods

	Without auxiliary constraints	With auxiliary constraints
Number of iterations	13	6
Number of feasibility cuts	1100	468

The operating cost comparison of fixed and optimized participation factors is shown in Table 3.9. Optimizing the participation factor can reduce the system operating costs by 2.91%.

A comparison of reserves determined by exogenous and endogenous methods is shown in Fig. 3.18, and the reserve cost comparison is shown in Table 3.10.

For the wind power and load reserve, the reserve capacity obtained by the endogenous method considering generator output constraints, network constraints and generator faults is larger than that of the exogenous method. For the contingency reserve, the endogenous method considers the actual generator dispatch. This is unlike the exogenous method, which simply takes the maximum online generator capacity as the contingency reserve, thus effectively reducing the required contingency reserve. The overall reserve cost is reduced by 36.45%.

To quantify the impact of the wind power uncertainty and wind power penetration rate on the wind power reserve and operating cost, under the condition of a 4500 MW load, and a [−90, 90] load uncertainty interval, the S&NCED model is solved under different wind power uncertainties and wind power penetration rates. The results of

Table 3.9 Operating cost comparison of fixed and optimized participation factors

	Fixed participation factors	Optimized participation factors
Operating cost ($)	475561.53	461724.66

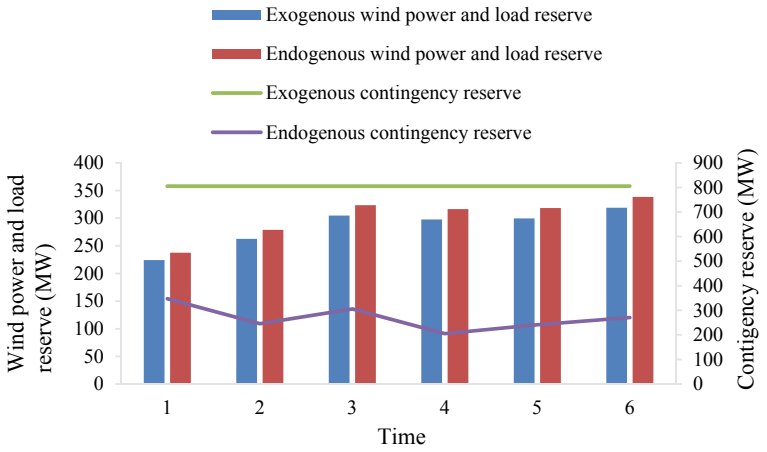
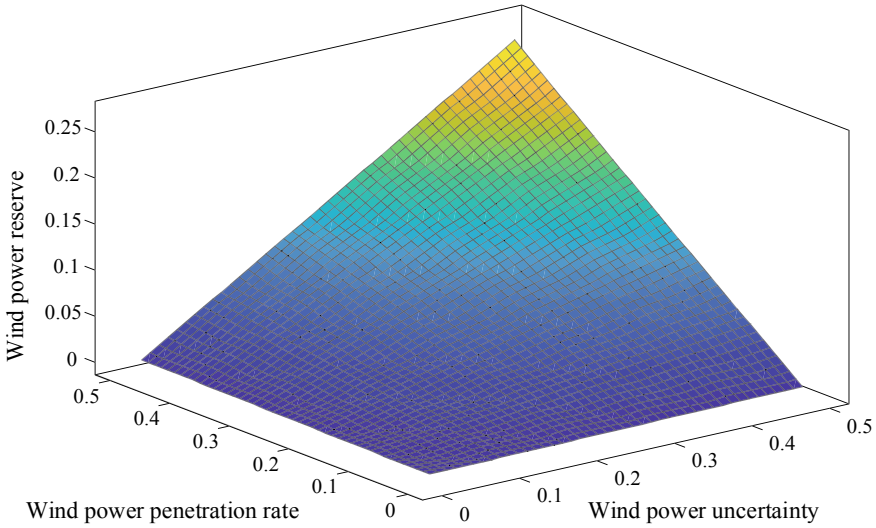


Fig. 3.18 The result comparison of the endogenous and exogenous reserve determination approaches

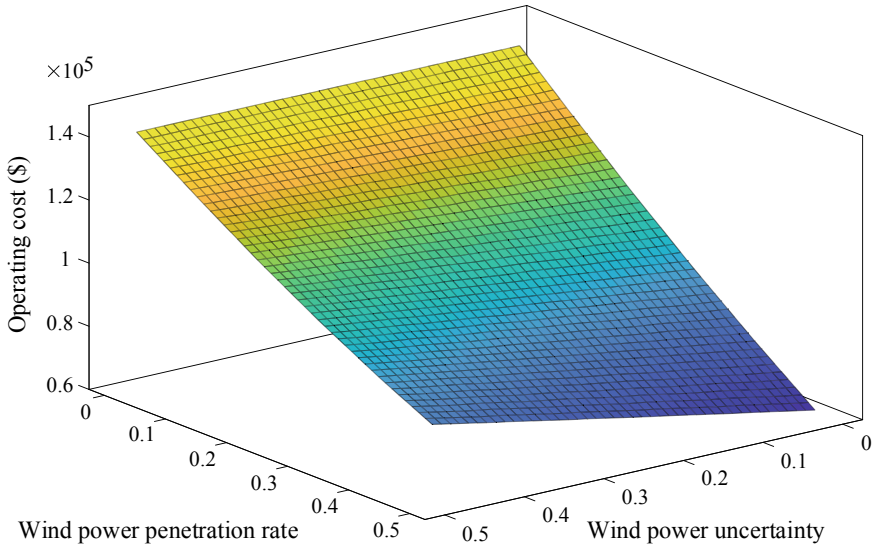
Table 3.10 Comparison of reserve costs between the endogenous and exogenous methods

	Exogenous method	Endogenous method
Reserve cost ($)	82465.06	52405.38

the wind power reserve and operating costs under different wind power uncertainties and wind power penetration rates are shown in Fig. 3.19.



(a) Wind power reserve



(b) Operating cost

Fig. 3.19 Comparison of different wind power uncertainties and different wind power penetration rates

Unlike case 1, there is no obvious inflection point in the linear relationship between the wind power reserve or operating cost and the wind power uncertainty in a relatively large-scale system, that's because the number of generators in case 2 is large and the overall technical limit is large. The wind power reserves and operating costs are linearly related to the wind power penetration rate and the wind power uncertainty. The higher the wind power penetration rate and the greater the wind power uncertainty, the larger the required wind power reserve. The higher the wind power penetration rate and the smaller the wind power uncertainty, the lower the operating cost and the better the economic efficiency. Additionally, the figure shows that the degree of influence of the wind power uncertainty and the wind power penetration rate on the wind power reserve are similar. On the other hand, for the operating cost, the wind power penetration rate has a greater impact because the wind power penetration rate mainly affects the generation scheduling. When the penetration rate increases, although the required wind power reserve and the reserve cost increase, the fossil fueled generator output decreases. Thus, the generation costs are reduced, and the overall operating costs are significantly reduced.

3.6 Conclusion

In this chapter, quantifying approaches on the flexibility requirements of high share RES penetrated power systems are proposed, then flexibility-based optimal scheduling methods are presented.

In order to quantify the system flexibility and provide operation instruction, ADWP is proposed as an assessment metric. ADWP is formulated as an M-dimensional polyhedron, and its conservativeness can be controlled by introducing the adjustment coefficient. Further, two approaches for flexible LAUC with different purposes are presented. Three-stage interval optimization approach is provided for seeking the most economical scheduling scheme that can reach the largest ADWP, which reflect the maximum flexibility potential of the system. In order to minimize the overall cost, the scenario-based stochastic optimization approach is provided to obtain the most economical ADWP, which exclude extreme scenarios that call for more expensive flexibility resources but have a very low probability of occurrence. Advantages in flexibility and economy of the proposed approaches compared with other two approaches are demonstrated on two cases.

For the issue of the reserve decision and coordinated optimization of generation and reserve of power system integrated with large-scale renewable energy, we proposed a day-ahead generation–reserve bilevel decision model. The flexibility requirement models of the load and wind power are built on the basis of historical data and forecast information, while the conventional unit are modeled by the output plan and the forced outage rate. Then the quantitative relationship between flexibility and operating reserve is established by the universal generating function theorm. Taking it as a constraint, the day-ahead power generation-reserve bilevel decision model is formulated. The generation-reserve joint dispatching scheme obtained from the

proposed model cannot only meet the load, but also ensure the flexibility to meet given requirements. In this way, the dispatching scheme ensures that there is sufficient reserve capability to cope with uncertainties and fluctuation within the system.

Aimed at the problem of wind power reserve quantification and determination, an endogenous reserve determination method for wind power integrated power systems considering network constraints is proposed. The conclusions are as follows: (1) The established S&NCED model based on AARO guarantees a 100% success of Monte Carlo trials, dominating the merits of robustness. Additionally, introducing the uncertainty budget can effectively reduce the conservativeness. (2) The proposed two-stage iterative solution method for the S&NCED model, which is based on the Benders decomposition and includes auxiliary constraints, can effectively reduce the computational complexity and accelerate the convergence. (3) The proposed endogenous reserve determination method can provide a more economical generation-reserve dispatch scheme under the premise of ensuring operational reliability with a relatively small solution time.

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Chapter 4
Distributed Dispatch Approach in AC/DC Hybrid Systems



The increasing integration of volatile renewables energy (VRE) has highlighted the importance of flexibility in power system operation. Due to the imbalance in development between large-scale VRE and electricity load centers, the transmission of large amounts of VRE has drawn substantial attention. Given that HVDC offers a unique capability in terms of the regulation of power flow, the hybrid AC/DC grid can operate more flexibly and cost-effectively. Considering computing efficiency, autonomy of each regional grid and information privacy, compared with centralized dispatch approach, distributed dispatch approach is more suitable. So, we propose a distributed, hierarchical and robust dispatch approach in AC/DC hybrid systems with high flexibility requirement. Multi-stage multi-area economic dispatch models are formulated with hybrid AC/DC links and VSC-MTDC networks respectively and are solved distributedly. Case studies based on hybrid AC/DC systems of 3-area 18-bus, 3-area 117-bus, 3-area 354-bus, 4-terminal with four 6-bus systems and 4-terminal with four 118-bus systems demonstrate the effectiveness and efficiency of the proposed mechanism and method.

4.1 Introduction

By the end of 2020, the global cumulative installed capacity of wind power had reached 743 GW [1]. The increasing integration of wind power has highlighted the importance of flexibility in power system operation. Wind power is mainly transmitted to the load centers through HVAC or HVDC lines. Since the AC grid expansion is limited by the legislation and the capacity of long-distance transmission, HVDC is being considered as an alternative solution. The bulk AC/DC hybrid system will be the most under construction. Besides, Connecting AC areas through the voltage source converter based multiterminal high voltage direct current (VSC-MTDC) grid can provide several advantages, which is becoming a very promising transmission mode

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M. Zhou et al., *Power System Flexibility*, Power Systems,
https://doi.org/10.1007/978-981-19-9075-5_4

in operation. Considering more and more hybrid AC/DC systems will be commissioned in the near future, the study about the transmission of wind power through AC/DC grid is of great importance.

From the perspective of bulk AC/DC hybrid system operation, current studies mainly focus on the optimal power flow (OPF) problem. In [2], the convex relaxation technique is used to transform the AC/DC OPF problem into a semidefinite program. In [3], an improved corrective security-constrained OPF is proposed for a voltage source converter-based multi-terminal HVDC system. In [4], a formulation for a multi-area DC OPF consisting of both HVAC and HVDC lines is introduced. However, to the best of the author's knowledge, there is hardly any literature examining the role of HVDC transmission systems in promoting inter-regional wind power accommodation.

One of the key challenges for the meshed AC/DC hybrid system with special structure is the interaction between AC systems and the MTDC system. In [5], the approximate Newton directions decomposition algorithm for the general continuous optimization problems with special structure is presented. In case of an overlay MTDC grid connecting separate AC areas, a hierarchical scheduling architecture is a preferred scheme. The recent work of Kargarian et al. employs the hierarchical analytical target cascading (ATC) [6, 7] to solve the operation problem for active distribution grid [8] and a transmission system encompassing numbers of active distribution grids [9]. Based on ATC, a hierarchical optimal power flow (OPF) model in AC/DC hybrid distribution grid and a multilevel OPF model in three-phase unbalanced distribution grid is proposed in [10] and [11], respectively. However, in these models, the uncertainty of renewable energy is ignored.

The existing decentralized or distributed dispatch approaches are mainly designed for AC system ED or OPF problemsm [12–17]. According to the decomposition method for the decision variables, the existing decomposition techniques can be classified into dual decomposition approaches and primal decomposition approaches.

The dual decomposition approaches [12–14] establish a relaxed local optimization problem, and the dual variables are utilized to coordinate the local problems. The most representative technique is Lagrangian relaxation (LR) [12], in which the coupling constraints are relaxed and the Lagrangian multipliers are iteratively updated by the sub-gradient method. To improve the convergence of LR, augmented Lagrangian relaxation (ALR) [13, 14] has been developed by adding quadratic terms corresponding to the coupling constraints into the objective function. The auxiliary problem principle [14] is also applied to make the objective function of ALR separable.

The primal decomposition techniques [15–17] divide the decision variables into local variables and coupling variables. The coupling variables are first fixed in the sub-problems and then solved iteratively in the master problem by an upper-level coordinator. The recent work of Kargarian et al. [18] employs the analytical target cascading (ATC) technique [19] in the multi-area security-constrained unit commitment (SCUC) problem for large-scale power systems. This kind of distributed optimization technique incorporates the advantages of both LR and the mixed integer programming method.

To fill the aforementioned gap, we propose the distributed, hierarchical and robust dispatch approach in two different AC/DC hybrid systems with high flexibility requirement: bulk AC/DC hybrid systems and VSC-MTDC meshed AC/DC hybrid systems. For bulk AC/DC hybrid systems, to fully coordinate the power generation and load of different regional grids and to mutually benefit multiple regions, this section considers the flexible and controllable nature of the HVDC tie-line to further explore its potential in promoting inter-regional wind power accommodation. Accordingly, the concept of the pseudo generator is introduced to finely model the flexible adjustment capability of the HVDC tie-line, and an improved ATC-based distributed dispatch approach for a bulk AC/DC hybrid system with high flexibility requirement is presented. For VSC-MTDC meshed AC/DC hybrid systems, to fully coordinate the AC grids, the DC grid and the VSC stations to mutually benefit multiple regions, hierarchical and robust scheduling (HRS) formulation is proposed in this section, which decomposes the entire system into three interacting levels (the low level is a two-stage adaptive robust SCUC problem for the AC grids, the middle level is a day-ahead power transmission optimization problem for the VSC stations, and the high-level is a multi-period OPF problem for the DC grid) and managed by an integrated ATC and C&CG (IAC) algorithm.

4.2 Distributed Dispatch Approach in Bulk AC/DC Hybrid Systems

In this section, to fully use the flexible adjustment capability of the HVDC tie-line to promote inter-regional wind power accommodation, an operation model for the HVDC tie-line is presented. Then, a distributed SCUC approach based on the analytical target cascading technique is proposed, where the day-ahead SCUC problem is decomposed into an upper-level master problem and parallel sub-problems of lower-level regional dispatch. The master problem is in charge of determining the day-ahead transmission plan for the HVDC tie-line, and the lower-level dispatch centers independently solve their SCUC problems in parallel in accordance with the hierarchical and partitioned power scheduling mode.

4.2.1 Distributed Scheduling Framework for Bulk AC/DC Hybrid Transmission Systems

The prototype of bulk AC/DC hybrid transmission systems used in this section is abstracted from the Gansu provincial wind power transmission project, where wind-thermal-bundled power is transmitted to two different receiving-side grids via HVAC and HVDC tie-lines. In Gansu, a province in western China, the ratio of the installed wind power capacity over the annual maximum load is expected to reach 100% by

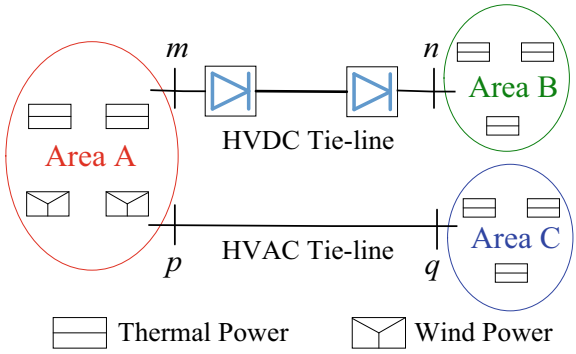


Fig. 4.1 Bulk AC/DC hybrid system with wind power

2020. This example is very representative in terms of the high share of renewable power generation and HVDC/HVAC hybrid transmission. Therefore, we take this example AC/DC hybrid transmission system, shown in Fig. 4.1, as the research object.

In this bulk AC/DC hybrid system, areas A, B and C generally belong to different dispatch entities, especially under power sector deregulation. The centralization of all regional grids will lead to operation by a single central super entity with complete knowledge and control of the entire system. This centralization may create substantial regulatory and political issues because the regional grids would have to give up their power, responsibility and control to the central entity. Hence, the distributed dispatch approach becomes an attractive option, which has the merits of preserving area scheduling independence and information privacy.

The proposed distributed dispatch approach for AC/DC hybrid systems is inspired by the ATC technique for multi-level hierarchical optimization problems [50], [51]. According to the traditional ATC formulation [49], the bulk AC/DC hybrid system is decomposed into the master problem and the sub-problems of the lower-level regional grids. In this framework, the only constraint for the master problem is the regional consistency constraint, and the operating constraints of the tie-lines are included in the corresponding sub-problems. That is, the day-ahead transmission plan of the HVDC tie-line is determined by the corresponding regional grids, while the master problem is responsible for only coordinating the HVDC tie-line transmission plan developed by the corresponding regional grids to ensure the feasibility of the operating point for the entire system. However, a large number of integer and continuous variables are introduced when modeling the flexible adjustment capability of the HVDC tie-line, as detailed in Section III-A. If the operating characteristics of the tie-lines are optimized by the regional grids, the computational complexity of the regional grids further increases, while the computing ability of the upper-level master problem is not fully utilized. In addition, for the hierarchical and partitioned power scheduling mode of multi-level dispatch centers in China, the upper-level dispatch center is better suited to determine the day-ahead transmission plan of the HVDC tie-line than the lower-level dispatch centers. Therefore, the traditional ATC-based distributed dispatch cannot be well adapted to bulk AC/DC hybrid systems.

Accordingly, we propose an improved ATC-based distributed dispatch approach for bulk AC/DC hybrid systems. Each lower-level dispatch center is responsible for only scheduling its generating units, and coordinating its interconnection with the upper-level dispatch center. The upper-level dispatch center can be regarded as the creation of a new independent system operating entity for bulk AC/DC hybrid systems, with no knowledge of the regional grids but complete knowledge of the tie-lines. As a result, the operating constraints of the tie-lines are included in the master problem rather than in the sub-problems. Thus, the transmission plan of the HVDC tie-line is formulated by the upper-level dispatch center. In this way, the regional consistency constraint in the master problem is no longer necessary. Moreover, the computing capability of the upper-level dispatch center can be better utilized to ease the computational burden of the lower-level dispatch centers.

The key to decomposing a large-scale system into regional subsystems lies in how to handle tie-lines. At present, the multi-area decentralized or distributed dispatch approaches all focus on the AC systems. In these systems, fictitious buses are added to create interactions between zones, and the voltage angle difference of the boundary buses is used to calculate the tie-line power. For a bulk AC/DC hybrid system, the HVDC tie-line has no border voltage angle, so the voltage angle difference of the boundary buses is not applicable to handle the power flow of the HVDC tie-line. In addition, the HVDC tie-line can flexibly control its power, that is, the transferred power is a controllable variable, which can be directly used as the shared variable. As a consequence, we employ the branch cutting method to decompose the AC/DC hybrid systems and take the tie-line power as a shared variable. That is, a bulk AC/DC hybrid system, as shown Fig. 4.1, is decomposed into separate areas, as shown in Fig. 4.2, where the power injection of a tie-line is regarded as a pseudo generator added to the tie-line boundary bus. In this way, the flexible and controllable nature of the HVDC tie-line can be fully modeled by the output characteristics of this pseudo generator.

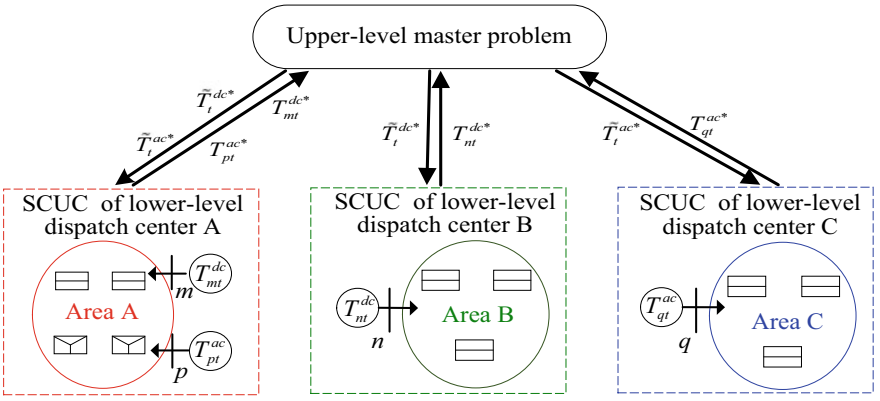


Fig. 4.2 Region decomposition for a bulk AC/DC hybrid system

4.2.2 Improved ATC-Based Distributed SCUC for a Bulk AC/DC Hybrid System

4.2.2.1 Operating Constraints of the HVDC Tie-Line

Since the pseudo generator is added to the tie-line boundary bus, the operating characteristics of the HVDC tie-line can be represented by the output characteristics of the pseudo generator, which is used to achieve joint optimization of the HVDC tie-line power and the regional generating units. In general, considering the operating reliability and lifetime of the DC converters, the output constraints of the HVDC tie-line pseudo generator are more complex than those of conventional generators.

(1) Lower/upper limits of the HVDC tie-line power

$$\underline{T}_{mn}^{dc} \le \tilde{T}_{t}^{dc} \le \overline{T}_{mn}^{dc}, \forall t, n \in \Omega_{dc}^{B} \tag{4.1}$$

(2) Power adjustment direction constraint

To protect the DC converter, the power adjustment direction of the HVDC tie-line cannot be reversed during adjacent periods. The binary variables x_t, x_t^+, and x_t^- are defined to represent whether the HVDC tie-line adjusts its power, adjusts its power upward, and adjusts its power downward at time t, respectively. Therefore, the constraint that the power adjustment direction cannot be reversed during adjacent periods can be expressed as:

$$\begin{cases} x_t^+ + x_t^- = x_t \le 1 \\ x_t^+ + x_{t+1}^- \le 1 \\ x_{t+1}^+ + x_t^- \le 1 \end{cases}, \forall t \tag{4.2}$$

(3) Number of adjustments to the HVDC tie-line power within one day

This constraint reflects that the number of adjustments of a DC converter within one day is limited to ensure its reliable operation, that is,

$$\sum_{t \in T} \left(x_t^+ + x_t^- \right) \le S \tag{4.3}$$

(4) Power adjustment continuity constraint

The HVDC tie-line power cannot be continuously adjusted upward or downward during adjacent periods, that is,

$$\begin{cases} x_t^+ + x_{t+1}^+ \le 1 \\ x_t^- + x_{t+1}^- \le 1 \end{cases}, \forall t \tag{4.4}$$

In addition, the values of x_t^+ and x_t^- can be expressed by the change in HVDC tie-line power. Here, M^+ and M^- are constants designed to make the inequalities valid.

$$\begin{cases} \tilde{T}_{t+1}^{dc} - \tilde{T}_t^{dc} \le M^+ x_{t+1}^+ \\ \tilde{T}_t^{dc} - \tilde{T}_{t+1}^{dc} \le M^- x_{t+1}^- \end{cases}, \forall t \tag{4.5}$$

(5) Lower/upper power adjustment rate limits

In contrast to the ramp-up/ramp-down rate restriction of a conventional generator, the lower power adjustment rate limit of the HVDC tie-line is set to be greater than a non-zero positive constant to avoid repeated small-amplitude adjustments of the HVDC tie-line power.

$$x_t \underline{\delta}^{dc} \le \left| \tilde{T}_{t+1}^{dc} - \tilde{T}_t^{dc} \right| \le x_t \overline{\delta}^{dc}, \forall t \tag{4.6}$$

(6) Stepwise operating constraint

To maintain the HVDC tie-line power in a stepwise state, once the HVDC tie-line power is adjusted, it should be kept constant for a minimum duration of N_T. We define the binary variables a_t^+ and a_t^- to represent whether the HVDC tie-line is to begin or end the power adjustment at time t.

$$\begin{cases} a_t^- + \sum_{\tau=t+1}^{\min(T,t+N_T)} a_\tau^+ \le 1 \\ a_t^+ \ge x_{t+1} - x_t \\ a_t^- \ge x_t - x_{t+1} \end{cases}, \forall t \tag{4.7}$$

In addition, the values of a_t^+ and a_t^- must satisfy the following constraint:

$$\begin{cases} a_t^+ + a_t^- \le 1 \\ a_t^+ + a_{t+1}^- \le 1 \\ a_t^- + a_{t+1}^- \le 1 \end{cases}, \forall t \tag{4.8}$$

(7) Transmission of electrical energy over a scheduling cycle

This constraint reflects that the HVDC tie-line transmission of electrical energy is determined by the day-ahead cross-border trading contract.

$$\left(1 - \rho^{dc}\right) Q^{dc} \le \sum_{t \in T} \tilde{T}_t^{dc} \le \left(1 + \rho^{dc}\right) Q^{dc} \tag{4.9}$$

The values of $\underline{\delta}^{dc}$, $\overline{\delta}^{dc}$, N_T and S in the above HVDC model can be determined by historical operating experience and the performance of the DC converter station.

4.2.2.2 Operating Constraints of the HVAC Tie-Line

(1) Upper/lower bounds of the HVAC tie-line power

$$0 \le \tilde{T}_t^{ac} \le \overline{T}_{pq}^{ac}, \forall t,\ p \in \Omega_{ac}^{A}, q \in \Omega_{ac}^{C} \tag{4.10}$$

(2) Transmission of electrical energy over a scheduling cycle

Similar to the constraint for the HVDC tie-line, this constraint reflects that the HVAC tie-line transmission of electrical energy is determined by the day-ahead cross-border trading contract.

$$\left(1-\rho^{ac}\right)Q^{ac} \le \sum_{t\in T} \tilde{T}_t^{ac} \le \left(1+\rho^{ac}\right)Q^{ac} \tag{4.11}$$

4.2.2.3 Master Problem of Upper-Level Coordination Dispatch

In the proposed improved ATC-based distributed SCUC (IATC-DSCUC), the upper-level master problem plays the role of a central coordinator for the entire AC/DC hybrid system. This central coordinator has no knowledge of the regional grid but complete knowledge of the tie-lines and is responsible for determining the day-ahead transmission plan of the tie-lines. The upper-level master problem receives the optimal outputs of the pseudo generators uploaded by lower-level dispatch centers, coordinates them to minimize the tie-line power deviations, updates the reference value of the tie-line transmission plan and sends back to the corresponding lower-level dispatch centers. The master optimization model of the proposed IATC-DSCUC is as follows:

$$\begin{aligned}\min \sum_{t\in T}\Bigg\{ & \sum_{m\in\Omega_{dc}^{A}}\left[\alpha_{mt}^{dc}\left(\tilde{T}_{t}^{dc}-T_{mt}^{dc*}\right)+\left[\beta_{mt}^{dc}\left(\tilde{T}_{t}^{dc}-T_{mt}^{dc*}\right)\right]^{2}\right] \\ & +\sum_{n\in\Omega_{dc}^{B}}\left[\alpha_{nt}^{dc}\left(\tilde{T}_{t}^{dc}-T_{nt}^{dc*}\right)+\left[\beta_{nt}^{dc}\left(\tilde{T}_{t}^{dc}-T_{nt}^{dc*}\right)\right]^{2}\right] \\ & +\sum_{p\in\Omega_{ac}^{A}}\left[\alpha_{pt}^{ac}\left(\tilde{T}_{t}^{ac}-T_{pt}^{ac*}\right)+\left[\beta_{pt}^{ac}\left(\tilde{T}_{t}^{ac}-T_{pt}^{ac*}\right)\right]^{2}\right] \\ & +\sum_{q\in\Omega_{ac}^{C}}\left[\alpha_{qt}^{ac}\left(\tilde{T}_{t}^{ac}-T_{qt}^{ac*}\right)+\left[\beta_{qt}^{ac}\left(\tilde{T}_{t}^{ac}-T_{qt}^{ac*}\right)\right]^{2}\right]\Bigg\}\end{aligned} \tag{4.12}$$

s.t.

$$(4.1)-(4.9)$$
$$(4.10)-(4.11)$$

The objective function (4.12) minimizes the tie-line power interaction errors between the sub-problems.

Compared to the traditional ATC-based distributed SCUC (TATC-DSCUC), the redesigned upper-level master optimization model does not require a regional consistency constraint but includes the operating constraints of the tie-lines. When the tie-line power interaction errors satisfy the proposed convergence criterion in Sect. 4.2.3, the day-ahead tie-line transmission plan determined by the upper-level master problem is the final solution. Then, this solution is issued to the corresponding lower-level dispatch center for execution.

4.2.2.4 Sub-problem of Lower-Level Regional Dispatch

Each lower-level dispatch center is responsible for the operation of the grid that falls within its area of responsibility and can independently solve its SCUC problem in parallel without communicating with its neighbors. In contrast to that of the centralized SCUC (CSCUC), the objective function of the IATC-DSCUC contains an augmented Lagrangian penalty term to minimize the tie-line power interaction errors. As a result, the tie-line power optimized by the lower-level dispatch centers can approximate the reference power issued by the upper-level master problem. The sub-optimization model of each lower-level dispatch center (e.g., area A) is as follows:

$$\min \sum_{t\in T}\Bigg\{\sum_{i\in\Omega_{\mathrm{G}}^{A}}\left[a_i\left(P_{it}^{\mathrm{G}}\right)^2+b_iP_{it}^{\mathrm{G}}+u_{it}c_i+v_{it}C_i^{\mathrm{SU}}\right]+\sum_{j\in\Omega_{\mathrm{W}}^{A}}C^{\mathrm{WS}}P_{jt}^{\mathrm{WS}} + \sum_{m\in\Omega_{dc}^{A}}\left[\alpha_{mt}^{dc}\left(\tilde{T}_t^{dc*}-T_{mt}^{dc}\right)+\left[\beta_{mt}^{dc}\left(\tilde{T}_t^{dc*}-T_{mt}^{dc}\right)\right]^2\right] + \sum_{p\in\Omega_{ac}^{A}}\left[\alpha_{pt}^{ac}\left(\tilde{T}_t^{ac*}-T_{pt}^{ac}\right)+\left[\beta_{pt}^{ac}\left(\tilde{T}_t^{ac*}-T_{pt}^{ac}\right)\right]^2\right]\Bigg\} \tag{4.13}$$

s.t.

$$-u_{it-1}+u_{it}-u_{ik}\le 0,\ k=t,\cdots,t+MU_i-1,\forall i\in\Omega_{\mathrm{G}}^{A},\forall t \tag{4.14}$$

$$u_{it-1}-u_{it}+u_{ik}\le 1,\ k=t,\cdots,t+MD_i-1,\forall i\in\Omega_{\mathrm{G}}^{A},\forall t \tag{4.15}$$

$$-u_{it-1}+u_{it}-v_{it}\le 0,\forall i\in\Omega_{\mathrm{G}}^{A},\forall t \tag{4.16}$$

$$u_{it-1}-u_{it}-w_{it}\le 0,\forall i\in\Omega_{\mathrm{G}}^{A},\forall t \tag{4.17}$$

$$\sum_{i\in\Omega_{\mathrm{G}}^{A}}P_{it}^{\mathrm{G}}+\sum_{j\in\Omega_{\mathrm{W}}^{A}}\left(P_{jt}^{\mathrm{W}}-P_{jt}^{\mathrm{WS}}\right)=\sum_{k\in\Omega_{\mathrm{D}}^{A}}D_{kt}+\sum_{m\in\Omega_{dc}^{A}}T_{mt}^{dc}+\sum_{p\in\Omega_{ac}^{A}}T_{pt}^{ac},\forall t \tag{4.18}$$

$$u_{it}\underline{P}_i^{\mathrm{G}}\le P_{it}^{\mathrm{G}}\le u_{it}\overline{P}_i^{\mathrm{G}},\forall i\in\Omega_{\mathrm{G}}^{A},\forall t \tag{4.19}$$

$$P_{it}^{\mathrm{G}}+r_{it}^{u}\le u_{it}\overline{P}_i^{\mathrm{G}},0\le r_{it}^{u}\le u_{it}UR_i,\forall t \tag{4.20}$$

$$P_{it}^{\mathrm{G}}-r_{it}^{d}\ge u_{it}\underline{P}_i^{\mathrm{G}},0\le r_{it}^{d}\le u_{it}DR_i,\forall i\in\Omega_{\mathrm{G}}^{A},\forall t \tag{4.21}$$

$$\sum_{i\in\Omega_{\mathrm{G}}^{A}}\left(P_{it}^{\mathrm{G}}+r_{it}^{u}\right)+\sum_{j\in\Omega_{\mathrm{W}}^{A}}\left(P_{jt}^{\mathrm{W}}-P_{jt}^{\mathrm{WS}}\right)\ge\sum_{k\in\Omega_{\mathrm{D}}^{A}}D_{kt}+\sum_{m\in\Omega_{dc}^{A}}T_{mt}^{dc}+\sum_{p\in\Omega_{ac}^{A}}T_{pt}^{ac}+R_t^{+A},\forall t \tag{4.22}$$

$$\sum_{i\in\Omega_{\mathrm{G}}^{A}}\left(P_{it}^{\mathrm{G}}-r_{it}^{d}\right)+\sum_{j\in\Omega_{\mathrm{W}}^{A}}\left(P_{jt}^{\mathrm{W}}-P_{jt}^{\mathrm{WS}}\right)\le\sum_{k\in\Omega_{\mathrm{D}}^{A}}D_{kt}+\sum_{m\in\Omega_{dc}^{A}}T_{mt}^{dc}+\sum_{p\in\Omega_{ac}^{A}}T_{pt}^{ac}-R_t^{-A},\forall t \tag{4.23}$$

$$R_t^{-A}=\omega_d\sum_{j\in\Omega_{\mathrm{W}}^{A}}\left(\overline{P}_j^{\mathrm{W}}-P_{jt}^{\mathrm{W}}\right),\forall t \tag{4.24}$$

$$R_t^{+A} = \omega_u \sum_{j \in \Omega_{\mathrm{W}}^A} P_{jt}^{\mathrm{W}} + \omega_D \sum_{k \in \Omega_{\mathrm{D}}^A} D_{kt}, \forall t \tag{4.25}$$

$$-u_{it} DR_i \le P_{it}^{\mathrm{G}} - P_{it-1}^{\mathrm{G}} \le u_{it-1} UR_i, \forall i \in \Omega_{\mathrm{G}}^A, \forall t \tag{4.26}$$

$$0 \le P_{jt}^{\mathrm{WS}} \le P_{jt}^{\mathrm{W}}, \forall j \in \Omega_{\mathrm{W}}^A, \forall t \tag{4.27}$$

$$\underline{T}_{mn}^{dc} \le T_{mt}^{dc} \le \overline{T}_{mn}^{dc}, \forall t, m \in \Omega_{dc}^A, n \in \Omega_{dc}^B \tag{4.28}$$

$$0 \le T_{pt}^{ac} \le \overline{T}_{pq}^{ac}, \forall t, p \in \Omega_{ac}^A, q \in \Omega_{ac}^C \tag{4.29}$$

$$\left| \sum_{i \in \Omega_{\mathrm{G}}^A} H_{li}^{\mathrm{G}} P_{it}^{\mathrm{G}} + \sum_{j \in \Omega_{\mathrm{W}}^A} H_{lj}^{\mathrm{W}} \left(P_{jt}^{\mathrm{W}} - P_{jt}^{\mathrm{WS}} \right) - \sum_{p \in \Omega_{ac}^A} H_{lp}^{ac} T_{pt}^{ac} - \sum_{m \in \Omega_{dc}^A} H_{lm}^{dc} T_{mt}^{dc} - \sum_{k \in \Omega_{\mathrm{D}}^A} H_{lk}^{D} D_{kt} \right| \le \overline{F}_l, \quad \forall l \in L^A, \forall t \tag{4.30}$$

The objective function (4.13) is designed to minimize the total generation cost of area A, including the conventional generation cost and the penalty cost due to tie-line power deviations. Constraints (4.14) and (4.15) represent the minimum up- and down-time restrictions of a conventional generator, respectively. Constraints (4.16) and (4.17) represent the start-up and shut-down operations of a conventional generator, respectively. Constraint (4.18) is the load balance constraint. Constraint (4.19) is the minimum/maximum output of each conventional generator. Constraints (4.20) and (4.21) are the upward/downward spinning reserve capacity restrictions of a conventional generator, respectively. Constraints (4.22)–(4.25) are the upward/downward spinning reserve requirements. Constraint (4.26) is the ramp-up/ramp-down rate of a conventional generator. Constraint (4.27) is the curtailed wind power. Constraints (4.28) and (4.29) are the minimum/maximum output of the HVDC and HVAC tie-line pseudo generators, respectively. Constraint (4.30) is the transmission capacity security of the internal lines.

Compared to the TATC-DSCUC, the redesigned lower-level sub-optimization model does not include the operating constraints of the tie-lines. After all the lower-level dispatch centers have acquired their optimal solutions, the optimal outputs of the tie-line pseudo generators are uploaded to the upper-level master problem for coordination.

4.2.3 Solution Procedure

According to the traditional ATC formulation [49], the regional grids interconnected by tie-lines develop separate day-ahead tie-line transmission plans. The convergence

criterion in the TATC-DSCUC must separately check whether the tie-line power interaction errors of all the scheduling periods between the tie-line transmission plans developed by the regional grids and the regional consistency coordination variables issued by the upper-level master problem satisfy the convergence precision. In contrast, in our improved ATC formulation, the transmission plans of the tie-lines are formulated by the upper-level dispatch center rather than by the lower-level regional grids. Thus, the regional consistency constraint is no longer necessary in the master problem, and the corresponding convergence criterion must be redesigned to better fit the proposed IATC-DSCUC.

The redesigned convergence criterion (4.31) in our IATC-DSCUC checks whether the mean value of the tie-line power deviations between the issued reference value of the tie-line transmission plan from the upper-level master problem and the uploaded feedback values from the lower-level sub-problems satisfies the convergence precision in the τ th iteration.

$$\begin{cases} \dfrac{\max\left\{\frac{1}{T}\sum\limits_{t\in T}\left|\tilde{T}_t^{dc*}(\tau)-T_{mt}^{dc*}(\tau)\right|,\ \frac{1}{T}\sum\limits_{t\in T}\left|\tilde{T}_t^{dc*}(\tau)-T_{nt}^{dc*}(\tau)\right|\right\}}{\overline{T}_{mn}^{dc}} \le \varepsilon_{dc} \\ \dfrac{\max\left\{\frac{1}{T}\sum\limits_{t\in T}\left|\tilde{T}_t^{ac*}(\tau)-T_{pt}^{ac*}(\tau)\right|,\ \frac{1}{T}\sum\limits_{t\in T}\left|\tilde{T}_t^{ac*}(\tau)-T_{qt}^{ac*}(\tau)\right|\right\}}{\overline{T}_{pq}^{ac}} \le \varepsilon_{ac} \end{cases} \tag{4.31}$$

Taking the HVDC tie-line bus m and the HVAC tie-line bus p of area A as an example, if the convergence criterion is not satisfied in the τ th iteration, the penalty weights will be updated according to (4.32) and (4.33), and the $\tau+1$ th iteration will then be performed to calculate a new update.

$$\begin{cases} \alpha_{mt}^{dc}(\tau+1)=\alpha_{mt}^{dc}(\tau)+2\beta_{mt}^{dc}(\tau)^2\left[\tilde{T}_t^{dc*}(\tau)-T_{mt}^{dc*}(\tau)\right], \forall t \\ \beta_{mt}^{dc}(\tau+1)=\gamma\beta_{mt}^{dc}(\tau) \end{cases} \tag{4.32}$$

$$\begin{cases} \alpha_{pt}^{ac}(\tau+1)=\alpha_{pt}^{ac}(\tau)+2\beta_{pt}^{ac}(\tau)^2\left[\tilde{T}_t^{ac*}(\tau)-T_{pt}^{ac*}(\tau)\right], \forall t \\ \beta_{pt}^{ac}(\tau+1)=\gamma\beta_{pt}^{ac}(\tau) \end{cases} \tag{4.33}$$

where γ is an algorithm factor. Here, $\gamma \ge 1$ is strictly necessary for the convex optimization problem [51] to ensure that the sequence of the quadratic term's penalty weights in (4.32) and (4.33) is nondecreasing.

The IATC-DSCUC for a bulk AC/DC hybrid system is solved using an iterative procedure to find the optimal solution for the entire system. Since the local lower-level dispatch centers are disconnected from each other, the sub-problems can be run on different processors in parallel without exchanging information with neighboring areas. The steps of the solution procedure are as follows, and the flowchart is shown in Fig. 4.3.

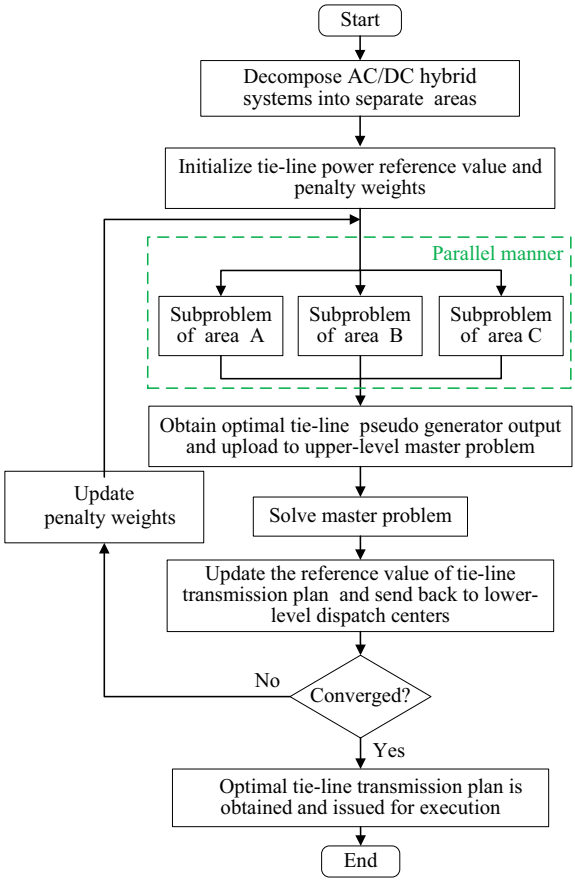


Fig. 4.3 Flowchart of the IATC-DSCUC solution procedure

Step (1) Decompose the bulk AC/DC hybrid system into separate areas.

Step (2) Set the iteration index $\tau = 1$ and initialize the penalty weights $\alpha_{mt}^{dc}(\tau)$, $\beta_{mt}^{dc}(\tau)$, $\alpha_{pt}^{ac}(\tau)$, $\beta_{pt}^{ac}(\tau)$, $\alpha_{nt}^{dc}(\tau)$, $\beta_{nt}^{dc}(\tau)$, $\alpha_{qt}^{ac}(\tau)$, and $\beta_{qt}^{ac}(\tau)$ and the tie-line power reference values $\tilde{T}_t^{dc*}(\tau - 1)$ and $\tilde{T}_t^{ac*}(\tau - 1)$.

Step (3) Solve the lower-level sub-problems in parallel to obtain the optimal tie-line pseudo generator outputs $T_{mt}^{dc*}(\tau)$, $T_{nt}^{dc*}(\tau)$, $T_{pt}^{ac*}(\tau)$, and $T_{qt}^{ac*}(\tau)$, and upload these outputs to the upper-level master problem for coordination.

Step (4) Solve the upper-level master problem to update the tie-line power reference values $\tilde{T}_t^{dc*}(\tau)$ and $\tilde{T}_t^{ac*}(\tau)$, and send these values back to the corresponding lower-level dispatch centers.

Step (5) Check whether convergence criterion (4.31) is achieved. If so, the optimal day-ahead tie-line transmission plan is obtained and issued for execution, and the solution procedure stops; otherwise, go to Step 6.

Step (6) Set $\tau = \tau + 1$ and update the penalty weights using (4.32)–(4.33), then go to Step 3.

4.3 Distributed Dispatch Approach in the VSC-MTDC Meshed AC/DC Hybrid Systems

In this section, considering the multilevel structure of the VSC-MTDC meshed AC/DC system, the scheduling problem is decomposed into three interactive levels according to the hierarchical analytical target cascading (ATC) technique. The low level is a two-stage adaptive robust security-constrained unit commitment problem for the AC system, which is solved by the column-and-constraint generation (C&CG) algorithm; the middle level is a day-ahead power transmission optimization problem for the VSC stations, where a day-ahead operation model for the VSC station is presented to fully use its flexible and controllable power adjustment capability to promote wind power accommodation; and the high level is a multi-period optimal power flow problem for the DC grid. An integrated ATC and C&CG algorithm is proposed to enable the hierarchical and robust scheduling. The parallel solution is used to accelerate the hierarchical scheduling.

4.3.1 Hierarchy of VSC-MTDC Meshed AC/DC Grid

4.3.1.1 VSC-MTDC Meshed AC/DC Hybrid System as a Three-Level Hierarchy

It is assumed an AC/DC grid consists of several VSCs that connect different AC areas through a MTDC network (see Fig. 4.4). This prototype is abstracted but extended from the Zhangbei four-terminal VSC-HVDC system in China, where the wind-thermal-bundled power is transmitted to the load center in North China via the MTDC network.

The scheduling of the above VSC-MTDC meshed AC/DC system in a centralized manner is typically a large-scale mixed-integer nonlinear optimization problem,

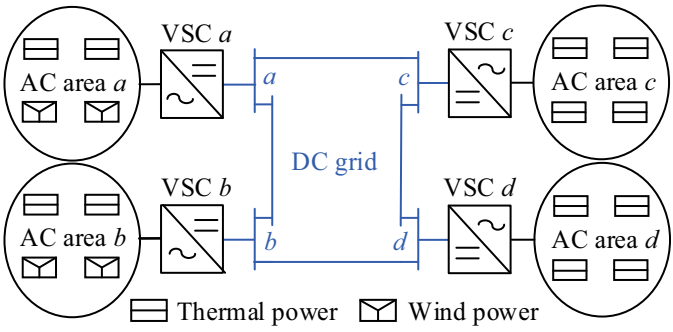


Fig. 4.4 VSC-MTDC meshed AC/DC grid

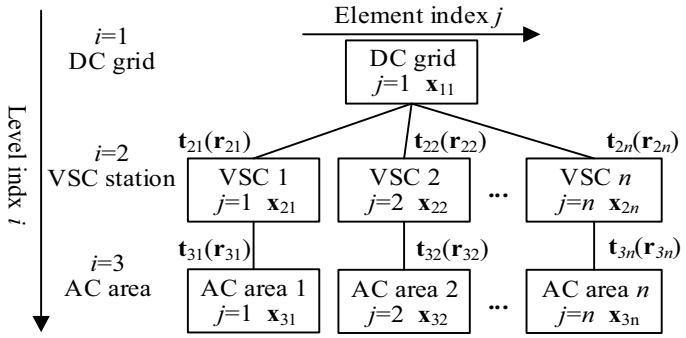


Fig. 4.5 Three-level hierarchy of the VSC-MTDC meshed AC/DC grid

leading to great computational complexity. Besides, the centralization will lead to operation by a single central super-entity with complete knowledge and control of the entire meshed AC/DC grid. This centralization may create substantial regulatory and political issues because the local system operators have to give up their governance and control to the central super entity. Thus, a hierarchical scheduling architecture is a preferred scheme, which can not only preserve the information privacy and decision independency, but also comply with the philosophy of electricity market operations.

If the above VSC-MTDC meshed AC/DC system is decomposed into two parts: the AC system and the MTDC system, the optimization of MTDC system is still a mixed-integer nonlinear optimization problem that is hard to solve. Thus, we decompose it into three parts: the AC areas, the DC grid, and the VSC stations coordinating the AC and DC grids. According to this meshed structure, a three-level hierarchy can be developed to fit the structure of the VSC-MTDC meshed AC/DC grid, as shown in Fig. 4.5. The element at level 1 represents the DC grid, the elements at level 2 represent the VSCs, and the elements at level 3 represent the AC areas. This structure indicates the adaptability of the hierarchical ATC technique to the VSC-MTDC meshed AC/DC grid.

4.3.1.2 General Description of the Hierarchical ATC Method

In this subsection, we formulate the HRS model within the ATC framework. Consider that the all-in-one (AIO) problem (4.34) expresses a centralized scheduling model for the entire VSC-MTDC meshed AC/DC hybrid system.

$$\begin{aligned} &\min \quad f(\mathbf{x}) \\ &s.t.\ \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ &\quad \mathbf{h}(\mathbf{x}) = \mathbf{0} \end{aligned} \tag{4.34}$$

where $\mathbf{x}$ is the vector of all variables, f is the objective function, $\mathbf{g}$ and $\mathbf{h}$ are respectively the inequality and equality constraints.

The ATC formulations belong to the class of augmented Lagrangian relaxation (ALR) methods. The feature that distinguishes ATC from other ALR methods is the representation and treatment of variables that couple subproblems using target-response pairings [37] [38]. The ATC procedure first decomposes the system into a multilevel hierarchical structure and recognizes parents and children. At the next step, penalty functions are introduced to model subproblems' interdependencies.

In ATC, subproblems (also called elements or autonomous systems) in the upper-levels are parents of subproblems in the lower-levels. By the same token, subproblems in the lower-levels are children of subproblems in the upper-levels. Although a child has only one parent, a parent could have multiple children. This hierarchical interconnection implies that subproblems at the same level do not share any connection/information with each other. If we assume the ATC structure as a graph, subproblems and tie-lines are respectively nodes and edges of the graph. As shown in Fig. 4.5, elements in the upper-level are hierarchically connected to elements in the lower-level. Thus, ATC is a suitable method to solve the scheduling of VSC-MTDC meshed AC/DC hybrid systems in a hierarchical manner.

The coordination in ATC is performed by communicating the targets and responses between parents and children [37]. Responses of components higher in the hierarchy depend on the responses of components lower in the hierarchy, but not vice versa. As shown in Fig. 4.5, assume each element has its own local variables $\mathbf{x}_{ij}$, and is coupled with other elements through target variables $\mathbf{t}_{ij}$. To make the objective functions and constraints in (4.34) separable, the response variables $\mathbf{r}_{ij}$ are introduced as copies of the target variables $\mathbf{t}_{ij}$, which leads to the modified AIO formulation with addition of the following consistency constraints.

$$\boldsymbol{\chi}_{ij} = \mathbf{t}_{ij} - \mathbf{r}_{ij} = 0 \tag{4.35}$$

These sets of consistency constraints are relaxed as the penalty function in the ATC subproblem. For the element j at level i, $\mathbf{f}_{ij}$ is the vector of local objective, and the vector functions $\mathbf{g}_{ij}$ and $\mathbf{h}_{ij}$ are the local inequality and equality constraints, respectively. $\mathbf{r}_{ij}$ is the vector of response variables to its parent at level $i-1$, $\mathbf{t}_{(i+1)j}$ is the vector of target variables to its children at level $i+1$, and R_i is the set of children for which level i sets targets and receives responses. Using method of multipliers [38] [52] with elimination of consistency constraints results in the relaxed AIO problem expressed as:

$$\begin{aligned}
&\min\ f_{ij}\left(\overline{\mathbf{x}}_{ij}\right)+\alpha_{ij}^{\mathbf{T}}\left(\mathbf{t}_{ij}-\mathbf{r}_{ij}\right)+\left\|\boldsymbol{\beta}_{ij}\circ\left(\mathbf{t}_{ij}-\mathbf{r}_{ij}\right)\right\|_{\mathbf{2}}^{\mathbf{2}}\\
&\quad+\sum_{j\in R_i}\left[\boldsymbol{\alpha}_{(i+1)j}^{T}\left(\mathbf{t}_{(i+1)j}-\mathbf{r}_{(i+1)j}\right)+\left\|\boldsymbol{\beta}_{(i+1)j}\circ\left(\mathbf{t}_{(i+1)j}-\mathbf{r}_{(i+1)j}\right)\right\|_{\mathbf{2}}^{\mathbf{2}}\right]\\
&s.t.\quad \mathbf{g}_{ij}\left(\overline{\mathbf{x}}_{ij}\right)\le\mathbf{0}\\
&\quad\mathbf{h}_{ij}\left(\overline{\mathbf{x}}_{ij}\right)=\mathbf{0}\\
&\text{with}\quad \overline{\mathbf{x}}_{ij}=\left[\mathbf{x}_{ij},\mathbf{r}_{ij},\mathbf{t}_{(i+1)j}\right]
\end{aligned}\tag{4.36}$$

4.3.1.3 Modeling Target and Response Variables

In this subsection, both target and response variables, as shared variables between the VSC-MTDC meshed AC/DC systems, are identified based on the physical connection among the AC areas, DC grid, and VSC stations. The three-level hierarchy of VSC-MTDC meshed AC/DC systems is illustrated in Fig. 4.6 to clearly show the flow of target and response variables. The AC areas, DC grid, and VSC stations are located in three different levels (DC grid is in the high level, VSC stations are in the middle level, and AC areas are in the low level). Because the VSC station is capable of controlling its power flexibly, the transferring power is the control variable which can be adopted directly as the target and response variables. The power exchange through the physical connection is the shared variable between these three independent systems, which links the SCUCs of AC areas, power transmission optimization of VSCs, and multi-period OPF of DC grid together.

The coupling interface between the AC and DC grid can be simply regarded as a pseudo generator or load added to the boundary bus. It should be noted that the pseudo generation and the pseudo load might be negative. The active power exchanged between the DC grid and VSC station is defined as the target variable $\mathbf{t}_{2j}$ in the DC grid's problem, and the response variable $\mathbf{r}_{2j}$ in the VSC station's problem, as shown in (4.37). Similarly, the active power exchanged between the VSC station and AC grid is defined as the target variables $\mathbf{t}_{3j}$ in the VSC station's problem, and the response variables $\mathbf{r}_{3j}$ in the AC grid's problem, as shown in (4.38). The superscripts in (4.37) and (4.38) indicate the problems where the sharing variables are calculated and the "*" represents the optimal values, the same below.

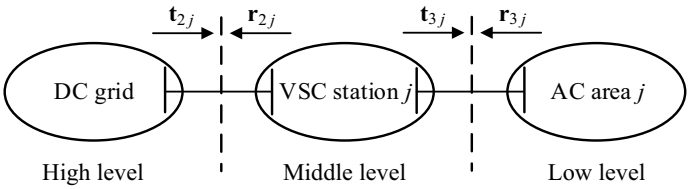


Fig. 4.6 Modeling target and response variables

$$\left[\mathbf{t}_{2j}\ \ \mathbf{r}_{2j}\right] = \left[\mathbf{P}_j^{DC*}\ \ \mathbf{P}_j^{VSC*}\right] \tag{4.37}$$

$$\left[\mathbf{t}_{3j}\ \ \mathbf{r}_{3j}\right] = \left[\mathbf{P}_j^{VSC*}\ \ \mathbf{P}_j^{AC*}\right] \tag{4.38}$$

4.3.2 Hierarchical and Robust Scheduling Formulation

4.3.2.1 Lower Level Robust SCUCs of AC Grids

The lower level is formulated as a two-stage adaptive robust SCUC problem with its own decision variables and coupling decision variables between the AC grid and neighboring VSC station. Under the hierarchical ATC framework, each lower level AC area is responsible for the scheduling of its own area, which can achieve the parallel calculation of robust SCUC problem in each AC area. In the first stage, the neighboring VSC power injection and the unit commitment decisions of thermal units are included as decision variables when the wind power is uncertain. Then in the second stage, the worst-case wind power scenario in the predefined uncertainty sets for each AC area is identified. Following the worst-case available wind power scenario, the security-constrained economic dispatch (SCED) decisions are made to minimize the total generation and wind power curtailment costs. Note that since the VSC power cannot be adjusted frequently in the real-time stage, the VSC power is optimized in the first stage instead of the second stage. The optimization model of the low level robust SCUC problem (e.g., AC area a) is as follows:

$$\begin{aligned}\min_{\mathbf{u},\mathbf{v},\mathbf{z},\mathbf{P}^{AC}} \sum_{t=1}^{T} \Bigg\{ & \sum_{g\in\Omega_a^G} \left(u_{gt}C_g^{ON} + v_{gt}C_g^{SU} + z_{gt}C_g^{SD}\right) \\ & + \alpha_{at}^{AC}\left(P_{at}^{VSC*} - P_{at}^{AC}\right) + \left[\beta_{at}^{AC}\left(P_{at}^{VSC*} - P_{at}^{AC}\right)\right]^2 \\ & + \max_{\tilde{\mathbf{P}}^W\in\Lambda^a} \min_{\mathbf{P}^G,\mathbf{P}^W} \left[\sum_{g\in\Omega_a^G}\sum_{k} c_{gk}P_{gkt}^G + \sum_{w\in\Omega_a^W} C^{WS}\left(\tilde{P}_{wt}^W - P_{wt}^W\right) \right] \Bigg\}\end{aligned} \tag{4.39}$$

s.t.

$$-u_{gt-1} + u_{gt} - u_{gj} \le 0,\ j = t,\cdots,t + MU_g - 1, \forall g \in \Omega_a^G, \forall t \tag{4.40}$$

$$u_{gt-1} - u_{gt} + u_{gj} \le 1,\ j = t,\cdots,t + MD_g - 1, \forall g \in \Omega_a^G, \forall t \tag{4.41}$$

$$-u_{gt-1} + u_{gt} - v_{gt} \le 0, \forall g \in \Omega_a^G, \forall t \tag{4.42}$$

$$u_{gt-1} - u_{gt} - z_{gt} \leq 0, \forall g \in \Omega_a^G, \forall t \tag{4.43}$$

$$\underline{P}_a^{VSC} \leq P_{at}^{AC} \leq \overline{P}_a^{VSC}, \forall t \tag{4.44}$$

$$\sum_{g \in \Omega_a^G} P_{gt}^G + \sum_{w \in \Omega_a^W} P_{wt}^W = \sum_{m \in \Omega_a^D} P_{mt}^D + P_{at}^{AC}, \forall t \tag{4.45}$$

$$u_{gt} \underline{P}_g^G \leq P_{gt}^G \leq u_{gt} \overline{P}_g^G, \forall g \in \Omega_a^G, \forall t \tag{4.46}$$

$$0 \leq P_{gkt}^G \leq u_{gt} \overline{P}_{gk}^G, \forall g \in \Omega_a^G, \forall t \tag{4.47}$$

$$P_{gt}^G = \sum_k P_{gkt}^G, \forall g \in \Omega_a^G, \forall t \tag{4.48}$$

$$P_{gt}^G - P_{gt-1}^G \leq u_{gt-1} UR_g + \left(u_{gt} - u_{gt-1}\right) \underline{P}_g^G + \left(1 - u_{gt}\right) \overline{P}_g^G, \forall g \in \Omega_a^G, \forall t \tag{4.49}$$

$$P_{gt-1}^G - P_{gt}^G \leq u_{gt} DR_g + \left(u_{gt-1} - u_{gt}\right) \underline{P}_g^G + \left(1 - u_{gt-1}\right) \overline{P}_g^G, \forall g \in \Omega_a^G, \forall t \tag{4.50}$$

$$0 \leq P_{wt}^{\mathrm{W}} \leq \tilde{P}_{wt}^{\mathrm{W}}, \forall w \in \Omega_a^W, \forall t \tag{4.51}$$

$$\left| \sum_{g \in \Omega_a^G} H_{lg}^G P_{gt}^G + \sum_{w \in \Omega_a^W} H_{lw}^W P_{wt}^W - \sum_{m \in \Omega_a^D} H_{lm}^D P_{mt}^D - H_l^{AC} P_{at}^{AC} \right| \leq \overline{L}_l^{AC}, \forall l \in \Omega_a^L, \forall t \tag{4.52}$$

The objective function in (4.39) contains the generation cost, wind curtailment cost and the augmented Lagrangian penalty term related to shared variables. Constraints (4.40) and (4.41) are the minimum up-/down-time limits of units, respectively. Constraints (4.42) and (4.43) are the start-up/shut-down limits of units, respectively. Constraint (4.44) is the power injection limits of neighboring VSC. Constraint (4.45) is the power balance constraint. Constraints (4.46)–(4.48) are the maximum/minimum capacity of units. Constraint (4.49)–(4.50) are the ramp-up/-down rate of units. Constraint (4.51) is the output limit of wind farms. Constraint (4.52) is the transmission security of AC lines.

The maximization problem in (4.39) seeks the worst-case available wind power scenario that jeopardizes the total operating costs. Since wind farms in different AC areas are geographically far apart, the correlation of wind energy among areas is not considered here, and the uncertainty of wind energy is specific to the area index. To overcome the conservativeness, we introduce the temporal and spatial uncertainty budgets. Then, without loss of generality, we assume the following cardinality constrained uncertainty set for wind power in each AC area (e.g., AC area a).

$$\begin{aligned}\Lambda_a = \Big\{ & \tilde{P}_{wt}^{W}, w \in \Omega_a^W, t \in T \mid \\ & \tilde{P}_{wt}^{W} = P_{wt}^{W0} + \varepsilon_{wt}^{+}\left(\overline{P}_{wt}^{W} - P_{wt}^{W0}\right) - \varepsilon_{wt}^{-}\left(P_{wt}^{W0} - \underline{P}_{wt}^{W}\right), \\ & \sum_{t=1}^{T}\left(\varepsilon_{wt}^{+} + \varepsilon_{wt}^{-}\right) \le \Gamma_w^{TB}, \sum_{w\in\Omega_a^W}\left(\varepsilon_{wt}^{+} + \varepsilon_{wt}^{-}\right) \le \Gamma_t^{SB}, \\ & \varepsilon_{wt}^{+} \in [0,1], \varepsilon_{wt}^{-} \in [0,1]\Big\}\end{aligned} \tag{4.53}$$

The choice of Γ_t^{SB} and Γ_w^{TB} can adjust the conservativeness of the robust solution. Specifically, if both Γ_t^{SB} and Γ_w^{TB} are 0, then there is no uncertainty in wind power output, while if Γ_w^{TB} is T and Γ_t^{SB} equals to the number of wind plants, all wind plants can reach their lower/upper limits in every period, corresponding to the most conservative case.

4.3.2.2 Middle Level Power Transmission Optimization of VSCs

The middle level is a power transmission optimization problem of VSC station. Due to the fast response characteristic of VSCs, the MTDC grid can provide operational support by redistributing power flow, thereby facilitating wind power accommodation. The output characteristics of the VSC station are fully modeled to use its flexible adjustment capability to promote wind power accommodation, which can achieve joint optimization of the VSC transmission power and the generating units in each AC area, thus ensuring the meshed AC/DC system to operate more flexibly and cost-effectively. The optimization function of the VSC station (e.g., VSC station a) is as follows:

$$\begin{aligned}\min \sum_{t=1}^{T} \Big\{ & \alpha_{at}^{AC}\left(P_{at}^{VSC} - P_{at}^{AC*}\right) + \left[\beta_{at}^{AC}\left(P_{at}^{VSC} - P_{at}^{AC*}\right)\right]^2 \\ & + \alpha_{at}^{DC}\left(P_{at}^{VSC} - P_{at}^{DC*}\right) + \left[\beta_{at}^{DC}\left(P_{at}^{VSC} - P_{at}^{DC*}\right)\right]^2 \Big\}\end{aligned} \tag{4.54}$$

Similar to the AC and DC grid, the objective function of VSC in (4.54) is the penalty function related to shared variables. Referring to our previous work in [53], the flexible and controllable power adjustment capability of the VSCs is modeled as below:

(1) Minimum/Maximum VSC Transmission Power Limits

$$\underline{P}_a^{VSC} \le P_{at}^{VSC} \le \overline{P}_a^{VSC}, \forall a \in \Omega^{VSC}, \forall t \tag{4.55}$$

(2) VSC Power Adjustment Direction Constraint

The binary variables γ_{at}, γ_{at}^{+} and γ_{at}^{-} are defined to represent whether the VSC adjusts its power, adjusts its power upward, and adjusts its power downward at hour

t, respectively. The constraint (4.56) ensures that the power adjustment direction of the VSCs cannot be reversed during adjacent periods.

$$\begin{cases} \gamma_{at}^{+} + \gamma_{at}^{-} = \gamma_{at} \le 1 \\ \gamma_{at}^{+} + \gamma_{at+1}^{-} \le 1 \\ \gamma_{at+1}^{+} + \gamma_{at}^{-} \le 1 \end{cases}, \forall a \in \Omega^{VSC}, \forall t \tag{4.56}$$

(3) Number of Adjustments to the VSC Transmission Power within a Scheduling Cycle

In this paper, the number of adjustments to the VSC power within a scheduling cycle is limited to increase its operating life, which can be determined by the practical operating experience and the performance of the VSCs.

$$\sum_{t=1}^{T} \left(\gamma_{at}^{+} + \gamma_{at}^{-}\right) \le S, \forall a \in \Omega^{VSC} \tag{4.57}$$

(4) VSC Power Adjustment Continuity Constraint

The VSC transmission power cannot be continuously adjusted upward or downward during the adjacent periods, that is,

$$\begin{cases} \gamma_{at}^{+} + \gamma_{at+1}^{+} \le 1 \\ \gamma_{at}^{-} + \gamma_{at+1}^{-} \le 1 \end{cases}, \forall a \in \Omega^{VSC}, \forall t \tag{4.58}$$

Additionally, the value of γ_{at}^{+} and γ_{at}^{-} can be expressed by the transmission power change in VSC station. Here, M^{+} and M^{-} are larger constants to make the following inequalities valid.

$$\begin{cases} P_{at+1}^{VSC} - P_{at}^{VSC} \le M^{+}\gamma_{at+1}^{+} \\ P_{at}^{VSC} - P_{at+1}^{VSC} \le M^{-}\gamma_{at+1}^{-} \end{cases}, \forall a \in \Omega^{VSC}, \forall t \tag{4.59}$$

(5) Minimum/Maximum VSC Power Adjustment Rate Limits

The upper power adjustment rate of VSC is less than its maximum power adjustment rate. However, the lower adjustment rate of VSC is set to be larger than a non-zero positive constant in this work to avoid the frequent small range power adjustment.

$$\gamma_{at}\underline{\delta}_{a}^{VSC} \le \left|P_{at+1}^{VSC} - P_{at}^{VSC}\right| \le \gamma_{at}\overline{\delta}_{a}^{VSC}, \forall a \in \Omega^{VSC}, \forall t \tag{4.60}$$

(6) Stepwise Operating Constraint of VSC Stations

To maintain the VSC transmission power in a stepwise state, once the transmission power is adjusted, it should be kept constant for a minimum duration of N_T. We

define the binary variables κ_{at}^{+} and κ_{at}^{-} to represent whether the VSC station is to start or end the power adjustment at hour t.

$$\begin{cases} \kappa_{at}^{-} + \sum_{\tau=t+1}^{\min(T,t+N_T)} \kappa_{a\tau}^{+} \le 1 \\ \kappa_{at}^{+} \ge \gamma_{at+1} - \gamma_{at} \\ \kappa_{at}^{-} \ge \gamma_{at} - \gamma_{at+1} \end{cases}, \forall a \in \Omega^{VSC}, \forall t \tag{4.61}$$

Additionally, the value of κ_{at}^{+} and κ_{at}^{-} is required to satisfy the following constraint.

$$\begin{cases} \kappa_{at}^{+} + \kappa_{at}^{-} \le 1 \\ \kappa_{at}^{+} + \kappa_{at+1}^{+} \le 1 \\ \kappa_{at}^{-} + \kappa_{at+1}^{-} \le 1 \end{cases}, \forall a \in \Omega^{VSC}, \forall t \tag{4.62}$$

4.3.2.3 High-Level Multi-period OPF of the DC Grid

The high-level is a multi-period OPF problem of the DC grid. The system operator of DC grid needs to increase its operational benefits by minimizing its power loss and following the connected VSC station's power injection requirements. Note that in this formulation, the VSCs are connected directly to each other without any middle point or DC bus. In the presence of any DC bus, it can be treated as a VSC node without power production or consumption. The nonlinear constrained optimization model of the DC grid is formulated as:

$$\begin{aligned} \min \sum_{t=1}^{T} \Bigg\{ & \sum_{a,b\in\Omega^{DC},b>a} \left(U_{at}^{DC} - U_{bt}^{DC}\right)^2 G_{ab} \\ & + \sum_{a\in\Omega^{DC}} \left[\alpha_{at}^{DC}\left(P_{at}^{VSC*} - P_{at}^{DC}\right) + \left[\beta_{at}^{DC}\left(P_{at}^{VSC*} - P_{at}^{DC}\right)\right]^2\right] \Bigg\} \end{aligned} \tag{4.63}$$

s.t.

$$P_{at}^{DC} = \sum_{b\in\Omega^{DC},b\neq a} \left(U_{at}^{DC} - U_{bt}^{DC}\right) G_{ab} U_{at}^{DC}, \forall a \in \Omega^{DC}, \forall t \tag{4.64}$$

$$\underline{U}_{a}^{DC} \le U_{at}^{DC} \le \overline{U}_{a}^{DC}, \forall a \in \Omega^{DC}, \forall t \tag{4.65}$$

$$\left|G_{ab}\left(U_{at}^{DC} - U_{bt}^{DC}\right)\right| \le \overline{I}_{ab}^{DC}, \forall a, b \in \Omega^{DC}, b \neq a, \forall t \tag{4.66}$$

$$\underline{P}_a^{VSC} \le P_{at}^{DC} \le \overline{P}_a^{VSC}, \forall a \in \Omega^{DC}, \forall t \tag{4.67}$$

The objective function in (4.63) is the power loss of DC lines plus augmented Lagrangian penalty term related to shared variables in all VSCs. Equation (4.64) is the DC power balance at converter nodes. Equations (4.65), (4.66) and (4.67) are the limit of dc voltage, dc current, and VSC station maximum capacity, respectively.

4.3.3 Solution Methodology

4.3.3.1 Integration of ATC and C&CG

The solution order of the HRS model is that low level sub-optimization problems (LSPs) of AC grids are solved first, followed by high level sub-optimization problem (HSP) of DC grid, and middle level sub-optimization problems (MSPs) of VSC stations are solved. All three level decisions are made one day in advance. Because each LSP is a two-stage RO model, it can be handled using the C&CG algorithm by iterative solution of the master problem (MP) and the subproblem (SP). An integrated ATC and C&CG algorithm is employed to solve the above HRS model, which is illustrated in Fig. 4.7.

In this solution framework, each AC grid and DC grid only needs to exchange the boundary power with its neighboring VSC station. Note that during the solution procedure, all the lower level AC grids can be solved in parallel, and all the middle level VSC stations can be solved in parallel. The steps of the solution procedure are as follows, and the flowchart is shown in Fig. 4.8.

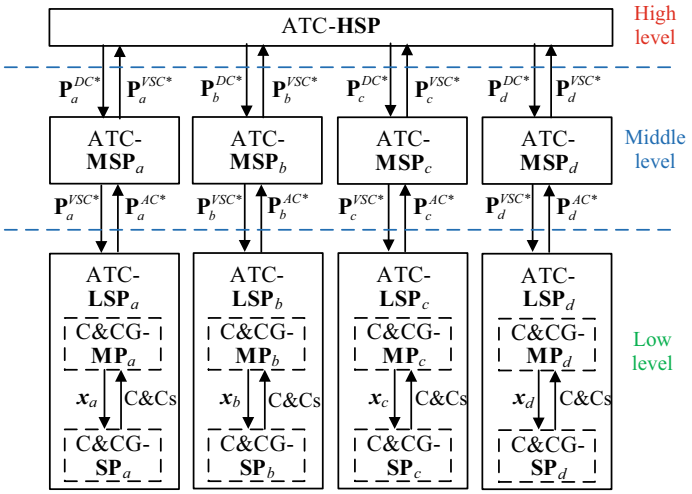


Fig. 4.7 An overview of the IAC algorithm

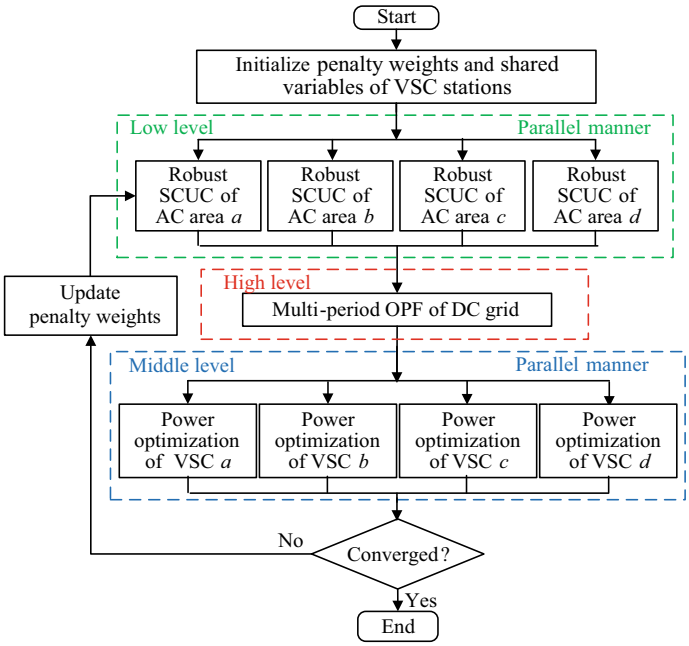


Fig. 4.8 Solution flowchart for the VSC-MTDC meshed AC/DC system

Step (1) Set the iteration index $\tau = 1$ and initialize the Lagrangian multipliers $\boldsymbol{\alpha}_t(\tau)$, $\boldsymbol{\beta}_t(\tau)$ and shared variable $\mathbf{P}_t^{VSC*}(\tau - 1)$ of VSCs.

Step (2) Solve the lower level problems in parallel for each AC area with $\mathbf{P}_t^{AC}(\tau)$ as decision variable and $\mathbf{P}_t^{VSC*}(\tau - 1)$ from the previous iteration (using C&CG algorithm in Section IV-B).

Step (3) Solve the high-level problem for DC grid with $\mathbf{P}_t^{DC}(\tau)$ as decision variable and $\mathbf{P}_t^{VSC*}(\tau - 1)$ from the previous iteration.

Step (4) Solve the middle level problems in parallel for each VSC station with $\mathbf{P}_t^{VSC}(\tau)$ as the decision variable and $\mathbf{P}_t^{AC*}(\tau)$, $\mathbf{P}_t^{DC*}(\tau)$ from Step 2 and Step 3.

Step (5) Check convergence criterion in (4.68). If satisfied, the optimal solution is obtained and the iteration is stopped; otherwise, go to Step 6.

$$\begin{cases} \dfrac{\frac{1}{T}\sum_{t=1}^{T}\left|P_{at}^{VSC*}(\tau) - P_{at}^{AC*}(\tau)\right|}{\overline{P}_a^{VSC}} \le \eta^{AC} \\ \dfrac{\frac{1}{T}\sum_{t=1}^{T}\left|P_{at}^{VSC*}(\tau) - P_{at}^{DC*}(\tau)\right|}{\overline{P}_a^{VSC}} \le \eta^{DC} \end{cases}, \forall a \in \Omega^{VSC} \tag{4.68}$$

Step (6) Set $\tau = \tau + 1$. Update Lagrangian multipliers and go to Step 2.

$$\begin{cases} \boldsymbol{\alpha}_t(\tau+1) = \boldsymbol{\alpha}_t(\tau) + 2\boldsymbol{\beta}_t(\tau) \circ \boldsymbol{\beta}_t(\tau) \circ \boldsymbol{\chi}_t(\tau) \\ \boldsymbol{\beta}_t(\tau+1) = \mu\boldsymbol{\beta}_t(\tau) \end{cases}, \forall t \tag{4.69}$$

where $\boldsymbol{\chi}_t$ represents the vector of consistency constraints, μ is the algorithm coefficient. Here, $\mu \geq 1$ is strictly necessary [38] to ensure that the sequence of the quadratic term's Lagrangian multipliers in the objective function is nondecreasing. The selection of μ is preferably small to avoid premature algorithm convergence and runs into the local optimum.

In fact, applying the combination of ALR and (4.69) is known as the method of multipliers [38] [52]. It has been proven that when the problem is a convex optimization, the ATC algorithm can converge to the globally optimal solution [38] [54]. Although there is no direct proof for the nonconvex MIP problems being studied in this work, for a nonconvex optimization problem, the non-convexity can be mitigated by ATC where quadratic penalty terms acting as a local convexifier [38] [54] are added to the objective function to improve the problem convexity.

Solution of Lower Level AC Grid

The lower level AC area is essentially a two-stage RO problem, which can be addressed by the C&CG algorithm [55] This algorithm exploits a master-subproblem iteration framework, and it identifies critical scenarios from the uncertainty set progressively and includes their recourse problems in the master problem. For simplicity, we write the two-stage RO of AC area a (omit the subscript a) in a matrix form:

$$\min_{\mathbf{x}} C(\mathbf{x}) + \max_{\mathbf{w}\in\Lambda}\left(\mathbf{b}^T\mathbf{w} + \min_{\mathbf{y}\in\Theta(\mathbf{x},\mathbf{w})} \mathbf{d}^T\mathbf{y}\right) \tag{4.70}$$

s.t.

$$\mathbf{A}\mathbf{x} \leq \mathbf{f} \tag{4.71}$$

$$\begin{aligned} \Lambda = \{\mathbf{w}\,|\mathbf{w} = \mathbf{w}^0 + \widehat{\mathbf{w}}^+\boldsymbol{\varepsilon}^+ - \widehat{\mathbf{w}}^-\boldsymbol{\varepsilon}^-, \\ \mathbf{D}\left(\boldsymbol{\varepsilon}^+ + \boldsymbol{\varepsilon}^-\right) \leq \boldsymbol{\Gamma}, \\ 0 \leq \boldsymbol{\varepsilon}^+ \leq 1, 0 \leq \boldsymbol{\varepsilon}^- \leq 1\} \end{aligned} \tag{4.72}$$

$$\Theta(\mathbf{x}, \mathbf{w}) = \{\mathbf{y}\,|\mathbf{F}\mathbf{y} \leq \mathbf{g} - \mathbf{B}\mathbf{x} - \mathbf{C}\mathbf{w}\} \tag{4.73}$$

In these expressions, vector $\mathbf{x}$ represents the first-stage decision variables, $\mathbf{w}$ denotes the uncertain parameters, and $\mathbf{y}$ represents the second-stage decision variables. Inequality (4.71) describes constraints (4.40)–(4.44). Inequality (4.72) includes constraint (4.53). Inequality (4.73) describes constraints (4.46)–(4.52).

(1) Problem Reformulation

To solve the above two-stage RO, we first find the dual of the innermost minimization problem by observing that it is a linear program and strong duality hold, then transform the second-stage problem into the following:

$$\max_{\mathbf{w},\boldsymbol{\lambda}} \mathbf{b}^T\mathbf{w} - \boldsymbol{\lambda}^T(\mathbf{g} - \mathbf{B}\mathbf{x} - \mathbf{C}\mathbf{w}) \tag{4.74}$$

s.t.

$$\begin{aligned} &\mathbf{d}^T + \boldsymbol{\lambda}^T\mathbf{F} = 0 \\ &\boldsymbol{\lambda} \geq 0, \mathbf{w} \in \Lambda \end{aligned} \tag{4.75}$$

Since the budgets Γ_t^{SB} and Γ_w^{TB} are integer, the worst-case scenario must be one in which wind power equals to its upper bound, lower bound or forecast value. In other words, the candidate values of variables $\boldsymbol{\varepsilon}^+$ and $\boldsymbol{\varepsilon}^-$ must be either zero or one. With this insight, the bilinear program is reformulated into a MILP using the big-M method as follows:

$$\Psi(x) = \max_{\mathbf{w},\boldsymbol{\lambda},\boldsymbol{\xi}^+,\boldsymbol{\xi}^-,\boldsymbol{\varepsilon}^+,\boldsymbol{\varepsilon}^-} \mathbf{b}^T\mathbf{w} - \boldsymbol{\lambda}^T\left(\mathbf{g} - \mathbf{B}\mathbf{x} - \mathbf{C}\mathbf{w}^0\right) + \mathbf{1}^T\boldsymbol{\xi}^+ + \mathbf{1}^T\boldsymbol{\xi}^-$$

$$\mathbf{SP}: \quad \begin{aligned} s.t.\ \ & \mathbf{w} = \mathbf{w}^0 + \widehat{\mathbf{w}}^+\boldsymbol{\varepsilon}^+ - \widehat{\mathbf{w}}^-\boldsymbol{\varepsilon}^- \\ & \mathbf{d}^T + \boldsymbol{\lambda}^T\mathbf{F} = 0 \\ & \boldsymbol{\xi}^+ \leq M\boldsymbol{\varepsilon}^+, \boldsymbol{\xi}^+ - \boldsymbol{\lambda}^T\left(\mathbf{C}\widehat{\mathbf{w}}^+\right) \leq M\left(1 - \boldsymbol{\varepsilon}^+\right) \\ & \boldsymbol{\xi}^- \leq M\boldsymbol{\varepsilon}^-, \boldsymbol{\xi}^- + \boldsymbol{\lambda}^T\left(\mathbf{C}\widehat{\mathbf{w}}^-\right) \leq M\left(1 - \boldsymbol{\varepsilon}^-\right) \\ & \mathbf{w} \in \Lambda, \boldsymbol{\lambda} \geq 0, \boldsymbol{\varepsilon}^+ \in \{0,1\}, \boldsymbol{\varepsilon}^-\boldsymbol{\varepsilon}\{0,1\} \end{aligned}$$

(2) Column-and-Constraint Generation Procedure

Step (1) Initialization: Set $LB = -\infty$, $UB = +\infty$, $n = 0$ and $O = \emptyset$. Set the convergence tolerance $\rho > 0$.

Step (2) Solve the **MP**:

$$\mathbf{MP}: \quad \begin{aligned} \min_{\mathbf{x},\alpha}\ \ & C(\mathbf{x}) + \sigma \\ s.t.\ \ & \mathbf{A}\mathbf{x} \leq \mathbf{f} \\ & \sigma \geq \mathbf{b}^T\mathbf{w}^{*(k)} + \mathbf{d}^T\mathbf{y}^{(k)},\ \forall k \in O \\ & \mathbf{F}\mathbf{y}^{(k)} \leq \mathbf{g} - \mathbf{B}\mathbf{x} - \mathbf{C}\mathbf{w}^{*(k)},\ \forall k \leq n \end{aligned}$$

Obtain the optimal solution $\left\{\mathbf{x}^{*(n+1)}, \sigma^{*(n+1)}, \mathbf{y}^{*(1)}, ..., \mathbf{y}^{*(k)}\right\}$ and update $LB = C\left(\mathbf{x}^{*(n+1)}\right) + \sigma^{*(n+1)}$.

Step (3) Solve the **SP**s: Solve the **SP**s and obtain the optimal solutions $\left\{\mathbf{w}^{*(n+1)}, \boldsymbol{\lambda}^{*(n+1)}\right\}$.

Step (4) Check for convergence: update $UB = \min\left\{UB, C\left(\mathbf{x}^{*(n+1)}\right) + \Psi\left(\mathbf{x}^{*(n+1)}\right)\right\}$. If $(UB - LB)/LB < \rho$, terminate; otherwise go to Step 5.

Step (5) Generate constraints and columns:
(5.1): If $\Psi\left(\mathbf{x}^{*(n+1)}\right) < +\infty$, set $O = O \cup \{n+1\}$ and $n = n+1$. Add the decision variables $\mathbf{y}^{(n+1)}$ and the following constraints to **MP**, then go to Step 2:

$$\begin{aligned} \sigma &\geq \mathbf{b}^T \mathbf{w}^{*(n+1)} + \mathbf{d}^T \mathbf{y}^{(n+1)} \\ \mathbf{F}\mathbf{y}^{(n+1)} &\leq \mathbf{g} - \mathbf{B}\mathbf{x} - \mathbf{C}\mathbf{w}^{*(n+1)} \end{aligned} \tag{4.76}$$

(5.2): If $\Psi\left(\mathbf{x}^{*(n+1)}\right) = +\infty$, set $n = n+1$. Add the decision variables $\mathbf{y}^{(n+1)}$ and the following constraint to **MP**, then go to Step 2.

$$\mathbf{F}\mathbf{y}^{(n+1)} \leq \mathbf{g} - \mathbf{B}\mathbf{x} - \mathbf{C}\mathbf{w}^{*(n+1)} \tag{4.77}$$

4.4 Case Studies

4.4.1 Distributed Dispatch Approach in Bulk AC/DC Hybrid Systems

4.4.1.1 Three-Area 18-Bus AC/DC Hybrid System

Case 1 consists of three identical 6-bus systems [56] (namely, areas A, B and C), as depicted in Fig. 4.9. One HVAC tie-line links bus 3 of area A to bus 5 of area C, and the reactance value is 0.15 p.u. One HVDC tie-line connects bus 5 of area A to bus 5 of area B. Two identical wind plants are located at buses 2 and 4 of area A. The daily load and the output of each wind plant of area A are shown in Fig. 4.10, and the wind power penetration ratio of area A is up to 82%. The line flow limit of area A is doubled to enable greater wind power integration. To differentiate the three areas, the cost coefficient of a conventional generator in areas A, B, and C is multiplied by 1.0, 2.0, and 2.0, respectively. The load of areas A, B, and C is multiplied by 1.0, 1.2, and 1.2, respectively. In this way, we force the power imports to area B and C to achieve inter-regional accommodation of wind power. No deviation is allowed in the exchanged electrical energy on a tie-line, that is, $\rho^{dc}=\rho^{ac}=0$. The HVDC and HVAC tie-line capacity limits are [50, 150] MW and [0, 100] MW, respectively. The daily planned electrical energy transmission of the HVDC and HVAC tie-lines is 2 GWh and 1.5 GWh, respectively. The power adjustment rate of the HVDC tie-line is [10, 30] MW, and the penalty cost of wind curtailment is 100$/MW. The allowable number of adjustments of the HVDC tie-line power per day is $S = 6$, the minimum duration time interval is $N_T = 2$, the convergence threshold is $\varepsilon_{ac} = \varepsilon_{dc} = 1\%$, the initial value of shared variables $\tilde{T}_t^{dc*}=\tilde{T}_t^{ac*}=0$, the initial value of the penalty weights is $\alpha_{mt}^{dc}=\beta_{mt}^{dc} = \alpha_{nt}^{dc}=\beta_{nt}^{dc} = \alpha_{pt}^{ac}=\beta_{pt}^{ac} = \alpha_{qt}^{ac}=\beta_{qt}^{ac}=0.2$, and the algorithm factor is $\gamma = 1.2$.

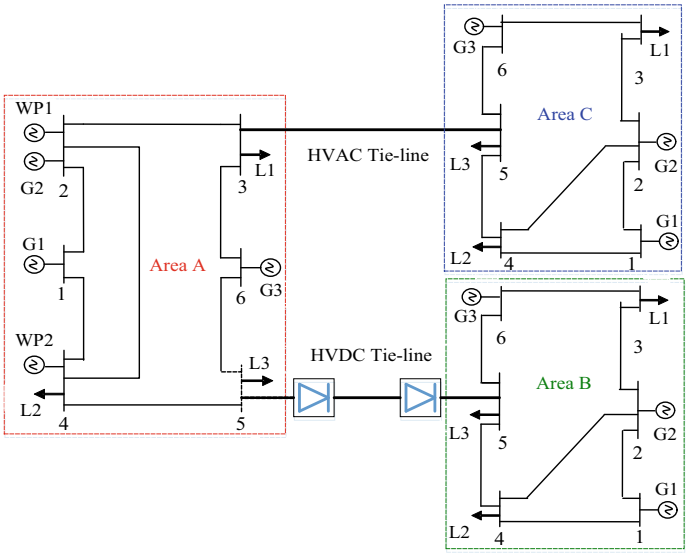


Fig. 4.9 Three-area 18-bus bulk AC/DC hybrid system

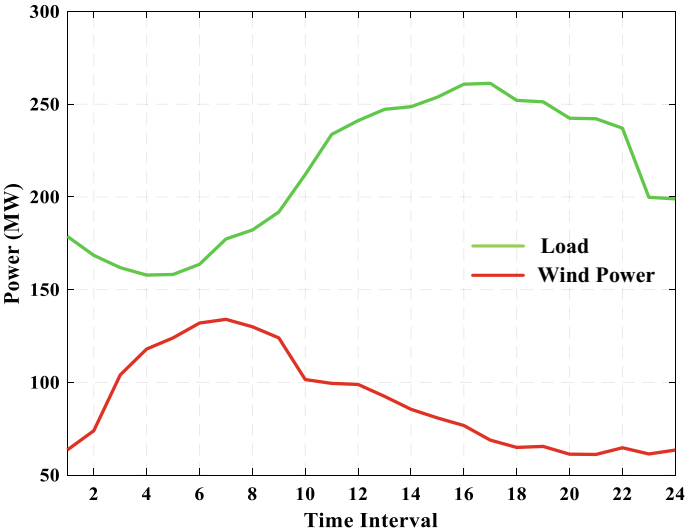


Fig. 4.10 Load and wind power forecasts

To verify the superiority of the proposed model considering the HVDC tie-line's flexible adjustment ability, four HVDC tie-line transmission modes with the same daily electrical energy transmission are selected for comparison.

Mode 1: Taking into account the flexible and controllable nature of the HVDC tie-line, the HVDC tie-line's flexible adjustment capability is fully modeled by a pseudo generator, and the tie-line transmission power is jointly optimized with the regional generating units.

Mode 2: Not considering the HVDC tie-line's flexible adjustment capability, a fixed and segmented power transmission mode of the HVDC tie-line is adopted, in which more power is transferred during the peak load period, and less power is transferred during the off-peak load period.

Mode 3: Similar to Mode 2, the fixed and segmented power transmission mode is adopted, in which more power is transferred during periods of high wind power, and less power is transferred during periods of low wind power.

Mode 4: Not considering the HVDC tie-line's flexible adjustment capability, a constant power transmission mode is adopted, in which the transmission power of the HVDC tie-line remains constant throughout the scheduling cycle.

For Mode 1, the proposed IATC-DSCUC is adopted, and for Mode 2–4, the CSCUC is adopted. The transmission plans of the HVDC tie-line and the optimized results of these four transmission modes are compared with each other in Fig. 4.11 and Table 4.1. The HVDC transmission power in Mode 1 increases during both the high wind power period and the peak load period, reflecting the flexible adjustment of the HVDC tie-line. As summarized in Table 4.1, although the peak-valley difference rate of the receiving side in Mode 2 is the smallest, the wind power curtailment rate and generation cost are much greater in Mode 2 than in Modes 1, 3 and 4. The wind power curtailment rate is greater in Mode 3 than in Mode 1, and the peak-valley difference rate of the receiving side is nearly 17% higher in Mode 3 than in Mode 1. In Mode 1, although the peak-valley difference rate of the receiving side is slightly greater than that of Mode 2, the wind power curtailment rate and generation cost are minimal. We can conclude that by making full use of the HVDC tie-line's flexible adjustment capability, the wind power inter-regional accommodation potential of the bulk AC/DC hybrid system can be exploited, the peak regulating pressure of the receiving side can be relieved, and the economics of the entire system can be improved.

The convergence performance of the proposed IATC-SCUC for the bulk AC/DC hybrid system is tested. As an example, for typical 3th, 8th, 11th, and 19th periods, the values of the shared variables among the regional sub-problems and the upper-level master problem are shown in Fig. 4.12. We can see that the tie-line power converges within an $\varepsilon_{ac} = \varepsilon_{dc} = 1\%$ tolerance after 14 iterations.

To verify the efficiency of the proposed IATC-DSCUC, the optimization results of CSCUC, IATC-DSCUC and TATC-DSCUC for Case 1 are compared in Table 4.2. In the solution of the CSCUC, a fictitious central entity with complete knowledge and control of the entire AC/DC hybrid system formulates the scheduling plan in a centralized manner. In the TATC-DSCUC, the operating constraints of the tie-lines are included in the corresponding sub-problems, and the only constraint for

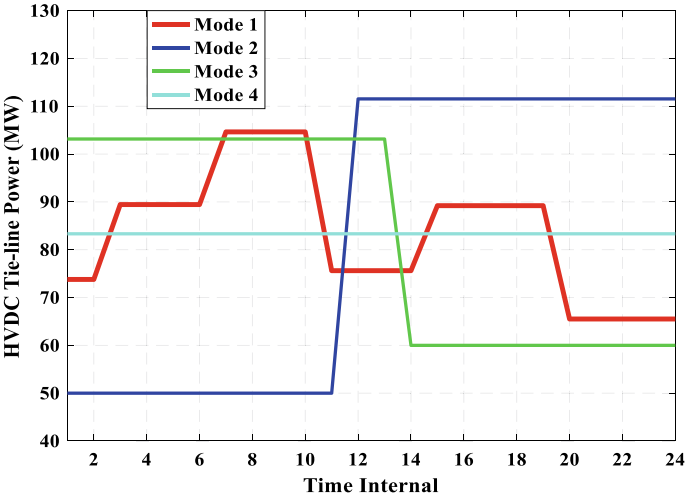


Fig. 4.11 HVDC tie-line transmission plans for case 1

Table 4.1 Comparisons of different transmission modes for case 1

	Mode 1	Mode 2	Mode 3	Mode 4
Wind power curtailment rate (%)	1.4	10.2	1.8	5.1
Peak-valley difference rate of receiving side (%)	49.8	44.8	64.6	53.9
Generation cost ($)	346,226	381,228	356,451	367,541

the master problem is the regional consistency constraint. The algorithm parameters and the convergence threshold of the TATC-DSCUC are the same as those in the proposed IATC-DSCUC. We can see that the generation cost of the IATC-DSCUC is 0.06% higher than that of the CSCUC. The generation cost of the TATC-DSCUC and the IATC-DSCUC are almost the same, but the number of iterations and the calculation time required for the IATC-DSCUC to converge are lower than that of the TATC-DSCUC. The IATC-DSCUC takes 53.2 s to converge, and the calculation time of the CSCUC is 27.3 s. The calculation time of the proposed IATC-DSCUC is higher than those of the CSCUC. This is because the scale of Case 1 is smaller and therefore the distributed dispatch would not be of great help to improving computing efficiency. Additionally, the distributed approach is not intended to improve computing efficiency but to achieve regional decomposition aiming for area scheduling independence and information privacy.

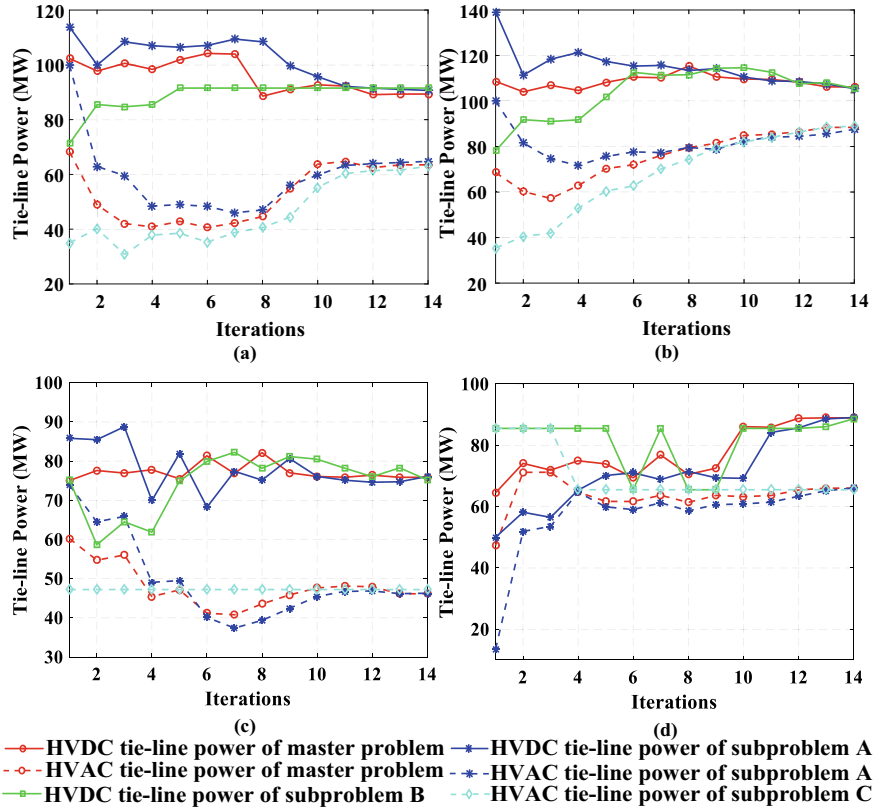


Fig. 4.12 Convergence curve of tie-line power for Case 1 in the **a** 3th period, **b** 8th period, **c** 11th period, and **d** 19th period

Table 4.2 Comparisons of different dispatch approaches for case 1

Dispatch approach	Iterations	Wind power curtailment rate (%)	Calculation time (s)	Generation cost ($)
CSCUC	–	1.4	27.3	346,021
IATC-DSCUC	14	1.4	53.2	346,226
TATC-DSCUC	18	1.4	79.4	346,219

4.4.1.2 Three-Area 117-Bus Bulk AC/DC Hybrid System

Case 2 consists of three New England 39-bus systems [57] (areas A, B and C). One HVAC tie-line connects bus 24 of area A to bus 24 of area C, and the reactance value is 0.025 p.u. One HVDC tie-line connects bus 9 of area A to bus 9 of area B. Three identical wind plants are located at buses 4, 21 and 27 of area A. The daily load of area A is 20 times the value in Case 1, and the output of one wind plant is 17 times

that in Case 1. The wind power penetration rate of area A is up to 55%. The load and cost coefficients of conventional generators are handled in the same way as in Case 1. The ramp-down/ramp-up rate is set to 30% of the maximum output, and the minimum continuous start up/off time of the conventional generators is consistently 3 h. The HVDC and HVAC tie-line capacity limits are [400, 1500] MW and [0, 1000] MW, respectively. The daily planned electrical energy transmission of the HVDC and HVAC tie-lines is 20 GWh and 15 GWh, respectively. The power adjustment rate of an HVDC tie-line is [100, 500] MW. The initial value of the penalty weights is $\alpha_{mt}^{dc} = \beta_{mt}^{dc} = \alpha_{nt}^{dc} = \beta_{nt}^{dc} = \alpha_{pt}^{ac} = \beta_{pt}^{ac} = \alpha_{qt}^{ac} = \beta_{qt}^{ac} = 0.02$, and the algorithm factor is $\gamma = 1.1$. The other parameters are the same as those in Case 1.

To further verify the superiority of the HVDC tie-line's flexible adjustment ability in promoting inter-regional wind power accommodation, similar to Case 1, four HVDC tie-line transmission modes with the same daily electrical energy transmission are adopted in Case 2, and the results are shown in Fig. 4.13 and Table 4.3. For Mode 1, which takes into account the controllable nature of the HVDC tie-line, the peak-valley difference rate of the receiving side is smaller, and the wind power curtailment rate and generation cost are minimal. Thus, the flexible adjustment of the HVDC tie-line's transmission power can again promote inter-regional wind power accommodation and improve the economics of the entire system.

Figure 4.14 depicts the convergence curve of the generation cost. The solutions of the proposed IATC-DSCUC nearly coincide with the results of the CSCUC after 16 iterations. The optimization results of the CSCUC, IATC-DSCUC and TATC-DSCUC for Case 2 are summarized in Table 4.4. The proposed IATC-DSCUC takes 114.5 s to converge with a generation cost of $4,194,989, and the CSCUC has a generation cost of $4,190,837 after 101.3 s. For Case 2, similar to the results in Case

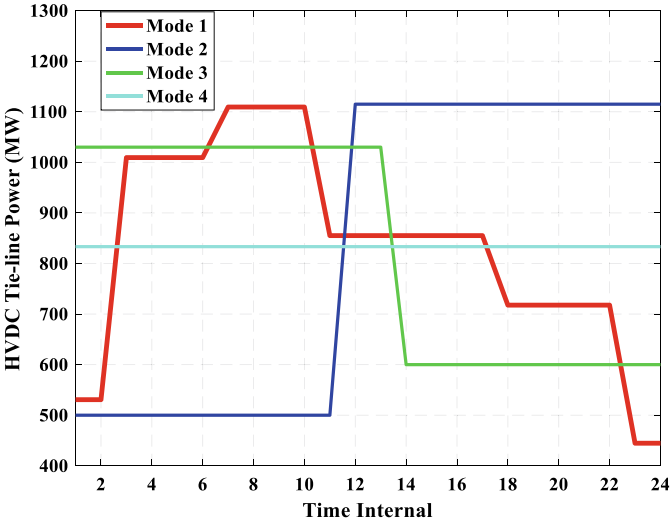


Fig. 4.13 HVDC tie-line transmission plans for case 2

Table 4.3 Comparisons of different transmission modes for case 2

	Mode 1	Mode 2	Mode 3	Mode 4
Wind power curtailment rate (%)	2.1	12.3	4.6	6.2
Peak-valley difference rate of receiving-side (%)	47.7	38.2	54.3	45.6
Generation cost (k$)	4195.0	4733.1	4360.9	4490.0

1, the generation costs of TATC-DSCUC and IATC-DSCUC are almost the same, but the number of iterations and the calculation time of IATC-DSCUC are lower than those of TATC-DSCUC.

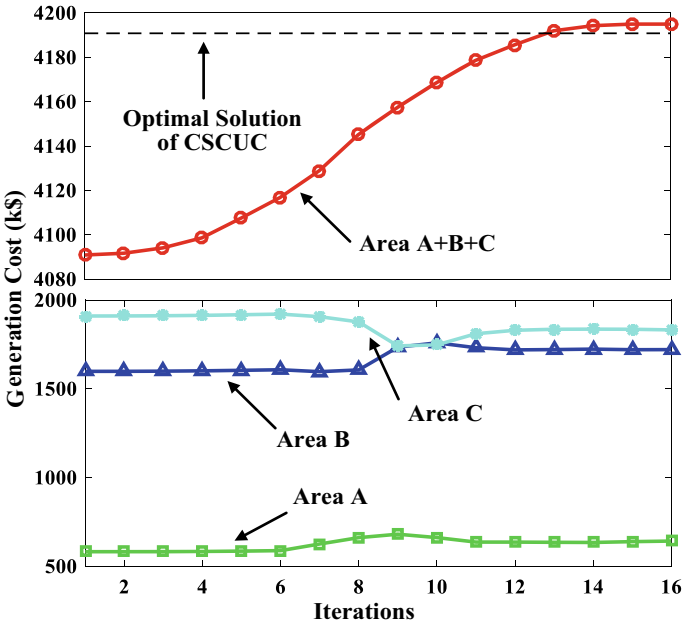


Fig. 4.14 Convergence curve of generation cost for case 2

Table 4.4 Comparisons of different dispatch approaches for case 2

Dispatch approach	Iterations	Wind power curtailment rate (%)	Calculation time (s)	Generation cost (k$)
CSCUC	–	2.1	101.3	4190.8
IATC-DSCUC	16	2.1	114.5	4195.0
TATC-DSCUC	21	2.1	186.1	4195.3

4.4.1.3 Three-Area 354-Bus Bulk AC/DC Hybrid System

Case 3 consists of 354 buses, 558 internal lines, 162 generators, 5 wind plants, 273 loads, 1 HVDC tie-line, and 1 HVAC tie-line and is created by three IEEE 118-bus systems [58]. The HVAC tie-line connects bus 65 of area A to bus 65 of area C, and the reactance value is 0.05 p.u. The HVDC tie-line connects bus 25 of area A to bus 25 of area B. Five identical wind plants are located at bus 36, bus 38, bus 63, bus 68, and bus 77 of area A. The daily load of area A is 26 times that in Case 1, and the output of one wind plant is 13 times that in Case 1. The wind power penetration rate of area A is up to 53%. The load and cost coefficients of conventional generators are handled in the same way as in Case 1 to differentiate the three areas. The HVDC and HVAC tie-line capacity limits are [400, 1500] and [0, 1500] MW, respectively. The day-ahead planned electrical energy transmission is 25 GWh for both the HVDC and HVAC tie-line. The other parameters are the same as those in Case 2.

Similar to Case 1, the four previously described HVDC tie-line transmission modes are adopted in Case 3, and the results are shown in Table 4.5. The wind power curtailment rate and the generation cost in Mode 1 are always the minimal values, since Mode 1 jointly optimizes HVDC tie-line transmission power with the regional generating units. Consequently, we can conclude that flexibly adjusting the HVDC tie-line transmission power can promote inter-regional wind power accommodation and improve the economics of the entire system for large-scale bulk AC/DC hybrid systems.

The comparison of the CSCUC, IATC-DSCUC and TATC-DSCUC for Case 3 is summarized in Table 4.6. The calculation times of the CSCUC and IATC-DSCUC are 1224.2 and 920.7 s, respectively. With increasing grid scale, the advantage in calculation time of the proposed IATC-DSCUC becomes outstanding, but the calculation time of the TATC-DSCUC is still greater than that of the CSCUC. This result occurs because in the TATC-DSCUC formulation, the complex operating constraints of tie-lines are included in the corresponding sub-problems, which increases the computing complexity of the regional grids, but fails to fully utilize the computing capability of the upper-level master problem. Besides, the convergence criterion of the TATC-DSCUC must separately check whether the tie-line power interaction errors of all the scheduling periods between the tie-line transmission plans developed by regional grids and the regional consistency coordination variables issued by the master problem satisfies the convergence precision, which will take more iterations to converge.

Table 4.5 Results of different VSC transmission modes for case 1

VSC transmission modes		Mode 1	Mode 2	Mode 3	Mode 4
Wind power curtailment rate	Area *a* (%)	1.68	6.45	4.78	4.99
	Area *b* (%)	7.21	18.13	13.90	16.01
Total generation cost ($)		282,632	339,446	301,123	311,451

Table 4.6 Comparisons of different scheduling models for case 1

Model	Wind power curtailment rate		DC grid energy loss rate (%)	Total generation cost ($)	Calculation time
	Area *a* (%)	Area *b* (%)			
CRS	1.61	6.97	4.2	280,350	>6 h
HRS	1.68	7.21	4.0	282,632	2,891 s

It is worth noting that in actual bulk AC/DC hybrid systems, the regional sub-optimization model can be simultaneously implemented by computers located in different regions, and the solutions are then uploaded to the upper-level dispatch center for coordination. With the ability to perform parallel computing, the proposed distributed dispatch approach has the potential to be computationally superior to the centralized dispatch approach, both in terms of solution speed and the maximum problem size that can be addressed. Furthermore, the advantages of the proposed IATC-DSCUC for bulk AC/DC hybrid systems reflected in the calculation time also show greater suitability for the hierarchical and partitioned power scheduling mode with multi-level dispatch centers. In short, the proposed distributed dispatch approach considering the flexible and controllable nature of the HVDC tie-line can not only promote inter-regional wind power accommodation, but also preserve the area scheduling independence for large-scale hierarchical and partitioned bulk AC/DC hybrid systems.

4.4.2 Distributed Dispatch Approach in VSC-MTDC Meshed AC/DC Hybrid Systems

4.4.2.1 Four-Terminal VSC-HVDC with Four 6-Bus AC Grids

As shown in Fig. 4.15, a four-terminal VSC-HVDC system with four identical 6-bus AC grids [59] is established in Case 1. The lengths between the terminals involved are $L_{ac} = L_{bd} = 300$ km, $L_{ab} = L_{cd} = 150$ km, and the resistance value is 0.01 Ω/km. The dc voltage limit is [90, 110] kV, and the dc current limit is [−5, 5] kA. The master station *c* keeps 100 kV to provide the constant dc voltage reference for the system. Bus 3 of each AC area is linked to the corresponding VSC. Two identical wind farms are located at buses 2 and 4 of AC areas *a* and *b*, respectively. The load and each wind farm output of area *a* are shown in Fig. 4.6. To differentiate the four AC areas, the load of areas *a*, *b*, *c,* and *d* is multiplied by 1.0, 1.0, 1.2, and 1.4, respectively. Each wind farm output in areas *a* and *b* is multiplied by 1.0 and 1.3, respectively. In this way, we force the power exports from the areas *a* and *b* to areas *c* and *d* to achieve the inter-regional accommodation of wind power. The ratio of total wind power and load in the entire system is up to 62%. The internal line flow limit of all AC areas is multiplied by 1.5 to enable greater wind power integration.

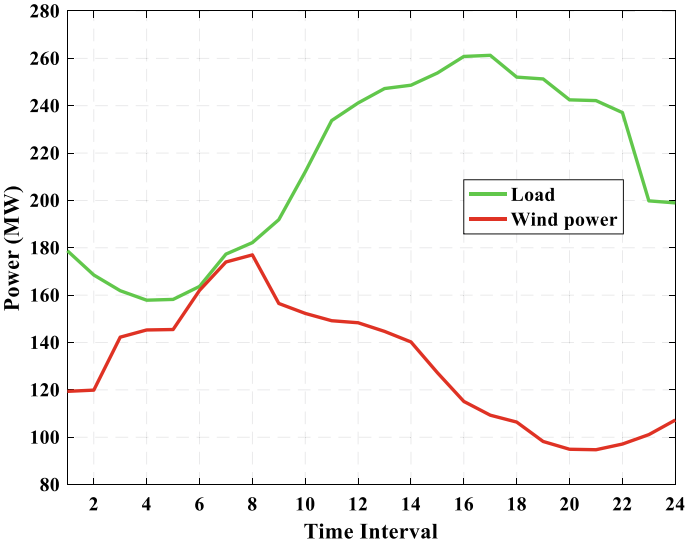


Fig. 4.15 Load and wind farm output

The wind output uncertainties are 20% of their forecast values, and the wind power curtailment penalty is 100$/MW. The minimum/maximum power limits of the VSCs are all [50, 200] MW, the power adjustment rates are all [20, 50] MWh^{-1}, and $S = 8$, $N_T = 2$. The initial values of the shared variables and Lagrangian multipliers are $\mathbf{P}_t^{VSC*}$=0 and $\boldsymbol{\alpha}_t = \boldsymbol{\beta}_t = 0.1$, respectively, and $\eta^{AC} = \eta^{DC} = 1\%$, $\mu = 1.1$. The gap tolerance of Gurobi is 0.1%, and the convergence tolerance of the C&CG algorithm is $\rho = 0.5\%$. The budgets of the areas a and b are both Γ_w^{TB}= 12, Γ_t^{SB}= 1.

(1) VSC Power Scheduling Results

To verify the superiority of the consideration of the VSCs' flexible adjustment capability, Case 1 is scheduled in four modes.

Mode 1: the flexible and controllable nature of VSCs is taken into account to achieve the joint optimization with the generating units in AC areas, which is the proposed HRS model.

Mode 2: a segmented VSC transmission mode is adopted, in which more power is transferred during the peak load period, and less power is transferred during the off-peak load period.

Mode 3: a segmented VSC transmission mode is adopted, in which more power is transferred during high wind power periods, and less power is transferred during low wind power periods.

Mode 4: a constant VSC transmission mode is adopted, in which the VSC power remains unchanged over the dispatching cycle.

It is noted that the total transmission energy of each VSC and the total DC grid energy loss over the scheduling cycle in Mode 2, 3 and 4 remain the same as Mode 1.

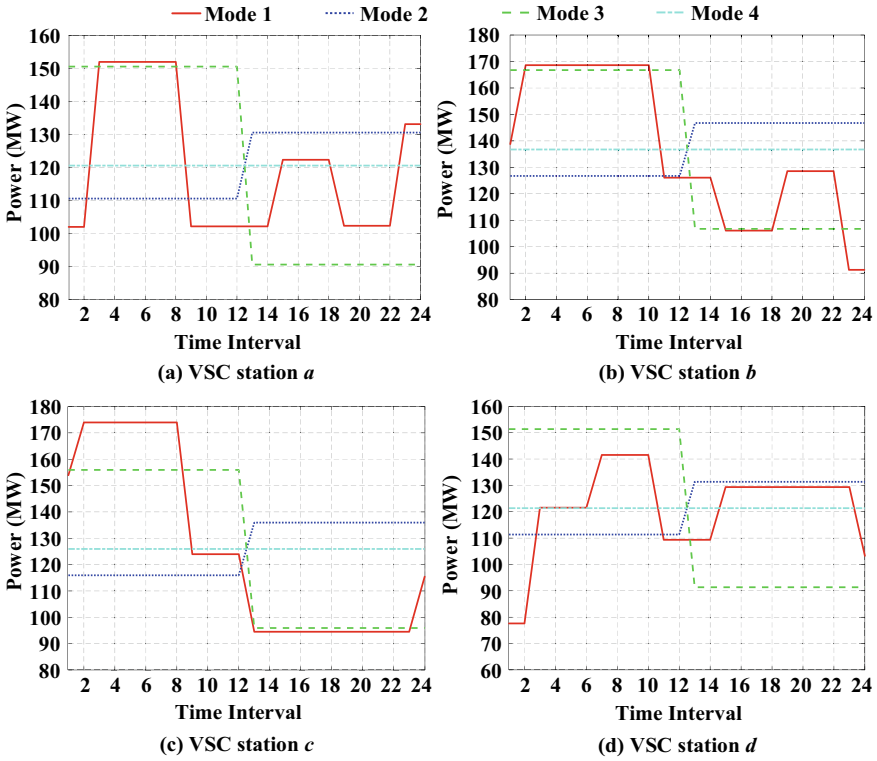


Fig. 4.16 Comparisons of different VSC transmission modes

The wind power curtailment rate in wind forecasting scenario of the robust solution is defined as the wind power curtailment rate in this work. Results are compared in Fig. 4.16 and Table 4.5.

It can be seen that the VSC power scheduled by Mode 1 is consistent with the potential wind variability in the AC areas *a* and *b*. In periods 2 through 10, as the potential wind power in areas *a* and *b* increases, the transmission power of VSCs *a* and *b* rises to enable more power to be exported from areas *a* and *b* to areas *c* and *d*. From period 14 to 22, the VSC power also rises to enable more power to be exported to areas *c* and *d* to support their higher load requirement. In contrast, Modes 2, 3 and 4 keep the VSC power invariant, which fails to follow the output variations of wind power in sending side (areas *a* and *b*) and the higher load demand in receiving side (areas *c* and *d*). As shown in Table I, the wind power curtailment rate in Mode 1 is the minimal, and the total generation cost in Mode 1 is reduced by up to 16.7%. It shows that the proposed Mode 1 can adapt the VSC power schedules to variations in wind power, and it helps to promote wind power accommodation, which ensures the meshed AC/DC system to operate more flexibly and cost-effectively.

(2) Convergence Performance.

The convergence performance of the proposed HRS model for the meshed AC/DC system is tested. As an example, for the typical 2nd, 9th, 16th, and 20th periods, the target and response variables among the low level, middle level, and upper level problems are shown in Fig. 4.17. We can see that the target and response variables converge to the same operating point after 36 iterations. Besides, the iterations needed for the C&CG convergence under each ATC iteration is basically three to eleven, showing that the C&CG algorithm converges fast.

To evaluate the quality of the solution, the solutions of the HRS model are compared with those by centralized robust scheduling (CRS) model with identical parameters, listed in Table 4.6. The optimization of the unit commitments in AC areas, the power flow in DC grid and the VSC power are included in the master problem, and the SCED of AC areas are included in the subproblem of the centralized C&CG algorithm. Then, the master problem and the subproblem are solved by the commercial solver Knitro and Gurobi, respectively.

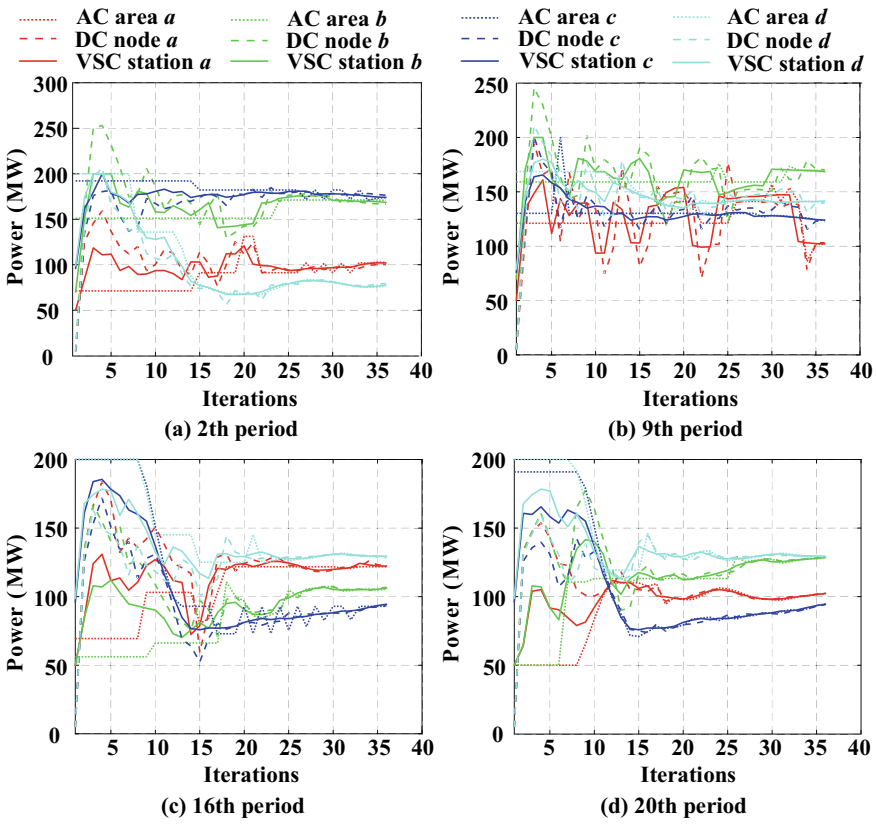


Fig. 4.17 Convergence curve of VSC transmission power for case 1

It is noted that the master problem in CRS is a large-scale nonlinear and nonconvex optimization problem, which makes it extremely challenged to be solved (more than 6 h). However, in the proposed HRS, the small-scale nonlinear and nonconvex DC grid multi-period OPF problem is separated from the linear AC grids and VSC stations problems, which can achieve the separation of nonlinear and linear optimization problems, thus greatly reducing the complexity of the scheduling model for the VSC-MTDC meshed AC/DC grid. The HRS takes 2891 s of solver time (the summation of the DC grid problem, the maximum time of the paralleled AC area and VSC station problems) to converge, which is far less than the CRS. The solution quality of the HRS model is also satisfactory. The total generation cost is only higher than that of CRS by 0.8%. The wind curtailment rate of areas *a* and *b* is 1.68% and 7.21%, respectively, and the total DC grid energy losses and energy loss rate over the scheduling cycle is 245.9MWh and 4.0%, respectively, which are close to the CRS model.

(3) Impact of Uncertainty Budgets.

Case 1 is scheduled using the proposed HRS model with varying budgets, the results are shown in Table 4.6. The budgets $\Gamma_w^{TB} = \Gamma_t^{SB} = 0$ degenerate into a deterministic problem with no parameter variation. We can see that the wind power curtailment rate basically increases with the increasing uncertainty budgets. This is because a larger uncertain budget corresponds to the more severe wind power fluctuations, which will lead to a more conservative scheduling solution. This will result in a reduction in the actual available wind power, and more thermal units being turned on to cope with the more severe wind power fluctuations. As a result, the VSC power is reduced, and the wind power curtailment rate is increased. Meanwhile, the total generation cost rises as the uncertainty budgets increase. This is because the larger uncertainty budgets will cause the thermal units to operate more frequently or generate more energy uneconomically to deal with the worst-case available wind power scenario. It should be noted that the robust solutions presented in Table 4.7 only reveals the generation cost in the worst-case scenario, the actual scheduling solutions can be better than the displayed results.

4.4.2.2 Four-Terminal VSC-HVDC with Four IEEE 118-Bus AC Grids

Similar to Case 1, a four-terminal VSC-HVDC system linked to four identical IEEE 118-bus AC grids [58] is established in Case 2. The AC line flow limit is tripled to enable greater wind power integration. Bus 89 of each AC area is linked to the corresponding VSC. Three identical wind farms are located at buses 38, 63 and 77 of AC areas *a* and *b*, respectively. Based on the load and wind output curve in Fig. 4.18, the load of areas *a*, *b*, *c* and *d* is multiplied by 20, 20, 24, and 28, respectively. Each wind farm output in areas *a* and *b* is multiplied by 10 and 13, respectively. The ratio of total wind power and load is up to 46%. The dc voltage limit is [450, 550] kV, and the master station *a* keeps 500 kV to provide the constant dc voltage reference. The minimum/maximum power limits of VSCs are all [500, 2000] MW, and the power

Table 4.7 Simulation results of HRS model with varying budgets

Budgets		Wind power curtailment rate		Total generation cost ($)
Γ_w^{TB}	Γ_t^{SB}	Area a (%)	Area b (%)	
0	0	1.88	6.52	235,984
6	1	2.03	7.11	266,988
	2	2.24	7.49	279,392
12	1	1.68	7.21	282,632
	2	2.21	7.44	310,749
18	1	1.73	7.24	282,149
	2	2.51	7.56	332,154
24	1	2.08	7.35	281,697
	2	2.91	8.10	345,271

adjustment rates are all [200, 500] MWh^{-1}. The gap tolerance of Gurobi is 1%, the convergence tolerance of C&CG is $\rho = 1\%$. The budgets of areas a and b are both $\Gamma_w^{TB} = 6$, $\Gamma_t^{SB} = 1$, and $\eta^{AC} = \eta^{DC} = 0.5\%$. The other parameters are the same as Case 1.

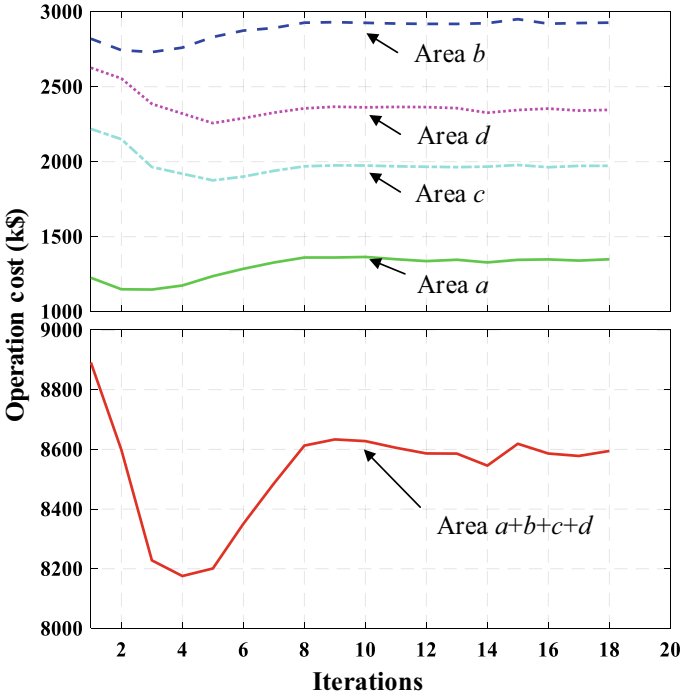


Fig. 4.18 Convergence curve of generation cost for AC areas in Case 2

Table 4.8 Comparisons of different VSC transmission modes for case 2

VSC transmission modes		Mode 1	Mode 2	Mode 3	Mode 4
Wind power curtailment rate	Area *a* (%)	3.58	6.13	4.69	4.73
	Area *b* (%)	10.84	18.02	15.87	17.33
Total generation cost (k$)		8 594	8 895	8 756	8 801

The previously described VSC transmission modes are also adopted in Case 2. Comparisons are illustrated in Table 4.8. Since Mode 1 jointly optimizes the VSC power with the generating units in AC areas, the wind power curtailment rate and total generation cost in Mode 1 are the minimal. Consequently, it can be concluded that by making full use of the VSCs' flexible adjustment capability, the wind power accommodation potential can be exploited, and the economics of VSC-MTDC meshed AC/DC grid can be improved. Figure 4.18 depicts the convergence curve for generation cost of AC areas, respectively. The proposed HRS model takes 18 iterations to converge with a total generation cost of 8 594 k$. The iterations needed for the C&CG convergence under each ATC iteration is basically three to eight.

The proposed HRS method is compared with traditional CRS method for Case 2. The calculation time of HRS method is about 5 h 44 min, however, the traditional CRS model fails to get optimal solution after running for more than 24 h. This is because the scale of AC grids in Case 2 is huge, which leads to a huge-scale nonlinear and nonconvex optimization problem for the master problem of the centralized C&CG, making the Knitro difficult to solve. In contrast, the small-scale nonlinear and nonconvex DC grid multi-period OPF is separated from the large-scale linear AC areas and VSCs problems in proposed HRS model. Thus, the small-scale nonlinear optimization problems and large-scale linear optimization problems can be effectively solved by the corresponding solver. By isolating the VSCs, AC areas, and DC grid, the proposed HRS model can greatly reduce computation complexity for the scheduling of large-scale VSC-MTDC meshed AC/DC grid. Meanwhile, the HRS formaultion can guarantee the decision independence of each system operator, which is under the operating philosophy of electricity market and the hierarchical and partitioned scheduling mode of China.

In the actual VSC-MTDC meshed AC/DC hybrid systems, the AC grids, DC grid and VSC stations optimization problems can be simultaneously implemented by computers located in different regions. With the ability to perform parallel computations, the proposed HRS approach have the potential to be computationally superior to centralized approach, both in terms of solution speed and the maximum problem size that can be addressed. Furthermore, the advantages of the proposed HRS approach reflected in the calculation time also show greater suitability for the hierarchical and partitioned power scheduling mode, which efficiently models interactions among AC areas, DC grid and VSC stations while exchanging a limited level of information and protecting privacy.

4.5 Conclusion

In this chapter, to fullfill the high flexibility requirement brought by the increasing integration of wind power, hybrid AC/DC systems are exploited to providing the flexibility. To fully coordinate the power flow of AC grids and the DC grid to mutually benefit multiple regions, the hierarchical and robust dispatch approach in two different AC/DC hybrid systems (bulk AC/DC hybrid systems and VSC-MTDC meshed AC/DC hybrid systems) with high flexibility requirement are proposed.

For bulk AC/DC hybrid systems, case studies based on 3-area 18-bus system, 3-area 117-bus system and 3-area 354-bus system indicate that the proposed distributed dispatch approach considering the flexible and controllable nature of the HVDC tie-line provides an effective dispatching tool for bulk AC/DC hybrid systems, promotes inter-regional wind power accommodation and improves the economics of the overall system.

For VSC-MTDC meshed AC/DC hybrid systems, case studies based on 4-terminal with four 6-bus systems and 4-terminal with four 118-bus systems demonstrate that the proposed HRS model has the potential to enable further wind power penetration for the VSC-MTDC meshed AC/DC grid, which can preserve the decision independency of each system operator, reduce the impact of wind power uncertainty on system operation, and promote wind power accommodation.

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Chapter 5
Exploring Operational Flexibility of AC/DC Power Grids



With the ever-increasing integration of renewable energy sources represented by wind and solar energy into power grids, its strong uncertainty and fluctuation present great challenges to the flexible operation of power grids. The traditional power regulation means of "generation-side" and "demand-side" are limited. With the rapid development of power electronics technology and other emerging control technology, the voltage source converter-based high voltage direct current transmission (VSC-HVDC) and the transmission switch (TS) provide new power regulation means on the "grid-side". In this chapter, the operational flexibility exploration of AC/DC power gird is researched. We try to describe the flexible operation mechanism of the power grid and propose the flexible operation improvement means. The case studies verify the proposed model.

5.1 Introduction

Due to the deficit of fossil energy and environmental concerns, large-scale renewables (represented by wind power and solar power) have been integrated into power systems during recent decades. However, the inherent uncertainty and volatility of renewable generations bring new challenges to the reliable operation of power systems. To ensure a real-time balance between supply and demand with the presence of uncertain renewable generations, more flexible regulation capability is required to accommodate fluctuant renewable generations. As the key technology of large-scale renewable energy collection and transmission, multi-terminal high voltage direct current transmission (MTDC) based on voltage source converter (VSC) has technical advantages in mitigating renewable fluctuation due to its flexible and fast power regulation capability. At present, there are several VSC-MTDC-based renewable energy transmission projects in operation in China, including the Nanao Project in Guangdong [1] and the Zhoushan Project in Zhejiang [2]. In particular, the world's first looped MTDC grid, the Zhangbei Project in Hebei Province [3], was put into operation at the

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M. Zhou et al., *Power System Flexibility*, Power Systems,
https://doi.org/10.1007/978-981-19-9075-5_5

end of 2019, collecting and transporting abundant renewable power in the Zhangbei District to the Beijing urban area. On the other hand, transmission switching (TS) of HVAC is also an available means in practice to improve the system performance [4] and is explored to be an effective control means for line overloading [5] and cost reduction [6]. Therefore, with large-scale renewables integration, the flexibility by controlling power flows through MTDC and HVAC TS should be fully utilized, which all belongs to flexible regulation methods of the gird-side. However, how the VSC-MTDC network flexibly operates is not studied, how to utilize the VSC power regulation capability is still to be researched, how to fully coordinate the HVAC TS and other flexible regulation means to improve the dispatch performance is also a research gap.

The organization of this chapter is as follows:

Section 5.2 introduces the flexible operation improvement of MTDC girds by VSC power regulation. In detail, the flexible operation mechanism of MTDC grids is researched in Sect. 5.2.2. A power droop regulation mode is designed and an optimal dispatch model of MTDC with multiple sources is built. By this optimal dispatch model, the influence factors and mechanism of flexible operation of the MTDC system are analyzed, and the flexible operation domain of the system is further formed. On this basis, a new power regulation mode of VSC station and MTDC grid is proposed to promote the flexible operation performance in Sect. 5.2.3. The VSC station can make self-adaptive regulations in response to the grid-connected renewable power fluctuation according to its actual available power margin, not by a fixed share. Further, the corresponding system-level regulation mode for the MTDC network is designed to exert the cooperative operation of multiple VSC stations.

Section 5.3 introduces the flexible operation exploitation of AC/DC hybrid power grids by HVAC TS control. In detail, a flexible generation-reserve joint dispatch method is developed in Sect. 5.3.1. To synergistically schedule them with generator units, a security-constrained economic dispatch (SCED) model is presented to realize the co-optimization of generation and TS. In Sect. 5.3.2, the model is transformed into a two-stage (normal state optimization and corrective dispatch verification) robust optimization (RO) model for the iterative solution with the aid of the column-and-constraint generation (C&CG) algorithm.

Section 5.4 introduces the case studies to prove the effectiveness of the proposed methods.

5.2 Improving Flexible Operation of MTDC Hybrid Networks by VSC Power Regulation

5.2.1 Problem Description

The operational flexibility of the MTDC hybrid system is to improve the friendliness of uncertain renewable energy by mobilizing multiple VSC stations to participate in

the coordinated operation of the MTDC power grid. Compared with the AC grid with natural power flow distribution and limited regulation capacity, MTDC can realize the coordinated control of multiple VSC stations by using the voltage modulation technology and power regulation strategy. Combined with the energy storage system, the complementary potential of wind energy, solar energy, storage energy, and other resources can be fully utilized, and maximize the operational flexibility of the hybrid system. It is necessary to design the power regulation model of the VSC station and study the operation flexibility mechanism of the MTDC girds.

For the power regulation mode of the VSC station, it traditionally applies the master–slave operation strategy or voltage margin strategy as the system-level strategy to arrange the operation of multiple VSC stations. Some studies apply the master–slave mode, obtaining the constant operation point of each VSC station by optimal power flow (OPF) [7, 8]. The scheduling plan by this operation strategy is premised on the ideal operation scenario and stable MTDC network, with the requirement of communication among the stations. The voltage margin control does not require inter-station communication though, but it still belongs to be the constant operation mode [9]. It is difficult for the above strategies to meet any power fluctuations or sudden grid failures, the flexible regulation capability of VSC stations cannot be exerted either. For this reason, the scholars focus on the power-voltage droop operation mode, an autonomous decentralized coordination strategy for multiple VSC stations, and no communication among the stations required. Mònica et al. [10] fixed the droop coefficient in droop mode and did not reflect the effect of power reference value on power distribution of VSC station. He et al. [11] only takes the power reference value as the optimization variable. The above methods are difficult to meet the control requirements of the MTDC network in coordinating and optimizing system power flow distribution and multi-resource complementarity.

Aim to this problem, a power droop regulation mode is designed and an optimal dispatch model of MTDC with multiple sources is built-in Sect. 5.2.2. By this optimal dispatch model, the influence factors and mechanism of flexible operation of the MTDC system are analyzed, and the flexible operation domain of the system is further formed.

What is more, optimizing the regulation behavior and parameters of the droop operation strategy for the VSC station is a challenge. The existing droop regulation mode should specify the power share of each VSC station in the event of power fluctuation, by presetting the fixed droop coefficient (FDC) on the reference points, with limited capacity to adapt to the continuous and dramatic renewables fluctuation, which leads to the frequent variations in operating power and dc voltage. The dc voltage stability is the core operational problem in the MTDC grid. The frequent variations of dc voltage threaten the power balance and stable operation of the MTDC grid due to the capacitive inertia of converter stations [12], moreover, working at the states that deviate from the rated dc voltage will adversely affect the safety and reliability of the converter station's equipment [13]. Hence, how to improve the VSC power regulation characteristics inflexible response to renewable power fluctuation and further enhance the dc voltage stability of the MTDC grid remains a critical task.

Recently, the idea of adaptive-coefficient droop idea is developed in the field of system control. A current influence factor is introduced into the V-I droop characteristic curve to reduce dc voltage deviation in [14], but making an indirect power allocation among VSC stations, which is not suitable for power allocation in the MTDC system. In [15], an acceleration factor is introduced into the V-P droop characteristic curve to adaptively regulate the power allocation, but this strategy cannot reduce the dc voltage deviation. They mainly focused on the control implementation of the proposed adaptive droop regulation mode, only paying attention to the dynamic control effect, but not trying to optimize the operation states and parameters of VSC stations from the perspective of power dispatch.

Aim to this problem, a novel power margin tracking (PMT) droop regulation mode for VSC station is proposed in Sect. 5.2.3. The VSC station can make self-adaptive regulations in response to the grid-connected renewable power fluctuation according to its actual available power margin, not by a fixed share. Further, the corresponding system-level regulation mode for the MTDC network is designed to exert the cooperative operation of multiple VSC stations.

5.2.2 Flexible Operation Mechanism and Model

5.2.2.1 Power Regulation Model of VSC Station

First, the steady-state power transmission model of VSC-MTDC is introduced.

Figure 5.1 shows a power transmission model without considering the transient process of VSC power devices. The renewable output is sent out through AC line i and connected to dc node i through VSC station i. The subscript s represents the electrical parameters in the grid-connection point. The subscript c and d respectively represent the ac and dc electrical parameters of the VSC station. The equivalent loss caused by converter transformer, converter reactor, conductor and other components is expressed in R_i, and its reactance value is expressed in jX_i.

Let:

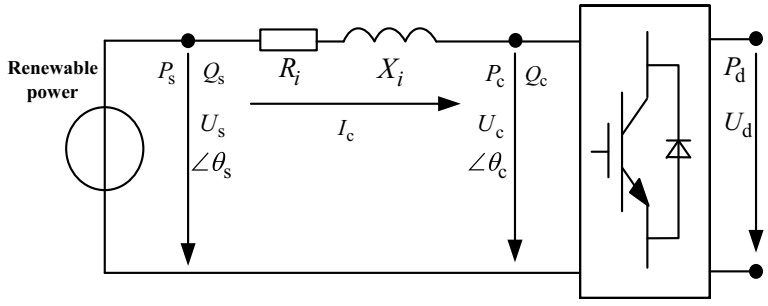


Fig. 5.1 Steady-state power transmission model of VSC station

$$\delta_i = \theta_{\mathrm{s},i} - \theta_{\mathrm{c},i} \tag{5.1}$$

By derivation, we can get the following:

$$\begin{cases} P_{\mathrm{s},i} = P_{\mathrm{c},i} = \dfrac{U_{\mathrm{s},i} \cdot U_{\mathrm{c},i}}{X_i} \sin \delta_i \\ Q_{\mathrm{s},i} = \dfrac{\left(U_{\mathrm{s},i}\right)^2}{X_i} - \dfrac{U_{\mathrm{s},i} \cdot U_{\mathrm{c},i}}{X_i} \cos \delta_i \\ Q_{\mathrm{c},i} = -\dfrac{\left(U_{\mathrm{c},i}\right)^2}{X_i} + \dfrac{U_{\mathrm{s},i} \cdot U_{\mathrm{c},i}}{X_i} \cos \delta_i \end{cases} \tag{5.2}$$

$$P_{\mathrm{d},i} = P_{\mathrm{c},i} - P_{\mathrm{loss},i}^{\mathrm{ac}} \tag{5.3}$$

$$P_{\mathrm{loss},i}^{\mathrm{ac}} = \mathrm{a}_i + \mathrm{b}_i \cdot I_{\mathrm{c},i} + \mathrm{c}_i \cdot \left(I_{\mathrm{c},i}\right)^2 + R_i \cdot \left(I_{\mathrm{c},i}\right)^2 \tag{5.4}$$

$$I_{\mathrm{c},i} = \sqrt{\left(P_{\mathrm{c},i}\right)^2 + \left(Q_{\mathrm{c},i}\right)^2} / U_{\mathrm{c},i} \tag{5.5}$$

where $P_{\mathrm{loss},i}^{\mathrm{ac}}$ is the total ac losses; X_i is the equivalent reactance of the AC line connected to the VSC station i. R_i represents the transmission loss of the collection line. a_i, b_i and c_i are respectively the coefficients of the constant term, first-order term, and second-order term of the operation loss formula of VSC station.

The power model of MTDC grid is as follows:

$$\begin{cases} P_{\mathrm{d},i} = U_{\mathrm{d},i} \cdot I_{\mathrm{d},i} \\ I_{\mathrm{d},i} = \sum\limits_{j \in n_{\mathrm{con}}} I_{\mathrm{d},ij} = \sum\limits_{j \in n_{\mathrm{con}}} g_{\mathrm{d},ij} \cdot \left(U_{\mathrm{d},i} - U_{\mathrm{d},j}\right) \end{cases} \tag{5.6}$$

$$U_{\mathrm{d},i} = \frac{M_i}{\sqrt{2}} U_{\mathrm{c},i} \tag{5.7}$$

$$M_i^{\min} \le M_i \le M_i^{\max} \tag{5.8}$$

where, n_{con} represents the dc nodes set which connect node i through dc line L_{ij}. $I_{\mathrm{d},i}$ and $g_{\mathrm{d},ij}$ are respectively the current and conductance of line L_{ij}. Equation (5.7) represents the voltage amplitude relationship between ac side and dc side of the VSC station adopting voltage modulation technology. Equation (5.8) represents the constraint of the voltage modulation ratio M_i.

Note that the power flow model of the VSC station is also applicable to the receiving-end station and the regulation station. Under the reference direction specified in Fig. 5.1, the off-grid power sent by the receiving-end station and the power absorbed by the regulation station from the MTDC grid are all negative values.

Next, the droop-based-VSC power regulation model is built.

Compared with conventional HVDC transmission technology, VSC-MTDC has flexible regulation capability, which is mainly reflected in voltage modulation technology and power regulation mode.

The voltage amplitude relationship between AC and DC sides of the VSC station is

$$U_{c,i} = M_i U_{d,i}/\sqrt{2} \tag{5.9}$$

where M_i is the voltage modulation ratio of VSC station i, which can be set according to the system operation requirements, that is, used as an optimization variable. Its constraint range is

$$M_i^{\min} \le M_i \le M_i^{\max} \tag{5.10}$$

For the power-voltage droop regulation mode, it specifies the power share of each VSC station in the event of power fluctuation by presetting the fixed droop coefficient k_i (FDC) on the reference point ($P_{d0,i}$, $U_{d0,i}$), combining the advantages of constant voltage mode and constant power mode. Without interconnection communication, it can locally realize the coordination of system imbalance among multiple VSC stations and maintain the relative stability of DC voltage. The power-voltage droop regulation mode is

$$U_{d,i} - U_{d0,i} + k_i(P_{d,i} - P_{d0,i}) = 0 \tag{5.11}$$

where the $P_{d0,i}$ and $U_{d0,i}$ are DC power reference value and DC voltage reference value, respectively. Figure 5.2 shows its operation characteristic curve, from which it can be seen it is jointly determined by droop coefficient and reference operation point. When multiple VSC interconnect, the system needs to coordinate and formulate the best operation mode for each VSC station according to the actual operation state ($P_{d,i}$, $U_{d,i}$). Then it is distributed to each VSC station to set its different droop parameters dispersedly.

The droop coefficient directly affects the unbalance power distribution. It traditionally determined the droop coefficient through the DC voltage limit and rated capacity of each VSC. This method makes maximum use of the transmission capacity

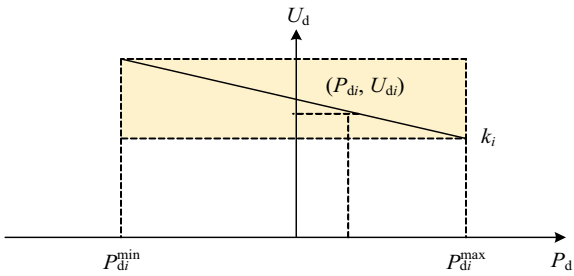


Fig. 5.2 Power-voltage droop operation characteristic curve

of the VSC station. However, it may increase the operating loss of MTDC, or lead to renewable abandonment. At the same time, it shall not be too big to prevent the DC voltage from being greatly affected when the system's active power fluctuates. Thus, the range of the droop coefficient is

$$0 \leq k_i \leq k_i^{\max} \tag{5.12}$$

where the minimum droop coefficient is set to 0, which refers to the constant DC voltage mode into the droop mode as a special case, and $k$max $i$is determined according to the actual situation.

For the reference operation point, three droop parameters cannot be determined together. Therefore, fix the DC voltage reference value take the DC power reference value as the optimization variable, and gives the corresponding constraints:

$$U_{\mathrm{d0},i} = U_{\mathrm{d}}^{\exp} \tag{5.13}$$

$$P_{\mathrm{d0},i}^{\min} \leq P_{\mathrm{d0},i} \leq P_{\mathrm{d0},i}^{\max} \tag{5.14}$$

where the $U^{\exp}{}_{\mathrm{d}}$ is the DC rated voltage of the grid. $P^{\min}{}_{\mathrm{d0},}$ and $P^{\max}{}_{\mathrm{d0},}$ are the minimum and maximum DC power reference values respectively, which can be set according to the transmission capacity of each VSC station.

5.2.2.2 Optimal Flexible Operation Model

Under the power fluctuation, the system tries to mobilize the cooperative operation of multi-VSC stations and the energy storage system to meet the load demand of the receiving-grid, while making full use of wind and solar power. For this purpose, the objective function is set to minimize the total curtailed renewable energy during the scheduling period:

$$\min f = \sum_{i \in n_{\mathrm{sen}}} P_{\mathrm{s},i}^{\mathrm{cur}} \tag{5.15}$$

where $P_{s,i}^{\mathrm{cur}}(t)$ is curtailed renewable power.

The constraints include:

(1) Power flow equations of ac/dc grids.
(2) Voltage modulation constraints of VSC stations.
(3) Power regulation model of VSC stations.

The sending-end stations apply the proposed droop mode. The regulation station and the receiving-end station apply the constant active power mode. The receiving-end power needs to be tracked to scheduling order:

$$P_{s,i} = P_{load} \quad i \in n_{rec} \tag{5.16}$$

where n_{rec} represents the receiving-end dc node. P_{load} is the power demand value provided by the receiving-end grid.

(4) The transmission capacity constraints of VSC stations:

$$0 \le \left|P_{d,i}\right| \le P_{MMC,i}^{cap} \tag{5.17}$$

where $P_{MMC,i}^{cap}$ is the rated capacity of VSC station i.

(5) Operation constraints of the system:

$$\begin{cases} U_i^{min} \le U_i \le U_i^{max} \\ \delta_i^{min} \le \delta_i \le \delta_i^{max} \\ U_{d,i}^{min} \le U_{d,i} \le U_{d,i}^{max} \\ I_{d,ij}^{min} \le I_{d,ij} \le I_{d,ij}^{max} \end{cases} \tag{5.18}$$

The above includes the amplitude and phase angle constraints of ac voltage, dc voltage constraints, and dc line current constraints.

(6) The power balance constraints of the system:

$$\sum_{i \in n_{dc}} P_{s,i} = \sum_{i \in n_{dc}} P_{loss,i}^{ac} + P_{loss}^{dc} \tag{5.19}$$

where, n_{dc} represents the set of total dc nodes. $P_{loss,i}^{ac}$ is the ac loss. P_{loss}^{dc} is the dc loss:

$$P_{loss}^{dc} = \sum_{i \in n_{dc}} \sum_{j \in n_{con}} g_{d,ij} \cdot \left(U_{d,j} - U_{d,i}\right)^2 \tag{5.20}$$

The optimal operation model of renewables generation MTDC collector system proposed above is a high-dimensional, non-convex and nonlinear programming problem. Therefore, we adopt the interior-point method suitable for solving large-scale nonlinear programming problems, invoking IPM solver to solve at Matlab 2017a platform.

5.2.2.3 Flexible Operation Domain

To verify the proposed model, this section refers to the actual Zhangbei four-terminal flexible DC project. The simulation parameters are set as follows: the reference power is 100 mva and the reference voltage is 500 kV. The rated power of the three wind power bases is 13.5, 13, and 13 respectively (Unless otherwise specified, the

numerical values adopted in this section are all in per unit); The predicted output value of the solar power base is 3. The installed capacity of the pumped storage power station is 18. The upper and lower limits of DC voltage are [0.9, 1.1]. The upper and lower limits of AC voltage are set as [0.67, 0.74]. The phase angle constraint of AC voltage is $[-\pi/2, \pi/2]$. DC line current constraint is [−20, 20]. Droop coefficient constraint is [0, 0.1]. The voltage modulation ratio constraint is (0, 1).

To explore the key influence factors of flexible operation performance, the system operation for multiple scenarios with different receiving-end power demands are tested. Figure 5.3 shows the optimization results of curtained renewable power and the regulated storage power. Among them, when the power demand of the receiving power grid changes within [13, 16], the system can avoid the curtained renewable power. When the power demand drops to 12 or below, the curtained renewable power occurs. The smaller the demand value, the greater curtained renewable power. Since the power regulation capacity of the storage system reaches the capacity limit, when the load power demand is too small, even the maximum power absorbed can not meet the power balance of the system, the renewable power will curtain. When the power demand exceeds 20, the system cannot provide the required power. Thus, the receiving-end acceptance range [13, 16] is the flexible regulation range under the current operation scenario for the system to deal with the uncertainty renewable power.

Based on the above flexible regulation range, the Fig. 5.3 further reveals the relationship between the system regulation capacity and the ratio of renewable outputs and storage capacity and form the flexible regulation domain of the VSC-MTDC network. The flexible regulation domain is surrounded by upper and lower surfaces. The upper surface reflects the maximum power support capability to the receiving-end grid. Once the receiving-end load demand exceeds this boundary, it will be

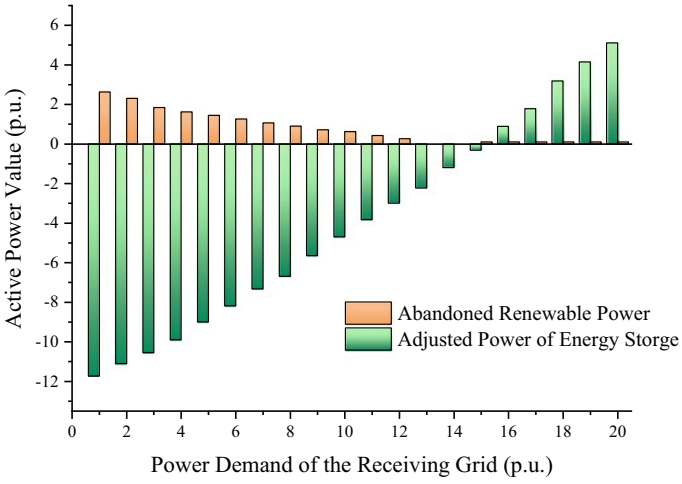


Fig. 5.3 Optimal scheduling results under different power demands

difficult for the system to meet the power balance even if all power regulation means are mobilized. The lower surface reflects the minimum absorbable power without any renewable curtailment. Once the receiving-end load demand below this boundary, the renewable curtailment will be inevitable.

For further analysis, the change rule of flexible regulation domain of the system under different capacity ratios of renewable power and energy storage described in Fig. 5.4 reflect the "grid-side" regulation capability of a given MTDC network to respond to the uncertainty renewable power and load demand. (1) There is a significant positive correlation between the flexible regulation capacity of the system and the energy storage capacity. If the energy storage capacity is too low, the system has very low power regulation. (2) With the increase of the installed capacity of renewable energy, the maximum support power and the minimum power acceptance without renewable curtailment are increased, but the flexible regulation range is slightly reduced. The reason is the renewable fluctuation expandation limits the power regulation of VSC-MTDC network. The enlightenment is that the power regulation of the network itself is crucial to the flexible operation of the whole system. For this, a novel power regulation mode for the multiple VSC stations is proposed in the next Sect. 5.2.3 to improve the flexible coordinated operation of the MTDC network itself.

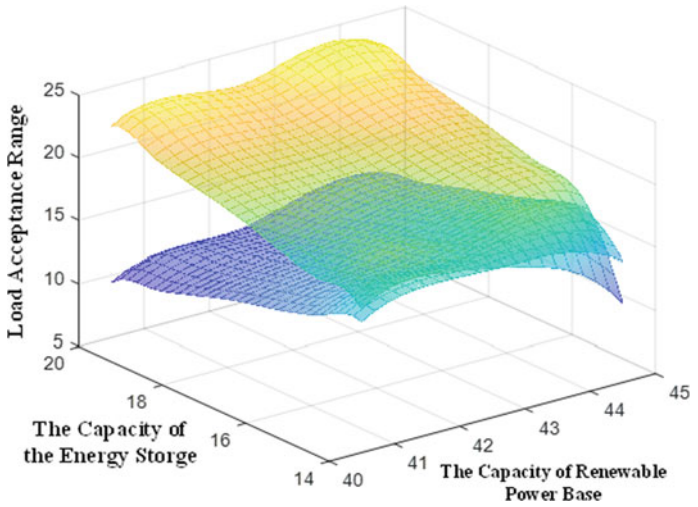


Fig. 5.4 Flexible regulation domain of the VSC-MTDC network

5.2.3 *Flexible Operation Improvement Mode for VSC Station*

5.2.3.1 Power Margin Tracking Droop Regulation (PMT) Mode

For the conventional FDC droop regulation mode, each converter station is designed to operate at its reference point ($P_{d0,i}$, $U_{d0,i}$) at the initial state. When there is any power fluctuation, each converter station will share this power deviation in proportion of its fixed droop coefficient, and automatically move to a new operation point ($P_{d,i}$, $U_{d,i}$). When presetting the droop coefficients of converter stations, to make the converter station with large rated capacity undertakes much more unbalanced power, the droop coefficient k_i is usually determined according to the inverse ratio of its rated capacity, as follows:

$$k_i = \left(U_{d,i}^{max} - U_{d0,i}\right)/\left(P_{d,i}^{max} - P_{d0,i}\right) \tag{5.21}$$

The formula (5.21) indicates that k_i is fixed by its preset reference point and its rated capacity of the converter station. The advantage of this regulation mode is to provide a cooperative operation for multi-converter stations without the need for communications. Moreover, it can achieve the differential regulation between DC power and DC voltage, maintain the approximate voltage stability in the MTDC grid, and have some certain regulation ability to renewables output fluctuation.

However, when the grid-connected power is in continuous and dramatic fluctuations, the FDC droop regulation mode has limitations, that is, no matter how large the power margin of converter remains (defined as the difference between the actual power and the regulation limit), or what dc voltage is, the converter stations can only share the unbalanced power according to their initial preset ratio. The cooperative operation capability of multiple stations is still limited. It may cause the operation of some stations to approach the power limit then lose the regulation ability to renewable fluctuations, unable to make the flexible response to power flow changes, resulting in serious deviations of dc voltage and affecting the operation stability of MTDC grid.

Aiming to the above problems, this paper proposes power margin tracking (PMT) droop regulation mode by introducing the power margin correction factor ξ_i to modify the fixed droop coefficient k_i:

$$\xi_i = \left|P_{d,i} - P_{d0,i}\right|/\left(P_{d,i}^{max} - P_{d0,i}\right) \tag{5.22}$$

It can be referred that $0 \le \xi_i \le 1$. The closer the real output $P_{d,i}$ to the limit, the smaller the power margin, and the greater the ξ_i is. Then, the proposed PMT droop mode is given by:

$$U_{d,i} = -\xi_i \cdot k_i\left(P_{d,i} - P_{d0,i}\right) + U_{d0,i} \tag{5.23}$$

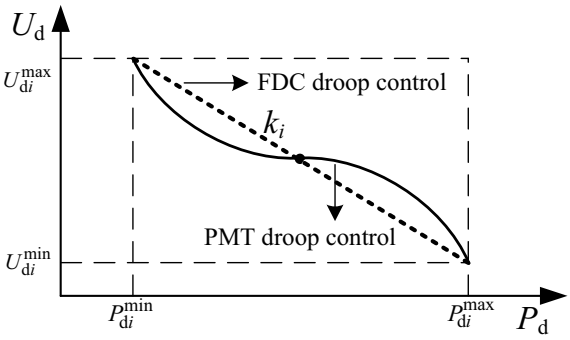


Fig. 5.5 Operational characteristic curve of power control strategies

Formula (5.23) indicates that the relationship between the real operation voltage $U_{d,i}$ and power $P_{d,i}$ is quadratic, no longer a linear one. Figure 5.5 shows the comparison between FDC droop mode and PMT droop mode.

Next, compare the proposed PMT droop mode with the conventional FDC droop mode from two aspects: power mitigation capability and dc voltage deviation.

a. The mitigation capability of power fluctuation

The slopes of FDC droop mode and the proposed PMT droop mode are respectively calculated by taking derivative of (5.11) and (5.23):

$$k_{\mathrm{FDC}} = dU_{\mathrm{d},i}/dP_{\mathrm{d},i} = -k_i \tag{5.24}$$

$$k_{\mathrm{PMT}} = dU_{\mathrm{d},i}/dP_{\mathrm{d},i} = -2k_i \cdot \left|P_{\mathrm{d},i} - P_{\mathrm{d0},i}\right| / \left(P_{\mathrm{d},i}^{\max} - P_{\mathrm{d0},i}\right) \tag{5.25}$$

Comparing (5.24) and (5.25), we can see:

(1) For PMT control, when converter station i operates at the reference point ($P_{\mathrm{d},i} = P_{\mathrm{d0},i}$), $k_{\mathrm{PMT}} = 0$, the station is actually operating at the state of constant voltage control, which is the same with FDC control.

(2) When the real-time operation state changes but is still near the reference point, that is when the dc power satisfies the following formula:

$$P_{\mathrm{d},i} \in \left(\left(3P_{\mathrm{d0},i} - P_{\mathrm{d},i}^{\max}\right)/2,\ \left(P_{\mathrm{d},i}^{\max} + P_{\mathrm{d0},i}\right)/2\right) \tag{5.26}$$

There is $k_{\mathrm{PMT}} < k_{\mathrm{FDC}}$. At this situation, the PMT droop mode tends to stabilize the dc voltage of the converter station.

(3) When the real-time operation state continues to deviate from the reference point and approach the regulation limit, that is the dc power satisfies the following formula:

$$P_{\mathrm{d},i}\varepsilon\left[\left(\mathrm{P}_{\mathrm{d,i}}^{\min}, \frac{3\mathrm{P}_{\mathrm{d0,i}} - \mathrm{P}_{\mathrm{d,i}}^{\max}}{2}\right) \cup \left(\frac{\mathrm{P}_{\mathrm{d,i}}^{\max} + \mathrm{P}_{\mathrm{d0,i}}}{2}, \mathrm{P}_{\mathrm{d,i}}^{\max}\right)\right] \tag{5.27}$$

There is $k_{\mathrm{PMT}} > k_{\mathrm{FDC}}$, the PMT droop mode tries to maintain the power stability of the converter station. Moreover, the closer the real operation power to the limit, the greater the droop coefficient, the smaller the power sharing ratio, that means to slow down approaching to the limit and retain its power regulation capability.

b. The dc voltage deviation

Denote the dc voltage deviation caused by the change of operation point in FDC droop mode as $\triangle u_1$, and $\triangle u_2$ for the proposed PMT droop mode, then:

$$\Delta u_1 = -k_i\left(P_{\mathrm{d},i} - P_{\mathrm{d0},i}\right) \tag{5.28}$$

$$\Delta u_2 = -\xi_i \cdot k_i\left(P_{\mathrm{d},i} - P_{\mathrm{d0},i}\right) \tag{5.29}$$

Due to $0 \leq \xi_i \leq 1$, there is:

$$|\Delta u_2| - |\Delta u_1| = (\xi_i - 1) \cdot k_i\left(P_{\mathrm{d},i} - P_{\mathrm{d0},i}\right) \leq 0 \tag{5.30}$$

It indicates that:

(1) When the converter station i operates at the reference point ($P_{\mathrm{d},i} = P_{\mathrm{d0},i}$): both control strategies have no dc voltage deviation, that is $|\triangle u_1| = |\triangle u_2| = 0$.
(2) When Station i reaches the limit capacity ($\xi_i = 1$), there is also $|\triangle u_1| = |\triangle u_2|$.
(3) As long as Station i deviates from the reference point, the dc voltage deviation is generated, but there is $|\triangle u_2| < |\triangle u_1|$, that is, the proposed PMT droop mode can reduce dc voltage deviations due to renewable output fluctuations.

Based on the above analysis, the proposed PMT droop regulation mode can track to power margin of the converter in real-time. So that each converter station can make self-adaptively regulations to respond to the fluctuation of wind and solar power according to its real-time operation state, moreover, participate in the cooperative operation of the system. This regulation mode has the merits in improving the mitigation capacity of power fluctuation, reducing the dc voltage deviation of VSC stations, and helping the operation stability of the MTDC grid.

5.2.3.2 Multi-period Optimal Flexible Operation Model

To further verify the flexible regulation capability of applying the PMT regulation mode in the generation of the renewable MTDC collector system, a day-ahead optimal operation model is built with the scheduling interval of 15 min.

A. The power flow model of MTDC grid with renewables generation.

There are three parts in the power flow model of the MTDC grid with renewables generation: the power generation model of wind and solar power, the power flow

model of the VSC station and its collection transmission line, and the power model of the MTDC grid.

The power generation model of wind and solar power is as follows:

$$\begin{cases} P_i^{\text{pre}}(t) = \sum\limits_{j \in qu_i} \left(P_{i,j}^{\text{cap}} \cdot \rho_{i,j}(t) \right) \\ P_{\text{s},i}(t) + P_{\text{s},i}^{\text{cur}}(t) = P_i^{\text{pre}}(t) \end{cases} i \in n_{\text{sen}} \tag{5.31}$$

where n_{sen} represents the dc node set of sending end. $P_i^{\text{pre}}(t)$ is the total predicted generation power of wind and solar power of node i at period t. qu_i is the number of renewables generation bases connected to node i. $P_{i,j}^{\text{cap}}$ is the rated capacity of the jth renewables generation base of node i. $\rho_{i,j}(t)$ is the predicted power normalized value of the corresponding renewables generation base. $P_{s,i}^{\text{cur}}(t)$ is the renewable curtailment power of node i at period t, to describe the power deviation between the actual output $P_{\text{s},i}(t)$ and the predicted power $P_i^{\text{pre}}(t)$.

The power flow model of the VSC station and MTDC grid please refer to the Sect. 5.2.2.1.

B. The power regulation model of energy storage station with multi-period coupling.

In a long-time operation scenario with the grid-connection output continuous fluctuation, the multi-period power coupling characteristics should be considered. This paper selects Pumped Storage (PS) station as the representative of the energy storage, its power regulation model is built as follows.

(1) Power regulation constraints

$$P_{\text{s},i}(t) = P_{\text{gen}}(t) - P_{\text{pum}}(t)\, i \in n_{\text{reg}} \tag{5.32}$$

$$\begin{cases} P^{\min} \le P_{\text{gen}}(t) \le P^{\max} \\ -P^{\max} \le P_{\text{pum}}(t) \le P^{\min} \end{cases} \tag{5.33}$$

$$P_{\text{gen}}(t) \cdot P_{\text{pum}}(t) = 0 \tag{5.34}$$

In (5.32), n_{reg} represents the dc node of regulating-end. In (5.33), $P_{\text{gen}}(t)$ represents the generation power in period t (positive value) while $P_{\text{pum}}(t)$ is the pumping power (negative value). $P^{\max}$ and $P^{\min}$ are the upper and lower limits of the regulated power respectively. Its lower limit is 0, while the upper limit is generally its rated capacity. Equation (5.34) describes that the PS station can only be in one working state at a single period.

(2) Reservoir capacity constraints

$$\begin{cases} Q_{\text{up}}^{\min} \le Q_{\text{up}}(t) \le Q_{\text{up}}^{\max} \\ Q_{\text{down}}^{\min} \le Q_{\text{down}}(t) \le Q_{\text{down}}^{\max} \end{cases} \tag{5.35}$$

where $Q_{\text{up}}(t)$ and $Q_{\text{down}}(t)$ are respectively the flow of the upper and lower reservoirs in period t. $Q_{\text{up}}^{\max}$, $Q_{\text{up}}^{\min}$, $Q_{\text{down}}^{\max}$ and $Q_{\text{down}}^{\min}$ are respectively the upper and lower capacity limits of the upper and lower reservoirs. Without considering the loss of water flow, the flow changes in the upper reservoir is equal to the lower reservoir, so that they can be simplified as one inequality constraint.

(3) The coupling constraints between multiple periods

$$Q_{\text{up}}(t) - Q_{\text{up}}(t-1) = -P_{\text{pum}}(t) \cdot \eta_{\text{p}} - P_{\text{gen}}(t) \cdot \eta_{\text{g}} \tag{5.36}$$

$$-\delta \le Q_{\text{up}}(96) - Q_{\text{up}}(1) \le \delta \tag{5.37}$$

Equation (5.36) represents the coupling relationship between the reservoir flow and regulating power in adjacent periods. η_{p} and η_{g} are respectively the flow/electricity-quantity conversion ratios in the pumping and generating working state. In (5.37), δ stipulates the maximum allowable flow variation between the initial and end time period of the scheduling cycle.

C. Multi-period optimal operation model.

a. Objective function

Under the continuous power fluctuation, the collector system tries to mobilize the cooperative operation of multi-VSC stations and the energy storage system to meet the load demand of the receiving-grid, while making full use of wind and solar power. For this purpose, the objective function is set to minimize the total curtailed renewable energy during the scheduling period:

$$\min f = \sum_{i \in n_{\text{sen}}} \sum_{t} P_{\text{s},i}^{\text{cur}}(t) \tag{5.38}$$

where $P_{s,i}^{\text{cur}}(t)$ is expressed in (5.31) above.

b. Constraints

(1) power flow equations of ac/dc grids.

(2) Voltage modulation constraints of VSC stations.

(3) Power regulation mode of VSC stations

The sending-end stations apply the proposed PMT droop mode, their constraints are (5.21)–(5.23). The regulation station and the receiving-end station apply the constant active power mode. The receiving-end off-grid power needs to be tracked to scheduling order:

$$P_{\text{s},i} = P_{\text{load}} \; i \in n_{\text{rec}} \tag{5.39}$$

where n_{rec} represents the receiving-end dc node. P_{load} is the power demand value provided by the receiving-end grid.

(4) The transmission capacity constraints of VSC stations:

$$0 \leq \left| P_{\text{d},i} \right| \leq P_{\text{MMC},i}^{\text{cap}} \tag{5.40}$$

where $P_{\text{MMC},i}^{\text{cap}}$ is the rated capacity of VSC station i.

(5) Wind and solar power constraints: (5.31)

(6) The power regulation constraints of energy storage system: (5.32)–(5.37).

This paper selects Pumped Storage (PS) as the energy storage system. Its regulation constraints contain the regulation power constraints, the reservoir capacity constraints, and the coupling constraints between multiple periods.

(7) Operation constraints of the collector system:

$$\begin{cases} U_i^{\min} \leq U_i \leq U_i^{\max} \\ \delta_i^{\min} \leq \delta_i \leq \delta_i^{\max} \\ U_{\text{d},i}^{\min} \leq U_{\text{d},i} \leq U_{\text{d},i}^{\max} \\ I_{\text{d},ij}^{\min} \leq I_{\text{d},ij} \leq I_{\text{d},ij}^{\max} \end{cases} \tag{5.41}$$

The above includes the amplitude and phase angle constraints of ac voltage, dc voltage constraints, and dc line current constraints.

(8) The power balance constraints of the collector system:

$$\sum_{i \in n_{\text{dc}}} P_{\text{s},i} = \sum_{i \in n_{\text{dc}}} P_{\text{loss},i}^{\text{ac}} + P_{\text{loss}}^{\text{dc}} \tag{5.42}$$

where, n_{dc} represents the set of total dc nodes. $P_{\text{loss},i}^{\text{ac}}$ is the ac loss. $P_{\text{loss}}^{\text{dc}}$ is the dc loss of MTDC grid:

$$P_{\text{loss}}^{\text{dc}} = \sum_{i \in n_{\text{dc}}} \sum_{j \in n_{\text{con}}} g_{\text{d},ij} \cdot \left(U_{\text{d},j} - U_{\text{d},i} \right)^2 \tag{5.43}$$

The multi-period optimal operation model of renewables generation MTDC collector system proposed above is a high-dimensional, non-convex and nonlinear programming problem. Therefore, this paper adopts the interior-point method suitable for solving large-scale nonlinear programming problems, invoking IPM solver to solve at Matlab 2017a platform.

5.3 Exploiting the Operational Flexibility of Wind Integrated Hybrid AC/DC Power Systems

5.3.1 SCED Model with TS for Hybrid AC/DC Grid

In this section, a SCED model with TS for wind integrated hybrid AC/DC grid considering multiple uncertainties (wind power fluctuation and generator failure) is established. Wherein, linear power flow model of DC grid is sketched, the uncertainty set is defined with wind power uncertainty described as intervals, and the endogenous reserve determination method is adopted to guarantee the accurate quantification of the reserve with the consideration of network constraints.

5.3.1.1 Linear Power Flow Model of the DC Grid

Inspired by the DC power flow model used for the AC grid, for the DC grid, a linear power flow approximation is used, which provides an acceptable trade-off between accuracy and computational complexity [17], the linearization process is as follows:

The quadratic power flow equation for the DC grid is:

$$p_{lnm}^{DC} = \frac{U_n(U_n - U_m)}{R_{nm}} \tag{5.44}$$

Assuming that the voltages are very close to the nominal voltage, defined as 1 p.u., the approximation can be obtained:

$$p_{lnm}^{DC} = \frac{u_n - u_m}{R_{nm}} \tag{5.45}$$

where $u_n = U_n - U_{ref}$.

5.3.1.2 Uncertainty Modeling

Two categories of uncertainties are considered in this paper. For discrete uncertainties (generator failure), they can be well represented by binary variables. For continuous uncertainties, i.e., the uncertain wind power outputs, they are described by intervals, expressed as:

$$\begin{aligned} w_{jt} &= w_{fjt} + [-\varepsilon_{jt}^{-}, \varepsilon_{jt}^{+}] \\ &= w_{fjt} + z_{wjt}^{+}\varepsilon_{jt}^{+} - z_{wjt}^{-}\varepsilon_{jt}^{-} \end{aligned} \tag{5.46}$$

Further, to control the conservativeness, a budget for the generator contingency μ_g is introduced. Considering the spatial smoothing effect of wind power fluctuations [18], a budget for wind power output uncertainty μ_w is employed. Then, the uncertainty set ζ representing all possible uncertainties is defined as:

$$\begin{aligned}\zeta = \{z = \{z_{git}, z^+_{wjt}, z^-_{wjt}\}| \\ \sum_{i=1}^{NG} z_{git} \geq NG - \mu_g, \sum_{j=1}^{NW} z^+_{wjt} + z^-_{wjt} \leq \mu_w \\ z_{git} \in \{0,1\}, z^+_{wjt} \in [0,1], z^-_{wjt} \in [0,1]\}\end{aligned} \tag{5.47}$$

5.3.1.3 SCED Model with TS

This paper focuses on spinning reserves, specifically up-spinning and down-spinning [19, 20]. The optimization goal of the proposed SCED model is to minimize the overall operating costs, including the generation costs and operating reserve costs:

$$\min \sum_{t=1}^{T} \left[\sum_{i=1}^{NG} C_{gi} + \sum_{i=1}^{NG} (u_i r^U_{it} + d_i r^D_{it}) \right] \tag{5.48}$$

where the fuel cost of the generator is in the form of the piecewise linear function, and C_{gi} can be calculated as:

$$C_{git} \geq a_{vi} p^0_{gbit} + b_{vi} \quad \forall i, \forall t, \forall v = 1, 2, \cdots, NV \tag{5.49}$$

Constraints of the dynamic SCED model are:

$$\theta^{min} \leq \theta^0_{nt} \leq \theta^{max} \quad \forall n, \forall t \tag{5.50}$$

$$\sum_{i \in G_n} p^0_{gi} + \sum_{j \in W_n} w_{ft} - \sum_{m \in N^{\infty}_0} p^{AC0}_{lrnt} - \sum_{s \in T_n} p^0_{Tt} = d_{nt} \quad \forall n, \forall t \tag{5.51}$$

$$\sum_{i \in G_n} p^0_{git} + \sum_{j \in \pi_n} w_{ft} - \sum_{m \in N^N_n} p^{DCO}_{ignnt} + \sum_{s \in T_n} p^0_{TIt} = d_{nt} \quad \forall n, \forall t \tag{5.52}$$

$$P^{\min}_{Gi} \leq p^0_{g_{it}} \leq P^{\max}_{Gi} \quad \forall i, \forall t \tag{5.53}$$

$$-R^{\text{aown}}_i \Delta T \leq p^0_{gbi(t+1)} - p^0_{gbi} \leq R^{up}_i \Delta T \quad \forall i, \forall t \tag{5.54}$$

$$r^U_{it} \leq R^{up}_i T_r \quad \forall i, \forall t \tag{5.55}$$

$$r_{it}^{D} \leq R_{i}^{down} T_{r} \quad \forall i, \forall t \tag{5.56}$$

$$p_{gbi}^{0} + r_{it}^{U} \leq P_{Gi}^{\max} \quad \forall i, \forall t \tag{5.57}$$

$$P_{Gi}^{min} \leq p_{gbi}^{0} - r_{it}^{D} \quad \forall i, \forall t \tag{5.58}$$

$$-M_{nm}(1 - q_{nmt}) \leq p_{lnmt}^{AC0} - \frac{1}{X_{nm}}(\theta_{nt}^{0} - \theta_{mt}^{0}) \leq M_{nm}(1 - q_{nmt}), \quad \forall (n, m), \forall t \tag{5.59}$$

$$-q_{nmt} F_{nm}^{AC} \leq p_{lnmt}^{AC0} \leq q_{nmt} F_{nm}^{AC} \quad \forall (n, m), \forall t \tag{5.60}$$

$$p_{lnmt}^{DC0} = \frac{1}{R_{nm}}(u_{nt}^{0} - u_{mt}^{0}) \quad \forall (n, m), \forall t \tag{5.61}$$

$$-F_{nm}^{DC} \leq p_{lnmt}^{DC0} \leq F_{nm}^{DC} \quad \forall (n, m), \forall t \tag{5.62}$$

$$-F_{s}^{T} \leq p_{Tst}^{0} \leq F_{s}^{T} \quad \forall s, \forall t \tag{5.63}$$

$$\theta^{min} \leq \theta_{nt}^{k} \leq \theta^{max} \quad \forall n, \forall k, \forall t \tag{5.64}$$

$$\sum_{i \in G_{\bar{n}}} p_{git}^{k} + \sum_{j \in W_{nj}} w_{jit}^{k} - \sum_{m \in N_{Ni}^{tic}} p_{Inmt}^{ACk} - \sum_{s \in \in_{\bar{n}}} p_{Ist}^{k} = d_{nt} \quad \forall n, \forall k, \forall t \tag{5.65}$$

$$\sum_{i \in G_{s}} p_{git}^{k} + \sum_{j \in W_{0}} w_{jt}^{k} - \sum_{m \in N^{UC}} p_{lmnt}^{DCk} + \sum_{S \in T_{n}} p_{Tst}^{k} = d_{nt} \quad \forall n, \forall k, \forall t \tag{5.66}$$

$$z_{gi}^{k} P_{Gi}^{\min} \leq p_{git}^{k} \leq z_{gi}^{k} P_{Gi}^{max} \quad \forall i, \forall k, \forall t \tag{5.67}$$

$$-M_{nm}(1 - q_{nmt}) \leq p_{lnmt}^{ACk} - \frac{1}{X_{nm}}(\theta_{nt}^{k} - \theta_{mt}^{k}) \leq M_{nm}(1 - q_{nmt}), \quad \forall (n, m), \forall k, \forall t \tag{5.68}$$

$$-q_{nmt} F_{nm}^{AC} \leq p_{lnmt}^{ACk} \leq q_{nmt} F_{nm}^{AC} \quad \forall (n, m), \forall k, \forall t \tag{5.69}$$

$$p_{lnmt}^{DCk} = \frac{1}{R_{nm}}(u_{nt}^{k} - u_{mt}^{k}) \quad \forall (n, m), \forall k, \forall t \tag{5.70}$$

$$-F_{nm}^{DC} \leq p_{lnmt}^{DCk} \leq F_{nm}^{DC} \quad \forall (n, m), \forall k, \forall t \tag{5.71}$$

$$-F_{s}^{T} \leq p_{Tst}^{k} \leq F_{s}^{T} \quad \forall s, \forall k, \forall t \tag{5.72}$$

$$p_{gi}^{k} - p_{git}^{0} \le r_{it}^{U} - \left(1 - z_{git}^{k}\right) P_{Gi}^{\min} \quad \forall (n,m), \forall k, \forall t \tag{5.73}$$

$$p_{gi}^{0} - p_{gii}^{k} \le r_{it}^{D} + \left(1 - z_{gii}^{k}\right) P_{Gi}^{\max} \quad \forall (n,m), \forall k, \forall t \tag{5.74}$$

Equations (5.50) and (5.64) are the bus angle constraints in the normal state and corrective dispatch, respectively. Equations (5.51), (5.52), (5.65) and (5.66) are the power balance constraints in the normal state and corrective dispatch for AC and DC bus, respectively. Equations (5.53) and (5.67) are the generator output constraints in the normal state and corrective dispatch, respectively. Equation (5.54) is the ramping rate constraint. Equations (5.55) and (5.56) indicate that the reserve capacity cannot exceed the ramping rate of the generator, and (5.57) and (5.58) are generation capacity constraints considering the reserve. Equations (5.59), (5.60), (5.68) and (5.69) are network constraints for AC transmission line considering TS in the normal state and corrective dispatch, respectively, where the parameter M_{nm} has to be greater than or equal to $|1/X_{nm}|max(|\theta_{nt} - \theta_{mt}|)$. To provide tighter constraints, it is better for M_{nm} to be as small as possible, so M_{nm} is taken as $\left|(1/X_{nm})(\theta^{max} - \theta^{min})\right|$. Equations (5.61), (5.62), (5.70) and (5.71) are network constraints for DC transmission line in the normal state and corrective dispatch, respectively. Equations (5.63) and (5.72) are the constraints of power flows through AC/DC terminals in the normal state and corrective dispatch, respectively. The flexibility of HVDC can be obtained by controlling and optimizing power flows through AC/DC terminals. Equations (5.73) and (5.74) indicate that the corrective dispatch after uncertainties occur is limited by the dispatch in the normal state and the upward and downward reserve.

5.3.2 Two-Stage RO Based on C&CG

The SCED model established in Section II is a mixed-integer linear programming (MILP) model containing uncertain variables, as there are too many combinations of uncertainties within the uncertainty set, solving with all constraints of all possible combinations of uncertainties is impractical, so how to find a method to represent the uncertainties and its related constraints is the key to the solution. In this paper, the RO method is adopted and addressed by the C&CG algorithm. RO solves the model by finding the worst-case scenario. By RO, the model has transformed into a two-stage (normal state optimization and corrective dispatch verification) problem, and the C&CG algorithm introduces variables and constraints related to the second-stage problem into the first-stage problem to provide tight and effective representations and achieve the solution. The SCED model established in Section II is transformed into the master problem of the normal state and the subproblem of the corrective dispatch. The optimization values of the first-stage variables obtained by solving the master problem are substituted into the subproblem, and then the subproblem finds the worst-case scenarios from the uncertainty set, and their recourse problems are

included in the master problem. The master problem and subproblem interact, and the model is iteratively optimized. The first-stage variables are the ones related to normal state optimization, which are:

$$\xi^f = \left\{q_{nnt}, p^0_{git}, r^D_{it}, r^D_{it}, \theta^0_{nt}, u^0_{nt}, p^{AC0}_{int}, p^{DC0}_{bunt}, p^0_{Tst}, C_{gi}\right\} \tag{5.75}$$

The second-stage variables are the ones related to corrective dispatch after uncertainties occur, which are:

$$\xi^s = \left\{p^k_{git}, \theta^k_{nt}, u^k_{nt}, p^{4Ck}_{bmnt}, p^{DCA}_{bmnt}, p^k_{Tst}\right\} \tag{5.76}$$

For simplicity, we rewrite the SCED model in the matrix form:

$$\min_{\xi^f} \; D(\xi^f) \tag{5.77}$$

$$\boldsymbol{F}\xi^f \leq \boldsymbol{h} \tag{5.78}$$

$$\forall z \in \zeta, \exists \xi^s \in \Omega = \{\xi^s | \boldsymbol{A}\xi^s \leq \boldsymbol{e} - \boldsymbol{B}\xi^f - \boldsymbol{C}z\} \tag{5.79}$$

To check the feasibility of the second-stage problem for a given ξ^{f*} and find the worst-case scenario, the following subproblem is defined:

(SP1)

$$\max_{z\in\zeta} \min_{\xi^s, v} \; 1^T \boldsymbol{v} \tag{5.80}$$

$$\boldsymbol{A}\xi^s - \boldsymbol{v} \leq \boldsymbol{e} - \boldsymbol{B}\xi^{f*} - \boldsymbol{C}z \tag{5.81}$$

$$\boldsymbol{v} \geq 0 \tag{5.82}$$

Taking the dual of the inner-level minimization problem with dual variables $\boldsymbol{\lambda}$, the above max–min problem is transformed as:

$$\max_{z\in\zeta, \boldsymbol{\lambda}} \; -\boldsymbol{\lambda}^T(\boldsymbol{e} - \boldsymbol{B}\xi^{f*} - \boldsymbol{C}z) \tag{5.83}$$

$$\boldsymbol{\lambda}^T \boldsymbol{A} = 0 \tag{5.84}$$

$$0 \leq \boldsymbol{\lambda} \leq 1 \tag{5.85}$$

Observing that for generator contingency, z_{git} is a binary variable which must be either 0 or 1, for wind power output uncertainty, the worst-case scenario must be one in which wind power equals to its upper bound, lower bound or forecast value,

which means the candidate values of variables z_{wjt}^{+} and z_{wjt}^{-} also must be either 0 or 1, based on that, the above bilinear problem can be reformulated into an equivalent MILP as follows:

(SP)

$$f = \max_{z,\boldsymbol{\lambda}} \ -\boldsymbol{\lambda}^T(\boldsymbol{e} - \boldsymbol{B}\boldsymbol{\xi}^{f*}) + 1^T\boldsymbol{\eta} \tag{5.86}$$

$$\boldsymbol{\lambda}^T\boldsymbol{A} = 0,\ 0 \le \boldsymbol{\lambda} \le 1,\ z\varepsilon\boldsymbol{\zeta} \tag{5.87}$$

$$\boldsymbol{\eta} \le \boldsymbol{C}^T\boldsymbol{\lambda} + M \cdot (1 - \boldsymbol{z}) \tag{5.88}$$

$$\boldsymbol{\eta} \le M \cdot \boldsymbol{z} \tag{5.89}$$

where big-M parameter M satisfies

$$M \ge \max_{\boldsymbol{\lambda}}\{\left\|\boldsymbol{C}^T\boldsymbol{\lambda}\right\|_{\infty} : s.t.\ \boldsymbol{\lambda}^T\boldsymbol{A} = 0,\ 0 \le \boldsymbol{\lambda} \le 1\} \tag{5.90}$$

After solving the subproblem SP, f and $\boldsymbol{z}^{*(\tau+1)}$ can be obtained, if $f > 0$, it means that the $\boldsymbol{\xi}^{f*}$ obtained by the master problem does not have a feasible corrective dispatch scheme after uncertainties occur. Then recourse decision variables $\xi^{s(\tau+1)}$ and the following constraints are added to the master problem:

$$A\xi^{z(\tau+1)} + B\xi^{f} \le e - Cz^{*(\tau+1)}. \tag{5.91}$$

After that, the master problem is updated as:

(MP)

$$\min_{\boldsymbol{\xi}^{f}} \ D(\boldsymbol{\xi}^{f}) \tag{5.92}$$

$$\boldsymbol{F}\boldsymbol{\xi}^{f} \le \boldsymbol{h} \tag{5.93}$$

$$A\xi^{s(\tau+1)} + B\xi^{f} \le e - Cz^{*(\tau+1)} \tag{5.94}$$

The C&CG procedure of the two-stage RO is:

(1) Solve the master problem MP and obtain the optimization values of the first stage variables $\boldsymbol{\xi}^{f*}$.
(2) Substitute $\boldsymbol{\xi}^{f*}$ into the subproblem SP for a solution; obtain f and $\boldsymbol{z}^{*(\tau+1)}$. If $f = 0$, terminate, otherwise, go to the next step.
(3) Update the master problem MP by generating recourse decision variables $\xi^{s(\tau+1)}$ and adding constraints (5.91) into the master problem, and return to step (1).

5.4 Case Studies

5.4.1 Verify of Power Margin Tracking Droop Regulation (PMT) Mode

5.4.1.1 Four-Terminal Simulation System

To verify the model proposed in this paper, referring to the Zhangbei DC Project, a four-terminal simulation model of the collector system as shown in Fig. 5.6 is built. In this model, the grid-connected renewable power imported into the sending-end VSC station 1 is provided by a wind power base WP1 and a solar power base PV1, while the grid-connected power imported into Station 2 is provided by WP2 and PV2. The regulation VSC station 3 is linked to a PS station. Station 4 is connected to the receiving-end grid.

The electrical parameters of the simulation grid are set as follows: The reference power is 100 MVA. The reference voltage is 500 kV. The conductance G_i and the susceptance B_i of the collection lines are respectively 20 and 200 (Unless otherwise specified, the numerical values adopted in this section are all in per unit). The other parameters of collection lines are shown in Table 5.1. The operation constraints are set as follows: The upper and lower limits of dc voltage are [0.9, 1.1]. The upper and lower limits of ac voltage are [0.66, 0.75], which considers the voltage amplitude relationship between the ac and dc side of the VSC station, and the ±6% voltage fluctuation range of the ac transmission lines. The upper and lower limits of ac

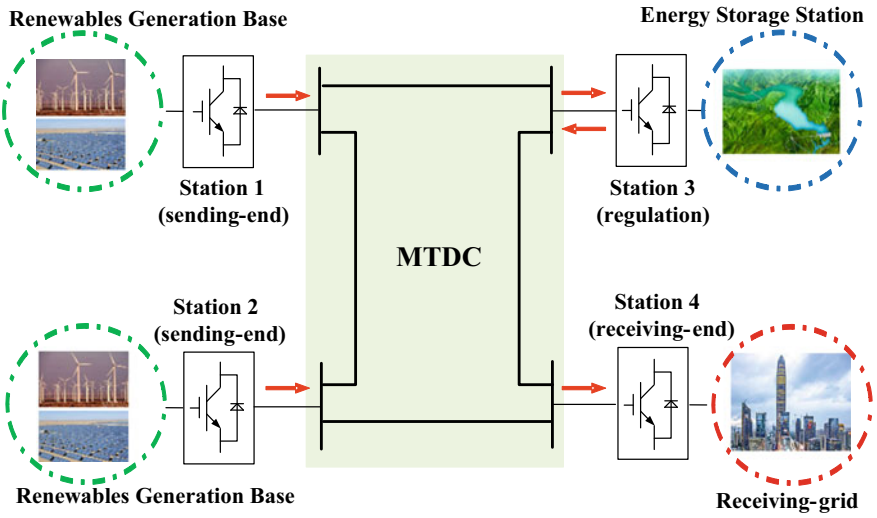


Fig. 5.6 The typical structure of collector system

Table 5.1 Parameter settings of dc transmission lines

Dc transmission lines	L_{12}	L_{13}	L_{24}	L_{34}
Length (km)	49.6	205.1	206.4	187.1
Conductance (p.u.)	5065.7	1225.0	1217.3	1342.9

Table 5.2 Parameter settings OF VSC stations

VSC station i	1	2	3	4
Rated voltage (p.u.)	1	1	1	1
Rated capacity (p.u.)	15	30	15	30
Dc voltage reference (p.u.)	1	1	–	–
Dc power reference (p.u.)	7.5	15	–	–
Droop coefficient (p.u.)	0.0133	0.00667	–	–

voltage phase angle are $[-\pi/2, \pi/2]$. The upper and lower limits of dc current in dc transmission lines are $[-30, 30]$, according to their current carrying capacity.

The operation parameters of VSC stations are set as follows: The basic parameters are shown in Table 5.2, where the droop coefficients are calculated by (5.21). The transmission constraints are determined by their rated capacity. The voltage modulation ratio constraint is (0, 1).

The power parameters of renewables generation bases are set as follows: the rated capacities (p.u.) of WP1, WP2, PV1, and PV2 are 13.5, 26, 9, and 19. Two wind power bases and two solar power bases are respectively set to have the same power fluctuation characteristics. Their normalized prediction curves of grid-connected power are given in Fig. 5.7, which are all sourced from the actual typical daily curves of renewables generation base in Zhangbei district, China. The reference power value of the receiving grid is set to 30, its normalized prediction curve of power demand is also given in Fig. 5.4, which is from the actual daily curves of the Beijing urban area.

The simulation parameters of the PS station are set as follows: The rated capacity is 18 (p.u.). The upper and lower limits of the upper reservoir capacity are $4.814 \times 107\ m^3$ and $7.53 \times 106\ m^3$. The initial water capacity is set to $2.784 \times 107\ m^3$. The conversion ratio of waterflow over electricity of generation mode and water-pumping mode are respectively $780\ m^3/(MWh)$ and $980\ m^3/(MWh)$. Set the PS station to work in the daily operation state, the maximum allowable flow variation between the initial and end period in a day is 20% of its adjustable storage capacity, that is $8.12 \times 106\ m^3$.

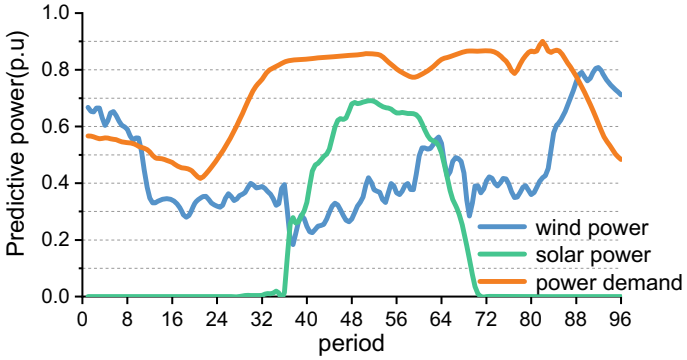


Fig. 5.7 Normalized prediction curves of renewables output and power demands

5.4.1.2 Operation Verification and Comparison of the Proposed Regulation Mode

Apply the proposed regulation mode to optimize operation of the case system. The optimal results show that the curtailed renewable energy is 5.74 (p.u.), the ac losses are 243.82 (p.u.), and the dc losses are 28.77 (p.u.). Figure 5.8 shows the actual grid-connected power of sending-ends (P_{s1} and P_{s2}) and the regulation power of PS station (P_{s3}). The given scheduling scheme can exert the complementary regulation characteristics of the energy storage.

In order to verify the effectiveness of the proposed PMT regulation mode, the conventional FDC droop regulation mode is used to optimize operation of the collector system for comparative analysis. At this time, Eq. (5.23) needs to be replaced by (5.11) described in Sect. 5.3.1.

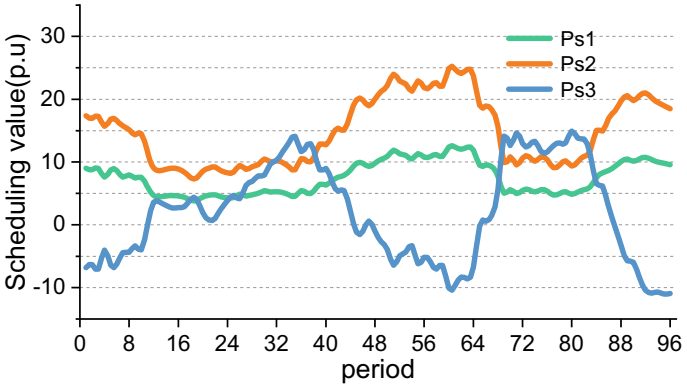


Fig. 5.8 Scheduling curves for multiple renewables

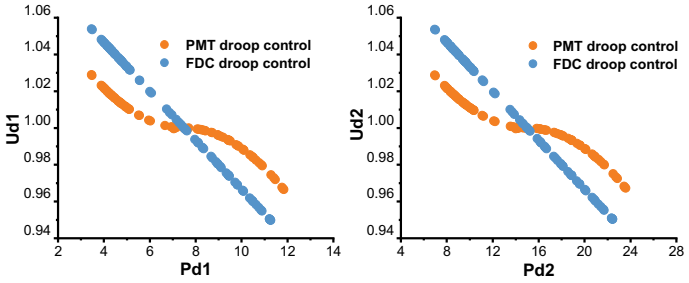


Fig. 5.9 Operation points of distribution of Station 1 and 2

Figure 5.9 presents the operation points distribution in different periods of Stations 1 and 2. It can be seen intuitively that when the operation status of VSC stations have to change due to the continuous fluctuation of renewable power: (1) Under the FDC mode, the VSC station can only search for a new operating point along the preset straight line, and share the fluctuating power by the fixed ratio. (2) Under the PMT mode, the VSC station can automatically search for a proper operating point in terms of its real operation status and available regulation margin.

This above-mentioned self-adaptive regulation and control process of the VSC station is realized by its droop coefficient tracking the power margin. The dynamic effect of the droop coefficient of Station 1 tracking its power margin under two control strategies in multiple periods is shown in Fig. 5.10 (for discrete operating points, the droop coefficients are expressed by the variation ratio of the dc operation voltage and power between two adjacent periods). The droop coefficient curves under two control strategies are appropriately numerically enlarged to be displayed for comparison with the corresponding power margin curve in the same picture. In addition, the Pearson correlation coefficient is adopted to quantitatively analyze the consistency of the variation trend between the droop coefficient and power margin. When applying FDC droop mode, the droop coefficient remains unchanged during the whole scheduling period. Its Pearson correlation coefficient is 0, indicating that the droop coefficient does not respond to the change of real-time operation states. When applying the proposed PMT droop mode, the droop coefficient follows the change of the power margin all the time. When the power margin is bigger, which means the larger regulation potential, thus giving a smaller droop coefficient to let it share more power fluctuation for a more stable dc voltage. When the power margin is smaller, which means approaching the regulation limit, thus gives a bigger droop coefficient to let it share fewer power fluctuations for stabilizing the operating power. The Pearson correlation coefficient exceeds 0.95, indicating the droop coefficient under the proposed mode tracking its power margin well.

Analyze the cooperative operation capability of stations 1 and 2. Table 5.3 shows the representative operation data of these two stations when the grid-connected power increased or decreased. (1) From period 3 to period 4, the grid-connected power is increased. So, the dc transmission power of the two stations is increased, their power

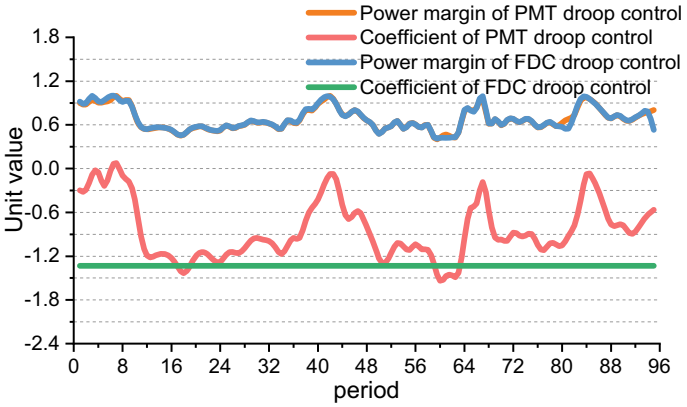


Fig. 5.10 Dynamic curves of droop coefficient and power margin of Station1

margins are reduced. However, since the power margin of Station 1 is smaller than Station 2 at period 3, the power increase proportion of Station 1 is less than that of Station 2, which is reflected in that the power margin decrease proportion of Station 1 is 0.25% less than that of Station 2. (2) From period 12 to period 13, the grid-connected power is decreased. So, both dc transmission power and power margins of the two stations are decreased. Since the power margin of Station 1 is bigger than Station 2 in period 12, the power decrease proportion of Station 1 is more than that of Station 2, which is reflected in that the power margin decrease proportion of Station 1 is 0.27% more than that of Station 2. The above analysis indicates that the application of the PMT droop regulation mode improves the cooperative operation capacity of multi-VSC stations for adaptively adjusting their power-sharing responsibility.

Furthermore, the optimal results of the renewables curtailment rate (RCR), network loss rate (NLR), and dc voltage average deviation rate (VADR) under the two control strategies are given in Table 5.4. Compared with the FDC droop mode, when applying the PMT droop mode, the NLR and the RCR are both slightly reduced, especially the VADR is significantly reduced by 56.8%. It proves the advantages

Table 5.3 Representative operation data of sending-end stations

Period	Grid-connection power (p.u.)		Power margin	
	Station 1	Station 2	Station 1 (%)	Station 2 (%)
3	8.80	16.95	90.00	92.30
4	8.98	17.29	87.94	89.99
Power increase from period 3 to 4			2.06↓	2.31↓
12	6.04	11.64	74.02	73.93
13	4.70	9.04	56.71	56.89
Power decrease from period 12 to 13			17.31↓	17.04↓

Table 5.4 The operation results of two control strategies

Indicator	FDC droop mode (%)	PMT droop mode
NLR	6.22	6.19
RCR	0.28	0.20
VADR	3.22	1.39

of the PMT droop mode in mitigating the grid-connected power fluctuations and enhancing the operation stability of the collector system.

5.4.2 Exploring Operational Flexibility of AC/DC Power Networks Using TS

5.4.2.1 14-Bus System

In case 1, based on the IEEE 14 bus system original data [16], one wind farm is connected to bus 3. The transmission capacity of each AC line adopts the data in [21], besides, to create congestion, line (4–5) is removed. The overlay HVDC grid of the IEEE 14 bus system is shown in Fig. 5.11, which is designed based on power configuration analysis to increase the transmission capacity between central buses [22]. All DC lines have the same resistance of 2.78 Ω, and F^{DC}= 50 MW, which matches the capacities of the rest of the grid [23]. The predicted load and wind power output is shown in Table A1 in the Appendix. For the wind power's uncertainty, we assume that the wind farm power output is a Gaussian distribution random variable with a mean of w_{fjt} and a standard deviation of $\sigma_{wjt} = 0.1{*}w_{fjt}$. For the proposed approach, $\varepsilon_{jt}{}^{+} = \varepsilon_{jt}{}^{-} = 3{*}\sigma_{wjt}$, which covers the 99.74% quantile of the wind farm output. (It should be mentioned that in practice, $\varepsilon_{jt}{}^{+}$ and $\varepsilon_{jt}{}^{-}$ can use the forecast error confidence interval provided by the prediction method [24].) The uncertainty budget $\mu_g = 1$, $\mu_w = 1$. The maximum and minimum bus voltage angles are ±0.6 radians. The original fuel cost of the generator is in the form of the quadratic function, to reduce computational complexity, it needs to be transformed to the piecewise linear one, of which parameters of each segment are calculated using the values of its boundary points obtained from the original function. In this paper, the number of segments of the piecewise linear function $NV = 3$. Upward and downward reserve unit costs are taken as half of the linear term fuel cost coefficients of the original generator fuel cost function.

In order to illustrate the effects of TS and HVDC on reducing the operating cost of the system, different HVDC transmission capabilities (expressed by the relative value compared to the designed HVDC transmission capacity) with and without TS are considered. The results are shown in Table 5.5.

From Table 5.5, HVDC transmission capability has an obvious impact on the operating cost. For example, when HVDC transmission capability is taking its rated value, that is 1, the total operating costs are the same with TS and without TS. This

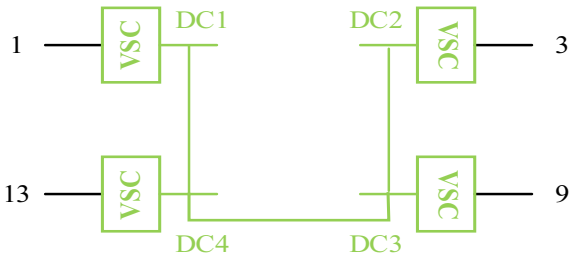


Fig. 5.11 Overlay HVDC grid of the modified IEEE 14 bus system

Table 5.5 Operating cost comparison of different HVDC transmission capabilities and with/without TS

HVDC transmission capability	w/ or w/o TS	Total cost ($)	Gen. cost ($)	Reserve cost ($)
1	w/	50,369.81	38,045.81	12,324.00
	w/o	50,369.81	38,045.81	12,324.00
0.75	w/	50,369.81	38,045.81	12,324.00
	w/o	50,392.46	38,104.66	12,287.80
0.5	w/	50,691.69	38,882.13	11,809.56
	w/o	51,415.45	38,852.06	12,563.39
0.25	w	Infeasible		
	w/o	Infeasible		

is because if HVDC had enough transfer capability, it can provide enough flexibility for transferring more power from the generating unit with low marginal cost, whether with or without TS has little impact on power flow distribution. When HVDC transmission capability decreases to 75% of its rated value, the total cost is 50369.81$ with TS, but 50,392.46$ without TS. The cost w/o TS is rising compared to the rated HVDC capability, and there is a difference between with TS and without TS, and this difference is getting larger as HVDC capability further decreases. Besides, when HVDC transmission capability decreases to 25% of its rated value, there is no feasible scheduling scheme, which means with 25% of rated HVDC transmission capacity, the network cannot provide enough flexibility to withstand multiple uncertainties, otherwise, wind curtailment or load shedding might be present. Furthermore, the reduction of the operating cost may either come from that of generation cost or that of reserve cost, proving the necessity of jointly dispatching generation and reserve. The above results show that employing TS can help HVDC provide flexibility for the system operation.

To further illustrate the effect of TS on power flow distribution and HVDC providing flexibility, we take the 5th period (with the largest net load) as an example, here HVDC transmission capacity is at half of its rated, the scheduled generation and reserves and operating cost with and without TS are shown in Table 5.6. Under one worst-case scenario where Generator 3 is out of service due to failure, and wind power

equals to its lower bound, when the output of each generator is taken as that with TS under this worst-case scenario, the corresponding power flow with and without TS is shown in Fig. 5.12.

From Fig. 5.12, with TS, line (7–9) and line (9–14) are optimized to switch off, and the power flow of each line is within its limit. Whereas if not adopting TS, the

Table 5.6 Scheduled generation and reserves and operating cost with and without TS at the 5th period

		With TS		Without TS	
		Power output (MW)	Upward reserve (MW)	Power output (MW)	Upward reserve (MW)
Generator	1	83.68	0.00	95.67	0.00
	2	0.00	83.68	0.00	67.25
	3	53.32	19.49	66.00	3.18
	4	66.00	9.61	41.33	54.34
	5	40.00	0.00	40.00	0.00
		Generation cost ($)	Reserve cost ($)	Generation cost ($)	Reserve cost ($)
		8534.85	1709.81	8339.90	2113.95
		Total cost ($)		Total cost ($)	
		10,244.66		10,453.85	

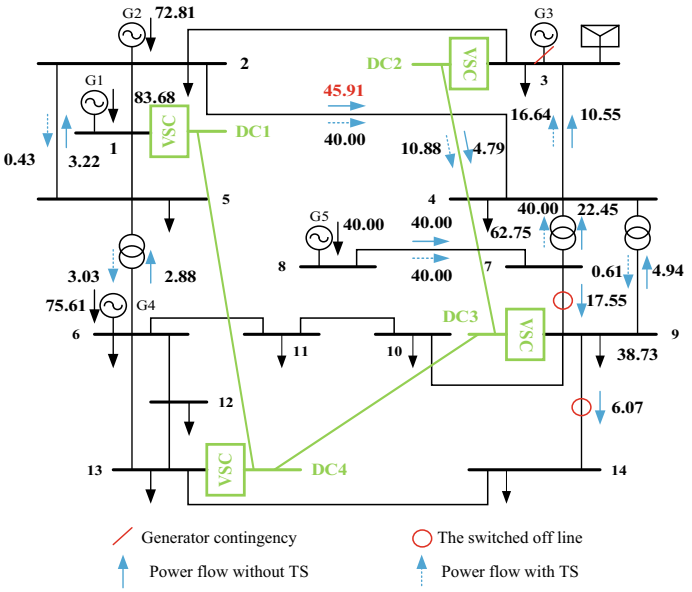


Fig. 5.12 Power flow of the hybrid system with and without TS (MW)

power flow of line (2–3) exceeds its limit. That is because when TS is adopted, line (7–9) and line (9–14) are switched off, power flow of line (4–7) increases from 22.45 to 40.00 MW, power deficiency of bus 4 which needs to be compensated by line (2–3) decreases; besides, power flows of line (2–5) and line (5–6) are reversed, so the transmission pressure of line (2–3) is relieved. As for power deficiency of bus 9 caused by switching off line (7–9), observing that the power flows of line (4–3) and DC line (DC2–DC3) increase, it is partly compensated through line (4–3) to DC line (DC2–DC3), showing that TS can enhance the performance of HVDC providing flexibility. Without TS, since power flow violation of line (2–3), Generator 2 must reduce its output to seek a feasible corrective dispatch scheme, causing the reduction of the reserve provided by Generator 2, since the reserve unit cost of Generator 2 is lower than that of other generators, the reserve cost increases, as shown in Table 5.5.

5.4.2.2 118-Bus System

Case 2 is based on the IEEE118-bus system original data (borrowed from [25]), 3 wind farms are connected to buses 59, 90 and 116. Because the original generator costs from [25] are abnormally low (Most of the generator costs are around 0.50$/MWh. A few generators have costs above 2.00$/MWh. These costs are about 50 times smaller than typical generator costs [26]), we modify them to 50 times the original ones. It should be noted that because the generator cost is in the form of the linear function, such modification would not affect the results and conclusions. An HVDC grid is formed by replacing 5 AC lines by DC lines as shown in Fig. 5.13. The terminals and DC lines are selected based on the betweenness analysis [27] to increase the transmission capacity between central buses [22]. All DC lines have the same resistance of 5.56 Ω, and $F^{DC} = 220$ MW, which is equal to the capacity of the original AC lines. The uncertainty budget $\mu_g = 1$, $\mu_w = 3$. The remaining data are determined in the same way as in Case 1.

Operating cost comparison of different HVDC transmission capabilities and with/without TS is shown in Fig. 5.14.

As HVDC transmission capability rises, the operating cost decrease, and the gap of the operating cost between different HVDC transmission capabilities is larger

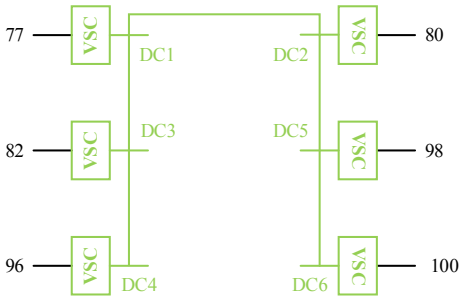


Fig. 5.13 HVDC grid of the modified IEEE 118 bus system

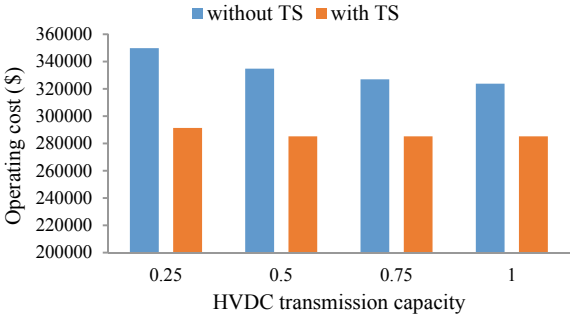


Fig. 5.14 Operating cost comparison of different HVDC transmission capabilities and with/without TS

Table 5.7 Comparison of operating costs of different uncertainty budgets

μ_w	3	2	1	0
Operating cost ($)	285,181.50	275,368.18	266,237.90	244,983.65

without TS than with TS. Unlike case 1, the operating cost with TS is always smaller than that without TS for all HVDC transmission capabilities, even though the gap of the operating cost between with TS and without TS narrows as HVDC transmission capability increase. Moreover, the benefits of TS are much more pronounced compared to case 1, making the reduction of the operating cost reach up to 16.72% when the HVDC transmission capability is small. The above results show that HVDC and TS cooperate to optimize power flows of the system, and combining HVDC with TS can make the system operation more flexible, resulting in lower operating cost.

In order to illustrate the effect of introducing the budget for wind power output uncertainty on controlling the conservativeness of robust optimization, different values for μ_w are considered. The results are shown in Table 5.7.

Introducing the budget for wind power output uncertainty has an obvious effect on reducing the operating cost. The budget for wind power output uncertainty is a representation of the spatial smoothing effect of wind power fluctuations, which means that total wind power fluctuations of a system with spatially distributed wind farms tend to be smaller than that of a system with wind farms of a single site. When μ_w equals to the number of wind farms, all wind farms can reach their lower/upper bound, corresponding to the most conservative case. The budget for wind power output uncertainty constrains total wind power fluctuations of the system, contributing to reduce the conservativeness of the robust optimization, and thus reducing the operating cost.

5.5 Conclusion

This chapter introduces the flexible operation mechanisms of MTDC/AC hybrid girds and two flexible dispatch methods by using the VSC regulation and AC-TS regulation techniques.

By the proposed VSC-MTDC flexible operation model, the flexible operation domain of the power grid is formed. On the one hand, the positive correlation between the flexible regulation capacity of the power grid and the energy storage capacity is significant to present. On the other hand, the power regulation of multiple VSCs limits the flexible operation of the power grid, which brings the enlightenment of the significance of improving the power regulation capability of the VSC station.

By the proposed novel power margin tracking (PMT) droop regulation mode and the corresponding flexible operation model, the VSC station can make a self-adaptive regulation according to its operational state for flexible response to wind and solar power fluctuation and improve the cooperative operation of multi-VSC stations. Compared to the traditional FDC droop mode, the proposed PMT mode can effectively reduce the DC voltage deviation, enhancing the operational stability of the MTDC grid.

By the proposed security-constrained robust economic dispatch model, TS can assist HVDC to release its flexibility by optimizing power flow distribution, which is more obvious when HVDC transmission capacity is relatively small. With the TS collaboration, the operating cost is reduced, and the transmission congestion is remitted to the MTDC/AC hybrid grids.

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Chapter 6
Demand Side Flexibility



With the continuous development of the types, functions, and controllability, demand-side resources begin to respond actively according to the operation state of power systems. The power demand side contains abundant adjustable resources, and have great potential in improving the flexibility of power systems. This chapter investigates the demand side flexibility from following three aspects: residential load side demand response, price incentive demand-side response, and integrated energy system demand-side response. In load-side demand response, a complete scheduling scheme is modeled based on the optimization of residential loads and distributed generation. The presented model reduces the cost of the user's electricity consumption and decreases the peak load and peak-valley difference of smart residential load without bringing discomfort to the users. As for price incentive demand-side response, an energy management model is established for intelligent temperature control appliances of commercial and residential customers which can optimize its electricity consumption according to power prices under the smart grid. And in the integer energy system side, a resilience-oriented stochastic integrated energy system configuration framework considering integrated demand response influence is proposed to improve the flexibility and resilience of the power system.

6.1 Introduction

As global climate issues become increasingly prominent, major countries in the world are actively promoting carbon neutrality. The power structure and system ecology of the new power system has undergone profound changes [1]. It is difficult to ensure the reliable supply and stable operation of the new power system only relying on the power-side regulation capability. In contrast, demand-side solutions are usually smaller in scale and more diversified in options, which promote the transformation of the power system from "source-load-driven" to "source-load interaction", and make full use of demand-side resources in new energy as the main body [2]. The role in the

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M. Zhou et al., *Power System Flexibility*, Power Systems,
https://doi.org/10.1007/978-981-19-9075-5_6

new power system is very urgent and necessary. With the application of technologies such as distributed power supply, energy storage, heat storage, and flexible regulation, the characteristics of load power consumption have undergone major changes. There are huge adjustable resources in the power load of industrial enterprises, commercial buildings, etc. But the current demand-side resource adjustment ability is insufficient, and it is difficult to form a stable load and grid coordination and interaction ability. So, its role in the new power system is very limited.

From the perspective of the development history of demand-side resource utilization, its utilization methods mainly include orderly power consumption, energy efficiency management, demand response, and precise real-time load control [3]. Orderly use of electricity refers to the management work of controlling part of the electricity demand by the law through administrative measures, economic means, and technical methods in the case of insufficient electricity supply, emergencies, etc. Energy efficiency management refers to a demand-side resource utilization method that adopts technical and management measures to stop waste, reduce power consumption, and achieve power conservation [4]. Demand response refers to the way that electricity users change demand-side resource utilization following price signals or incentive measures. Accurate real-time load control refers to the demand-side resource utilization method of the flexible load that can be quickly responded to the grid operating agency to accurately and real-time control.

At present, demand response practice is mainly concentrated in the areas with dense load and large peak-to-valley differences in the central and eastern regions in China [5]. It is aimed at the utilization of new energy and the balance of supply and demand in the scenarios of the transmission and receiving end power grids in the winter heating, summer peaking load, extreme grid failures, etc., Accident support and other requirements have formed three specific implementation methods of peak cutting, valley filling, and precise real-time load control. In terms of business model, the business model of demand-side resource utilization is relatively traditional and single [6]. Most of the business models are organized by power grid companies, with independent participation of users, and finally compensated by winning bid capacity and actual response effects. In recent years, new demand-side resource libraries represented by distributed power sources and adjustable loads have continued to expand, and new business models have been continuously tried [7].

Based on the above considerations, this chapter focuses on the operational flexibility of the power system with new energy as the mainstay, starting from the utilization method and practice of demand-side resources, and separately introduces the demand response of residential load, the interactive operation of intelligent temperature control load and the system that takes the impact of electricity prices into account. Three models of demand response are presented in this chapter to improve the flexibility of the new power system operation.

First, this chapter presents a demand response scheduling model for smart residential communities incorporating the current circumstances and the future trends of demand response programs. In this section, smart residential loads are firstly classified into different categories according to different demand response programs. A complete scheduling scheme is modeled based on the optimization of residential

loads and distributed generation. The presented model reduces the cost of the user's electricity consumption and decreases the peak load and peak-valley difference of smart residential load without bringing discomfort to the users, through which the residential community can participate in demand response efficiently.

The second section focuses on exploiting the demand-side flexibility resources in unit scheduling. In this section, an energy management model (EMM) is established for intelligent temperature control appliances (ITCAs) of commercial and residential customers which can optimize its electricity consumption according to electricity prices under the smart grid, and the calculation model on the price elasticity of demand (PED) of the ITCA is built on the EMM through the mathematical derivation, by which the calculation method of ITCAs' flexibility is proposed, namely, the PED of the ITCA is obtained with the consideration of appliance operation characteristics and customer's electricity consumption behaviors. Subsequently, taking the time-of-use (TOU) electricity price into account, a flexibility calculation method for ITCAs is proposed. Then, a unit scheduling model considering the flexibility of ITCAs is established to explore the influence of TOU power price on the power system operation.

The third section proposes a resilience-oriented stochastic integrated community energy system (ICES) configuration framework considering integrated demand response (IDR) influence. Firstly, generalized IDR models are set up in detail with elaborate fuzzy feature analysis of price-responsive multi-energy loads. Then, the vulnerability indicators of tie lines and converting devices of ICES are first introduced to demonstrate the occasional outage in normal operation and blackout in the emergent case. In addition, the worst-case conditional value-at-risk (WCVaR) theory is innovatively integrated into the traditional risk-neutral model, which is formulated as a two-stage stochastic chance-constrained programming problem, aiming to combine portfolio with minimizing the worst-case cost caused by a disaster. The models are then transformed into mixed-integer linear programming problems via several linearization techniques.

6.2 Residential Load Demand Response Model

In addition to electric energy-consuming load, renewable energy DG and ESS are also integrated into the smart residential community. On those of demand sides where the photovoltaic solar panel is installed, solar power consumption takes precedence, and the residual load is supplied by both ESS and grid. The objective function of this model is to minimize the purchased electricity cost of residents.

$$\text{Min} \sum_t \rho_t P_t^g - \sum_t \sum_j \rho_j^l P_{j,t}^l \tag{6.1}$$

where ρ_j^l is interruptible load curtailment tariff.

The power of both demand side load and grid side supply should meet the power balance constraint.

$$\sum_i P_{i,t}^V + \sum_i P_{i,t}^S + P_t^g = \sum_j (P_{j,t}^L - P_{j,t}^l) + \sum_j P_{j,t}^a + \sum_k P_{k,t}^a + \sum_j P_{j,t}^b + \sum_k P_{k,t}^b \tag{6.2}$$

where P_t^g is power purchased from grid at hour t, $P_{i,t}^V$ is solar power generation of unit i at hour t, $P_{i,t}^S$ is battery change/discharge power of battery i at hour t.

The purchased grid power is restricted by (6.3), thus preventing a large portion of load to be scheduled to low price hours, which may create new peaks.

$$P^{g,\min} \leq P_t^g \leq P^{g,\max} \tag{6.3}$$

where $P^{g,\min}$, $P^{g,\max}$ is minimum/maximum purchased power from grid at each hour.

Roof-top solar panels are uncontrollable DG. Since total residential load is considerably higher than solar power generation, this paper assumes all solar power is only used by the community instead of exporting to the grid.

ESS battery constraints are shown as follows:

$$P_{i,t}^S = P_{d,i,t} - P_{c,i,t} \tag{6.4}$$

$$I_{d,i,t} + I_{c,i,t} \leq 1 \tag{6.5}$$

$$I_{c,i,t} P_{c,i}^{\min} \leq P_{c,i,t} \leq I_{c,i,t} P_{c,i}^{\max} \tag{6.6}$$

$$I_{d,i,t} P_{d,i}^{\min} \leq P_{d,i,t} \leq I_{d,i,t} P_{d,i}^{\max} \tag{6.7}$$

$$E_{i,t} = E_{i,(t-1)} - (P_{d,i,t} \cdot \frac{1}{\eta_i^S} - \eta_i^S P_{c,i,t}) \Delta t / E_i^{\max} \tag{6.8}$$

$$E_i^{\min} / E_i^{\max} \leq E_{i,t} \leq 1 \tag{6.9}$$

$$E_{i,0} = E_{i,NT} \tag{6.10}$$

where $P_{d,i,t}$, $P_{c,i,t}$ is discharge power/charge power of battery i at hour t, $I_{d,i,t}$, $I_{c,i,t}$ is discharge/Charge binary status indices of battery i at hour t, $E_{i,t}$ is SOC of battery i at hour t, $P_{c,i}^{\min}$, $P_{c,i}^{\max}$ is minimum/maximum charge power of battery i at each hour, $P_{d,i}^{\min}$, $P_{d,i}^{\max}$ is minimum/maximum discharge power of battery i at each hour, η_i^S is discharge/charge inverter efficiency, $E_i^{\min}$, $E_i^{\max}$ is minimum/Maximum stored energy of battery I, NT is total schedule hours.

When the battery discharges, its performance as a power source. On the other hand, when it charges, it acts as a load (6.4). The battery cannot charge or discharge

at the same time (6.5). The battery hourly charge and discharge power are restricted by (6.6), (6.7) respectively. Equations (6.8) and (6.9) are battery SOC constraints, namely, the battery cannot charge when its SOC reaches 1, or discharge when its SOC is 0. The initial SOC is equal to the SOC at the end of the scheduling hour (6.10).

To sum up, LA schedules the residential community load, and DG intends to minimize total power purchase cost under different DR programs. This model optimizes the day-ahead (24 h) power scheduling of residential communities considering TOU, CPP, RTP, and IL. The complete model is as follows:

$$\begin{aligned} &\text{Min} \sum_t \rho_t P_t^g - \sum_t \sum_j \rho_j^l P_{j,t}^l \\ &s.t. \begin{cases} (2.19) - (2.38) \\ (6.2) - (6.10) \end{cases} \end{aligned} \tag{6.11}$$

6.3 Price-Based Demand Response Model

At present, the enhancement of the thermal power unit flexibility through reducing the minimum technical output and an increasing the ramp rate of the thermal power units on the generation side is an effective measure to improve the flexibility of the power system. Meanwhile, information and communication technology has developed, which makes it possible for demand response (DR) resources to provide flexibility by reducing or shifting demand.

Temperature control appliances (TCAs), represented by air conditioning, account for a large share of electricity consumption. For instance, the electricity consumption proportion of residential air conditioning reaches up to 40% during summer peak load periods. Therefore, TCAs have a great potential to be controlled to provide schedulable resources. Moreover, it is widely accepted that TCAs can be used to provide flexibility to the grid since TCAs have large thermal inertia. There are some researches on the aggregation of TCAs to provide auxiliary service (i.e., flexible thermostatically controlled loads). Price-based DR, by which TCAs can be applied to provide auxiliary service, uses market price signals to change the original electricity consumption behavior of the customers who participate in the system with frequency modulation or emergency regulation considering customers' willingness.

To address the aforementioned concerns, this chapter introduces a concept of the intelligent temperature control appliance (ITCA) which can optimize its electricity consumption according to power prices under the smart grid. Different from the existing studies on the aggregation of TCAs and investigating the influence of ITCAs on the power system operation at the same time, this chapter explores the flexibility of ITCAs from the appliance level and evaluates the influence of these flexibility resources on unit scheduling.

6.3.1 Energy Management Model of the ITCA

Energy management for commercial and residential customers has become an important research area under the smart grid concept, which also provides new ideas for exploiting demand-side flexibility. Under these conditions, it is possible to optimize ITCAs' electricity consumption according to power prices under the smart grid. Thus, this study first establishes the EMM for the ITCA and then analyzes the demand characteristics of the ITCA based on this model. Finally, the flexibility of customers' ITCAs is explored.

The energy management of the customer's ITCA considers the electricity cost and comfort level of the customer, and the operation of the ITCA is affected by appliance operation characteristics and the customer's electricity habits. Specifically, commercial central air-conditioning and residential air-conditioning are considered in this work.

This study aims to minimize the electricity cost and discomfort of the customer on the ITCA, as expressed in Eq. (6.12):

$$\min f = C_a + wU_a \tag{6.12}$$

Noting that it is not easy to provide a systematic way to determine the appropriate weight because they depend on each customer's personal preference and condition. The weights of customers who care more comfort is greater than that of customers who pay more attention to electricity cost. However, each customer can find an appropriate weight for himself/herself easily by several trial-and-errors. This study assumes that owners of ITCAs have the same preference for comfort and electricity cost according to expert advice. The weight for commercial customers follows the uniform distribution of the interval while that for resident customers follows the uniform distribution of the interval. This study determines weights of commercial and resident customers by generating random numbers according to these two distributions, respectively.

C_a is the electricity cost of the customer on the ITCA, which can be calculated as Eq. (6.13)

$$C_a = \sum_{t \in T} \lambda_n^t P_a^t \tag{6.13}$$

This work uses the negative benefit to evaluate the customer's discomfort and establishes a quadratic convex function to quantify the customer's discomfort when appliance a is running, and it is normalized. The discomfort function is expressed in Eqs. (6.14)–(6.15):

$$U_a = \frac{1}{D_{a,\max}} \sum_{t \in T_a} \left(T_{a,in}^t - T_{a,in,best}\right)^2 \tag{6.14}$$

$$D_{a,\max} = \left(T_{a,in,\max} - T_{a,in,best}\right)^2 |T_a| \tag{6.15}$$

where $|T_a|$ is the norm of set T_a, indicating the length of appliance running time, Eq. (6.14) represents the normalized discomfort level, and Eq. (6.15) represents the maximum temperature deviation. It is evident that the greater the temperature deviation, the more uncomfortable the customer is; moreover, the discomfort gradually deepens, and $U_a \in [0, 1]$.

There are some differences between the environment of commercial central air conditioning and that of the household ITCA, and thus, the modeling methods are different. The thermodynamic first-order equivalent model of the commercial ITCA is presented in Eq. (6.16).

$$T_{a,in}^t = \frac{Z_a^t}{W_a}\left(1 - e^{-\frac{W_a}{Y_a}\Delta t}\right) + \frac{P_a^t}{W_a}\left(1 - e^{-\frac{W_a}{Y_a}\Delta t}\right) - T_{a,in}^{t-1} e^{-\frac{W_a}{Y_a}\Delta t}, t \in T_a \tag{6.16}$$

where W_a is the heat transfer of the roof, external walls and windows of the commercial building, Y_a is the heat storage of commercial building walls and air, and ${Z_a}^t$, related to the outdoor temperature and the calculated temperature of the external wall cooling load at sub-interval (SI) t, represents the effects of equipment, lighting, human body heat dissipation and solar radiation on indoor temperature. The calculation methods of W_a, Y_a, and ${Z_a}^t$, are shown in the (6.17) to (6.20).

In (6.16), W_a, Y_a and ${Z_a}^t$ can be obtained as (6.17) to (6.19), where parameters are calculated as (6.20).

$$W_a = A_{a,r}G_r + A_{a,ew}G_{ew} + A_{a,w}G_w \tag{6.17}$$

$$Y_a = C_a V_{a,co}\rho_a + A_{a,iw}G_{iw} \tag{6.18}$$

$$Z_a^t = A_a T_a^{out} + B_a + C_a + D_a \tag{6.19}$$

$$\begin{cases} A_a = A_{a,w}G_w + A_{a,r}G_r \\ B_a = A_{a,ew}G_{ew} + (T_e + T_w + T_s + T_n) \\ C_a = A_{a,co}\left(C_e N_e + C_l N_l + n_k u\left(C_p q_s + q_l\right)\right) \\ D_a = A_{a,w}\left(q_{fe}K_{cle} + q_{fw}K_{clw} + q_{fs}K_{cls} + q_{fn}K_{c\ln}\right) \end{cases} \tag{6.20}$$

where G_r, G_{ew}, G_w and Giw are heat transfer coefficients of the roof, external walls, windows and internal walls of the commercial building, respectively; A_{ar}, $A_{a,ew}$, $A_{a,w}$ and $A_{a,n}$ are areas of the roof, external walls, windows and internal walls of the commercial building, respectively; C_a and r_a are the specific heat capacity and density of air, respectively; $A_{a,co}$ and $V_{a,co}$ is the area and volume of the commercial building; T_e, T_w, T_s and T_n are east, west, south and north calculation temperatures of the cooling load, respectively; q_s and q_l are sensible heat and latent heat dissipating

heat of human body, respectively; n_k is the number of people per unit area; Ne and N_l are heat dissipated per unit area of equipment and lighting, respectively; q_{fe}, q_{fw}, q_{fs} and q_{fn} are the maximum radiant heat of the east, west, south and north of the building, respectively; K_{cle}, K_{clw}, K_{cls} and K_{cln} are cooling load parameters of east, west, south and north exterior window of the building, respectively; C_e, C_l and C_p are cooling load parameters of equipment, lighting and human body, respectively; u is the Coefficient of aggregation. It is noting that A_a, B_a, C_a and D_a are the intermediate quantities for calculating ${Z_a}^t$, which are only related to commercial building parameters, and are not time-varying.

The thermodynamic first-order equivalent model of the household ITCA is expressed in (6.21), in which the first, second and third items are the influence of ${T_{a,out}}^t$, ${Q_a}^t$ and ${T_{a,in}}^{t-1}$ on ${T_{a,in}}^t$.

$$T_{a,in}^t = T_{a,out}^t\left(1 - e^{-\frac{\Delta t}{R_a C_a}}\right) + Q_a^t R_a\left(1 - e^{-\frac{\Delta t}{R_a C_a}}\right) + T_{a,in}^{t-1} e^{-\frac{\Delta t}{R_a C_a}},\ Q_a^t = \eta_a P_a^t,\ t \in T_a \tag{6.21}$$

The operation of the ITCA is affected by the rated power and the customer's production and business or family living habits. Thus, the constraint condition can be obtained as Eq. (6.22):

$$\begin{cases} 0 \le P_a^t \le x_a^t P_{a,rate} \\ x_a^t = 0, \forall t \in T \backslash T_a, \forall a \in A \\ T_a = \{S_a, S_{a+1}, \ldots, F_{a-1}, F_a\} \end{cases} \tag{6.22}$$

When the room temperature is between the minimum and maximum temperatures set by the customer, which is acceptable for the customers. Therefore, the temperature constraint of the appliance a can be obtained as (6.23):

$$T_{a,in,\min}^t \le T_{a,in}^t \le T_{a,in,\max}^t, \forall t \in T_a \tag{6.23}$$

6.3.2 *Flexibility of ITCAs*

Considering that the flexibility of ITCAs originates from the flexible regulation of each ITCA in the power system, this study first derives and analyzes the demand characteristics of an ITCA through the mathematical derivation to obtains the PED of the ITCA. Then, The ITCAs' flexibility can be obtained based on the PED. Finally, a calculation model of ITCAs' flexibility under TOU power price is built.

According to the mathematical derivation presented, we can obtain the demand characteristics of the ITCA, as shown in Fig. 6.1. In view of these characteristics, this study defines the PED as (6.24):

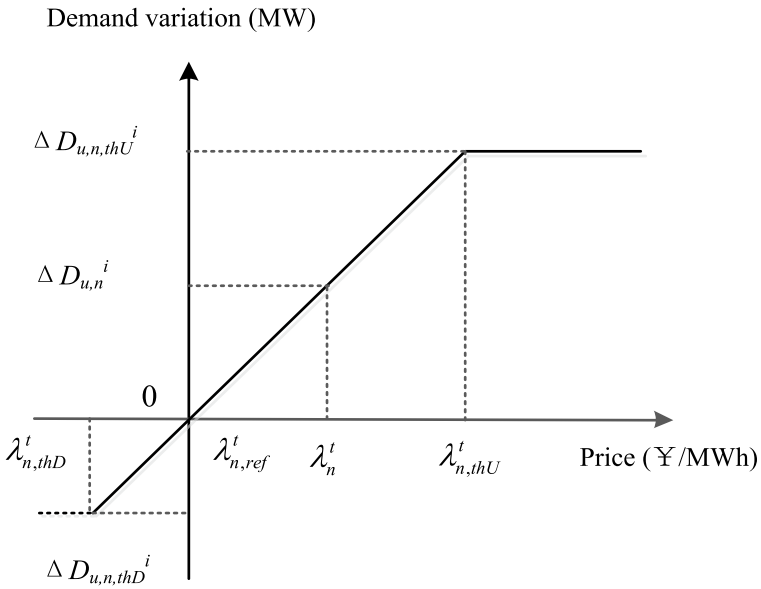


Fig. 6.1 Curve of demand variation with power prices

$$\alpha_{n,u}^{t,i} = \frac{\Delta D_{n,u}^{i}}{\Delta \lambda^{t}} \tag{6.24}$$

As we can see, when $\lambda_{n,thD}^{t} \le \lambda_{n}^{t} \le \lambda_{n,thU}^{t}$, the slope of the curve is $\alpha_{n,u^{t,i}}$. Evidently, when $i = t$, $\alpha_{n,u}^{t,i} \le 0$. Furthermore, the demand at SI t decreases with the increase in the power price at SI t, which is also known as self-elasticity of the PED and it can be calculated by (6.25). When $i \ne t$, $\alpha_{n,u}^{t,i} \ge 0$. Moreover, the demand at SI i increases with the increase in the power price at SI t, which is called cross-elasticity of the PED and it can be calculated by Eq. (6.26). The demand characteristics of the ITCA are as follows. (1) When $\lambda_{n,thD}^{t} \le \lambda_{n}^{t} \le \lambda_{n,thU}^{t}$, there is an linear relationship between demand variation and power price. (2) When $\lambda_{n}^{t_n} > \lambda_{n_1 thU}$ and $\lambda_{n}^{t_n} < \lambda_{n,thD}^{t_n}$ the demand variation reaches the limit. The former is due to the rated power of the appliance or the highest demand from the customer, which constitute an upper limit on the increase in the demand, and the latter is because the lowest demand of the customer results in an upper limit on the reduction in the demand. (3) It is obvious that cross elasticity only exists between adjacent SIs from the mathematical derivation (i.e., $a_{n,u}^{t,i} = 0$, when $i \ne t+1$ and $i \ne t-1$) and the self-elasticity is the same except for the starting sub-interval.

$$\alpha_{n,u}^{t,t} = \begin{cases} -\dfrac{D_{a,\max}}{2w(k_3)^2}, t = S_a \\ -\dfrac{\left(1+(k_3)^2\right)D_{a,\max}}{2w(k_3)^2}, t \in T_a \backslash S_a \end{cases} \tag{6.25}$$

$$\alpha_{n,u}^{t,i} = \begin{cases} \dfrac{k_1 D_{a,\max}}{2w(k_3)^2}, i \in \{t+1, t-1\} \\ 0, otherelse \end{cases} \tag{6.26}$$

It is worth noting that k_3 of commercial ITCA and residential ITCA are different, which can be obtained by Eqs. (6.26) and (6.27), respectively. In addition, k_3 depends on the ITCA and $D_{a,max}$ depends on the customer's electricity consumption habits. Thus, this study reasonably assumes that the PED of the ITCA will not change in a short time for a customer.

$$k_{3,com} = (1 - k_1) \cdot \frac{1}{W_a}, k_1 = e^{\frac{-W_a \Delta t}{Y_a}} \tag{6.27}$$

$$k_{3,res} = (1 - k_1) \cdot \eta_a R_a, k_1 = e^{\frac{-\Delta t}{R_a C_a}} \tag{6.28}$$

If the power price continues to increase after the power price is increased to $\lambda_{n,thU}^t$, the demand transfer or reduction will not increase, but will damage the interests of customers. Similarly, when the power price is reduced to $\lambda_{n,thD}^t$, further reduction of the power price will not increase the demand transfer or reduction, but will hurt the interests of the grid or independent system operator (ISO). Therefore, here, the power price change interval of node n is set as $\left[\lambda_{n_2 thD}, \lambda_{n,thU}^t\right]$.

The PED represents the influence of power price on demand, namely, ITCA's flexibility. Thus, the ITCAs' flexibility of node i can be obtained through the summation of PEDs. Based on the PED of the ITCA, this study can obtain the demand variation of customer u of node n, as expressed in:

$$\Delta D_{u,n}^t = \alpha_{u,n}^{t,t} \cdot \left(\lambda_n^t - \lambda_{n,ref}^t\right) + \sum_{i \in T \setminus t} \alpha_{u,n}^{t,i} \cdot \left(\lambda_n^i - \lambda_{n,ref}^i\right) \tag{6.29}$$

where the first term is the demand variation (demand reduction) under the influence of the self-elasticity, and the second term is the demand variation (demand transfer from other SIs) under the influence of cross-elasticity.

To motivate customers to participate in this project, this study gives them rewards based on the customer's ITCA demand variation and the power price, combined with the incentive coefficient γ. In detail, the flexibility resource cost of the ITCA can be calculated as:

$$C_{\Delta D,u,n}^t = -\gamma \cdot \sum_{t \in T} \Delta D_{u,n}^t \cdot \lambda_n^t \tag{6.30}$$

Thus, the flexibility of ITCAs of node n and its corresponding cost can be denoted by (6.31) and (6.32), respectively.

$$\Delta D_n^t = \sum_{u_n \in U_n} \Delta D_{u,n}^t \tag{6.31}$$

$$C^t_{\Delta D,n} = \sum_{u \in U_n} C^t_{\Delta D,u,n} \tag{6.32}$$

6.3.3 ITCAs' Flexibility Under TOU Power Price

Based on the calculation method of ITCAs' flexibility, this study can obtain the demand variation of node i as long as parameters of power price are set. Therefore, a model of ITCAs' flexibility under TOU power price is proposed because of the wide implementation of TOU power price for commercial and residential customers.

In this work, the influence of the ratio of peak to valley of price (RPtVP) of the TOU power price on the flexibility of ITCAs and the power system operation will be quantitatively analyzed. Thus, we quantify the ITCAs' flexibility combined with the TOU power price. The TOU power price model is presented in (6.33) to (6.35):

$$\lambda^{TOU}_n = \begin{cases} \lambda^{TOU}_{n,op}, t \in T_{n,op} \\ \lambda^{TOU}_{n,mp}, t \in T_{n,mp} \\ \lambda^{TOU}_{n,of}, t \in T_{n,of} \end{cases} \tag{6.33}$$

$$\lambda^{TOU}_{n,op} = \frac{K_n(\beta_n + 1)}{K_n + 1} \lambda^{TOU}_{n,mp} \tag{6.34}$$

$$\lambda^{TOU}_{n,of} = \frac{(\beta_n + 1)}{K_n + 1} \lambda^{TOU}_{n,mp} \tag{6.35}$$

It is worth noting that in this study, except for the RPtVP, all other parameters of TOU power price are predetermined. Combining (6.29) to (6.35), the flexibility of the ITCA and its corresponding cost under the TOU power price can be obtained as presented in (6.36) to (6.43):

$$\Delta D^t_n = \sum_{u_n \in U_n} \left[\left(\alpha^{t,t}_{u_n} + \sum_{i \in T_{n,op} \backslash t} \alpha^{t,i}_{u_n} \right) \cdot \left(\lambda^{TOU}_{n,op} - \lambda^{TOU}_{n,op,ref} \right) + \left(\lambda^{TOU}_{n,of} - \lambda^{TOU}_{n,of,ref} \right) \cdot \sum_{i \in T_{n,of}} \alpha^{t,i}_{u_n} \right], t \in T_{n,op} \tag{6.36}$$

$$C^t_{\Delta D,n} = -\lambda^{TOU}_{n,op} \cdot \sum_{u_n \in U_n} \left[\left(\alpha^{t,t}_{u_n} + \sum_{i \in T_{n,op} \backslash t} \alpha^{t,i}_{u_n} \right) \cdot \left(\lambda^{TOU}_{n,op} - \lambda^{TOU}_{n,op,ref} \right) + \left(\lambda^{TOU}_{n,of} - \lambda^{TOU}_{n,of,ref} \right) \cdot \sum_{i \in T_{n,of}} \alpha^{t,i}_{u_n} \right], t \in T_{n,op} \tag{6.37}$$

$$\Delta D^t_n = \sum_{u_n \in U_n} \left[\left(\alpha^{t,t}_{u_n} + \sum_{i \in T_{n,of} \backslash t} \alpha^{t,i}_{u_n} \right) \cdot \left(\lambda^{TOU}_{n,of} - \lambda^{TOU}_{n,of,ref} \right) + \left(\lambda^{TOU}_{n,op} - \lambda^{TOU}_{n,op,ref} \right) \cdot \sum_{i \in T_{n,op}} \alpha^{t,i}_{u_n} \right], t \in T_{n,of} \tag{6.38}$$

$$C^t_{\Delta D,n} = -\lambda^{TOU}_{n,of} \cdot \sum_{u_n \in U_n} \left[\left(\alpha^{t,t}_{u_n} + \sum_{i \in T_{n,of} \setminus t} \alpha^{t,i}_{u_n} \right) \cdot \left(\lambda^{TOU}_{n,of} - \lambda^{TOU}_{n,of,ref} \right) + \left(\lambda^{TOU}_{n,op} - \lambda^{TOU}_{n,op,ref} \right) \cdot \sum_{i \in T_{n,op}} \alpha^{t,i}_{u_n} \right], t \in T_{n,of} \tag{6.39}$$

$$\Delta D^t_n = \sum_{u_n \in U_n} \left[\left(\lambda^{TOU}_{n,op} - \lambda^{TOU}_{n,op,ref} \right) \cdot \sum_{i \in T_{n,op}} \alpha^{t,i}_{u_n} + \left(\lambda^{TOU}_{n,of} - \lambda^{TOU}_{n,of,ref} \right) \cdot \sum_{i \in T_{n,of}} \alpha^{t,i}_{u_n} \right], t \in T_{n,mp} \tag{6.40}$$

$$C^t_{\Delta D,n} = -\lambda^{TOU}_{n,mp} \cdot \sum_{u_n \in U_n} \left[\left(\lambda^{TOU}_{n,op} - \lambda^{TOU}_{n,op,ref} \right) \cdot \sum_{i \in T_{n,op}} \alpha^{t,i}_{u_n} + \left(\lambda^{TOU}_{n,of} - \lambda^{TOU}_{n,of,ref} \right) \cdot \sum_{i \in T_{n,of}} \alpha^{t,i}_{u_n} \right], t \in T_{n,mp} \tag{6.41}$$

$$\lambda^{TOU}_{n,op,ref} = \frac{K_{n,ref}\left(\beta_{n,ref} + 1\right)}{K_{n,ref} + 1} \lambda^{TOU}_{n,mp} \tag{6.42}$$

$$\lambda^{TOU}_{n,of,ref} = \frac{\left(\beta_{n,ref} + 1\right)}{K_{n,ref} + 1} \lambda^{TOU}_{n,mp} \tag{6.43}$$

Considering that a single customer has little impact on the power system operation, the load aggregator (LA) is designed and selected to integrate the demand response resources of a group of customers to take full advantage of DR capacity. That is to say, the LA acts as the agent between the ISO and customers, managing the end-users with the DR program. The LA buys electricity from the grid and sells electricity to customers, and at the same time, the LA can provide ancillary services to the ISO. In other words, LAs can control demand variation and form a certain scale of regulation resources to participate in system scheduling. Therefore, LAs are in charge of providing the information of ITCAs' flexibility of each load node for the ISO in this study.

6.3.4 Unit Scheduling Model Considering the Flexibility of ITCAs

This work considers the ITCAs' flexibility of commercial and residential customers into the unit scheduling and studies the influence of the demand-side ITCAs' flexibility resource on the power system operation. The model is described in detail from the two aspects of the objective function and constraint conditions.

6.3.4.1 Objective Function

$$\min C = C_1 + C_2 + C_3 + C_4 \tag{6.44}$$

$$C_1 = \sum_{t \in T} \sum_{g \in G} C_{g,start} \cdot x^t_g \left(1 - x^{t-1}_g\right) \tag{6.45}$$

$$C_2 = \sum_{t\in T}\sum_{g\in G}\left(a_g \cdot \left(P_g^t\right)^2 + b_g \cdot P_G^t + c_g\right) \tag{6.46}$$

$$C_3 = \sum_{n\in N}\sum_{t\in T} C_{\Delta D,n}^t \tag{6.47}$$

$$C_4 = W_{punish} \cdot \sum_{w\in W}\sum_{t\in T}\left(P_{w,o}^t - P_w^t\right) \tag{6.48}$$

Equation (6.44) represents the total operating cost of the system. C_1 and C_2 represent the start-up and variable costs of all thermal power units, respectively. C_3 is the cost of the flexibility resource. Considering that the integration of large-scale wind power to the grid has brought about accommodation problem. Wind power is curtailed due to the insufficient flexible regulation capability or the network constraint, so it is appropriate to include the punishment to wind curtailment in the total cost in order to schedule the flexible resources in an economic way. In addition, it also helps to analyze the influence of flexible resources on wind curtailment, so C_4 is designed as the penalty of wind curtailment.

In order to facilitate problem solving, this study linearizes C_1 as follows:

$$\begin{aligned} &C_1 = \sum_{t\in T}\sum_{g\in G} C_1\left(P_g^t\right) \\ &C_1\left(P_g^t\right) \geq 0 \\ &C_1\left(P_g^t\right) \geq C_{g,start} \cdot \left(x_g^t - x_g^{t-1}\right), t \in T\backslash 1 \end{aligned} \tag{6.49}$$

6.3.4.2 Constraint Condition

(1) Power system power balance and reserve constraints

$$\sum_{g\in G} P_g^t + \sum_{w\in W} P_w^t - \sum_{n\in\Omega} \Delta D_n^t = \sum_{n\in\Omega} P_{n,L}^t, \forall t \tag{6.50}$$

$$\sum_{g\in G} x_g^t \cdot P_{g,\max} + \sum_{w\in W} P_w^t - \sum_{n\in\Omega} \Delta D_n^t \geq (1 + r_{load}) \cdot \sum_{n\in\Omega} P_{n,L}^t, \forall t \tag{6.51}$$

$$\sum_{g\in G} x_g^t \cdot P_{g,\min} + \sum_{n\in\Omega} P_{n,wind}^t - \sum_{n\in\Omega} \Delta D_n^t \leq (1 - r_{load}) \cdot \sum_{n\in\Omega} P_{n,L}^t, \forall t \tag{6.52}$$

Equation (6.50) is the power system power balance constraint. Equations (6.51) and (6.52) represent the upward spinning reserve constraint and the downward spinning reserve constraint of the power system, respectively.

(2) Operating constraints of thermal power units

$$\begin{cases} x_{g,t} - x_{g,t-1} - x_{g,w} \le 0, \forall w \in \{t, t+1, ..., T_{g,st} + t - 1\} \\ x_{g,t-1} - x_{g,t} + x_{g,w} \le 1, \forall w \in \{t, t+1, ..., T_{g,sd} + t - 1\} \end{cases}, \forall t \tag{6.53}$$

$$P_{g,\min} x_g^t \le P_g^t \le P_{g,\max} x_g^t, \forall t \tag{6.54}$$

$$\begin{cases} P_g^t - P_g^{t-1} \le r_{g,up} x_g^{t-1} + P_{g,\max}\left(1 - x_g^{t-1}\right) \\ P_g^{t-1} - P_g^t \le r_{g,dn} x_g^t + P_{g,\max}\left(1 - x_g^t\right) \end{cases}, \forall t \tag{6.55}$$

Equation (6.53) is the minimum up and down time constraint of the thermal power unit g. Equation (6.54) is the technical output constraint of the thermal power unit g. Equation (6.55) is the thermal power unit ramp constraint.

(3) Operating constraints of wind power

$$0 \le P_w^t \le P_{w,o}^t, \forall t \tag{6.56}$$

(4) Transmission network constraints

$$\sum_{w \in \Omega_n^W} P_w^t - \sum_{l \in \Omega_n^L(+)} F_l^t + \sum_{l \in \Omega_n^L(-)} F_l^t - \Delta D_n^t = P_{n,L}^t, \forall n, \forall t \tag{6.57}$$

$$F_l^t = B_l\left(\theta_{l(+)}^t - \theta_{l(-)}^t\right), \forall l, \forall t \tag{6.58}$$

$$-F_{l,\max} \le F_l^t \le F_{l,\max}, \forall l, \forall t \tag{6.59}$$

$$-\pi \le \theta_n^t \le \pi, \forall n, \forall t \tag{6.60}$$

Equation (6.57) is the node power balance constraint. Referring to [8, 9], this work adopts the DC power flow to calculate the transmission line power, as in (6.58) to (6.60).

Based on the proposed unit scheduling model, this study aims to analyze the influence of TOU power price on the flexible operation of power system considering flexibility of ITCAs and provide a reference for the formulation of RPtVP of regional TOU power price. It is noting that demand variation (i.e., ΔD_n^t) only depends on the RPtVP under other parameters (i.e., the mid-peak hours price ${\lambda_{n,mp}}^{TOU}$ and ratio of the distance of on-peak to mid-peak distance to that of mid-peak to off-peak for price β_n) of TOU power price unchanged. That is to say, the demand variation ${\Delta D^u}_{n,t}$ is a function of TOU power price, while the RPtVP of TOU power price is a decision variable. However, in order to analyze the RPtVP of TOU power price influence on the power system operation, the RPtVP is more like a predetermined value. In other words, it is precise because of the research purpose of this study that the model is not like a simultaneous co-optimization. In addition, it is possible to achieve a

simultaneous co-optimization by really setting the RPtVP of TOU power price as a decision variable.

6.4 Integrated Energy System Demand Response Model

Energy is the basis for human survival, universal urbanization, and the lifeblood of modern economic prosperity. The past few decades witnessed a remarkable growth in multi-dimensional energy demands and increasingly worldwide anxiety for the energy crisis and global warming. Most recently, the term smart energy system (SES) [10] has been put forward to meet the aforementioned challenges considering environmental impact, techno-economic analysis, energy efficiency, and other vital issues for achieving eventual sustainability.

SES was first introduced by Lund et al. aiming at providing a novel understanding of the future energy system. Previous to this, several concepts such as smart grid and microgrid and nearly/net-zero energy buildings are validated effective in tackling efficient and environmentally friendly energy use. In contrast, SES is capable to facilitate reliability and resilience in the overall multi-energy complementary provision in each aspect of energy utilization. Besides, SES is expected to serve as an economic and efficient energy management pattern from the viewpoint of cascade utilization of broader energy domains instead of only or primarily focusing on the electricity sector.

Among the existing definitions of SES, the most frequently mentioned are the integrated energy system (IES) or integrated community energy system (ICES) [11]. Interconnected cross-region ICESs, smart city, national energy systems and Energy Internet, etc. are gaining increasing popularities in reshaping a more sustainable society. Smart energy hub (SEH) plays a core role in describing energy allocation and consumption of ICES. In general, SEH contains seven fundamental components: (1) coupling energy network integration, (2) renewable energy sources (RESs) accommodation, (3) distributed energy resources; (4) smart multi-energy demands; (5) information and communication technologies; (6) demand response (DR) availability and (7) Energy management system (EMS). EMS enables absorption, coordination and optimum allocation of energy from electricity, natural gas, distributed heating and electrified transportation networks, to satisfy diversified loads via energy converting devices (CDs) and energy storage systems (ESs) inside the SEH. Moreover, SEH provides versatile, scalable and adaptive options for modeling and analyzing economic dispatch and expansion planning of ICES in the form of simple combined heat and power (CHP) units or dozens of structures. In this regard, the rational configuration planning of CDs and ESs in SEH, which is the focus of this paper, is particularly essential and urgent for improving the cost-effectiveness, multi-energy synergy and full-scale energy utilization efficiency of ICES.

DR programs provide pivotal virtual resources to mitigate system deficiencies as well as benefit-friendly supply–demand interactions. What's more, the coupling characteristics of multi-energy flexible loads make SEH an excellent platform for

implementing supply–demand interactions, which enable ICES to differ from the traditional energy system with "smart" features. With the gradual maturity of energy markets, traditional electrical DR is extended to integrated DR (IDR), namely interplays of smart energy demands in synthetic energy carriers in response to price or incentive mechanisms.

6.4.1 Typical Topology

The topological diagram of the typical SEH-based ICES in this chapter is illustrated in Fig. 6.2. Generally, tasks of the EMS center in ICES consist of energy purchase, reallocation, consumption, data metering, and information sharing. For the sake of computational simplicity, three basic assumptions are worth mentioning hereby: (1) losses in power transformer, NG compressor, and power electronic devices like DC/AC inverter at the output port of the photovoltaic (PV) panels are omitted; (2) except purchasing energy from supply networks, ICES has the potential to sell surplus electricity to the power grid, while sales of NG and heating energy are unaccounted due to transmission wastage and pricing limitations in practical energy markets; (3) EMS center has access to each device and consumer in gathering information and giving orders.

As shown in Fig. 6.2, SEH plays a central role in the multi-energy transaction, replacement and IDR implementation among upstream distributed energy networks, ICES, and end-users. The electricity generated by the power grid, PV and CHP are collected in the power bus and then sent to heat pump (HP), electric chiller (EC) as well as meet the electricity demand. Thermal energy from the network, solar heater (SH), CHP, gas boiler (GB) and HP is gathered in the heating bus in

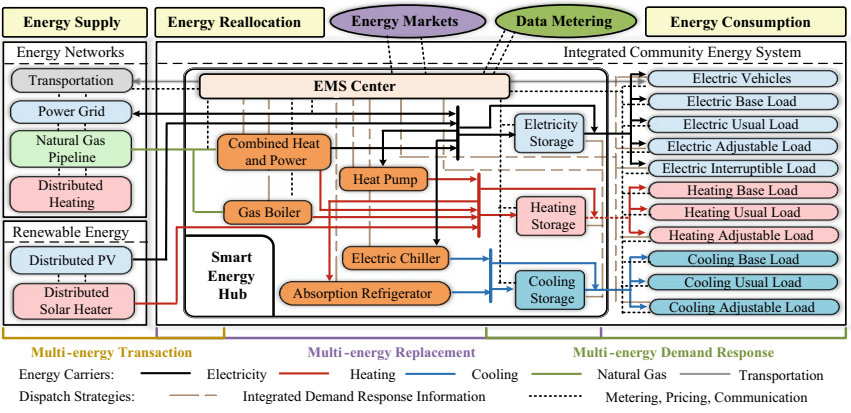


Fig. 6.2 Topological diagram of the typical SEH-based ICES

response to heating and absorption refrigerator (AR) needs. Cooling loads are satisfied by EC and AR. There are high uncertainties in PV and SH outputs because of the volatile and stochastic features such as solar radiation intensity and ambient temperature, which also lead to seasonal fluctuation of multiple loads. Inside the SEH, energy coupling is implemented through CDs including CHP, GB, HP, EC, and AR. Meanwhile electric, heating and cooling ESs (EESs, HESs, and CESs) increase the energy reallocation flexibility. Electric vehicles (EVs) serve as mobile hubs connecting power grid and transport network due to the vehicle to grid technology; ALs and ILs provide friendly interactions depending on the energy prices, incentive payments, and regulation orders from EMS center.

6.4.2 Integrated Demand Response Model

Benefited from the virtual energy storage equipment such as ice storage air conditioner, the elasticity of energy consumptions like peak load shifting are enhanced. In addition, the improvement of comprehensive EMS level in SEH and energy market mechanism further exploits the DR potential in even unchanged loads. Therefore, IDR modeling is divided into Multi-energy Demand Response (MEDR) and Multi-energy Replacement (MER) in this section to characterize disparate DR capabilities of smart loads and reciprocal conversions in multi-energy carriers, respectively.

6.4.2.1 Multi-energy Demand Response (MEDR) Model

(1) Base load and usual load

Base load and usual load remain non-sensitive to energy price changes, nor do they participate in MEDR projects. The typical base loads are those electric demands, hot and cooling water in vital industries such as hospitals, stations, and data centers. Energy supply for base load is required to be guaranteed in any operational scenario of the SEH. Usual loads are representatives of common daily energy consumptions which could withstand short-term partial cut off to release supply pressure.

(2) Adjustable load

AL is responsive to multi-energy price changes and varies in correlation with consumer habits changes, e.g. adjusting the air conditioner setting temperature, switching the heating pipe valve, transferring periods of washing machine usage, etc. This sort of load thereby has the flexibility to shift energy consumption from high price periods to low price ones to reduce energy costs. Due to the difficulty in accurately investigating the elastic factors, the responsive AL is expressed as the sum of deterministic and fuzzy components as shown in (6.61).

$$\begin{cases} \tilde{P}_{t,s}^{al} = P_{t,s}^{al0} + \Delta P_{t,s}^{al} + \Delta \tilde{P}_{t,s}^{al} \\ \tilde{H}_{t,s}^{al} = H_{t,s}^{al0} + \Delta H_{t,s}^{al} + \Delta \tilde{H}_{t,s}^{al} \\ \tilde{C}_{t,s}^{al} = C_{t,s}^{al0} + \Delta C_{t,s}^{al} + \Delta \tilde{C}_{t,s}^{al} \end{cases} \tag{6.61}$$

where t, s are the indexes for time interval (for example, 1 h) and scenarios, and have the same meanings in the rest of this paper; $\tilde{P}_{t,s}^{al}$, $\tilde{H}_{t,s}^{al}$, $\tilde{C}_{t,s}^{al}$ and $P_{t,s}^{al0}$, $H_{t,s}^{al0}$, $C_{t,s}^{al0}$ denote the value of electric, heating, cooling AL after and before MEDR, respectively. $\Delta P_{t,s}^{al}$, $\Delta H_{t,s}^{al}$, $\Delta C_{t,s}^{al}$, and $\Delta \tilde{P}_{t,s}^{al}$, $\Delta \tilde{H}_{t,s}^{al}$, $\Delta \tilde{C}_{t,s}^{al}$ represent the expected and fluctuant amount of change in AL. In our work, fuzzy quantities are described with wavy line superscript.

Further, AL variations are expressed in (6.62) with respect to sensitivity to price changes. Taking electric AL as an example, heating and cooling ALs have similar forms to (6.62).

$$\begin{cases} \Delta P_{i,s}^{al} = \sum_{j=1}^{T} \varepsilon_{i,j}^{ef} \Delta \lambda_{j,s}^{e} P_{i,s}^{al0} \Big/ \lambda_{j,s}^{e} \\ \Delta \tilde{P}_{i,s}^{al} = \sum_{j=1}^{T} \tilde{\varepsilon}_{i,j}^{e} \Delta \lambda_{j,s}^{e} P_{i,s}^{al0} \Big/ \lambda_{j,s}^{e} \end{cases} \tag{6.62}$$

where $\varepsilon_{i,j}^{ef}$ and $\tilde{\varepsilon}_{i,j}^{e}$ refer to the forecasted and deviation value of the elasticity coefficients of electric AL, valuing negative with same time slots i and j, while positive if $i \neq j$ [40]. $\Delta\lambda_{j,s}^{e}$ and $\lambda_{j,s}^{e}$ are varied and initial electricity prices. T means the set of time intervals in a day, namely 24 h.

It is clear that the nondeterminacy of ALs derives from the fuzzy uncertainty of their endogenous elastic coefficients. Therefore, the fuzzy set theory is an appropriate tool to better understand the practical pattern of elasticity. Here trapezoidal fuzzy variables are applied to express $\tilde{\varepsilon}_{i,j}^{e}$, and the corresponding tetrad is:

$$\tilde{\varepsilon}_{i,j}^{e} = \left(\varepsilon_{i,j}^{e1}, \varepsilon_{i,j}^{e2}, \varepsilon_{i,j}^{e3} \varepsilon_{i,j}^{e4}\right) = \varepsilon_{i,j}^{ef}\left(r_1^e, r_2^e, r_3^e, r_4^e\right) \tag{6.63}$$

where $\varepsilon_{i,j}^{e1}$ to $\varepsilon_{i,j}^{e4}$ and r_1^e to r_4^e represent the parameters of membership function and scaling factors, respectively.

The implementation of MEDR should not overly affect the normal production and life of price-based consumers. On this premise, this paper adopts fuzzy chance constraint (FCC) to avoid possible dissatisfaction of ALs. Therefore, restrictions are set in (6.64) so that the total energy consumption of ALs throughout the day is approximately unchanged and within bounds under certain confidence.

$$\begin{cases} \mathrm{Cr}\left\{\left|\sum_{t}^{T}\left(\Delta P_{t,s}^{al} + \Delta \tilde{P}_{t,s}^{al}\right)\right| \leq \Delta P_{\max}^{al}\right\} \geq \alpha_e^{al} \\ \mathrm{Cr}\left\{\left|\sum_{t}^{T}\left(\Delta H_{t,s}^{al} + \Delta \tilde{H}_{t,s}^{al}\right)\right| \leq \Delta H_{\max}^{al}\right\} \geq \alpha_h^{al} \\ \mathrm{Cr}\left\{\left|\sum_{t}^{T}\left(\Delta C_{t,s}^{al} + \Delta \tilde{C}_{t,s}^{al}\right)\right| \leq \Delta C_{\max}^{al}\right\} \geq \alpha_c^{al} \end{cases} \tag{6.64}$$

where $\mathrm{Cr}\{\cdot\}$ refers to the credibility of $\{\cdot\}$; α_e^{al}, α_h^{al}, and α_c^{al} are the confidence levels; $\Delta P_{\max}^{al}$, $\Delta H_{\max}^{al}$, and $\Delta C_{\max}^{al}$ separately indicate the maximum allowed total transfer amount limits of AL during and after MEDR event.

Meanwhile, the energy prices announced to users should be limited as shown in (5).

$$\begin{cases} \left(1-\phi_{\max}^{e}\right)\lambda_{t,s}^{eb} \le \lambda_{t,s}^{eb} + \Delta\lambda_{t,s}^{e} \le \left(1+\phi_{\max}^{e}\right)\lambda_{t,s}^{eb} \\ \left(1-\phi_{\max}^{he}\right)\left(\lambda_{t,s}^{eb}+\Delta\lambda_{t,s}^{e}\right) \le \lambda_{t,s}^{eb} + \Delta\lambda_{t,s}^{h} \le \left(1+\phi_{\max}^{he}\right)\left(\lambda_{t,s}^{eb}+\Delta\lambda_{t,s}^{e}\right) \\ \left(1-\phi_{\max}^{ce}\right)\left(\lambda_{t,s}^{eb}+\Delta\lambda_{t,s}^{e}\right) \le \lambda_{t,s}^{eb} + \Delta\lambda_{t,s}^{c} \le \left(1+\phi_{\max}^{ce}\right)\left(\lambda_{t,s}^{eb}+\Delta\lambda_{t,s}^{e}\right) \\ \sum_{t}^{T}\left(\lambda_{t,s}^{eb}+\Delta\lambda_{t,s}^{e}\right) \Big/ T \le \sum_{t}^{T}\lambda_{t,s}^{eb} \Big/ T \end{cases} \tag{6.65}$$

The price received by AL is the sum of the electricity tariff from power grid ($\lambda_{t,s}^{eb}$) and varied electric, heating and cooling parts set by EMS center ($\Delta\lambda_{t,s}^{e}$, $\Delta\lambda_{t,s}^{h}$, and $\Delta\lambda_{t,s}^{c}$). The first three lines of (6.65) are set to ensure AL response neither to fall into dead zone nor to enter the saturation zone, denoted by $\phi_{\max}^{e}$, $\phi_{\max}^{he}$, and $\phi_{\max}^{ce}$, which describe the largest floating ratios of ALs in consideration of correlations of different energy categories. The fourth item shows that the formulated average electricity price cannot be higher than the purchase price in order to protect the benefit of consumers.

(3) Interruptible load

In the electricity market environment, IL participates in MEDR projects according to the pre-signed contract and obtains certain economic compensation to improve the reliability of system operation. IL is limited by constraints of capacity, maximum interruption time period and the number of interruptions, as shown in (6.66).

$$\begin{cases} 0 \le P_{t,s,k}^{il} \le u_{t,s,k}^{il} P_{k,\max}^{il} \\ \sum_{t=n}^{n+T_{k,\max}^{il}} u_{t,s,k}^{il} \le T_{k,\max}^{il}, \quad n = 1, 2, \ldots, T - T_{k,\max}^{il} \\ \tau_{t,s,k}^{on} - \tau_{t,s,k}^{off} = u_{t,s,k}^{il} - u_{t-1,s,k}^{il} \\ 0 \le \tau_{t,s,k}^{on} + \tau_{t,s,k}^{off} \le 1, \quad \sum_{t=1}^{N_T-1} \tau_{t,s,k}^{on} \le M_k \end{cases} \tag{6.66}$$

where $P_{t,s,k}^{il}$, $P_{k,\max}^{il}$, $T_{k,\max}^{il}$ and M_k denote the interrupted amount, the upper limit of interruption, maximal single interrupt duration and the largest number of interruptible times in one scheduling cycle of k-th IL. $u_{t,s,k}^{il}$ is the binary variable reflecting interrupt status of k-th IL, i.e., valuing 1 of $u_{t,s,k}^{il}$ indicates the occurrence of interruption. N_T refers to the total number of T. Additional binary variables $\tau_{t,s,k}^{on}$ and $\tau_{t,s,k}^{off}$ are introduced in the last three parts of (6.66) to linearly describe whether IL at the end of time slot t is interrupted or interrupt is terminated.

(4) Electric vehicle

EV is crucial distributed energy storage element, especially in community-based SEH. For the l-th EV, its load $P_{t,s,l}^{ev}$ ought to obey constraints in (6.67).

$$\begin{cases} 0 \le P_{t,s,l}^{evc} \le u_{t,s,l}^{evc} P_{l,\max}^{ev},\ 0 \le P_{t,s,l}^{evd} \le u_{t,s,l}^{evd} P_{l,\max}^{ev} \\ P_{t,s,l}^{ev} = P_{t,s,l}^{evc} - P_{t,s,l}^{evd},\ 0 \le u_{t,s,l}^{evc} + u_{t,s,l}^{evd} \le 1 \end{cases} \tag{6.67}$$

where $P_{t,s,l}^{evc}/P_{t,s,l}^{evd}$ and $u_{t,s,l}^{evc}/u_{t,s,l}^{evd}$ express the power and the mutually-exclusive 0–1 status variables of charging/discharging. $P_{l,\max}^{ev}$ is the power limit of EV.

The state of charge (SOC) of EV, $S_{t,s,l}^{ev}$ is calculated with charging/discharging efficiencies $\eta_l^{evc}/\eta_l^{evd}$ and rated battery capacity e_l^{batt} and is limited to lower/upper bounds $S_{l,\min}^{ev}/S_{l,\max}^{ev}$ (setting as 0.2/0.9) in (6.68).

$$\begin{cases} S_{t+1,s,l}^{ev} = S_{t,s,l}^{ev} + \left(P_{t,s,l}^{evc}\eta_l^{evc} - P_{t,s,l}^{evd}/\eta_l^{evd}\right)/e_l^{batt} \\ S_{l,\min}^{ev} \le S_{t,s,l}^{ev} \le S_{l,\max}^{ev} \end{cases} \tag{6.68}$$

Additional constraints are set in (6.69) to stipulate the feasibility of EV scheduling, and to make sure the residual battery level is enough to meet the travel demand.

$$\begin{cases} P_{t,s,l}^{ev} = 0, \quad t \in \left[T_{l,l}^{ev}, T_{l,r}^{ev}\right] \\ S_{dep,s,l}^{ev} \ge S_{l,\min}^{ev} + d_l^{dri}/d_{l,\max}^{dri} \end{cases} \tag{6.69}$$

where the first line of (6.69) means that EV load is considered zero between time nodes when EV leaves ($T_{l,l}^{ev}$) and reaches ($T_{l,r}^{ev}$) the charging area in SEH. d_l^{dri} and $d_{l,\max}^{dri}$ represent the planned trip distance and allowed maximum driving kilometers of l-th EV.

6.4.2.2 Multi-energy Replacement (MER) Model

MER process aims at offering alternative energy paths to link integrated resources and diversified demands oriented by economic objectives set by EMS. In this way, base load and usual load can be satisfied through optimized energy sources without altering self-consumption to gain equivalent DR ability. MER can be formulated as FCCs of energy balance:

$$\begin{aligned} &\mathrm{Cr}\left\{\begin{array}{l} P_{t,s}^{pv} + P_{t,s}^{buy} - P_{t,s}^{sell} + \sum_r P_{t,s,r}^{chp} + \sum_q \left(P_{t,s,q}^{dis} - P_{t,s,q}^{ch}\right) = \\ P_{t,s,l}^{bl} + P_{t,s}^{ul} + \tilde{P}_{t,s}^{al} + \sum_l P_{t,s,l}^{ev} - \sum_k P_{t,s,k}^{il} + \sum_m P_{t,s,m}^{hp} + \sum_p P_{t,s,p}^{ec} - P_{t,s}^{lc} \end{array}\right\} \ge \alpha_e^{al} \\ &\mathrm{Cr}\left\{\begin{array}{l} H_{t,s}^{sh} + \sum_r H_{t,s,r}^{chp} + \sum_h H_{t,s,h}^{gb} + \sum_v \left(H_{t,s,v}^{dis} - H_{t,s,v}^{ch}\right) + \sum_m H_{t,s,m}^{hp} = \\ H_{t,s}^{bl} + H_{t,s}^{ul} + \tilde{H}_{t,s}^{al} - \sum_o H_{t,s,o}^{ar} - H_{t,s}^{lc} \end{array}\right\} \ge \alpha_h^{al} \\ &\mathrm{Cr}\left\{\sum_p C_{t,s,p}^{ec} + \sum_o C_{t,s,o}^{ar} + \sum_w \left(C_{t,s,w}^{dis} - C_{t,s,w}^{ch}\right) = C_{t,s}^{bl} + C_{t,s}^{ul} + \tilde{C}_{t,s}^{al} - C_{t,s}^{lc}\right\} \ge \alpha_c^{al} \end{aligned} \tag{6.70}$$

where $P_{t,s}^{pv}$ and $H_{t,s}^{sh}$ are electric and heating output of PV and SH; $P_{t,s}^{buy}$ and $P_{t,s}^{sell}$ means the electric power purchased from and sold to power grid; $P_{t,s,r}^{chp}/H_{t,s,r}^{chp}$ and $H_{t,s,h}^{gb}$ denote the electric/heating output of the r-th CHP and heating output of h-th

GB; $P^{dis}_{t,s,q}/P^{ch}_{t,s,q}$, $H^{dis}_{t,s,v}/H^{ch}_{t,s,v}$ and $C^{dis}_{t,s,w}/C^{ch}_{t,s,w}$ are the discharging/charging energy rate of q-th EES, v-th HES and w-th CES; $P^{bl}_{t,s}/P^{ul}_{t,s}$, $H^{bl}_{t,s}/H^{ul}_{t,s}$ and $C^{bl}_{t,s}/C^{ul}_{t,s}$ represent base loads/usual loads of three energy forms; $P^{hp}_{t,s,m}/H^{hp}_{t,s,m}$, $P^{ec}_{t,s,p}/C^{ec}_{t,s,p}$ and $H^{ar}_{t,s,o}/C^{ar}_{t,s,o}$ refer to input/output energy rate of m-th HP, p-th EC, and o-th AR; $P^{lc}_{t,s}$, $H^{lc}_{t,s}$ and $C^{lc}_{t,s}$ express the involuntary curtailment of usual loads; Consequently, (6.70) means that the supply–demand balances of electricity, heating, and cooling are guaranteed with probabilities of no less than α^{al}_e, α^{al}_h and α^{al}_c.

6.4.3 Two-Stage Stochastic Chance-Constrained Programming Model

6.4.3.1 The Worst-Case Conditional Value-at-Risk (WCVaR)

CVaR, as a coherent risk measurement, are widely attractive in risk-based linear programming of energy management of ICES. Generally, with confidence level φ, $CVaR_\varphi(\boldsymbol{x})$ can be defined as the expected value of loss that exceeds the value-at-risk ($VaR_\varphi(\boldsymbol{x})$):

$$CVaR_\varphi(\boldsymbol{x}) \triangleq \min_{VaR_\varphi(\boldsymbol{x})} \left(VaR_\varphi(\boldsymbol{x}) + \frac{1}{1-\varphi} \int_{\boldsymbol{y}} \left[f(\boldsymbol{x}, \boldsymbol{y}) - VaR_\varphi(\boldsymbol{x}) \right]^+ p(\boldsymbol{y}) d\boldsymbol{y} \right) \tag{6.71}$$

where $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{y} \in \mathbb{R}^m$ are the n-dimensional decision and m-dimensional random vectors, respectively; $p(\boldsymbol{y})$ is the distribution of $\boldsymbol{y}$; $f(\boldsymbol{x}, \boldsymbol{y})$ is the loss function; $[\boldsymbol{x}]^+$ refers to the non-negative value of $\boldsymbol{x}$.

The values of CVaR and VaR often require calculation and estimation based on historical data associated with certain distributions. However, the uncertainties are usually intractable to capture specific functions in practical scenarios. Deviations of simulation should be accounted as well faced with unanticipated severe disasters. Therefore, worst-case CVaR (WCVaR) is put forward as an effective philosophy to increase the resilience of ICES in portfolio selection. $WCVaR_\varphi(\boldsymbol{x})$ is defined as the upper bound of $CVaR_\varphi(\boldsymbol{x})$:

$$WCVaR_\varphi(\boldsymbol{x}) \triangleq \sup_{p(\boldsymbol{y})} CVaR_\varphi(\boldsymbol{x}) \tag{6.72}$$

For discrete random variables, WCVaR minimization problem can be formulated in (6.73):

$$\min WCVaR_\varphi(\boldsymbol{x}) = \min_{(\boldsymbol{x},a,\theta)\in\mathbb{R}^n\times\mathbb{R}\times\mathbb{R}} \max_{\boldsymbol{\pi}^\sigma\in\mathbb{R}^{S^\sigma}} \left\{ \theta : a + \frac{1}{1-\varphi} \sum\nolimits_{\tau=1}^{S^\sigma} \pi^\sigma_\tau \left[f(\boldsymbol{x}, \boldsymbol{y}^\sigma_\tau) - a \right]^+ \le \theta, \quad \sigma = 1, 2, \cdots, L \right\} \tag{6.73}$$

where θ and a are values of $WCVaR_{\varphi}(\boldsymbol{x})$ and $CVaR_{\varphi}(\boldsymbol{x})$; π_{τ}^{σ} denotes the probability of τ-th sample $\boldsymbol{y}_{\tau}^{\sigma}$ with respect to the σ-th likelihood distribution; S^{σ} and L are the numbers of corresponding samples and likelihood distributions; $\boldsymbol{\pi}^{\sigma}$ is an array of π_{τ}^{σ}: $\boldsymbol{\pi}^{\sigma} = \left(\pi_1^{\sigma}, \pi_2^{\sigma}, \cdots \pi_{S^{\sigma}}^{\sigma}\right)^{\mathrm{T}} \in \mathbb{R}^{S^{\sigma}}$.

By introducing auxiliary vectors $\boldsymbol{\mu}^{\sigma} = \left(\mu_1^{\sigma}, \mu_2^{\sigma}, \cdots, \mu_{S^{\sigma}}^{\sigma}\right)^{\mathrm{T}} \in \mathbb{R}^{S^{\sigma}}$ and $\boldsymbol{\mu} = \left(\boldsymbol{\mu}^1; \boldsymbol{\mu}^2; \cdots; \boldsymbol{\mu}^L\right)^{\mathrm{T}} \in \mathbb{R}^m$ where $m = \sum_{\sigma=1}^{L} S^{\sigma}$, we can rewrite (6.73) as the following tractable model.

$$\begin{cases} \min\limits_{(\boldsymbol{x},a,\theta,\boldsymbol{\mu})\in\mathbb{R}^n\times\mathbb{R}\times\mathbb{R}\times\mathbb{R}^m} \theta \\ \text{s.t.} \begin{cases} a + \frac{1}{1-\varphi}(\boldsymbol{\pi}^{\sigma})^{\mathrm{T}}\boldsymbol{\mu}^{\sigma} \le \theta, \ u_{[\tau]}^{\sigma} \ge f\left(\boldsymbol{x}, \boldsymbol{y}_{\tau}^{\sigma}\right) - a \\ \boldsymbol{\mu}^{\sigma} \ge 0, \boldsymbol{\pi}^{\sigma} \ge 0, \ \forall\sigma, \sigma = 1, \ldots, L, \ \tau = 1, \ldots, S^{\sigma} \end{cases} \end{cases} \tag{6.74}$$

6.4.3.2 Risk-Neutral Model

The risk-neutral model (TSCCP model) for ICES configuration is designed with minimum cost based on multiple scenarios of solar and load forecasting for the planning year. In the first stage, the types, numbers, and capacities of candidate CDs and ESs are optimally determined. The second stage focuses on exploring economic operation, IDR pricing, and influence as well as supply reliability by scenario-related daily simulation aiming to give risk-neutral feedback and correct the optimal results in stage one. The model is formulated as follows.

(1) Objective function

The objective function of TSCCP model is defined in (6.75)–(6.76) to minimize total cost C_T composed of annualized investment cost C_I, expected annual operating expense C_O and penalty for insufficient supply (C_P) in which terms of energy purchase (C_{ep}), device maintenance in MER (C_{mer}) and compensation in MEDR (C_{medr}) are included simultaneously.

$$\min C_T = C_I + \sum\nolimits_{s}^{S} \sum\nolimits_{t}^{T} n\pi_s (C_O + C_P) \tag{6.75}$$

$$\begin{cases} C_I = \sum\nolimits_{g}^{G} Cap_g c_g^I x_g r_g \left(1 + r_g\right)^{Y_g} \Big/ \left(\left(1 + r_g\right)^{Y_g} - 1\right) \\ C_O = C_{ep} + C_{mer} + C_{medr} \\ C_{ep} = \lambda_{t,s}^{eb} P_{t,s}^{buy} - \lambda_{t,s}^{es} P_{t,s}^{sell} + \lambda_t^{gas} \sum\nolimits_{g} \left(V_{t,s,r}^{gas_chp} + V_{t,s,h}^{gas_gb}\right) \\ C_{mer} = \sum\nolimits_{g}^{G} c_g^M \left(Out_{t,s,g}^{CD} + Ch_{t,s,g}^{ES} + Dis_{t,s,g}^{ES}\right) \\ C_{medr} = \sum\nolimits_{k} c_k^{IL} P_{t,s,k}^{il} + \sum\nolimits_{l} c_l^{EV} P_{t,s,l}^{evd} \\ C_P = c_e^{voll} P_{t,s}^{lc} + c_h^{voll} H_{t,s}^{lc} + c_c^{voll} C_{t,s}^{lc} \end{cases} \tag{6.76}$$

where S is the set of scenarios, n and π_s denote the number of days in a year and probability of scenario s. Subscript g and G refer to the index and set of the candidate CD or ES, and for equipment g, Cap_g is the rated capacity, x_g means the binary variable that equals to 1 if installed, r_g and Y_g are the interest rate and payback period of investment to calculate C_I; c_g^I and c_g^M refer to the unit costs of installation and maintenance; $V_{t,s}^{gas_chp}$ and $V_{t,s}^{gas_gb}$ represent the total NG volume needed by CHPs and GBs; $Out_{t,s,g}^{CD}$ and $Ch_{t,s,g}^{ES}/Dis_{t,s,g}^{ES}$ are generic forms of the output energy rate of CD and charging/discharging rate of ES, respectively. λ_t^{es} and λ_t^{gas} reflect the prices of selling electricity and purchasing NG with energy networks which are considered consistent in each scenario in this paper. c_k^{IL} and c_l^{EV} are compensation cost for k-th IL and battery loss cost which is related to battery discharge depth, cycle number and unit capacity for l-th EV. c_e^{voll}, c_h^{voll} and c_c^{voll} denote the unintended load curtailment cost (value of loss load) of electricity, heating and cooling energy.

(2) Constraints

For $\forall t \in N_T$ and $\forall s \in N_S$, the aforementioned constraints (6.61) to (6.70) plus following technical and economic constraints (6.77) to (6.83) are required to be satisfied.

$$u_g \le x_g \tag{6.77}$$

$$\begin{cases} P_{t,s,r}^{chp} = V_{t,s,r}^{gas_chp} \rho^{gas} \eta_r^{chp_e} \\ H_{t,s,r}^{chp} = V_{t,s,r}^{gas_chp} \rho^{gas} \eta_r^{chp_h} \\ Out_{r,\min}^{chp} u_{t,s,r}^{chp} \le P_{t,s,r}^{chp} \le Out_{r,\max}^{chp} u_{t,s,r}^{chp} \\ \left| P_{t,s,r}^{chp} - P_{t-1,s,r}^{chp} \right| \le dP_{r,\max}^{chp} \\ on_{t,s,r} - off_{t,s,r} = u_{t,s,r}^{chp} - u_{t-1,s,r}^{chp} \\ on_{t,s,r} - off_{t,s,r} \le 1 \\ on_{t,s,r} + \sum_{\tau=t+1}^{\min\left(N_T, t+T_{r,\min}^{off}-1\right)} off_{\tau,s,r} \le 1 \\ off_{t,s,r} + \sum_{\tau=t+1}^{\min\left(N_T, t+T_{r,\min}^{on}-1\right)} on_{\tau,s,r} \le 1 \end{cases} \tag{6.78}$$

$$\begin{cases} H_{t,s,h}^{gb} = V_{t,s,h}^{gas_gb} \rho^{gas} \eta_h^{gb} \\ Out_{h,\min}^{gb} u_{t,s,h}^{gb} \le H_{t,s,h}^{gb} \le Out_{h,\max}^{gb} u_{t,s,h}^{gb} \end{cases} \tag{6.79}$$

$$\begin{cases} H_{t,s,m}^{hp} = COP_m^{hp} P_{t,s,m}^{hp} \\ Out_{m,\min}^{hp} u_{t,s,m}^{hp} \le H_{t,s,m}^{hp} \le Out_{m,\max}^{hp} u_{t,s,m}^{hp} \\ C_{t,s,p}^{ec} = COP_p^{ec} P_{t,s,p}^{ec} \\ Out_{p,\min}^{ec} u_{t,s,p}^{ec} \le C_{t,s,p}^{ec} \le Out_{p,\max}^{ec} u_{t,s,p}^{ec} \\ C_{t,s,o}^{ar} = COP_o^{ar} H_{t,s,o}^{ar} \\ Out_{o,\min}^{ar} u_{t,s,o}^{ar} \le C_{t,s,o}^{ar} \le Out_{o,\max}^{ar} u_{t,s,o}^{ar} \end{cases} \tag{6.80}$$

$$\begin{cases} 0 \le Ch_{t,s,g}^{ES} \le Ch_{g,\max}^{ES} u_{t,s,g}^{ESc} \\ 0 \le Dis_{t,s,g}^{ES} \le Dis_{g,\max}^{ES} u_{t,s,g}^{ESd} \\ 0 \le u_{t,s,g}^{ESc} + u_{t,s,g}^{ESd} \le 1 \\ E_{t+1,s,g}^{ES} = \left(1 - \kappa_g^{ES}\right) E_{t,s,g}^{ES} + Ch_{t,s,g}^{ES} \eta_g^{ESc} - Dis_{t,s,g}^{ES} / \eta_g^{ESd} \\ E_{g,\min}^{ES} \le E_{t,s,g}^{ES} \le E_{g,\max}^{ES} \\ E_{0,s,g}^{ES} = E_{T,s,g}^{ES} \end{cases} \tag{6.81}$$

$$\begin{cases} 0 \le P_{t,s}^{lc} \le P_{\max}^{lc} \\ EENS^e = \sum\nolimits_s^S \pi_s \sum\nolimits_t^T P_{t,s}^{lc} \le EENS_{\max}^e \end{cases} \tag{6.82}$$

$$\begin{cases} 0 \le P_{t,s}^{buy} \le \psi_{t,s}^{e_buy} P_{\max}^{tran}, \ 0 \le P_{t,s}^{sell} \le \psi_{t,s}^{e_sell} P_{\max}^{tran} \\ 0 \le \psi_{t,s}^{e_buy} + \psi_{t,s}^{e_sell} \le 1 \\ \left| P_{t,s}^{buy} - P_{t-1,s}^{buy} \right| \le dP_{\max}^{tran}, \ \left| P_{t,s}^{sell} - P_{t-1,s}^{sell} \right| \le dP_{\max}^{tran} \\ 0 \le \sum\nolimits_r V_{t,s,r}^{gas_chp} + \sum\nolimits_h V_{t,s,h}^{gas_gb} \le \psi_{t,s}^{gas_buy} V_{\max}^{gas} \end{cases} \tag{6.83}$$

where u_g is the binary variable representing the operating status of equipment g and is limited in (6.77) to ensure that the operation of each CD or ES is feasible ($u_g = 1$) only in case of being invested ($x_g = 1$). ρ^{gas} is the heat value of NG (valuing 9.7 kWh/m^3 in this book). η_g^{CD} and COP_g^{CD} denote the efficiency and coefficient of performance of g-th CD. $Out_{g,\min}^{CD}$ and $Out_{g,\max}^{CD}$ refer to the minimal/maximal output of the equipment. $dP_{r,\max}^{chp}$ is the maximum ramp rate of r-th CHP. $on_{t,s,r}$ and $off_{t,s,r}$ are auxiliary binary variables to ensure minimum continuous startup/shutdown periods $T_{r,\min}^{on}/T_{r,\min}^{off}$ of CHP. Therefore, (6.77) to (6.80) show property, capacity and operation limitations of r-th CHP, h-th GB, m-th HP, p-th EC, o-th AR, respectively. For g-th ES, $\eta_g^{ESc}/\eta_g^{ESd}$, $Ch_{g,\min}^{ES}/Dis_{g,\min}^{ES}$ and $Ch_{g,\max}^{ES}/Dis_{g,\max}^{ES}$ describe the efficiency, minimum and maximum energy rate of charging/discharging process; $E_{t,s,g}^{ES}$ and $E_{g,\min}^{ES}/E_{g,\max}^{ES}$ are the stored energy as well as its top/bottom limits (setting 90%/20% of the nominal capacity in this work); κ_g^{ES} implies the self-discharge rate. Taking electric load curtailment as an example in (6.82), the meanings of variables hereafter also apply to heating and cooling forms. $P_{\max}^{lc}$ is the largest value of allowed supply insufficiency. The reliability performance of the system is measured by $EENS^e$

and restricted by $EENS^{e}_{\max}$ in (6.82). EENS is one of the most important reliability indices to quantify the expected unsupplied energy in a year due to generation inadequacy. Furthermore, (6.83) is introduced to ensure the transaction of electricity and purchase of NG are within respective capacity limits $P^{tran}_{\max}$, $V^{gas}_{\max}$ and ramp rate $dP^{tran}_{\max}$; Besides, binary variables $\psi^{e_buy}_{t,s}/\psi^{e_sell}_{t,s}$ and $\psi^{gas_buy}_{t,s}$ are adopted to avoid simultaneous electricity buying/selling and imply NG pipeline status (equals 1 if normal status).

As mentioned above, the risk-neutral model (TSCCP model) aims to find solutions in view of both economic and risky aspects and can be concluded in (6.84).

$$\begin{cases} \min\limits_{\mathbb{Z}_{\mathrm{I}}=\left\{x^{chp}_{r},x^{gb}_{h},x^{hp}_{m},x^{ec}_{p},x^{ar}_{o},x^{ees}_{q},x^{hes}_{v},x^{ces}_{w}\right\}} \quad C_I+\sum_s^S\sum_t^T n\pi_s(C_O+C_P) \\ \mathbb{Z}_{\mathrm{II}}=\left\{\begin{array}{l} P^{buy}_{t,s},P^{sell}_{t,s},P^{chp}_{t,s,r},P^{dis}_{t,s,q},P^{ch}_{t,s,q},P^{ev}_{t,s,l}, \\ P^{il}_{t,s,k},P^{lc}_{t,s},H^{gb}_{t,s,h},H^{dis}_{t,s,v},H^{ch}_{t,s,v},H^{hp}_{t,s,m}, \\ H^{lc}_{t,s},C^{dis}_{t,s,w},C^{ch}_{t,s,w},\Delta\lambda^{e}_{t,s},\Delta\lambda^{h}_{t,s},\Delta\lambda^{c}_{t,s} \end{array}\right\} \\ \text{s.t. } (6.61)-(6.71),(6.76)-(6.83) \end{cases} \tag{6.84}$$

where $\mathbb{Z}_{\mathrm{I}}$ implies the set of specific x_g of r-th CHP, h-th GB, m-th HP, p-th EC, o-th AR, q-th EES, v-th HES and w-th CES which is decided in stage one before random parameters in stage two are observed. $\mathbb{Z}_{\mathrm{II}}$ is the adaptive decision spaces in the second phase.

(3) Risk-averse Model

In normal operating situations, energy infrastructure that directly connected to networks, like CHPs and GBs suffer from contingent outages. While in a catastrophe, power tie line and NG pipeline that directly exposed to the environment, are more likely to be destroyed and more difficult to be replaced rapidly than CDs and ESs located in solid surroundings. Therefore, in this research, component unavailability is counted as risk sources and simulated as chance constraints:

$$\begin{cases} \Pr\left\{\psi^{e_buy}_{t,s},\psi^{e_sell}_{t,s}\le\delta^{tran}_{t}\right\}\ge\beta^{tran} \\ \Pr\left\{\psi^{gas_buy}_{t,s}\le\delta^{gas}_{t}\right\}\ge\beta^{gas} \qquad ,\ \forall s\in N_S \\ \Pr\left\{u^{chp}_{t,s,r}\le\delta^{chp}_{t,r}\right\}\ge\beta^{chp},\ \Pr\left\{u^{gb}_{t,s,h}\le\delta^{gb}_{t,h}\right\}\ge\beta^{gb} \end{cases} \tag{6.85}$$

where $\Pr\{\cdot\}$ is the probability that chance constraint $\{\cdot\}$ is established; δ^{tran}_{t}, δ^{gas}_{t}, $\delta^{chp}_{t,r}$, and $\delta^{gb}_{t,h}$ are the defined vulnerability indicators, which are binary variables introduced to signify the offline state of electricity transmission line, NG pipeline, r-th CHP, h-th GB while valuing 0 refers to disconnection of the component. $\beta=[\beta^{tran},\beta^{gas},\beta^{chp},\beta^{gb}]$ are corresponding confidence level set.

In this way, by substituting WCVaR methodology (6.74) into (6.84), the risk-averse model (WCVaR-TSCCP model) with confidence level φ and resilience variable set $\mathbb{Q}_R$ representing above risk elements is formulated in (6.86). Other than the

risk-neutral model, the target of the risk-averse model is to diminish the VaR of the worst scenario to avoid risks.

$$\begin{cases} \min\limits_{\substack{\mathbb{Z}_{\mathrm{I}},\mathbb{Z}_{\mathrm{II}},\\ \mathbb{Q}_R=\left\{\delta_t^{tran},\delta_t^{gas},\delta_{t,r}^{chp},\delta_{t,h}^{gb},\delta_t^{al},\delta_{t,k}^{il},\delta_{t,l}^{ev}\right\}}} \theta \\ \text{s.t.} \begin{cases} \theta \geq VaR_\varphi(\boldsymbol{x}) + \frac{1}{1-\varphi}\sum\limits_{s=1}^{S}\pi_s\mu_s \\ \mu_s \geq C_I + \sum_{t=1}^{T} n(C_O + C_P) - VaR_\varphi(\boldsymbol{x}) \\ \boldsymbol{x} \in \mathbb{Z}_{\mathrm{II}}, \mu_s \geq 0, \forall s \in S \\ (1)-(11), (16)-(23), (25) \end{cases} \end{cases} \tag{6.86}$$

6.4.3.3 Flowchart of the Approach

In a word, the strategy of the proposed approach can be summarized as follows.

Step 1: Input the parameters of ICES, including prediction data, properties, prices, and technical bounds.

Step 2: Generate the initial scenario set S^0 which combines the typical seasonal PV/SH outputs, demands and EV charging of planning year, as well as vulnerability indicators ($\mathbb{Q}_R$) in ICES by MCM method.

Step 3: Cut S^0 to target scenario set S with a total scenario number of N_S.

Step 4: For normal operation scenarios, solve the risk-neutral model with converting TSCCP model (6.84) to a MILP problem with crisp equivalent conversion of (6.64) and (6.70). As a result, the type, capacity, and number of CDs and ESs ($\mathbb{Z}_{\mathrm{I}}$) are decided in stage one to configure the ICES. Operating variables ($\mathbb{Z}_{\mathrm{II}}$) are optimized in stage two to repeatedly correct results of stage one to achieve the economic goal.

Step 5: Consequently, for disaster scenarios, solve the risk-averse model with converting the WCVaR-TSCCP model (6.86) to a MILP problem by linearizing the additional constraints in (6.85) using the big-M method. The optimal values of WCVaR and cost under different confidence levels iteratively give feedback to adjust investment and operation results of ICES to acquire both economy and resilience.

The procedures of the proposed approach are presented in the form of a flowchart as illustrated in Fig. 6.3.

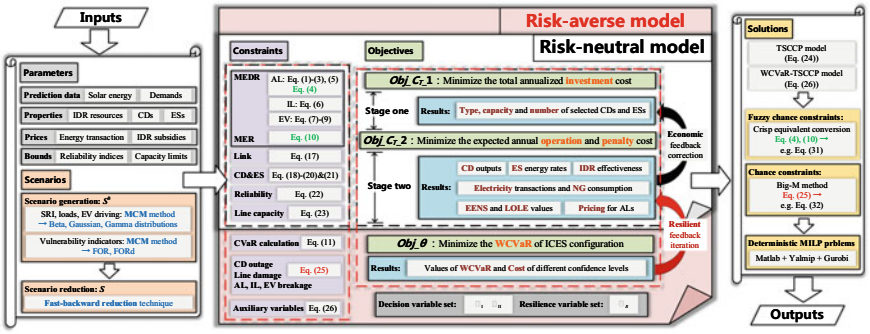


Fig. 6.3 Components of smart residential community load and distributed generation

6.5 Case Studies

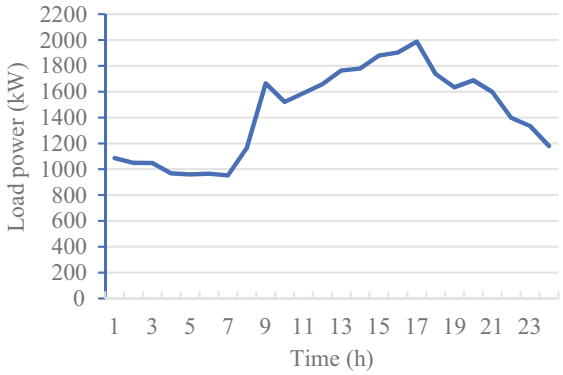
6.5.1 Residential Load Demand Response

The case study data is mainly acquired from a smart residential community of smart grid application demonstration project in Suzhou, Jiangsu province, China. The DR result of the complete residential community is analyzed and discuss in three cases of IL with TOU, CPP, and RTP respectively.

Residential load curve without DR program is shown in Fig. 6.4.

Case 1: Residential load curve with TOU and IL programs is shown in Fig. 6.5. After participating in TOU and IL programs, the residential peak load decreases from 1988.21 to 1909.90 kW by 3.94%. Besides, the time of load peak appearance is shifted from 17 to 19 h, and the peak-valley deference decreases from 1036.59 to 958.28 kW by 7.56%. The total energy consumption reduces by 0.87%, and the total power purchase cost is 10505.12¥.

Fig. 6.4 Residential load without DR program



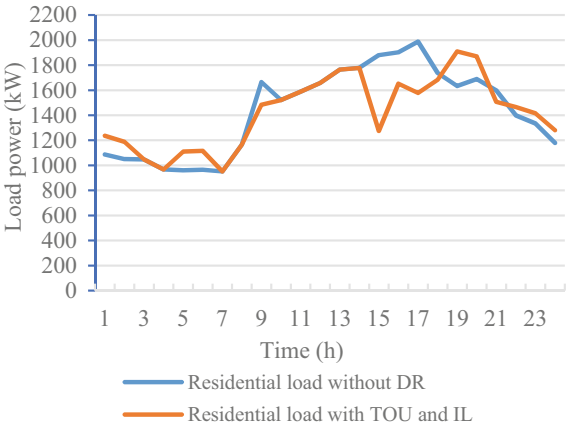


Fig. 6.5 Residential load with TOU and IL programs

Case 2: Residential load curve with CPP and IL programs is shown in Fig. 6.6. After participating in CPP and IL programs, the peak load decreases to 1909.90 kW by 3.94%, and the time of load peak appearance is shifted from 17 to 19 h. The peak-valley deference decreases to 958.28 kW by 7.56%. The community total energy consumption reduces by 1.65%, and the total power purchase cost is 15784.35¥.

Case 3: Residential load curve with RTP and IL programs is shown in Fig. 6.7. It is the same with case 1 and 2 that the peak load decreases to 1909.90 kW, the time of load peak appearance is shifted to 19 h, and the peak-valley deference decreases to 958.28 kW. However, in comparison to case 1 and 2, the community total energy consumption reduces by 2.11%, and the total power purchase cost is 12511.16¥.

It can be observed from the case result above that the peak load and peak-valley difference are dramatically reduced in 3 cases respectively at the same level. The load peak appearance time is shifted to 19 h, which is beneficial to alleviate the power load of the grid in daylight. Figure 6.8 shows the comparison of the total

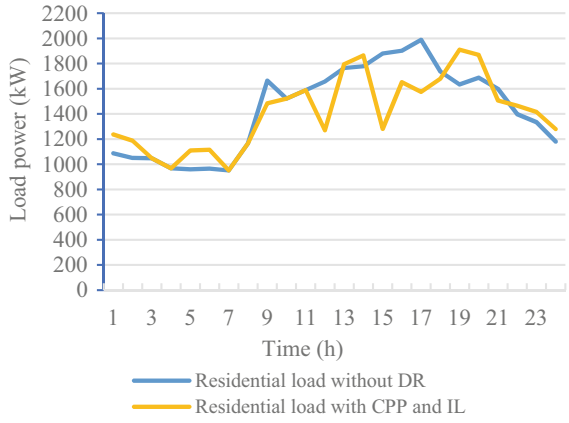


Fig. 6.6 Residential load with CPP and IL programs

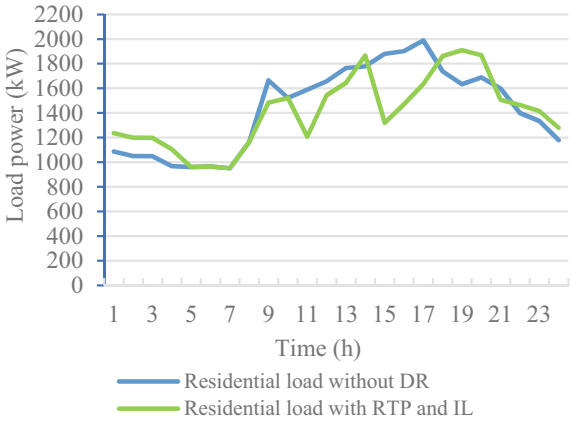


Fig. 6.7 Residential load with RTP and IL programs

energy consumption and the total power purchase cost among 3 cases. According to the 3 cases result, although the peak load and peak-valley deference decrement are the same, the total energy consumption in case 3 is less than in other cases. However, customers' power purchase cost in case 1 is the lowest among the 3 cases. Besides, in comparison to case 1, the load curves in cases 2 and 3 are more volatile, which is not conducive for the generation with a low ramping rate to respond to the load variation. In summary, the CPP and RTP programs in this case study need improvement to achieve superior DR results. The TOU combined with IL is the optimal DR program in this case, and the residential community optimally responds under this program.

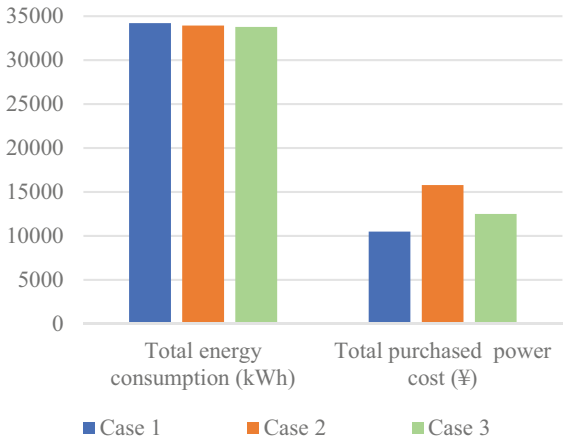


Fig. 6.8 Total energy consumption and the total power purchase cost of 3 cases

6.5.2 Price-Based Demand Response Model

The case is based on the IEEE RTS79 system including 26 thermal power units. To quantitatively analyze the influence of the flexibility of ITCAs on the power system operation, this section first analyzes the flexibility of customers' ITCAs by the influence of the RPtVP of the TOU power price on the flexibility of ITCAs. Then, based on the proposed unit scheduling model considering the flexibility of ITCAs, the influence of the RPtVP on the power system operation is analyzed through thermal power generation, wind curtailment, and total cost. Finally, the relationship between the flexibility of ITCAs and wind curtailment improvement is analyzed.

6.5.2.1 Influence of the RPtVP of TOU Power Price on the Flexibility of ITCAs

To analyze the influence of the RPtVP, the day is divided into 24 SIs. In detail, $T_f = [11, 12, 13, 14, 15, 19, 20, 21]$, $T_p = [8, 9, 10, 16, 17, 18, 22, 23]$, $T_g = [1, 2, 3, 4, 5, 6, 7, 24]$, the mid-peak hours price $\lambda_{n,mp}$ TOU $= 0.8745 * 10^3$¥/MWh, RPtVP $K_{n,ref} = 3.736$, and ratio of the distance of on-peak to mid-peak distance to that of mid-peak to off-peak for price $\beta_n = 1.052$. Among them, the smart customers involve 72 commercial buildings and 300,000 residential customers. Under the condition that $\lambda_{n,mp}^{TOU}$ and β_n are constant, the flexibility of ITCAs is simulated and analyzed by changing the RPtVP. The variation range of RPtVP is [4, 14], and the step value is 1.

The simulation results are shown in Fig. 6.8. As the RPtVP increases, the demand during on-peak hours gradually decreases, whereas the demand during mid-peak hours and off-peak hours continues to increase. This is mainly because when $\lambda_{n,mp}^{TOU}$ and β_n are constant, increasing the RPtVP means that the on-peak hours price increases and the off-peak hours price decreases, which will lead to a reduction in on-peak hours demand while some demand is transferred to mid-peak hours and off-peak hours. In addition, the ITCA has a temperature constraint, that is, the customer has minimum comfort requirements for the ITCA, which leads to a gradual slowdown trend in the demand variation of ITCAs during off-peak, mid-peak, and on-peak hours (Fig. 6.9). When the RPtVP increases to 7, the change rate of the demand at each SI is less than 20%.

The flexibility resource cost of ITCAs is shown in Fig. 6.10. As the RPtVP increases, the flexibility resource cost of ITCAs gradually increases, but the rate of increase gradually slows down, which is also caused by the customers' minimum demand for ITCAs. Similarly, after the RPtVP increases to 7, the rate of change in the flexibility resource cost of ITCAs is less than 25%.

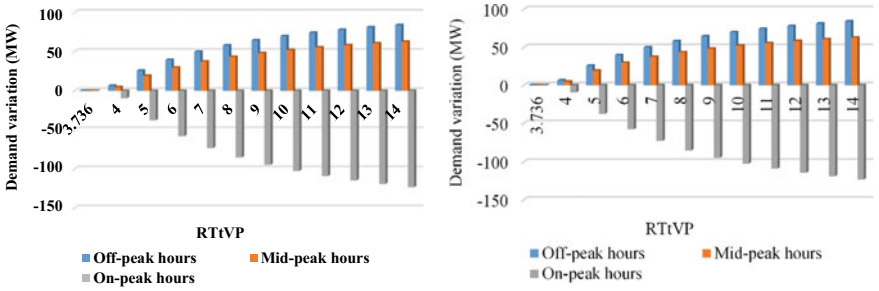


Fig. 6.9 Demand variation of customers during off-peak, mid-peak, and on-peak hours with the increase in the RPtVP

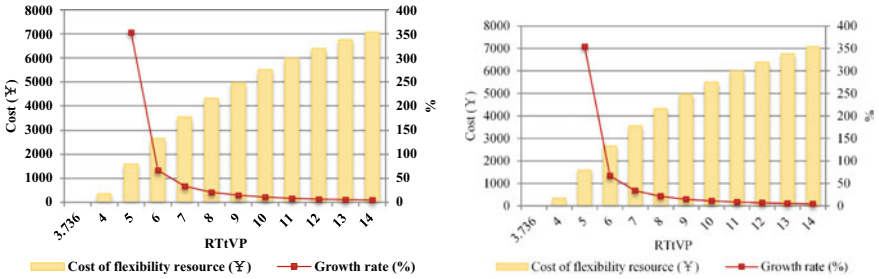


Fig. 6.10 Cost of flexibility resource with the increase in the RPtVP

6.5.2.2 Influence of the RPtVP of TOU Power Price on Power System Operation

(1) Thermal power output

The simulation results show that with the increase in the RPtVP, the output of thermal power during on-peak hours has decreased significantly, and the overall power generation has also been continuously reduced, as shown in Fig. 6.11. With the increase in the RPtVP, the on-peak hour's price increases, and the off-peak hour's price decreases, which leads to a decrease in demand during on-peak hours and an increase in demand during the mid-peak hours and off-peak hours.

In addition, combined with the flexibility resource of ITCAs, there is the characteristic of "with the increase in the RPtVP, the change of demand gradually slows down," and regardless of the decrease in thermal power output during on-peak hours or the decrease in total thermal power generation, it is gradually slowed down.

(2) Wind curtailment

The simulation results show that as the RPtVP increases, the wind curtailment rate continues to decrease, but the decline rate gradually decreases, and when the RPtVP is greater than 7, the change rate of the wind curtailment rate is less than 0.1% point,

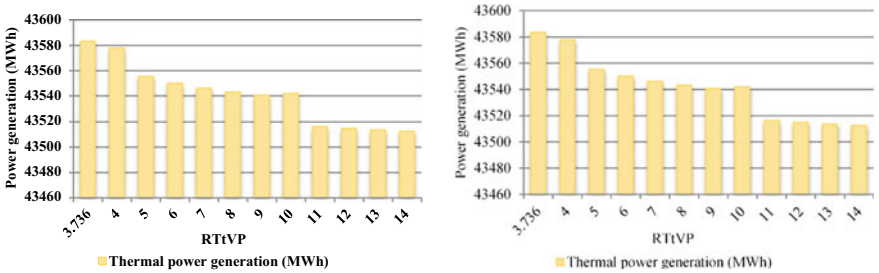


Fig. 6.11 Thermal power generation with the increase in the RPtVP

as shown in Fig. 6.12. This is because, under this wind power condition, the wind curtailment mainly occurs in the mid-peak and off-peak hours, and as the RPtVP increases, the demand will shift to mid-peak and off-peak hours, which will improve the wind curtailment and increase the wind power utilization. Taking SI 18 as an example, as shown in Fig. 6.13, the demand variation gradually increases with the increase in the RPtVP, and the wind curtailment gradually decreases. This is because wind power replaces thermal power to apply part of the increased demand during the off-peak hours. In addition, after the RPtVP reaches 11, also owing to the flexibility resource characteristics of ITCAs, the decline rate of the wind curtailment rate will gradually decrease.

(3) Total cost

The simulation results show that with the increase in the RPtVP, the total cost decreases, and the overall reduction rate gradually slows down. When the RPtVP is greater than 7, the change rate of the total system cost is less than 0.01%, as shown in Fig. 6.14. The total cost comprises the thermal power operation cost, wind abandonment penalty, and flexibility resource cost of ITCAs. Although the flexibility resource cost increases with the increase in RPtVP, the thermal power operation cost and wind abandonment penalty are more significantly reduced, resulting in a gradual reduction in the total cost of the power system.

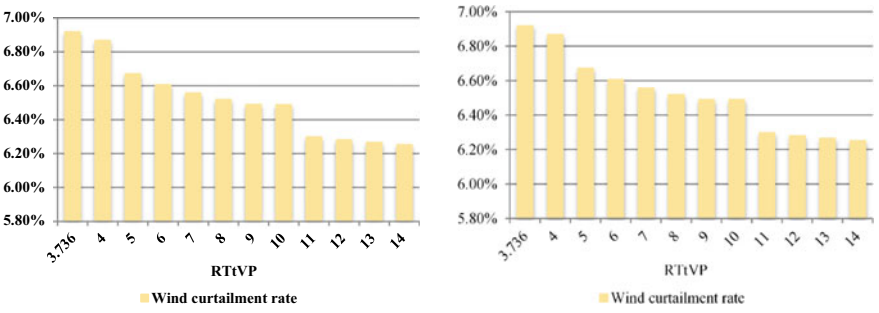


Fig. 6.12 Wind curtailment rate with the increase in the RPtVP

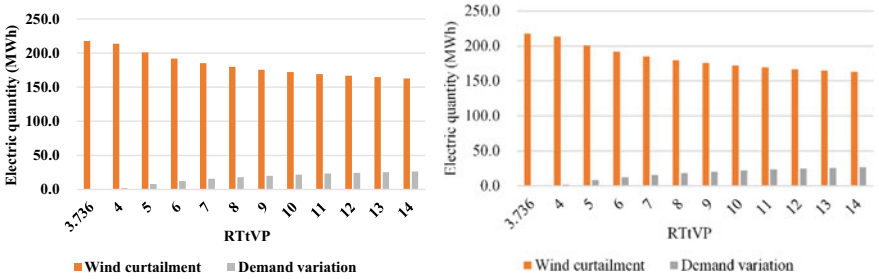


Fig. 6.13 Wind curtailment and demand variation at SI 18 with the increase in the RPtVP

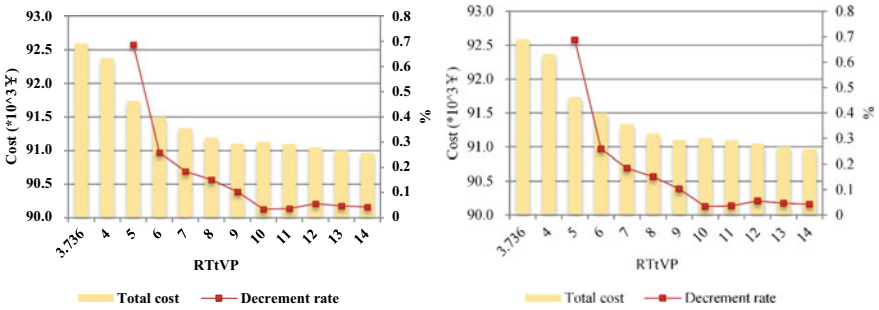


Fig. 6.14 Total cost with the increase in the RPtVP

6.5.3 Integrated Energy System Demand Response

6.5.3.1 Scenario and Case Settings

(1) Initial scenario data settings

The solar energy and multi-energy loads have obvious seasonality and are hereby characterized by load patterns for three typical days representing the transition (spring and autumn), summer and winter season, respectively. In this way, solar radiation and temperature of typical days for the planning year are predicted via wavelet neural network based on the historical local meteorological report. Electricity demands are estimated on the basis of the maximum capacity of power transformers at the park port and load patterns extracted from building clusters. Heating and cooling demands, referring in particular to space heating and cooling, are calculated through differences between indoor/outdoor temperature (National ministry of housing of China requires indoor temperature to be no less than 18 °C in winter and cooling is needed when outdoor temperature surpasses 26 °C in summer). A total of 5,000 m^2 rooftop solar panels are planned in the park, where PV electrical efficiency is 34% and SH thermal efficiency is 60%.

Therefore, the initial scenario set S^0 which includes 200 scenarios of loads and outputs of PV and SH in transition, summer and winter days is illustrated in Fig. 6.15. S^0 is generated by MCM method. The forecasted values are set as mean values while standard deviations are assumed to be linearly increasing, which are 10% and 15% of mean values at the start and end of the day. There are roughly 70% of the consumers in the region expressed willingness to attend AL program through a questionnaire, namely, 70% of the multi-energy loads shown in Fig. 6.16 are ALs, the rest are accordingly usual loads. Base loads are constant in the same season based on maximum demands of the data center of the park.

S^0 is further reduced to the representative scenario set S with corresponding probabilities, as demonstrated in Fig. 6.16. The randomness of each typical day is described by 3 scenarios. In a summer day, PV and SH outputs, electric load and base load, cooling load and base load have the largest values (4.69 and 8.28 MW, 5.13 and 1.77 MW, 13.76 and 2.62 MW, respectively). Heating load and base load reach a peak of 14.63 and 2.61 MW in winter day.

(2) Case settings

To validate the effectiveness and applicability of the proposed approach, 6 cases are set with details listed in Table 6.1. Case 1–3 apply risk-neutral models: case 1 is the reference case where ALs have no elasticity, EVs charges naturally, no IL responses

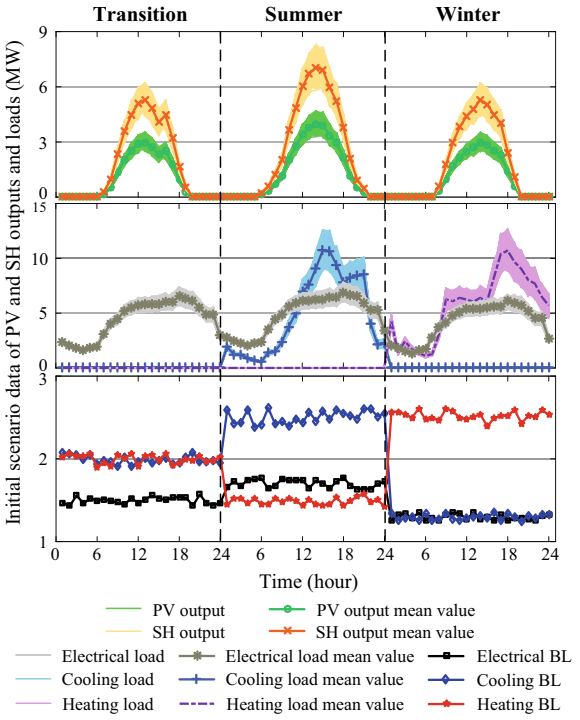


Fig. 6.15 Initial scenario data of solar energy and multi-energy loads in 3 typical days

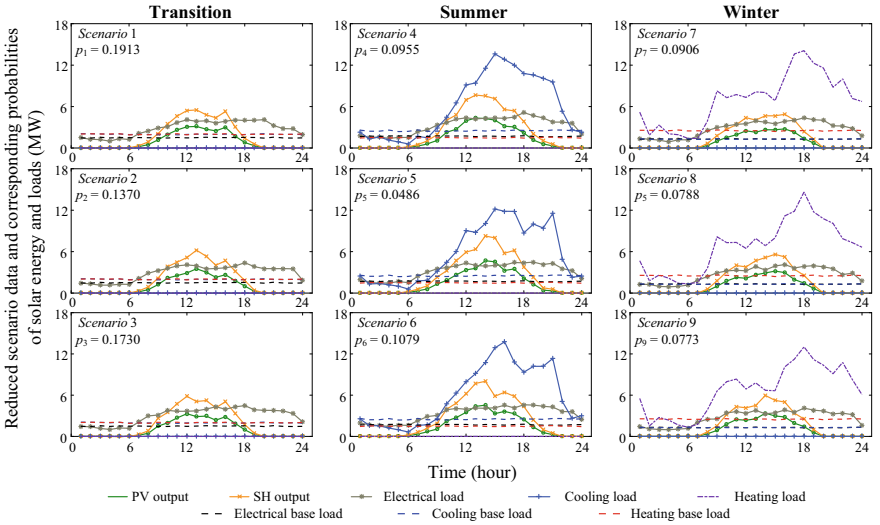


Fig. 6.16 Reduced scenarios and corresponding probabilities of solar energy and loads in 3 typical days

and no vulnerability indicators are considered; on this basis, case 2 and 3 are set to separately examine the influence of IDR and fuzzy feature. It is clear that in case 2, all the scaling factors of AL are equal to 1. Case 4–6 test risk-averse models: case 4 examines the portfolio results of ICES in the normal state in presence of fuzzy IDR and common FOR; case 5 cares about the most severe disaster like a hurricane. Hurricane usually occurs in summer and is more inclined to severely (measured by FORd) affect the overhead power line which is exposed in the air rather than NG pipeline, CHPs, and GBs. Case 6 is set to investigate the effects of IDR in WCVaR-TSCCP models by comparing with case 5. Confidence levels β and φ are set 90% in all the cases.

Table 6.1 Comparision of 6 cases

	IDR	Fuzzy feature	Vulnerability indicators		Objective	
			Normal	Emergency	Cost	WCVaR
Case 1	✗	✗	✗	✗	✓ (15)	✗
Case 2	✓	✗	✗	✗	✓ (15)	✗
Case 3	✓	✓	✗	✗	✓ (15)	✗
Case 4	✓	✓	✓	✗	✗	✓ (26)
Case 5	✓	✓	✗	✓ (hurricane)	✗	✓ (26)
Case 6	✗	✗	✗	✓ (hurricane)	✗	✓ (26)

6.5.3.2 Effect Analysis of MER

To discuss the MER effect in normal and emergency states, we compared the expected daily energy transaction between ICES and networks and outputs of CDs and ESs in case 1 and 6. Scenario 1, 4 and 9 are selected as typical season days. The results are shown in Fig. 6.17 with negative values representing power sale in Fig. 6.17a and ES discharge in Fig. 6.17b.

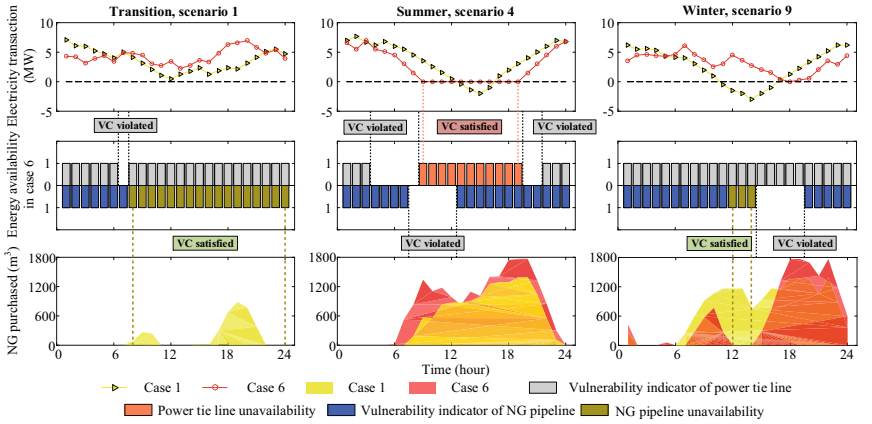
In normal cases, demands of transition seasons are relatively lower and more balanced. The multi-energy needs are satisfied mainly by power grid, HP and EC in valley periods and PV, CHP, SH, and AR in rush hours. Cooling or Heating loads soar more greatly than electric loads in summer or winter, therefore CHPs and GBs contribute more to provide thermal energy, which also enables ICES to sell surplus electricity to power grid to reduce cost during 12:00–16:00 when PV reaches biggish outputs. HP, EC, and AR work at high levels to ensure energy supply. ESs are utilized frequently in smoothing load curves.

It is manifest that compared to case 1, emergency case 6 chose two more CHP#2 in case of outages of already selected CHPs. GBs are therefore not necessary in providing heating energy. Besides, the installed CDs and ESs in case 6 are capable to cope with deficiencies of outer energy. As shown in Fig. 6.17a, the simulated vulnerability indicators of power tie line and NG pipeline are denoted by gray and blue bars. The red and green bars in "VC satisfied" areas express that the actual states of corresponding lines are unavailability. In scenario 1, NG supply is shut down due to its FOR during 8:00–24:00, when electricity supply is normal. This results in the shut off of CHPs, more outputs of HPs and ARs and further power purchase. The situation is similar in scenario 9, except that the system has to buy extra gas for CHPs to generate enough heating energy after 18:00. Power line of scenario 4 experiences the worst breakage caused by the hurricane from 4:00 to 21:00. In this period, MER process renders ICES to rely predominately on NG-fired CHPs for supply so that system can operate even when there is no electricity from outside in most of the daytime (9:00–19:00). Moreover, power sale of Case 6 is cancelled in summer and winter to mitigate the potential risk from tie line disruption.

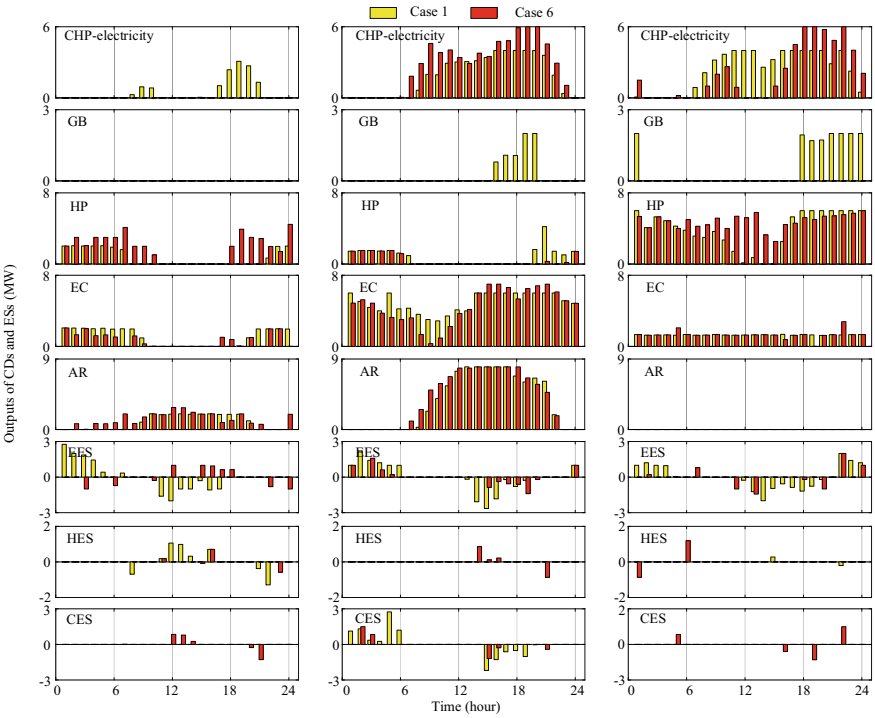
6.5.3.3 Effect Analysis of MEDR

MEDR resources are able to transfer part of energy use to provide virtual energy to relieve supply pressure.

IL response is not required in Case 2–3 because of the costly compensation fees but is needed in risk-included cases. The response of each IL in Cases 4 and 5 is presented in Fig. 6.18. ILs exert influences only in transition day when the lack of CHP and GB production makes purchased power insufficient to support the demands. In Case 4, ILs serves as an all-important power supply to avoid power shortage and reach a daily sum of 8.16 MW. This amount is dropped to 1.99 MW in Case 5, where the electricity gap is filled by alternative CHP outputs. IL#1 can work multiple times



(a) Energy transaction



(b) Energy outputs of CDs and ESs

Fig. 6.17 Expected daily energy transaction and outputs of CDs and ESs in different scenarios of case 1 and 6

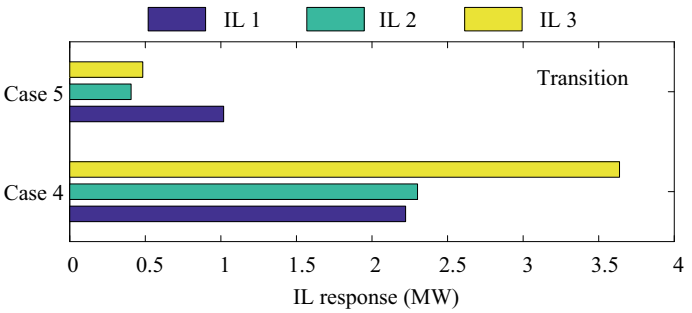


Fig. 6.18 IL response in a transition day of case 4 and 5

and hours and are needed more in an emergency case for more frequent and stochastic risks.

The average daily total power and mean SOC of EVs are illustrated in Fig. 6.19, where negative power reflects discharging. The DR potential of EV acquires further exploitation in the proposed approach. In contrast to Cases 1 and 6, the charging behaviors in other cases are extended to the whole rush hours, thus decreasing the peak value of natural charging power dramatically. Plus, the excess energy of various EVs is released at departure times (17:00–20:00). What's more, both two types of EVs could adjust the charging process in different risk preferences. For instance, in Case 2 or 3 EVs act as source role: they discharge a total of 0.48 MW in a day to fill power gaps on the premise of meeting minimal travel needs, but in Case 4 and 5 EVs mainly absorb extra energy (daily total of 2.89 and 2.65 MW, respectively) generated by CHPs. It is also reasonable that EV clusters have a higher compensation allowance and lower mean SOC upon leaving the park in cases 2 and 3 than that in Cases 4 and 5.

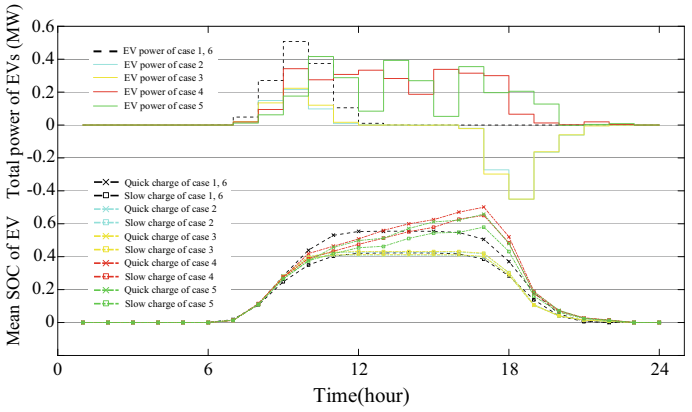


Fig. 6.19 Averaged daily total power and mean SOC of EVs

The expensive CHP lifts the installation cost despite decreased numbers of ES. The condition is similar in Case 5, where another CHP#2 is selected to proactively provide backup supply and consequently lower MEDR participations significantly. The total penalty of loss of load was reduced by 34.78% compared to case 4. In this way, WCVaR values are nearly the same in cases 4 and 5, meaning that the risk of the hurricane has been reduced to the normal level by the proposed approach.

6.6 Conclusion

In the traditional power system, the demand side resources usually only accept the instructions of the power grid dispatching department passively as the power load, and the active response is rarely considered. With the continuous development of the types, functions, and controllability, demand-side resources begin to respond actively according to the actual operation state of power systems. The power demand side contains abundant adjustable resources, which has great potential in improving the flexibility of power systems. This chapter studies the demand side flexibility from three aspects: residential load side demand response, price incentive demand-side response, and integrated energy system demand-side response. The specific conclusions are as follows:

First, a demand response scheduling model for smart residential communities incorporating the current circumstances and the future trends of demand response programs is presented in this chapter. A complete scheduling scheme is modeled based on the optimization of residential loads and distributed generation. The presented model reduces the cost of the user's electricity consumption and decreases the peak load and peak-valley difference of smart residential load without bringing discomfort to the users, through which the residential community can participate in demand response efficiently.

The second section focuses on exploiting the demand-side flexibility resources in unit scheduling. An energy management model is established for intelligent temperature control appliances of commercial and residential customers which can optimize its electricity consumption according to power prices under the smart grid, and the calculation model on the price elasticity of demand of the ITCA is built on the EMM through the mathematical derivation, by which the calculation method of ITCAs' flexibility is proposed, namely, the PED of the ITCA is obtained with the consideration of appliance operation characteristics and customer's electricity consumption behaviors. The price-based demand response model analyzes the demand characteristics and PED of the ITCA from the physical appliance level and combines TOU power price to propose a calculation method of the flexibility of the customers' ITCAs.

A resilience-oriented stochastic integrated community energy system configuration framework considering integrated demand response influence is proposed in the third section. Generalized IDR models are set up in detail with elaborate fuzzy feature analysis of price-responsive multi-energy loads. The vulnerability indicators

of tie lines and converting devices of ICES are first introduced to demonstrate the occasional outage in normal operation and blackout in the emergent case. In addition, the worst-case conditional value-at-risk (WCVaR) theory is innovatively integrated into the traditional risk-neutral model, which is formulated as a two-stage stochastic chance-constrained programming problem, aiming to combine portfolio with minimizing the worst-case cost caused by a disaster. The models are then transformed into mixed-integer linear programming problems via several linearization techniques. At last, case studies are carried out to test the effectiveness and sensitivity of the proposed approach.

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Chapter 7
Large-Scale Distributed Flexible Resources Aggregation



This chapter studies the aggregation of large-scale distributed flexibility resources, and aggregates a large number of flexible loads into a small number of aggregation load models. This chapter mainly considers large scale demand response aggregation and distributed energy storage aggregation. From the study of large-scale demand response and distributed energy storage, it is found that they need effective clustering and aggregation equivalence methods to give play to their flexibility value. Demand response contains a large number of transferable and interruptible loads, when large-scale interruptible and shiftable load is connected to the power system and its operation mode is optimized, the system operation cost will be reduced and greater flexibility will be provided for the system. The most representative of distributed energy storage is electric vehicle (EV), the randomness of large-scale electric vehicles will significantly influence the reliable and economic operation of power systems, while integrating a large population of EVs into the system-level operation and market bidding makes sense, so it is essential to construct an aggregate model to capture their available flexibility.

7.1 Introduction

A large number of distributed flexibility resources from demand-side provide great flexibility potential. However, the most critical problem is that if the scheduling is based on individuals, the computational complexity is high and difficult to deal with. Effective aggregation is needed to embed the optimal scheduling at the system level. The large-scale flexible load will introduce hundreds of millions of variables and constraints into the power system optimization model, resulting in a sharp increase in the dimension of the model, which makes it almost impossible to solve the optimization model in a limited time. In addition, flexible load model usually contains discrete variables, which further increases the difficulty of flexible load participating in power system optimization calculation. Therefore, it is necessary to find new

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M. Zhou et al., *Power System Flexibility*, Power Systems,
https://doi.org/10.1007/978-981-19-9075-5_7

models and methods to solve the large-scale transferable load optimization problem. This chapter starts from two important representatives of large-scale distributed flexible load, namely large-scale interruptible and shiftable load, and large-scale distributed energy storage.

In the research of distributed computing on flexible resources, distributed control is widely applied in the scheduling of the micro-grid [1, 2], and for scheduling with larger-scale flexible loads, the distributed algorithms based on the Alternating Direction Method of Multipliers and game theory are verified to be effective. Rivera et al. [3] formulated a versatile and scalable distributed convex optimization framework, which in simulation scheduled a million flexible loads to fill system valleys within 30 min. For multi-period optimal power flow optimization with flexible loads, Fan et al. [4] developed a distributed algorithm based on alternating direction method of multipliers, updating steps by alternating iterations. The scalability and effectiveness of the algorithm is also tested. Mohsenian-Rad et al. [5] presented an autonomous and distributed demand-side energy management system among users through game theory and proved the optimality and convergence of the model when power of the loads is continuous. De Paola et al. [6] extended the work in [7], coordinated the continuous or ON/OFF loads operating in the electricity market with the framework of game theory. A distributed iterative algorithm based on Nash equilibrium is designed and tested with the large-scale loads. Kumar et al. [6] proposed an aggregative game for flexible loads in the day-ahead electricity market, and tested with 100,000 loads; however, the calculation time is not illustrated. Chen et al. [8] applied game theory to study a large number of users to provide operational reserves to the power system through load aggregators. In addition, the impact of communication on distributed computing has also been studied [9, 10].

According to the characteristics of distributed flexible resources, this chapter divides them into two categories: large scale interruptible and shiftable load aggregation and distributed energy storage aggregation. Interruptible and shiftable load and distributed energy storage are two very important distributed flexibility resources. Interruptible and transferable load can flexibly arrange the operating power for a long time, reduce the peak load and fill the valley load, which makes it more suitable for one day in advance and day scheduling. At the same time, EV is selected to represent distributed energy storage because of its mobility and more application potential. What's more, the model is easy to be extended to ordinary household energy storage and has strong applicability.

The existing literature on the modeling EV aggregation can be divided into two categories. The first is to model the population dynamics of the charging loads, and characterize the set of admissible aggregate load profiles [11–13]. However, this kind of method does not explicitly express the analytical formulations that EV aggregation can offer to the grid. The second is to geometrically capture the set of admissible aggregate load profiles without violating any operational constraints, e.g., Minkowski sum and Fourier-Motzkin elimination [14]. However, these methods are computationally expensive and the storage capability of EVs are temporally correlated. Efficient and accurate dispatchable region formation is the basis of V2G application in system operation. On the one hand, a system operator can achieve optimal internal

operation and earn profits in market bidding with the help of the virtual storage that EV aggregation forms. On the other hand, the EV aggregation can serve as non-wire alternatives to capacity expansion when forming dispatchable region [15]. To the best of our knowledge, few studies have investigated the feasible region of the charging and vehicle-to-grid (V2G) flexibility in market bidding. Therefore, this paper aims to explore forming dispatchable region approach of plug-in electric vehicle (PEV) aggregation, which is then applied in MG bidding to verify prediction accuracy and computation efficiency of the dispatchable region.

As a kind of flexible load, interruptible and shiftable load can flexibly adjust its operation period in a long time, cut peak and fill valley, which is more suitable for participating in the day ahead and day in day optimization of power system. When large-scale interruptible and shiftable load is connected to the power system and its operation mode is optimized, the system operation cost will be reduced and greater flexibility will be provided for the system. In Sect. 7.2, focusing on the scheduling of the power system with large-scale demand response, this chapter proposes an equivalent aggregated model of interruptible and shiftable load based on grouping equivalence, proves the equivalence between the models, estimates the upper bound of equivalent deviations, and establishes a scheduling model applying the equivalent aggregated model.

The technology based on the participation of electric vehicles in distributed energy storage power supply can provide huge energy storage potential for the power system and can be used as the main regulation means of "peak cutting and valley filling" at the power grid side, to alleviate the pressure of the power system on the investment, operation and maintenance of power facilities. To integrate a large population of PEVs into the system-level operation and market bidding, it is essential to construct an aggregate model to capture their available flexibility. In Sect. 7.3, for the aggregation of distributed energy storage, electric vehicles are selected as the research object, a dispatchable region formation method of the EV aggregation is proposed for MG bidding considering both the day-ahead markets and the balancing markets. A detailed model for EV fleets is formulated to depict the dispatchable region in market bidding. While the aggregation of large-scale EVs contributes to a more flexible power system and serves as a flexibility asset, several challenges remain to be addressed. First, it is crucial to accurately depict the analytic expression of the dispatchable region of EV aggregation to investigate the charging and V2G flexibility in market bidding. Second, it is difficult to forecast the behaviors of EV owners in practice, so it is of vital importance to analyze the effect of the EV uncertainty on the dispatchable region parameters. Third, microgrid (MG) operator may expose to market risks from various uncertainties, for example, renewable generation and spot market prices, it would be desirable to investigate the effect of this joint volume-price risk as well as the uncertainty of the EV aggregation dispatchable region in MG bidding. Section 7.3 aims to address these three challenges.

7.2 Large Scale Interruptible and Shiftable Load Aggregation

7.2.1 Equivalent Aggregated Model for Large-Scale Interruptible and Shiftable Loads

7.2.1.1 Model of Interruptible and Shiftable Loads

The Interruptible and Shiftable load studied in this paper is the load that can flexibly adjust its power consumption in a certain period, while the total energy consumed is fixed. According to power adjustment modes, flexible loads can be divided into continuous loads and ON/OFF loads. The former can continuously adjust its power consumption, whereas the latter can only control power through ON/OFF.

In order to schedule flexible loads, a day is divided into T equal time intervals, and the duration of a time interval is Δt. In addition, for a flexible load i with rated power ${P_i}^{rate}$ and required energy E_i, we can calculate its necessary work intervals T_i by $T_i = E_i/({P_i}^{rate}{}_{\Delta t})$, and then use T_i instead of E_i as its parameter.

For a continuous flexible load i with the schedulable period $[\alpha_i, \beta_i]$, rated power Pirate and necessary work intervals T_i, its individual model is as follow:

$$0 \le P_i(t) \le P_i^{rate}, \forall t \in [\alpha_i, \beta_i] \tag{7.1}$$

$$P_i(t) = 0, \forall t \notin [\alpha_i, \beta_i] \tag{7.2}$$

$$\sum_{t \in [\alpha_i, \beta_i]} P_i(t) = T_i P_i^{rate} \tag{7.3}$$

$$\mathcal{P}_i = \{P_i(\cdot) | P_i(t) \text{ subject to } (7.1) - (7.3)\} \tag{7.4}$$

where $P_i(t)$ is the power of flexible load i at time t; $P_i(\cdot)$ is $[P_i(1), P_i(2), P_i(3), \ldots, P_i(T)]$, that is the power vector composed of $P_i(t)$ at all times; P_i is the set of all the feasible power vector $P_i(\cdot)$. Equations (7.1) and (7.2) ensure flexible load i can only work in its schedulable period and its power will not exceed its rated power. Equation (7.3) guarantees the working time is enough for its task.

If the flexible load i is an ON/OFF load, its individual model is:

$$P_i(t) = u_i(t) P_i^{rate}, u_i(t) \in \{0, 1\}, \ \forall t \in [\alpha_i, \beta_i] \tag{7.5}$$

$$\mathcal{P}_i = \{P_i(\cdot) | P_i(t) \text{ subject to } (2), (3), (5))\} \tag{7.6}$$

where $u_i(t)$ is a binary variable, describing the operating state of flexible load i at time t. For the ON/OFF load, the T_i needs to be an integer to ensure $P_i \neq \emptyset$. Therefore,

we assume that each T_i appearing in this paper is a positive integer, whether for a continuous load or an ON/OFF load.

Afterwards, the aggregated model of flexible loads is established based on the individual models. For a set Ω including millions of flexible loads, its power should be equal to the sum of the power of all flexible loads in the set:

$$\mathcal{P}_\Omega = \left\{ P_\Omega(\cdot) \middle| P_\Omega(\cdot) = \sum_{i\in\Omega} P_i(\cdot),\ P_i(\cdot) \in \mathcal{P}_i \right\} \tag{7.7}$$

where $P_\Omega(\cdot)$ is the power vector of set Ω at all times; P_Ω is the set of all feasible $P_\Omega(\cdot)$; and P_i is the feasible domain of each load in set Ω, based on its individual model.

Then for the power of set Ω at time t, $P_\Omega(t)$, there should clearly be

$$P_\Omega(t) = \sum_{i\in\Omega} P_i(t), \quad \forall t \tag{7.8}$$

7.2.1.2 Grouping Equivalence

Computational complexity increases rapidly with the size of the problem, leading to the need for an equivalent model. As for a set of flexible loads as described by (7.7), there could be hundreds of millions of variables and constraints in that aggregated model, creating problems for load scheduling. Therefore, we hope to find an equivalent alternative model. However, due to the diversity of the parameters of flexible loads, it is difficult to use a simple mathematical model to equate the aggregated model. Hence, a method of grouping equivalence is proposed to solve it.

The key to grouping equivalence is clustering flexible loads of the same or similar parameters into a group, and then establishing an equivalent model for each group. In this way, the aggregated model for a set of flexible loads is equivalent to the sum of the equivalent models of each group. Figure 7.1 illustrates this process.

The criterions for grouping are the parameters of flexible loads. According to the individual model, the parameters of flexible load i are P_i^{rate}, α_i, β_i, T_i, by which the flexible loads could be divided into groups. After grouping, the loads in each group have the same or similar parameters. Then the equivalent model for a group is established, as discussed in the next section.

7.2.2 Equivalent Model for a Single Group

This section presents the equivalent model for a group of flexible loads with the same or similar parameters. First, the mathematical definition of the equivalent model is

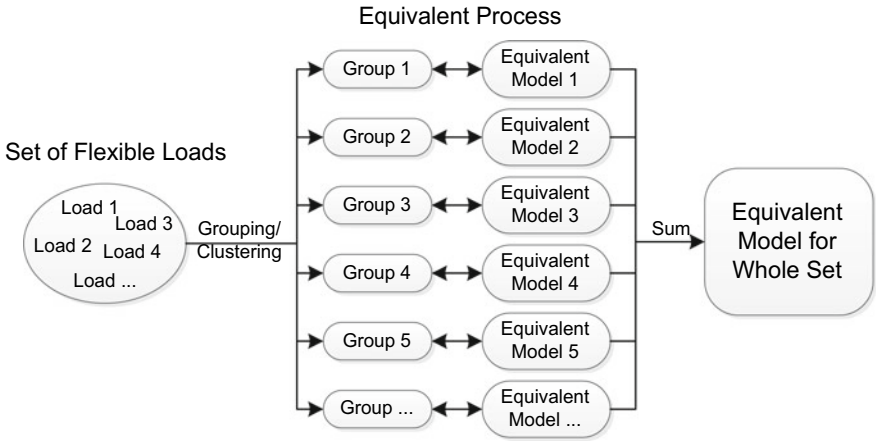


Fig. 7.1 Process of grouping equivalence

described by set theory. Then the equivalent models for the groups with continuous loads or ON/OFF loads are discussed. Last, we analyze how to merge and reduce the number of groups. Finally we establish the equivalent model for a merged group.

7.2.2.1 Definition of the Equivalent Model

The equivalent model is established when the loads in a group are thought of as a whole and what's concerned is their total power. The group can then be replaced by a model that has the same total power; this is the equivalent model. Considering the total power of the group is actually a feasible domain, equivalence means the two models have the same feasible domain of the total power. Therefore, the equivalent model can be defined by set theory as follows:

Definition 7.1 Consider load model A, B of the following form:
Load A:

$$\mathcal{P}_{\mathrm{A}} = \{P_{\mathrm{A}}(\cdot) | H_{\mathrm{A}}(P_{\mathrm{A}}(\cdot), Y_{\mathrm{A}}) = 0, G_{\mathrm{A}}(P_{\mathrm{A}}(\cdot), Y_{\mathrm{A}}) \leq 0\} \tag{7.9}$$

Load B:

$$\mathcal{P}_{\mathrm{B}} = \{P_{\mathrm{B}}(\cdot) | H_{\mathrm{B}}(P_{\mathrm{B}}(\cdot), Y_{\mathrm{B}}) = 0, G_{\mathrm{B}}(P_{\mathrm{B}}(\cdot), Y_{\mathrm{B}}) \leq 0\} \tag{7.10}$$

where $P_A(\cdot)$, $P_B(\cdot)$ is the total power vector of load model A, B; P_A, P_B is the set of all the feasible power vector $P_A(\cdot)$, $P_B(\cdot)$; H_A, H_B are the equality constraints of load model A, B, and G_A, G_B are the inequality constraints; and Y_A, Y_B are the variables in constraints except $P_A(\cdot)$, $P_B(\cdot)$.

If $P_B = P_A$, the load model B is called the equivalent model of load model A, and load model B is said to be equivalent to load model A.

In Definition 7.1, the equivalence relationship between load models is symmetrical. If B is equivalent to A, A must be equivalent to B.

However, the equivalence relationship in Definition 7.1 is so strict that it allows for no deviation in equivalence, which may lead to difficulties in establishing the equivalent model. However, an equivalent model with a small deviation is actually acceptable. Hence, an equivalence relationship with a certain deviation is proposed as follows:

Definition 7.2 Consider load model A, B of the form in (7.9)–(7.10). Their feasible domains of the total power are *PA* and *PB*. If for some $\varepsilon \geq 0$ the following condition (7.11)–(7.12) is fulfilled:

$$\forall x \in \mathcal{P}_{\mathrm{A}}, \exists y \in \mathcal{P}_{\mathrm{B}}, \|y - x\| \leq \varepsilon \tag{7.11}$$

$$\forall y \in \mathcal{P}_{\mathrm{B}}, \exists x \in \mathcal{P}_{\mathrm{A}}, \|x - y\| \leq \varepsilon \tag{7.12}$$

Then load model B is called the ε-equivalent model of load model A, and load model B is said to be ε-equivalent to load model A.

Also, the ε-equivalence relationship is symmetrical. If $\varepsilon = 0$, the ε-equivalence will turn into an equivalence relationship in Definition 7.1, which indicates that Definition 7.2 is an extension of Definition 7.1. In addition, the $\|\cdot\|_\infty$ is used for Definition 7.2 in this paper, because it represents the upper limit of absolute deviations at all times.

7.2.2.2 Equivalent Model for Continuous Loads

For a group of *N* continuous flexible loads with the same parameters, the equivalent model is established by the following theorem:

Theorem 7.1 *For a group of N continuous flexible loads with the same parameters of* P_g^{rate}, α_g, β_g, T_g, *the model* P_g *is in the following form:*

$$\mathcal{P}_g = \left\{ P_g(\cdot) \left| \begin{array}{c} P_g(t) = \sum_{i=1}^{N} P_i(t), \ \forall t \\ 0 \leq P_i(t) \leq P_g^{rate}, \forall i, \forall t \\ P_i(t) = 0, \ \ t \notin [\alpha_g, \beta_g], \forall i \\ \sum_t P_i(t) = T_g P_g^{rate}, \forall i \end{array} \right. \right\} \tag{7.13}$$

The model P_g *could be equivalent to the model* $P_{g,e}$, *whose form is as follows:*

$$\mathcal{P}_{g,e} = \left\{ P_{g,e}(\cdot) \left| \begin{array}{c} 0 \le P_{g,e}(t) \le N P_g^{rate}, \forall t \\ P_{g,e}(t) = 0, \quad t \notin [\alpha_g, \beta_g] \\ \sum_t P_{g,e}(t) = N T_g P_g^{rate} \end{array} \right. \right\} \tag{7.14}$$

Proof According to Definition 7.1, P_g is equivalent to the model $P_{g,e}$, meaning:

$$\mathcal{P}_g = \mathcal{P}_{g,e} \tag{7.15}$$

Proving (7.15) is equivalent to proving (7.16) and (7.17):

$$\forall P_g(\cdot) \in \mathcal{P}_g, \quad \exists P_{g,e}(\cdot) = P_g(\cdot), \quad P_{g,e}(\cdot) \in \mathcal{P}_{g,e} \tag{7.16}$$

$$\forall P_{g,e}(\cdot) \in \mathcal{P}_{g,e}, \quad \exists P_g(\cdot) = P_{g,e}(\cdot), \quad P_g(\cdot) \in \mathcal{P}_g \tag{7.17}$$

① To prove (7.16), we can sum the constraints for $P_i(t)$ in P_g, and then the constraints for $P_g(t)$ can be obtained. The constraints for $P_g(t)$ are the same as the constraints in $P_{g,e}$. Hence, for any $P_g(\cdot) \in P_g$, just let the $P_{g,e}(\cdot) = P_g(\cdot)$, and the $P_{g,e}(\cdot)$ must fulfill the constraints in $P_{g,e}$, so there is always $P_g(\cdot) = P_{g,e}(\cdot) \in P_{g,e}$. Equation (7.16) is fulfilled.

② To prove (7.17), for any $P_{g,e}(t)$ in $P_{g,e}$, we can let every $P_i(t)$ is equal to $P_{g,e}(t)$/N. It's easy to verify that every $P_i(t)$ satisfies the constraints in P_g. Therefore, for the sum of these $P_i(t)$s, the $P_g(\cdot)$, there is $P_g(\cdot) \in P_g$. And meanwhile, $P_g(\cdot) = P_{g,e}(\cdot)$. Equation (7.17) is fulfilled.

Theorem 7.1 is thereby verified, and establishes an accurate equivalent model for continuous loads. The model has linear constraints on energy and power, which makes it approximate to a hydropower plant model.

7.2.2.3 Equivalent Model for ON/OFF Loads

The equivalent model for ON/OFF flexible loads has a wider range of applications. Loads with continuously adjustable power usually need to be controlled by electronic power devices, whereas devices needed for ON/OFF loads can be controlled with cheaper and simpler smart switches. Continuous loads can also operate in ON/OFF mode, indicating that the equivalent model for ON/OFF loads can be applied to continuous loads.

(1) Discrete equivalent model

For a group of N ON/OFF flexible loads with the same parameters, a discrete equivalent model is established by the following theorem:

Theorem 7.2 *For a group of N ON/OFF flexible loads with the same parameters of* P_g^{rate}, α_g, β_g, T_g, *if its model* P_g *is in the following form:*

$$\mathcal{P}_g = \left\{ P_g(\cdot) \left| \begin{array}{c} P_g(t) = \sum_{i=1}^{N} P_i(t), \forall t \\ P_i(t) = u_i(t) P_g^{\text{rate}}, \forall t, \forall i \\ u_i(t) \in \{0, 1\}, \forall t, \forall i \\ P_i(t) = 0, t \notin [\alpha_g, \beta_g], \forall i \\ \sum_t P_i(t) = T_g P_g^{\text{rate}}, \forall i \end{array} \right. \right\} \tag{7.18}$$

Then the model P_g could be equivalent to the model $P_{g,e}$ with a discrete power range, whose form is as follows:

$$\mathcal{P}_{g,e} = \left\{ P_{g,e}(\cdot) \left| \begin{array}{c} P_{g,e}(t) \in \{0, P_g^{vars}, 2P_g^{\gamma ae}, 3P_g^{var}, \ldots, NP_g^{vae}\} \\ P_{g,e}(t) = 0, t \notin [\alpha_g, \beta_g] \\ \sum_t P_{g,e}(t) = NT_g P_g^{rae} \end{array} \right. \right\} \tag{7.19}$$

Proof P_g equivalent to the model $P_{g,e}$ means (7.16) and (7.17) are fulfilled.

Just like the proof in Theorem 7.1, Eq. (7.16) is easy to verify by summing the constraints for P_i(t) in P_g. Therefore, we focus on the proof of (7.17) as follows.

The key to proving (7.17) is that for any $P_{g,e}(\cdot)$ in $P_{g,e}$, if there is an algorithm that can always find a $P_g(\cdot)$ that belongs to P_g and is equal to $P_{g,e}(\cdot)$, Eq. (7.17) must be fulfilled. To find the $P_g(\cdot)$, the main function of the algorithm is to disaggregate the $P_{g,e}(\cdot)$ into a sum of the P_i(t)s with their constraints fulfilled. If it is possible, the $P_g(\cdot)$, as the sum of the P_i(t)s must be belong to P_g and be equal to $P_{g,e}(\cdot)$.

Algorithm 1 is thus proposed to find the $P_g(\cdot)$. First the group of ON/OFF loads and $P_{g,e}(\cdot) \in P_{g,e}$ are described. Consider a group of N ON/OFF loads with the same parameters of P_g^{rate}, α_g, β_g, T_g, the energy that the group of loads need for working are described as the form of Fig. 7.2.

In Theorem 7.2, N and T_g can be any positive integers, but in Fig. 7.2, $N = 5$ and $T_g = 4$, are used as the example integers to illustrate Algorithm 1. The horizontal axis represents necessary work intervals for loads in the group, and they are the same as T_g based on conditions of Theorem 7.2. The vertical axis represents the power of

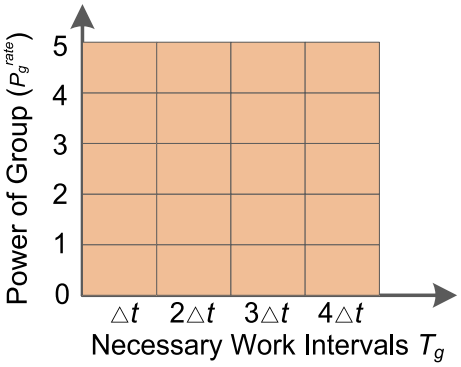


Fig. 7.2 Energy of ON/OFF flexible loads for operation

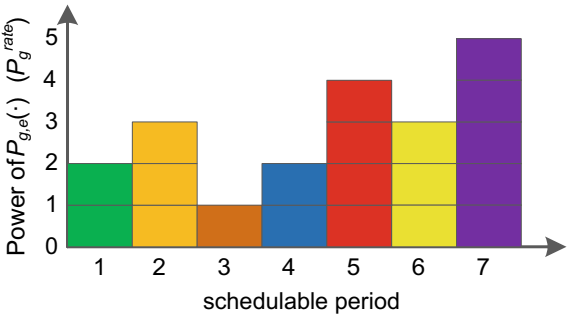


Fig. 7.3 A power curve of the discrete equivalent model

the group, with the unit of P_g^{rate}. Energy blocks of all loads are shown in Fig. 7.2, and each row of blocks represents the energy used by one load.

The $P_{g,e}(\cdot) \in P_{g,e}$ can be described in Fig. 7.3, as an illustrative example of Algorithm 1.

In Fig. 7.3, the horizontal axis represents schedulable time intervals, and the vertical axis represents the power of the $P_{g,e}(\cdot)$. The $P_{g,e}(\cdot)$ has the same energy as Fig. 7.2, and its power is discrete and no more than $N P_g^{rate}$. For clarity, energy blocks at each time are marked by a unique color.

In order to find the $P_g(\cdot)$ that fulfills (7.17), the objective of Algorithm 1 is to allocate energy blocks in Figs. 7.3 to 7.2 in an operational way for loads, which means the allocation must fulfill the constraints for the loads. Since the numbers of energy blocks in Figs. 7.2 and 7.3 are the same, after allocation, the number of energy blocks allocated to each load is T_g; hence (7.3) is always fulfilled. Besides, since the power of each load is 0 or P_g^{rate}, this requires the energy blocks at a certain time cannot allocate a load of more than one block. If the allocation is completed, the $P_i(t)$ for every load can be found, and then the $P_g(\cdot) = P_{g,e}(\cdot)$ can be found.

Algorithm 1 for allocation is as follows:

① Choose the energy blocks of a certain time that has not been allocated in Fig. 7.3.
② Start the allocation for Fig. 7.2 from the first unallocated column of energy blocks on the right side of the figure, in order from top to bottom. If the column is partly allocated, start the allocation from the top of the unallocated energy blocks of the column. If the energy blocks in this column are all allocated during the allocation, turn to the left column next to this column and continue allocation from top to bottom.

Repeat steps one and two until all the energy blocks in Fig. 7.3 are allocated. Because the number of energy blocks in Figs. 7.2 and 7.3 are the same, the allocation will be finished for Figs. 7.2 and 7.3 at the same time. The result of allocation is shown in Fig. 7.4

It's easy to verify that in Algorithm 1, energy blocks at a certain time do not allocate to a load more than one block. In Algorithm 1, allocation is carried out from right to left and from top to bottom. Under this order of allocation, only when $P_{g,e}(t)$ is greater than $N P_g^{rate}$, will there be a load obtaining two energy blocks of $P_{g,e}(t)$.

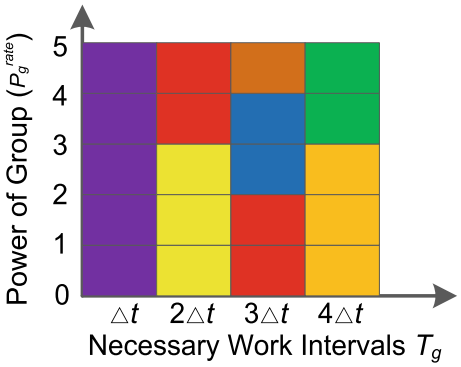


Fig. 7.4 Result of allocation from power curve of the discrete equivalent model to energy of ON/OFF flexible loads

However in (7.19) $P_{g,e}(t)$ can never be greater than NP_g^{rate}, which ensures a load will not obtain more than one energy block at a certain time; therefore the allocation is operational.

Thus, (7.17) in Theorem 7.2 is verified by Algorithm 1, and then Theorem 7.2 is verified.

Theorem 7.2 establishes an accurate equivalent model for ON/OFF loads, and hence reveals their flexibility. The Theorem indicates that even if the flexible loads can only work in ON/OFF mode, a group of loads with the same parameters can work like a hydropower plant model with a discrete power range, which means the group of ON/OFF loads and the hydropower plant have similar flexibility.

(2) Continuous equivalent model

Through the ε-equivalence in Definition 7.2, a continuous equivalent model for ON/OFF flexible loads is established by the following Theorem:

Theorem 7.3 *For a group of N ON/OFF flexible loads with the same parameters of* P_g^{rate}, α_g, β_g, T_g, *if its model* P_g *is in the form of (7.18), there is a model* $P_{g,e}$ *that is* ε*-equivalent to* P_g, *and in* ε*-equivalence, the* $\|\cdot\| = \|\cdot\|\infty$ *and* $\varepsilon = P_g^{rate}$. *The form of* $P_{g,e}$ *is (7.14).*

Proof $P_{g,e}$ ε-equivalent to the model P_g means:

$$\forall P_g(\cdot) \in \mathcal{P}_g, \quad \exists P_{g,e}(\cdot) \in \mathcal{P}_{g,e}, \quad \left\| P_{g,e}(\cdot) - P_g(\cdot) \right\|_\infty \leq \varepsilon = P_g^{rate} \tag{7.20}$$

$$\forall P_{g,e}(\cdot) \in \mathcal{P}_{g,e}, \quad \exists P_g(\cdot) \in \mathcal{P}_g, \quad \left\| P_g(\cdot) - P_{g,e}(\cdot) \right\|_\infty \leq \varepsilon = P_g^{rate} \tag{7.21}$$

For (7.20), it is obvious that $P_g \subseteq P_{g,e}$, hence:

$$\forall P_g(\cdot) \in \mathcal{P}_g, \quad \exists P_{g,e}(\cdot) \in \mathcal{P}_{g,e}, \quad P_{g,e}(\cdot) = P_g(\cdot) \tag{7.22}$$

$$\therefore \left\| P_{g,e}(\cdot) - P_g(\cdot) \right\|_\infty = 0 \leq \varepsilon = P_g^{rate} \tag{7.23}$$

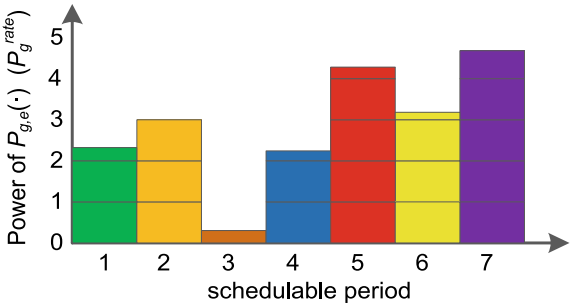


Fig. 7.5 A power curve of the continuous equivalent model

Therefore, (7.20) is verified.

The key to proving (7.21) is the same as in Theorem 7.2, which is to propose an algorithm that can always find the $P_g(\cdot)$ which fulfills (7.21). In order to find the $P_g(\cdot)$, Algorithm 2, as an extension of Algorithm 1, is proposed as follows.

First the group of ON/OFF loads and the $P_{g,e}(\cdot) \in P_{g,e}$ need to be described. In Theorem 7.3, the group of ON/OFF loads is the same as that in Theorem 7.2, so the energy they need is also described by Fig. 7.2. Then, the $P_{g,e}(\cdot) \in P_{g,e}$ can be described in Fig. 7.5. The $P_{g,e}(\cdot)$ in Fig. 7.5 has the same energy as Fig. 7.2, but its power range is a continuous interval $[0, N P_g^{rate}]$.

Algorithm 2 for allocation is as follows:

① Choose the energy blocks of a certain time that has not been allocated in Fig. 7.5.
② Start the allocation for Fig. 7.2 from right to left and from top to bottom. Because $P_{g,e}(t)$ is real number, there will be an energy block in Fig. 7.2 partly occupied by $P_{g,e}(t)$. The next allocation starts from an unoccupied part of this energy block. The results of allocation are shown in Fig. 7.6.

The energy blocks in Fig. 7.6 have more than one color, representing the need for loads to operate at different times with partial power of P_g^{rate}, which is inoperable for an ON/OFF load. Therefore, the next step is to reallocate this energy block to one of the $P_{g,e}(t)$s that occupy it.

③ Reallocate the energy blocks that have more than one color based on the following rules:

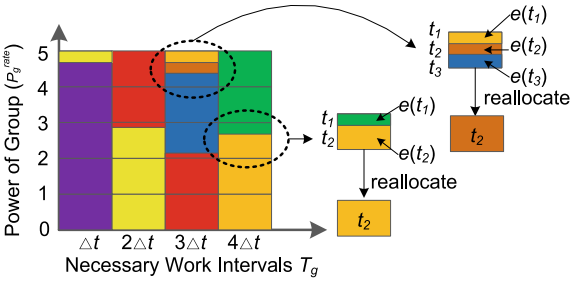


Fig. 7.6 Allocation and reallocation from power curve of the continuous equivalent model to energy of ON/OFF flexible loads

Rule A: If the energy block has two colors, from top to bottom, record the times corresponding to these two colors as t_1, t_2; and the energy they occupy in the energy block as $e(t_1)$, $e(t_2)$. If $e(t_1) \geq 0.5\ P_g^{rate}\ \Delta t$, the energy block is reallocated to t_1, otherwise the energy block is reallocated to t_2. $P_g^{rate}\ \Delta t$ is the energy of this block.

Rule B: If the energy block has K colors, $K \geq 3$, from top to bottom, record the times corresponding to these K colors as t_1, t_2, …, t_{K-1}, t_K; and the energy they occupy in the energy block as $e(t_1)$, e(t_2), …, $e(t_{K-1})$, $e(t_K)$. If $e(t_1) \geq 0.5\ P_g^{rate}\ \Delta t$, the energy block is reallocated to t_1. Otherwise, if $e(t_K) > 0.5\ P_g^{rate}\ \Delta t$, the energy block is reallocated to t_K. If the above two conditions are not satisfied, find the maximum of $e(t_2)$, …, $e(t_{K-1})$, and then the energy block is reallocated to the time corresponding to the maximum.

The reallocating rules are both shown in Fig. 7.6 with two different examples. After reallocating, the P_i(t) for every load can be found, and then the $P_g(\cdot)$ is found.

After $P_g(\cdot)$ is obtained, deviation between $P_{g,e}$(t) and P_g(t) in Algorithm 2 must be analyzed. The deviation appears in step three, reallocating. In step three, if $P_{g,e}$(t) at a certain time appears in at least two energy blocks, it will have two part deviations, separately at the start and end of $P_{g,e}$(t). According to Rule A and Rule B, the deviation of each part is no more than $0.5\ P_g^{rate}$, so the total deviation for $P_{g,e}$(t) is no more than P_g^{rate}. For $P_{g,e}(t)$ appearing in only one energy block, there must be $P_{g,e}(\mathrm{t}) \leq P_g^{rate}$. If the energy block is reallocated to time t, the deviation is $|P_g^{rate} - P_{g,e}(\mathrm{t})| \leq P_g^{rate}$; otherwise, the deviation is $|P_{g,e}(t) - 0| \leq P_g^{rate}$. Therefore, for any $P_{g,e}$(t), the deviation between $P_{g,e}(t)$ and $P_g(t)$ is no more than P_g^{rate}.

Then the operability for $P_g(\cdot)$ obtained in Algorithm 2 needs to be verified. Here the counter-evidence method is used for analysis. If there is a certain time t, two energy blocks of $P_g(t)$ are allocated to a load i. Then for the $P_{g,e}$(t) corresponding to P_g(t), based on the allocation order in step two, each load except for load i would have an energy block fully occupied by $P_{g,e}(t)$ for a total of $N - 1$ energy blocks. Meanwhile, for two energy blocks of $P_g(t)$ that are allocated to load i, in the block of the left column, $P_{g,e}(t)$ must occupy no less than half of the block, and in the block of the right column, $P_{g,e}(t)$ must occupy more than half of the block. Add these energy blocks together, and then the whole $P_{g,e}(t)$ is larger than $N P_g^{rate}$, which is against the constraints for $P_{g,e}(t)$ in (7.20). Therefore, for any $P_g(t)$, its energy blocks cannot allocate more than one block to a load. $P_g(\cdot)$ is operational and $P_g(\cdot) \in P_g$.

Through analysis of deviation and feasibility, a $P_g(\cdot)$ that fulfills (7.21) is found by Algorithm 2. Hence, (7.21) and Theorem 7.3 are verified.

Theorem 7.3 establishes a continuous equivalent model for ON/OFF loads. Although there are some deviations in the equivalent model, the deviations would be negligible if N is large enough. Besides, the equivalent model established is linear and contains only continuous variables, which makes it easy to add to optimization models without increasing their complexity.

7.2.2.4 Group Merger

Through merging some of the groups with similar parameters, the number of groups is reduced, as well the numbers of variables and constraints. Based on the above proof process, P_g^{rate} has the least impact on the proof. Therefore, the groups with different P_g^{rate} s could be merged into a larger group, in which the rate powers of loads are diverse. The equivalent model of the merged group is established based on Theorems 7.4 and 7.5, respectively for continuous loads and ON/OFF loads.

Theorem 7.4 *Consider a group of N continuous flexible loads with same parameters of α_g, β_g, T_g, and rate power Pirate for any load i is a positive number as the parameter. If its model P_g is in the following form:*

$$\mathcal{P}_g = \left\{ P_g(\cdot) \left| \begin{array}{c} P_g(t) = \sum_{i=1}^{N} P_i(t), \forall t \\ 0 \le P_i(t) \le P_i^{\text{rate}}, \forall i, \forall t \\ P_i(t) = 0, t \notin \left[\alpha_g, \beta_g\right], \forall i \\ \sum_t P_i(t) = T_g P_i^{rat}, \forall i \end{array} \right. \right\} \tag{7.24}$$

The model $\mathcal{P}_g$ could be equivalent to the model $\mathcal{P}_{g,e}$, whose form is as follows:

$$\mathcal{P}_{g,e} = \left\{ P_{g,e}(\cdot) \left| \begin{array}{c} 0 \le P_{g,e}(t) \le \sum_i P_i^{rate}, \forall t \\ P_{g,e}(t) = 0, \quad t \notin [\alpha_g, \beta_g] \\ \sum_t P_{g,e}(t) = \sum_i T_g P_i^{rate} \end{array} \right. \right\} \tag{7.25}$$

Proof $\mathcal{P}_g$ equivalent to the model $\mathcal{P}_{g,e}$ means the (7.16) and (7.17) are fulfilled. The proof for (7.16) is exactly the same as in Theorem 7.1. For (7.17), let every $P_i(t)$ equal to $P_{g,e}(t)\ P_g^{rate}/\sum_i P_g^{rate}$, and then it can be verified that Eq. (7.17) is fulfilled, same as in Theorem 7.1. Hence the Theorem 7.4 is verified.

Theorem 7.5 *Consider a group of N ON/OFF flexible loads with the same parameters of α_g, β_g, T_g, while P_i^{rate} for load i can be different. If its model $\mathcal{P}_g$ is in the following form:*

$$\mathcal{P}_g = \left\{ P_g(\cdot) \left| \begin{array}{c} P_g(t) = \sum_{i=1}^{N} P_i(t), \quad \forall t \\ P_i(t) = u_i(t) P_i^{rate}, \forall t, \forall i \\ u_i(t) \in \{0, 1\}, \quad \forall t, \forall i \\ P_i(t) = 0, \quad t \notin [\alpha_g, \beta_g], \forall i \\ \sum_t P_i(t) = T_g P_i^{rate}, \forall i \end{array} \right. \right\} \tag{7.26}$$

There is a model $\mathcal{P}_{g,e}$ that is ε-equivalent to $\mathcal{P}_g$, and in the ε-equivalence, the $\|\cdot\| = \|\cdot\|_\infty$ and $\varepsilon = max(P_i^{rate})$. The form of $\mathcal{P}_{g,e}$ is as (7.25).

Proof $\mathcal{P}_g$ equivalent to the model $\mathcal{P}_{g,e}$ means that (7.20) and (7.21) are fulfilled. The proof for (7.20) is the same as in Theorem 7.3. To prove (7.21), for any $P_{g,e}(t)$ in $\mathcal{P}_{g,e}$, just apply proposed Algorithm 2, and then obtain the $P_i(t)$ for every load, as well as the $P_g(\cdot)$.

After $P_g(\cdot)$ is obtained, deviation analysis between $P_{g,e}(t)$ and $P_g(t)$ and operability analysis for $P_g(\cdot)$ are needed. Theorem 7.3 proves that if $P_{g,e}(t)$ appears in at least two energy blocks, it will have two parts of deviations, and the deviation of each part is no more than $0.5P_i^{rate}$, where P_i^{rate} is the rate power of the reallocated energy block. Due to P_i^{rate} being $\leq \max(P_i^{rate})$ for any load i, the total deviation for $P_{g,e}(t)$ is no more than $\max(P_i^{rate})$. For $P_{g,e}(t)$ appearing in only one energy block, Theorem 7.3 proves that the deviation is no more than P_i^{rate}, therefore the deviation is no more than $\max(P_i^{rate})$. Hence, for any $P_{g,e}(t)$, the deviation between $P_{g,e}(t)$ and $P_g(t)$ is no more than $\max(P_i^{rate})$. The operability analysis for $P_g(\cdot)$ is exactly the same as in Theorem 7.3. Theorem 7.5 is thus verified.

In Theorem 7.5, as long as $\max(P_i^{rate})$ is not too large, the continuous equivalent model can significantly reduce the number of groups, and ensure a certain limit of equivalence deviations.

7.2.3 Scheduling with Equivalent Aggregated Model

The key to scheduling with the equivalent aggregated model is to reasonably group a set of flexible loads. There are two factors that should be considered for loads grouping, namely the distribution of load parameters and the scheduling requirements of the power system.

According to the Theorems in Section 7.3, the equivalent model is established based on a group of loads with the same or similar parameters. Therefore, the distribution of load parameters is very important for grouping and the parameters of the equivalent model.

The requirements of system scheduling also need to be taken into account. In general, a flexible load may have a wide scheduling period, but after system scheduling, the load will only work in the valley of the system load. Hence, the schedulable period outside the valley of the system load has little effect on scheduling of the load. Thus, when building a group, the $[\alpha_g, \beta_g]$ of the group can be set in the valley of the system load, instead of the whole schedulable period. A group built in this way will have a smaller schedulable period, which allows more loads to work in the period, and to be assigned to this group. Then, as each group becomes larger, the number of groups becomes smaller. Therefore, the specific range in which the flexible loads mainly work needs to be calculated, as the Reference Scheduling Period used for grouping criteria.

In the rest of this Section, the process to calculate the Reference Scheduling Period is illustrated, along with the method to group flexible loads based on the Reference

Scheduling Period, and load parameters distribution is proposed. The approach to scheduling with the equivalent models of groups is also presented.

7.2.3.1 Reference Scheduling Period

In order to obtain a possible load-operating period as the Reference Scheduling Period, a relaxation model of the set of flexible loads is used for preliminary scheduling. Consider a set Ω of flexible loads in form of (7.5)–(7.7), where α_i, β_i separately obey a probability distribution, and the range of α_i is $[\alpha_{p1}, \alpha_{p2}]$, while the range of β_i is $[\beta_{p1}, \beta_{p2}]$. The relaxation model $\mathcal{P}_r$ of the set is described by (7.27):

$$\mathcal{P}_r = \left\{ P_r(\cdot) \left| \begin{array}{c} 0 \le P_r(t) \le \sum\limits_i P_i^{rate}, \forall t \\ P_r(t) = 0, \quad t \notin [\alpha_r, \beta_r] \\ \sum\limits_t P_r(t) = \sum\limits_i T_i P_i^{rate} \end{array} \right. \right\} \tag{7.27}$$

where $\alpha_r = \alpha_{p1}, \beta_r = \beta_{p2}$. Without loss of generality, it's assumed that $\alpha_{p1} < \alpha_{p2}, \beta_{p1} < \beta_{p2}, \alpha_{p1} < \beta_{p2}$; and if the domain of α_i or β_i includes $+\infty$ or $-\infty$, the quantiles can be used to obtain the range in which most of the probability is concentrated.

Consider a scheduling optimization with a set Ω of flexible loads, described by (7.28):

$$\min = F\left(\sum_{i\in\Omega} P_i(\cdot), Y\right), \quad s.t.\ Y \in \mathcal{Y},\ P_i(\cdot) \in \mathcal{P}_i, \forall i \tag{7.28}$$

where Y is the other optimization variable in scheduling, and its feasible domain is $\mathcal{Y}$.

The sum of $P_i(\cdot)$ can be replaced with the relaxation model, as in (7.29):

$$\min = F(P_r(\cdot), Y), \quad s.t.\ Y \in \mathcal{Y},\ P_r(\cdot) \in \mathcal{P}_r \tag{7.29}$$

The scheduling optimization (7.29) can be solved by conventional algorithms because the relaxation model is a linear model which does not increase complexity. After the $P_r(\cdot)$ is obtained, the period of $P_r(\cdot) > 0$ is the best operating time for flexible loads based on the scheduling, which means the Reference Scheduling Period.

7.2.3.2 Grouping

After obtaining the Reference Scheduling Period, the parameters of groups need to be determined, and then the loads in set Ω can be grouped.

(1) Setting of Groups

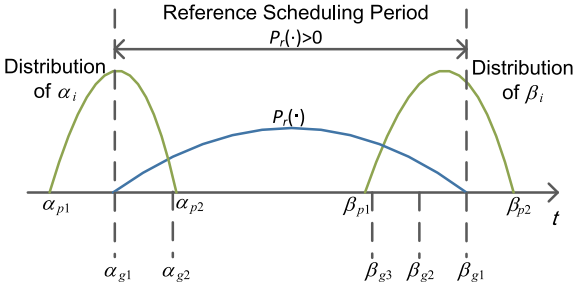


Fig. 7.7 Decision grouping criteria based on Reference Scheduling Period

According to Section 7.3, the key parameters for a group are α_g, β_g, T_g. Hence, the main goal of groups setting is to find the range of α_g, β_g, T_g.

The range of T_g is made of the unique T_i values of all the loads; it consists of some discrete values. The method to decide the range of α_g, β_g is shown in Fig. 7.7.

For a range of α_g, obtain the value from the start of the Reference Scheduling Period, then rearward at regular intervals, until α_{p2}, the end of distribution of α_i. For a range of β_g, obtain the value from the end of Reference Scheduling Period, then forward at regular intervals, until β_{p1}, the start of distribution of β_i. With these rules, in Fig. 7.7, the range of α_g is $\{\alpha_{g1}, \alpha_{g2}\}$, and the range of β_g is $\{\beta_{g1}, \beta_{g2}, \beta_{g3}\}$.

The parameters for groups can thus be set based on the range of α_g, β_g, T_g. After setting groups, the group whose $T_g > \beta_g - \alpha_g + 1$ needs to be deleted, because these parameters will cause the internal loads to have insufficient time to complete their work. In addition, an Unclassified Group must be established to accommodate the loads that cannot be assigned to the above groups.

(2) Grouping Loads

For each load i with α_i, β_i, T_i in set Ω, if the parameters of the group it belongs to are $\alpha_{i,g}$, $\beta_{i,g}$, $T_{i,g}$, the following steps can be used for grouping:

① As for $\alpha_{i,g}$, it's the value that fulfills: $\alpha_{i,g} \in$ range of α_g, $\alpha_{i,g} \geq \alpha_i$, and $\alpha_{i,g}$ is as small as possible.
② As for $\beta_{i,g}$, it's the value that fulfills: $\beta_{i,g} \in$ range of β_g, $\beta_{i,g} \leq \beta_i$, and $\beta_{i,g}$ is as big as possible.
③ As for $T_{i,g}$, $T_{i,g} = T_i$.
④ If there is no $\alpha_{i,g}$ or $\beta_{i,g}$ or $T_{i,g}$ that fulfills the above conditions, the load i is assigned to the Unclassified Group, or if $\alpha_{i,g}$, $\beta_{i,g}$ and $T_{i,g}$ all exist, find the group with the parameters of $\alpha_{i,g}$, $\beta_{i,g}$, $T_{i,g}$ in groups that are set in Section IV.B. If the group exits, the load i is assigned to this group. Otherwise, the load i is assigned to the Unclassified Group.

The condition, $\alpha_{i,g} \geq \alpha_i$, $\beta_{i,g} \leq \beta_i$ are used to ensure the $[\alpha_{i,g}, \beta_{i,g}]$ is the subinterval of $[\alpha_i, \beta_i]$, so the load i can work in $[\alpha_{i,g}, \beta_{i,g}]$. And, the $[\alpha_{i,g}, \beta_{i,g}]$ should be as big as possible to maintain the flexibility of load i, so $\alpha_{i,g}$ should small and $\beta_{i,g}$ should be big. Afterwards, the load is classified into the one of groups established in Section IV.B

based on $\alpha_{i,g}$, $\beta_{i,g}$ and $T_{i,g}$, and the ungrouped loads are placed in the Unclassified Group.

After loads grouping, groups having only a few loads can be deleted to reduce the number of groups. A threshold can be set for deleting, and the group whose number of loads is less than threshold can be deleted. The loads of deleted groups are assigned to the Unclassified Group.

Optimization with Groups

After grouping, the loads in scheduling optimization can be replaced by equivalent models of groups. Through replacing, the scheduling optimization in (7.28) can be described as follows:

$$\min = F\left(\sum_{g\in G} P_{g,e}(\cdot) + \sum_{i\in U} P_i(\cdot), Y\right) \tag{7.30}$$

$$s.t.\ Y \in \mathcal{Y},\ \ P_{g,e}(\cdot) \in \mathcal{P}_{g,e}, \forall g \in \mathcal{G},\ \ P_i(\cdot) \in \mathcal{P}_i, \forall i \in \mathcal{U} \tag{7.31}$$

where $P_{g,e}(\cdot)$ is the power vector of the equivalent model of group g; $\mathcal{G}$ is the set of all groups except the Unclassified Group; and $\mathcal{U}$ is the Unclassified Group.

In order to further simplify the scheduling model, the loads in the Unclassified Group are scheduled first through a heuristic algorithm, and then the scheduling optimization is solved. Although the loads in the Unclassified Group occupy a very small part of the total flexible loads, they bring many more variables and constraints than the other groups, because these loads are described by individual models. Hence, a heuristic algorithm is used to schedule these loads first, and then scheduling optimization is solved with equivalent models of groups. Although the result of the heuristic algorithm may have some deviations from the optimal solution in the first step, the deviations can be largely eliminated when scheduling with equivalent models of groups in the next step. In this way, the heuristic algorithm can reduce the number of variables and constraints, and the optimality of the solution is guaranteed by scheduling with equivalent models. The heuristic algorithm used in this paper is to place each load just in the valley of the total load. Then the scheduling optimization with equivalent models can be solved by conventional algorithms.

After obtaining the $P_{g,e}(\cdot)$ of the group, the $P_{g,e}(\cdot)$ can be disaggregated to the $P_i(\cdot)$ of each load inside. Then the actual power of the group $P_g(\cdot)$ can also be calculated as the sum of $P_i(\cdot)$. Finally, the actual value of objective function could be calculated with $P_i(\cdot)$ of every load.

The key to scheduling with the equivalent aggregated model is to reasonably group a set of flexible loads. There are two factors that should be considered for loads grouping, namely the distribution of load parameters and the scheduling requirements of the power system.

According to the Theorems in Sect. 7.3, the equivalent model is established based on a group of loads with the same or similar parameters. Therefore, the distribution of load parameters is very important for grouping and the parameters of the equivalent model.

The requirements of system scheduling also need to be taken into account. In general, a flexible load may have a wide scheduling period, but after system scheduling, the load will only work in the valley of the system load. Hence, the schedulable period outside the valley of the system load has little effect on scheduling of the load. Thus, when building a group, the $[\alpha_g, \beta_g]$ of the group can be set in the valley of the system load, instead of the whole schedulable period. A group built in this way will have a smaller schedulable period, which allows more loads to work in the period, and to be assigned to this group. Then, as each group becomes larger, the number of groups becomes smaller. Therefore, the specific range in which the flexible loads mainly work needs to be calculated, as the Reference Scheduling Period used for grouping criteria.

7.3 Large Scale EV Aggregation

7.3.1 Market Framework

A microgrid is chosen as the platform for PEVs optimal bidding into electricity market in this paper, as shown in Fig. 7.8. The microgrid incorporates the dispatchable region of EV aggregation, conventional and renewable energy resources, controllable loads, and micro turbines to meet the users' demand. By forming such dispatchable region, the EV aggregator does not need to expose the detailed information to MG, but only provide a projection-based feasible region depicting the available charging and discharging capability.

In this paper, in order to fully develop PEVs' flexible regulation capability, we consider the MG operator participating in a two-settlement electricity market, which includes two successive trading floors: the day-ahead market and the balancing market. Considering the relatively small capacity, the MG is assumed as a price-taker. In the day-ahead market, MG operator is supposed to submit hourly energy non-priced (quantity-only) bids according to forecasted market prices of day-ahead market and balancing market. In the balancing market, MG operator inform the system operator of their deviation from the contracted energy in the day-ahead market, which is settled at the imbalance price.

In this paper, a dual pricing mechanism is applied for the balancing market, which has been widely used in many European countries, such as Denmark, Spain, etc. The pricing mechanism of the dual pricing is presented as follows:

$$\lambda^{+} = \begin{cases} \lambda_{DA}, & \text{if } \lambda_{RT} \geq \lambda_{DA} \\ \lambda_{RT}, & \text{otherwise} \end{cases} \tag{7.32}$$

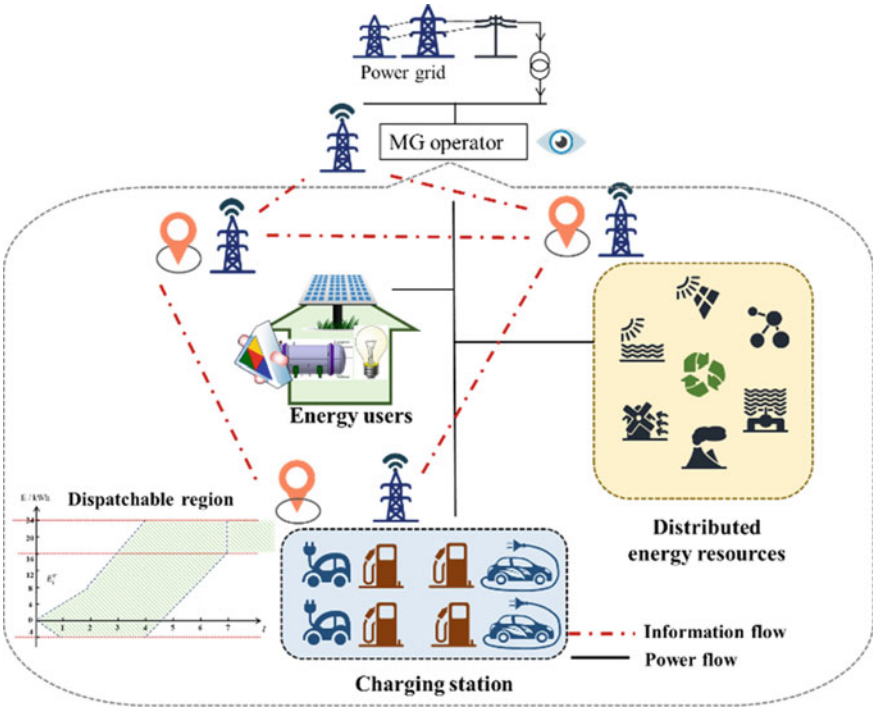


Fig. 7.8 The framework of an microgrid

$$\lambda^{-} = \begin{cases} \lambda_{RT}, & \text{if } \lambda_{RT} \geq \lambda_{DA} \\ \lambda_{DA}, & \text{otherwise} \end{cases} \tag{7.33}$$

Under a dual pricing mechanism, the energy deviation of the selected bids in day-ahead market is priced differently depending on the imbalance sign. If the imbalance created by MG operator is in the same direction of the overall system, which exacerbates the system imbalance, is priced at the balancing market price. In contrast, deviation in the opposite direction as the overall system imbalance is settled at the day-ahead market price. More details of the dual pricing mechanism can be found in [6].

7.3.2 Aggregate Model of Electric Vehicle Fleets

In this section, we establish the dispatchable region of large-scale EVs aggregation based on the individual EV model.

7.3.2.1 Individual Model of Electric Vehicle

The behavior of each PEV can be characterized by a set of stochastic parameters, including arrival time, departure time, initial state of charge (SOC) of battery and the required energy before departure. The individual charging and discharging model of EV i is formulated as follows:

$$0 \le P_t^{EV,C} \le \varphi_{i,t}^{EV} P_{\max}^{EV,C}, \forall t, \tag{7.34}$$

$$0 \le P_t^{EV,D} \le \varphi_{i,t}^{EV} P_{\max}^{EV,D}, \forall t, \tag{7.35}$$

$$\varphi_{i,t}^{EV} = \begin{cases} 0, & t < t_i^a \text{ or } t > t_i^d, \\ 1, & t_i^a \le t \le t_i^d. \end{cases} \tag{7.36}$$

$$SOC_{i,t}^{EV} = SOC_{i,t-1}^{EV} + \frac{1}{E_{i,capacity}^{EV}} \left(\psi_i^{EVC} P_{i,t-1}^{EVC} - P_{i,t-1}^{EVD} / \psi_i^{EVD} \right), \forall t, \tag{7.37}$$

$$SOC_{i,t_i^a}^{EV} = SOC_i^a, \forall t, \tag{7.38}$$

$$SOC_{i,t_i^d}^{EV} \ge SOC_i^d, \forall t, \tag{7.39}$$

$$SOC_{i,\min}^{EV} \le SOC_{i,t}^{EV} \le 1, \forall t, \tag{7.40}$$

where constraints (7.34)–(7.36) limit the charging and discharging power of an EV. In constraints (7.37), the SOC of an EV is related to the charging and discharging energy during a time slot. Constraints (7.38)–(7.40) show the initial SOC of an EV is set to the SOC at arrival time, and the SOC in departure time must satisfy the required energy of the EV battery. To preserve battery life, a safe minimum threshold of EV battery $SOC_{i,min}^{EV}$ is introduced.

Therefore, the dispatchable region of an individual EV can be characterized by its maximum charging and discharging power, and cumulative energy limits, which is shown in Fig. 7.9.

In practice, when the expected parking duration for an EV is not long enough to reach the target SOC, there is no dispatchable region for this kind of EV, and this EV is regarded as an uncontrollable load with the charging power equal to $P_{i,max}^{EV,C}$.

7.3.2.2 Dispatchable Region Modeling of Electric Vehicle Fleets

The total capacity of the EV aggregation is related to the number of EVs parked, which is time-varying. Besides, the SOC of individual EV cannot be simply added to depict the SOC of EV aggregation. These will make it difficult for MG operator to calculate the equivalent SOC of the EV aggregation in the market bidding. To address

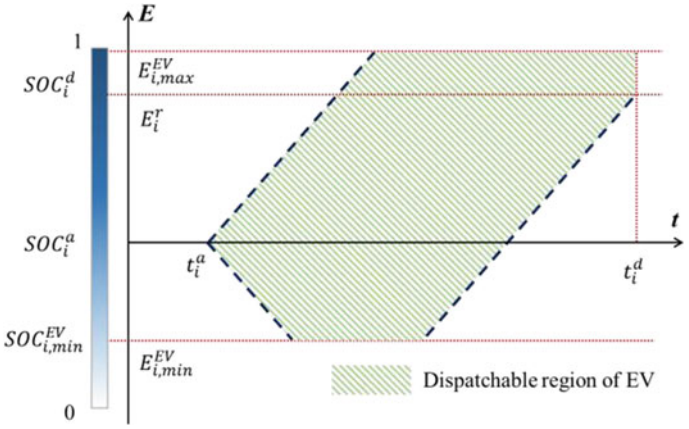


Fig. 7.9 The dispatchable region of an individual EV

this problem, firstly, the maximum energy consumption and production as well as the required energy of each EV are evaluated, and then we replace the SOC constraints in individual PEV model by the cumulative energy constraints, the aggregate model of EV aggregation is then formulated as follows:

$$0 \le P_t^{EVA,C} \le \sum_{i \in \Phi^{EV}} \varphi_{i,t}^{EV} P_{i,\max}^{EV,C}, \forall i, t, \tag{7.41}$$

$$0 \le P_t^{EVA,D} \le \sum_{i \in \Phi^{EV}} \varphi_{i,t}^{EV} P_{i,\max}^{EV,D}, \forall i, t, \tag{7.42}$$

$$\varphi_{i,t}^{EV} = \begin{cases} 0, \ t < t_i^a \ or \ t > t_i^d, \\ 1, \ t_i^a \le t \le t_i^d. \end{cases}, \forall i, \tag{7.43}$$

$$E_t^{EVA} = E_{t-1}^{EVA} + P_{t-1}^{EVA,C} - P_{t-1}^{EVA,D}, \forall i, t, \tag{7.44}$$

$$\sum_{i \in \Phi^{EV}} \underline{E}_{i,t}^{EV} \le E_t^{EVA} \le \sum_{i \in \Phi^{EV}} \overline{E}_{i,t}^{EV}, \forall i, t, \tag{7.45}$$

$$\overline{E}_{i,t}^{EV} = \begin{cases} 0, \ 0 \le t \le t_i^a \\ \min(P_{\max}^{EV,C}(t - t_i^a), E_{i,\max}^{EVC}), t_i^a < t \end{cases}, \forall i, \tag{7.46}$$

$$\underline{E}_{i,t}^{EV} = \begin{cases} 0, \ 0 \le t \le t_i^a \\ \max(P_{\max}^{EV,D}(t_i^a - t), E_{i,\max}^{EVD}, E_i^r - P_{\max}^{EV,C}(t_i^d - t), t_i^a < t \le t_i^d, \forall i, \\ E_i^r, t_i^d < t \end{cases} \tag{7.47}$$

$$E_{i,\max}^{EVC} = (1 - SOC_i^a) E_{i,capacity}^{EV} / \psi_i^{EVC}, \forall i, \tag{7.48}$$

Table 7.1 Parameters of electric vehicles

EV	t^a	t^d	SOC_i^a	$P_{i,max}^{EV,C}$ (kW)	$P_{i,max}^{EV,D}$ (kW)	$E_{i,capacity}^{EV}$ (kWh)
1	0	6	0.2	4	4	20
2	2	7	0.6	4	4	20

$$E_{i,\max}^{EVD} = (SOC_i^a - SOC_{i,\min}^{EV}) \cdot E_{i,capacity}^{EV} \cdot \psi_i^{EVD}, \forall i, \tag{7.49}$$

$$E_i^r = (SOC_i^d - SOC_i^a)/\psi_i^{EVC}, \forall i, \tag{7.50}$$

In this model, constraints (7.41)–(7.43) show that the charging and discharging power of the EV aggregation are restricted summation of maximum power of all parked EVs. In constraints (7.44), the cumulative energy of EV aggregation is related to the charging and discharging energy during a time slot. Constraints (7.45) the cumulative energy limit of the EV aggregation. In constraints (7.46) and (7.47), the analytical expression of the lower and upper boundaries of EV i are presented by piecewise functions. The maximum energy consumption and production as well as the required energy are calculated in constraints (7.48)–(7.50). Note that the power and cumulative energy limits in the proposed PEV aggregation model can consider the charging and discharging efficiency of different EVs.

In practice, the charging station can collect the information of each individual PEV, including arrival and departure time as well as the initial and target SOC. Based on the proposed aggregate model, the aggregated information of the EV aggregation including power and cumulative energy limit is then sent to the MG operator. Note that, the aggregated information is independent of the numbers of PEVs, it is much simpler and more precise to predict the dispatchable region of the EV aggregation than to predict the behaviour of individual EV.

Illustrative Example

A toy example is presented in this subsection to better illustrate the proposed dispatchable region formation approach. We analyze the dispatchable region of the EV aggregation consisting of two electric vehicles, the parameters of the EVs are shown in Table 7.1. For simplicity, $SOC_{i,min}^{EV}$ and SOC_i^d of both EVs are assumed to be 0 and 0.8, respectively.

The dispatchable region of the individual EV is shown in Fig. 7.10a, b, respectively. Based on the proposed aggregated dispatchable region formation approach, the dispatchable region of the aggregation including two EVs is shown in Fig. 7.11.

As can be seen in Fig. 7.11, the lower cumulative energy bound of the EV aggregation is higher than that of the EV2 at the time slot 4. This is because, the EV aggregation needs to meet the driving demand of EV1, as a result, the lower bound of the cumulative energy becomes higher in Fig. 7.11. In this way, the dispatchable

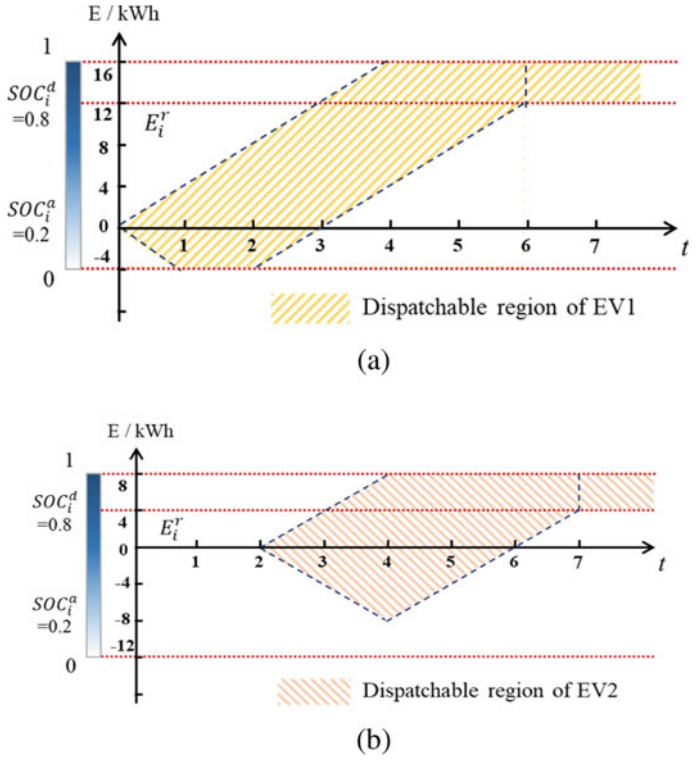


Fig. 7.10 The dispatchable regions of the individual EV1 and EV2

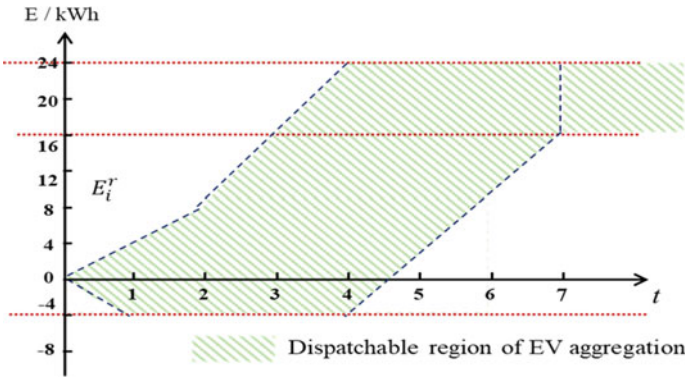


Fig. 7.11 The dispatchable region of the EV aggregation (EV1, EV2)

region of the two individual EVs is represented by an equivalent aggregation region, and the analytical expression can be derived from the aggregate model (7.41)–(7.47).

This small example shows that the proposed aggregate model can effectively represent the dispatchable regions of EV aggregation with a single set of constraints, which is independent of the EV population size. In addition, the example also indicates that the charging and discharging capacity of the aggregated EVs can be comprehensively considered in the dispatchable region of EV aggregation.

7.3.3 Model of Optimal Bidding Strategy of Microgrid

To verify the accuracy and effectiveness of the EV aggregation dispatch region formed in the previous section, an optimal market bidding strategy of an MG is formulated by a two-stage risk-constrained stochastic programming model considering the participation in both day-ahead and balancing markets.

7.3.3.1 Objective Function

The objective of the risk-constrained bidding strategy for an MG consists of four blocks: expected revenue in day-ahead and balancing market, operation cost of MG, and conditional value-at-risk (CVaR), shown as follows:

$$\underset{\mathbf{X}_{DA},\mathbf{X}_{RT}}{\text{Max}} \quad R_{DA} + R_{RT} - C^{OP} + \beta \text{CVaR}, \tag{7.51}$$

$$R_{DA} = \sum_{t \in \Phi^{\text{T}}} \lambda_{DA,t} P_{DA,t}^{MG}, \forall t, \tag{7.52}$$

$$R_{RT} = \sum_{t \in \Phi^{T}} \sum_{s \in \Phi^{S}} \gamma_s (\lambda_{s,t}^{+} P_{s,t}^{\text{MG}+} - \lambda_{s,t}^{-} P_{s,t}^{MG-}), \forall s, t, \tag{7.53}$$

where the decision variables are denoted by $\mathbf{X}^{DA}$ and $\mathbf{X}^{RT}$, representing the DA- and RT-related variables, respectively, including DA energy bids of MG $P_{DA,t}^{MG}$, positive and negative imbalance of MG in balancing market $P_{s,t}^{MG+}$, $P_{s,t}^{MG-}$. The expected revenue in day-ahead and balancing market is presented in (7.52) and (7.53). $\beta \in [0, \infty)$ is the weighting factor to materialize the tradeoff between expected revenue and CVaR. As β increases, the MG operator becomes more risk aversion, that is, the expected revenue term becomes less significant with respect to the CVaR term.

Constraints

(1) The constraints of renewable generators and micro turbines

$$0 \le P_{DA,i,t}^{\mathrm{WT}}, P_{DA,i,t}^{\mathrm{WT}} \le P_{i,\max}^{\mathrm{WT}}, \forall i, t, \tag{7.54}$$

$$0 \le P_{DA,i,t}^{\mathrm{PV}} \le P_{i,\max}^{\mathrm{PV}}, \forall i, t, \tag{7.55}$$

$$0 \le P_{DA,i,t}^{\mathrm{MT}} \le P_{i,\max}^{\mathrm{MT}}, \forall i, t, \tag{7.56}$$

$$0 \le P_{s,i,t}^{\mathrm{MT}} \le P_{i,\max}^{\mathrm{MT}}, \forall s, i, t, \tag{7.57}$$

$$-\delta_i^{ADJ} P_{i,\max}^{MT} \le P_{DA,i,t}^{MT,DA} - P_{s,i,t}^{MT} \le \delta_i^{ADJ} P_{i,\max}^{MT}, \forall s, i, t, \tag{7.58}$$

where constraints (7.54)–(7.57), the bidding power of the WT, PV, MT are restricted by its installed capacity. The real-time adjustments of MTs are bounded by (7.58).

(2) The constraints of local load demands

The local power demand of MG is classified into two categories: inelastic loads and elastic loads. The elastic loads are controlled by the demand response program.

$$\sum_{t \in \Phi^{\mathrm{T}}} P_{DA,i,t}^{L} = E^{MG,L}, \forall i, t, \tag{7.59}$$

$$P_{t,\min}^{L} \le P_{DA,i,t}^{L} \le P_{t,\max}^{L}, \forall i, t, \tag{7.60}$$

$$P_{t,\min}^{L} \le P_{s,i,t}^{L} \le P_{t,\max}^{L} \forall s, i, t, \tag{7.61}$$

where constraints (7.59) correspond to the prescribed total energy requirement. Constraints (7.60) and (7.61) show the upper and lower bound of the elastic loads.

(3) The constraints of the EV aggregation

$$0 \le P_{DA,t}^{EVA,C} \le P_{DA,t,\max}^{EVA,C}, \forall t, \tag{7.62}$$

$$0 \le P_{DA,t}^{EVA,D} \le P_{DA,t,\max}^{EVA,D}, \forall t, \tag{7.63}$$

$$0 \le P_{s,t}^{EVA,C} \le P_{s,t,\max}^{EVA,C}, \forall s, t, \tag{7.64}$$

$$0 \le P_{s,t}^{EVA,D} \le P_{s,t,\max}^{EVA,D}, \forall s, t, \tag{7.65}$$

$$E_{DA,t}^{EVA} = E_{DA,t-1}^{EVA} + P_{DA,t-1}^{EVA,C} - P_{DA,t-1}^{EVA,D}, \forall t, \tag{7.66}$$

$$E_{s,t}^{EVA} = E_{s,t-1}^{EVA} + P_{s,t-1}^{EVA,C} - P_{s,t-1}^{EVA,D}, \forall t, \tag{7.67}$$

$$E_{DA,t,\min}^{EVA} \le \sum_{1}^{t} E_{DA,t}^{EVA} \le E_{DA,t,\max}^{EVA}, \forall t, \tag{7.68}$$

$$E_{s,t,\min}^{EVA} \le \sum_{1}^{t} E_{s,t}^{EVA} \le E_{s,t,\max}^{EVA}, \forall t, \tag{7.69}$$

where constraints (7.62)–(7.65) show the limit of the charging and discharging power of the EV aggregation, respectively. In (7.66) and (7.67), the energy consumption of the EV aggregation is related to the charging and discharging energy during a time slot. Constraints (7.68) and (7.69) show the energy limit of the EV aggregation at time slot t in scenario s.

(4) The constraints of the microgrid

$$\begin{aligned} P_{DA,t}^{MG} = & \sum_{i\in\Phi^{\mathrm{WT}}} P_{DA,i,t}^{\mathrm{WT}} + \sum_{i\in\Phi^{\mathrm{PV}}} P_{DA,i,t}^{\mathrm{PV}} + \sum_{i\in\Phi^{MT}} P_{DA,i,t}^{\mathrm{MT}} \\ & - P_{DA,t}^{\mathrm{EVA,C}} + P_{DA,t}^{\mathrm{EVA,D}} - \sum_{i\in\Phi^{L}} P_{DA,i,t}^{\mathrm{L}} \end{aligned}, \forall i, t, \tag{7.70}$$

$$\begin{aligned} P_{s,t}^{MG} = & \sum_{i\in\Phi^{\mathrm{WT}}} P_{s,i,t}^{\mathrm{WT}} + \sum_{i\in\Phi^{\mathrm{PV}}} P_{s,i,t}^{\mathrm{PV}} + \sum_{i\in\Phi^{MT}} P_{s,i,t}^{\mathrm{MT}} \\ & - P_{s,t}^{\mathrm{EVA,C}} + P_{s,t}^{\mathrm{EVA,D}} - \sum_{i\in\Phi^{L}} P_{s,i,t}^{\mathrm{L}} \end{aligned}, \forall s, i, t, \tag{7.71}$$

$$P_{DA,t}^{MG}, P_{s,t}^{MG} \in [-P_{\max}^{\mathrm{MG}}, P_{\max}^{\mathrm{MG}}], \forall s, t, \tag{7.72}$$

$$P_{s,t}^{\mathrm{MG}} - P_{DA,t}^{\mathrm{MG}} = P_{s,t}^{\mathrm{MG+}} - P_{s,t}^{\mathrm{MG-}}, \forall s, t, \tag{7.73}$$

$$0 \le P_{s,t}^{\mathrm{MG+}} \le P_{s,t}^{\mathrm{MG}}, \forall s, t, \tag{7.74}$$

$$0 \le P_{s,t}^{\mathrm{MG-}} \le P_{\max}^{\mathrm{MG}}, \forall s, t, \tag{7.75}$$

$$C^{OP} = \sum_{t\in\Phi^{T}} \sum_{s\in\Phi^{S}} \left(\sum_{i\in\Phi^{MT}} \gamma_s c_i^{MT} P_{s,i,t}^{MT} + c^{bdc} P_{s,t}^{\mathrm{EVA,D}} \right), \forall s, i, t. \tag{7.76}$$

The day-ahead bidding of MG for energy is supported by the DERs in the MG, as shown in (7.70) and (7.71). In (7.72), the bidding of MG is restricted by the capacity of tie-line between MG and main grid. Constraints (7.73)–(7.75) correspond to the positive and negative imbalance of MG. Constraints (7.76) shows the operational costs of the MG with the fuel costs of the MTs and the battery degradation cost of EVs.

(5) The constraints of the CVaR

CVaR is the expected value of $(1-\alpha)\%$ of the lowest profits at a given confidence interval α. CVaR can be calculated by the following constraints (7.77)–(7.79).

$$\mathrm{CVaR} = \xi - \frac{1}{1-\alpha}\sum_{s\in\Phi^S}\gamma_s\eta_s, \forall s, \tag{7.77}$$

$$\xi - (R_{DA} + R_{RT} - C^{OP}) - \eta_s \geq 0, \forall s, \tag{7.78}$$

$$\eta_s \geq 0, \forall s. \tag{7.79}$$

7.4 Case Study

7.4.1 *Large Scale Interruptible and Shiftable Load Aggregation*

In this study, a case of day-ahead scheduling for the power system with large-scale flexible loads is analyzed. The scheduling period is set as 24 h and the duration of a time interval $\Delta t =$ is 0.25 h. The day-ahead load forecast of base load comes from historical data of the UK power system. Then 1×106 electric vehicles that need to charge at night are considered as the flexible loads that participate in the day-ahead scheduling. These flexible loads work in ON/OFF mode, and make up the set Ω. The distributions of their parameters are shown in Table 7.2.

In Table 7.2, the α_i, β_i and E_i follow the Gaussian distribution and P_i^{rate} follows the Uniform distribution. With parameters of E_i and P_i^{rate}, the necessary work intervals T_i can be calculated and rounded to an integer.

In terms of power supply, a quadratic function is used to express the total cost of system's power supply:

$$C = \sum_{t=1}^{T}(aP_{all}(t)^2 + bP_{all}(t) + c) \tag{7.80}$$

Table 7.2 Distribution of flexible loads' parameters

Parameters	Distributions
α_i	N (18:30 h, (1 h)^2)
β_i	N (7:30 h, (1 h)^2)
E_i	N (21 kWh, (3 kWh)^2)
P_i^{rate}	U (6 kW, 12 kW)

where $P_{all}(t)$ is the total load of the power system, and $a = 2 \times 10^{-4}$, $b = 0.3$, $c = 15{,}000$. The goal of day-ahead scheduling is to minimize the total cost of the power supply by scheduling flexible loads. All calculations are completed in MATLAB 2014a, on a computer with Intel i7 8750H and 16G RAM.

7.4.1.1 Result of Scheduling

This section presents the process of scheduling with the equivalent aggregated model, and compares its scheduling result with two other scenarios.

In order to schedule with the equivalent aggregated model, first the Reference Scheduling Period needs to be calculated. Since the relaxation model needs to cover almost all of the loads, its schedulable period is set as [15:30, 10:30], based on the 3σ criterion. Its upper limit of rate power is the sum of ${P_i}^{rate}$ of all flexible loads, and its energy is the sum of $T_i {P_i}^{rate} \Delta t$ of all the loads. Solve day-ahead scheduling with the relaxation model, and then a relaxed solution $P_r(\cdot)$ can be obtained as Fig. 7.12 shows.

The time for $P_r(\cdot) > 0$ is [23:45, 7:45], which is the Reference Scheduling Period. Afterwards, the grouping criteria are established based on the Reference Scheduling Period, as shown in Table 7.3.

Because almost every $\alpha_i \leq$ 23:45, the range of α_g, as the start of schedulable period for all the groups is 23:45. However, distribution for β_i intersects with the Reference Scheduling Period. Hence, the range of β_g, as the end of the schedulable period for groups, starts from 7:45 and ends at 4:45, for more than 99% $\beta_i \geq$ 4:45.

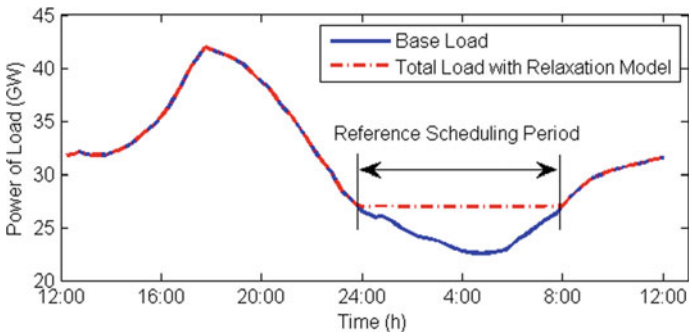


Fig. 7.12 Total load after scheduling with relaxation model

Table 7.3 Grouping criteria

Criteria	Value
Range of α_g	{23:45}
Range of β_g	{7:45, 7:15, 6:45, 6:15, 5:45, 5:15, 4:45}
Range of T_g	{4, 5, …, 21, 22}

The time interval in the range of β_g is set as $2\Delta t$ to reduce numbers of groups. The range of T_g consists of the unique values of T_i.

Hence, there are 133 groups initially, with an Unclassified Group, and then the flexible loads are assigned into the groups. In this case, the threshold for deleting the group is 0.05% number of flexible loads. After grouping, 83 groups are obtained, and they cover 991,029 flexible loads, leaving 8,971 flexible loads in the Unclassified Group. The parameters of some groups are listed in Table 7.4.

After grouping, a heuristic algorithm is applied to the Unclassified Group, and then conventional algorithms can solve scheduling with equivalent models of groups. Then the $P_i(\cdot)$ of each load and the $P_g(\cdot)$ of each group can be obtained based on Theorem 7.5, and the power of the total load can be also be acquired.

The result of scheduling with an equivalent aggregated model is compared with the following two scenarios:

Scenario 1: Early finish. In this scenario, all the loads will start working at the start of their schedulable period, and work continuously until their works are completed. Hence all the loads finish their works as early as possible.

Table 7.4 Parameters of some groups

Index	α_g	β_g	T_g	Number of loads	$\sum P_i^{rate}$ (MW)
1	23:45	7:45	5	1137	12.72
2	23:45	7:45	6	8594	95.19
3	23:45	7:45	7	30,711	334.10
4	23:45	7:45	8	55,268	579.21
5	23:45	7:45	9	61,839	608.80
...	...	...	...	...	...
31	23:45	6:45	8	28,957	303.43
32	23:45	6:45	9	32,094	316.62
33	23:45	6:45	10	28,120	253.89
34	23:45	6:45	11	22,738	187.69
35	23:45	6:45	12	18,107	138.34
...	...	...	...	...	...
51	23:45	6:15	16	2136	13.84
52	23:45	6:15	17	945	6.01
53	23:45	5:45	6	1836	20.38
54	23:45	5:45	7	6622	72.05
55	23:45	5:45	8	12,402	130.01
...	...	...	...	...	...
81	23:45	4:45	12	1185	9.05
82	23:45	4:45	13	893	6.40
83	23:45	4:45	14	610	4.17

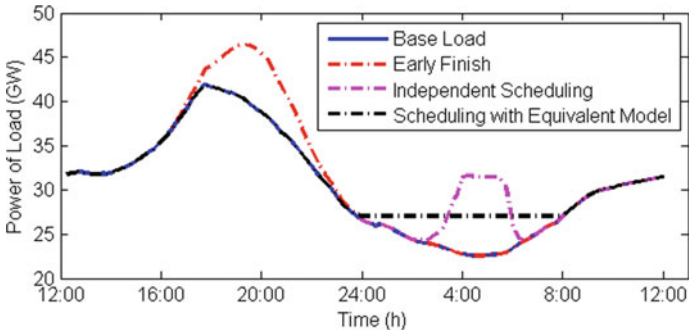


Fig. 7.13 Total load in different scenarios of scheduling

Scenario 2: Independent scheduling. In this scenario, the system operator broadcasts an electricity price to all the flexible loads, and the electricity price is positively related to the forecast of the base load. Then each load is scheduled independently to minimize its own electricity costs.

Figure 7.13 shows the total load of the system in different scenarios.

Scenario 1 reflects the natural energy demand of the flexible loads. Each load finishes its work as early as possible, which is most beneficial to the user's need. However, the scheduling mode leads the flexible loads to be superimposed on the peak period of the base load, and then results in an increase in peak load. The scheduling result will not only increase the total cost of the power system, but also propose higher requirements on the generation capacity of the system. Scenario 2 represents the operation of flexible loads under non-cooperative independent scheduling. Each flexible load will independently schedule its operation to minimize its cost, according to the electricity price from the system operator. Hence, all loads would work at the valley of the electricity price, and as a result, it forms a load peak at the valley. The problem is hard to solve by adjusting electricity prices. Regardless of how the electricity price is set, as long as the prices received by all flexible loads are the same, then their responses will be similar, and the loads will be concentrated at the valley of the electricity price, where a peak load will be formed. The way to solve the problem is coordinated scheduling. Scheduling with the equivalent aggregated model described in this paper is an approach to load coordinated scheduling. In Fig. 7.9, a scheduling model coordinates the flexible loads well, and the valley of the total load is almost filled into a horizontal line by flexible loads. It illustrates that the proposed equivalent aggregated model is effective in a scheduling model.

7.4.1.2 Deviations of Equivalence

This section analyzes the equivalence deviation of each group. The equivalence deviation is the deviation between $P_g(\cdot)$ of the group and $P_{g,e}(\cdot)$ of its equivalent model. Theorem 7.5 draws a conclusion that, for a group of ON/OFF loads, the

deviation between $P_g(\cdot)$ and $P_{g,e}(\cdot)$ will not exceed the maximum rated power of the loads at any time. Therefore, the equivalence deviation of each group is calculated to verify that conclusion.

Groups with the most loads and the least loads are analyzed as examples, and their equivalence deviations are shown in Figs. 7.14 and 7.15 Deviations of the groups only occur at the time of $P_{g,e}(\cdot) \neq 0$, which is consistent with the proving process in Section 7.3. Further, the absolute deviations of all times do not exceed 12 kW, the maximum rated power of the loads, which verifies the conclusion of Theorem 7.5. Considering the total power of a group is in the range of MW to hundreds of MW, the equivalence deviation is almost negligible.

In order to display the deviations of all groups, the maximum absolute deviation at all times, $\|P_{g,e}(\cdot) - P_g(\cdot)\|_\infty$, is calculated and shown in Fig. 7.16. It's clear that maximum absolute deviations of all groups do not exceed 12 kW, the maximum rated power of the loads. Hence, the result verifies the conclusion of Theorem 7.5.

Deviations between the $\sum P_g(\cdot)$, the total power of groups and $\sum P_{g,e}(\cdot)$, the total power of all equivalent models are shown in Fig. 7.13. According to Theorem 7.5, since the upper limit of the deviation of each group is max (P_i^{rate}), 12 kW, then the upper limit of the total deviation is the max (P_i^{rate}) multiplied by number of groups, 996 kW. But, because the deviations in disaggregation are random, the equivalence deviations of groups cancel each other, and the actual total deviation of all groups is much smaller than 996 kW, as shown in Fig. 7.17. Considering the total power of all groups up to several GW, the total deviation of all equivalent models is almost negligible. Besides, the $P_i(\cdot)$ of each flexible load has been verified to meet its constraints, which illustrates the result for each load is operable.

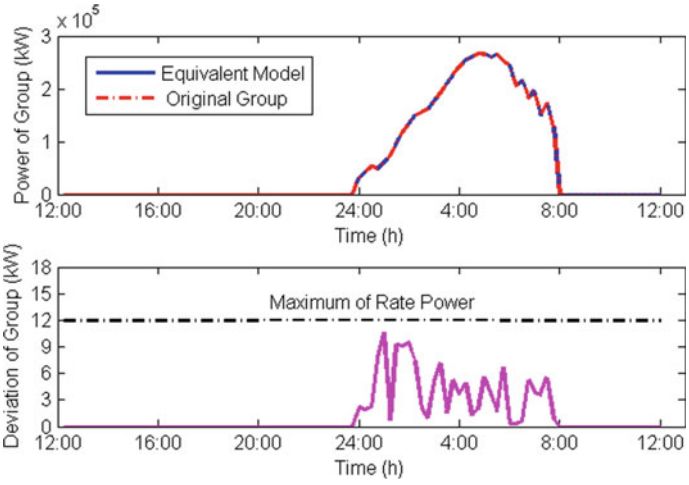


Fig. 7.14 Power and deviation of the group with most loads

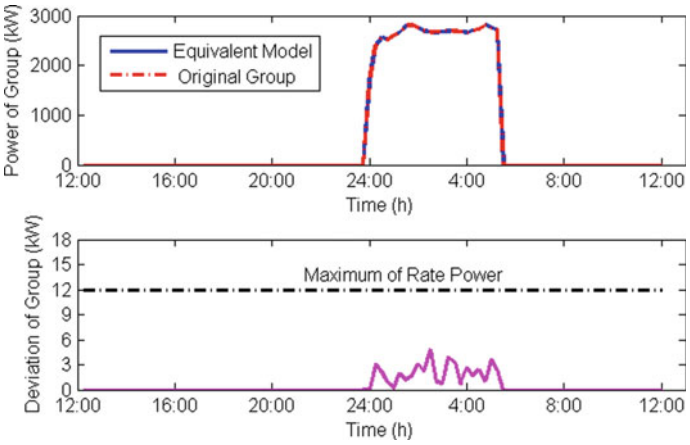


Fig. 7.15 Power and deviation of the group with least loads

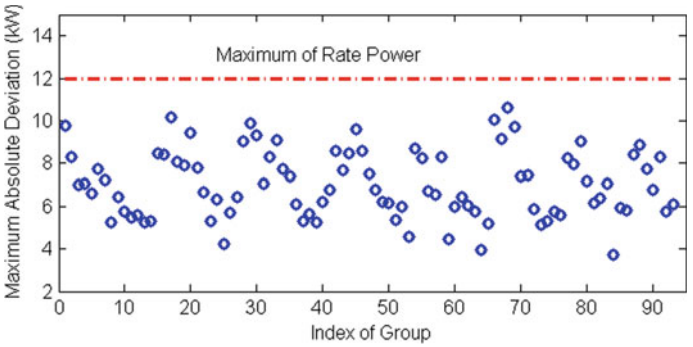


Fig. 7.16 Maximum absolute deviation of all groups

7.4.1.3 Calculation Performance

This section analyzes the calculation performance of the equivalence algorithm, which schedules the equivalent aggregated model, and compares it with the distributed algorithm in [7]. The distributed algorithm sequentially schedules the working time of the load, and repeatedly iterates until the scheduling model converges. Reference [7] applied the distributed algorithm to large-scale flexible loads scheduling and carries out a case study; hence it serves as a comparison algorithm for this paper. The equivalence algorithm and the distributed algorithm are tested on the same hardware and software platform. Besides, in the step of optimization with equivalent models, YALMIP/IPOPT is used as the optimization tool. Other steps are implemented by MATLAB code.

Table 7.5 displays the calculation time for the two algorithms to schedule with one million flexible loads. The calculation time of each step in the equivalence algorithm

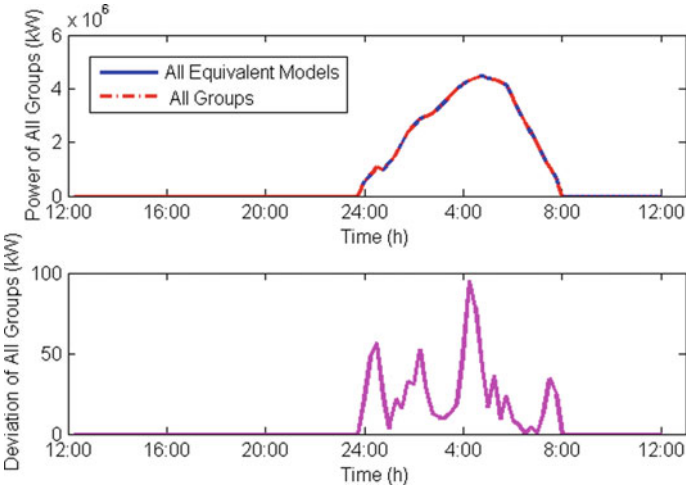


Fig. 7.17 Power and deviation of all groups

is described separately. Overall, the calculation speed of equivalence algorithm is nearly 154 times faster than the distributed algorithm, and the calculation efficiency has increased by more than two-orders of magnitude. The most time-consuming step of the equivalence algorithm is grouping, which accounts for 68% of the time used by the entire algorithm. The reason for the time-consuming step is that a large number of loop calculations are required to group each load. However, since the process of load grouping is independent, the step of grouping can be accelerated by parallel computing. Afterwards, the key step of scheduling, optimization with groups, is solved in about one second. The reason that this step can be solved so quickly is that a large number of loads with discrete variables are equivalent to an equivalent load model with continuous variables and linear constraints, which makes it possible to use conventional algorithms and existing solvers, and hence solve the model efficiently. By contrast, the distributed algorithm needs to repeatedly iterate all the loads in sequence, and the iterative process for the load is performed serially, which results in a relatively slow solving speed.

Figure 7.18 displays the calculation time of the two algorithms under different scales of loads. Under different scales of flexible loads, the proposed equivalence algorithm presents a good calculation speed, and performs scheduling in just tens of seconds, which is much faster than the distributed algorithm. During the process of scheduling, the step of grouping generally takes up more than 60% of the total calculation time, whereas other steps just take a few seconds. The step of grouping can be accelerated by parallel computing, so the time consumption of the equivalence algorithm can be further reduced. In addition, since the equivalent model of the group is linear and continuous, it can be easily embedded in various scheduling models without increasing their complexity, which is also difficult for the distributed algorithm to accomplish.

Table 7.5 Calculation time of two algorithms

Criteria	Time(s)	
	Equivalence algorithm	Distributed algorithm
Reference scheduling period	0.510	\
Grouping	8.070	
Heuristic algorithm for other group	0.665	
Optimization with groups	0.989	
Group power disaggregated	1.580	
Total	11.814	1829.442

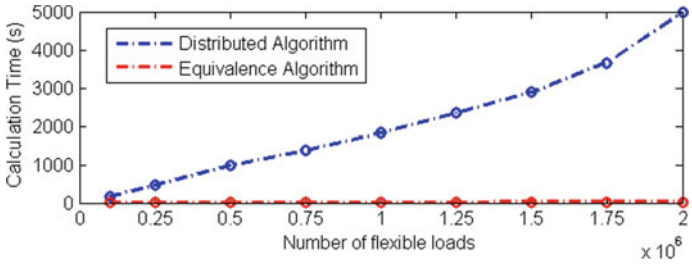


Fig. 7.18 Calculation time of two algorithms

7.4.2 Large Scale Distributed Energy Storage Aggregation

7.4.2.1 Basic Data

A microgrid with 1000 EVs is considered to validate the proposed model. The stochastic variables of PEVs are assumed to obey truncated Gaussian distributions, the parameters are presented in Table 7.6. Note that the arrival time should be earlier than the departure time for each PEV. The wind and solar data are from the NREL database [17]. We adopt 10 price scenarios for both the day-ahead and balancing markets, which are derived from [18]. A confidence level α is assumed to be 0.95 to compute CVaR in all periods. The risk aversion coefficient β is assumed to be 1.0 unless otherwise stated.

To demonstrate the effectiveness and benefits of the proposed framework, four cases are designed for comparisons:

(1) M1, which is the proposed framework with dispatchable region of PEV fleets;
(2) M2, an ideal situation: compared to M1, the charging station has the perfect information of each EV, that is, there is no forecast error of PEVs.

Table 7.6 Parameters of the stochastic variables

Parameters	Mean	Standard deviation	Min	Max
Initial SOC	0.5	0.2	0.2	0.9
Arrival time	7 h	4 h	1 h	18 h
Departure time	18 h	4 h	10 h	24 h

(3) M3, compared to M1, PEVs are modeled in individual-level.
(4) M4, as the benchmark, in which an uncoordinated charging strategy is applied, indicating that PEV fleets are charged at the maximum rate when they arrive.

7.4.2.2 Performance of Different Methods

To evaluate the benefits of the dispatchable region formation of the EV aggregation, the expected revenue of the optimal bidding strategy of MG under four cases are shown in Table 7.7.

As compared in Table 7.7, the MG in M1 can significantly increase its market revenue compared with in M3 and M4. Compared to the M2, which is the ideal situation, the revenue of M1 reached 95.6% of that of M2. In the day-ahead market, the dispatchable characters of PEVs in M1, M2, M3 play a better role, which leads to accommodate more renewable energy, especially for photovoltaic, as a result MG gets a higher revenue, compared with M4. In the balancing market, forming a dispatchable region of EV aggregation in MG bidding can maximize the flexibility potential of EVs due to the high prediction accuracy. Hence, the MG in M1 has a better performance in the balancing market than that in M3 and M4.

Compared to the case M3, in which PEVs are modeled and scheduled in individual level, the dispatchable region of EV aggregation for MG bidding significantly reduces the number of variables as well as the scale of the problem, the computation time is shown in Table 7.8. In this way, both the calculation efficiency and prediction accuracy are largely improved. also shows the expected real time imbalance of M1 and M3 considering different EV numbers in the MG. Note that, the real time imbalance refers to the sum of positive and negative imbalance energy.

As one can observe, with the increase of the EV numbers, for one aspect: the computation time in M3 increases exponentially, while the computation time in M1 hardly changes. This is because, in M1, the dispatchable region of the EV aggregation is restricted by a set of equivalent constraints of charging/discharging power and

Table 7.7 Expect market revenue of the four cases

Revenue\cases	M1	M2	M3	M4
Day-ahead market ($)	777.62	805.41	725.33	596.63
Balancing market ($)	−29.81	−22.85	−60.14	−125.4
Total expected revenue ($)	747.81	782.56	665.19	471.23

Table 7.8 Real-time imbalance and computation time of M1 and M3

EV numbers		100	200	1000	2000
M1	Real-time imbalance (MWh)	3.5	3.2	2.8	3.2
	Computation time (s)	13.52	13.25	15.33	13.92
M3	Real-time imbalance (MWh)	3.7	3.6	3.81	4.7
	Computation time (s)	127.8	394.5	3825.5	9632.0

cumulative energy, which is independent of the EV population size. As a result, the required computation time is short and does not change with the number of EVs. For another, the expected real time imbalance in both cases first decrease and then increase along with the increasing of the EV numbers, and the imbalance energy in M1 is 15% lower than M2 on average. Since the cluster forecasting of the EV fleets guarantees high accuracy in M1, so the real-time imbalance is relatively low. When the numbers of PEVs is small, the uncertainties in MG bidding mainly from the stochastic resources, as the number of PEVs increases, by aid of PEVs providing greater flexibility, real-time imbalance is reduced. However, when there are too many EVs, the uncertainty in MG bidding at this time mainly comes from the stochastic behavior of PEVs. As a result of this, the real time imbalance rises again, and this phenomenon is more significant in M3.

7.4.2.3 Impact of the Risk Aversion Coefficient

Generally speaking, if market members become more risk averse, they prefer to bid in day-ahead markets conservatively. In this subsection, we conduct sensitivity analysis on the risk aversion coefficient β to evaluate the effect of the proposed method on the MG bidding. The evolution of the bidding strategies and expected revenue with risk aversion in M1 and M3 are shown in Fig. 7.8.

By looking at Fig. 7.19, it can be said that the variation of the bidding strategies and the expected market revenue with risk aversion is more pronounced in M3 than in M1. With the increase of β the MG bidding in M3 is more conservative and the expected revenue decline faster, which means that MG bidding in M3 is more influenced by risk aversion. This is because the PEVs in M1 are scheduled based on the dispatchable region, which can guarantee high accuracy and provide greater flexibility for MG operator. MG operator is supposed to make tradeoff between expected market revenue and the CVaR, the efficient frontier of CVaR in M1 and M3 is shown in Fig. 7.20.

The efficient frontier corresponding to case M1 is displaced at the upper right portion, with a higher expected revenue and CVaR compared with the case M3. One can observe that, the variation of both the expected revenue and CVaR with risk aversion is more pronounced for the case M3, the different points of the efficient frontier being far. This confirms that, the MG operator suffer less risk impact in market bidding in M1.

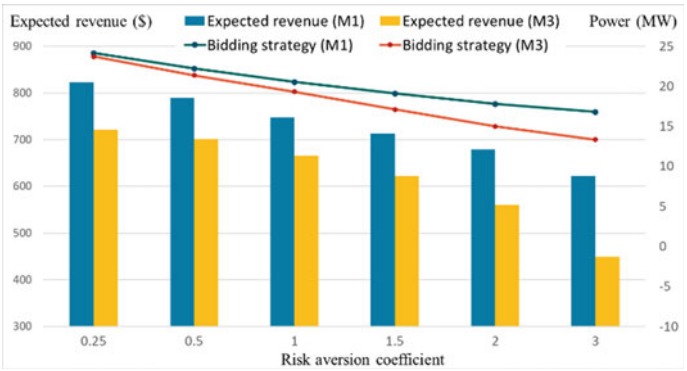


Fig. 7.19 Evolution of the bidding strategies and expected revenue with risk aversion in M1 and M3

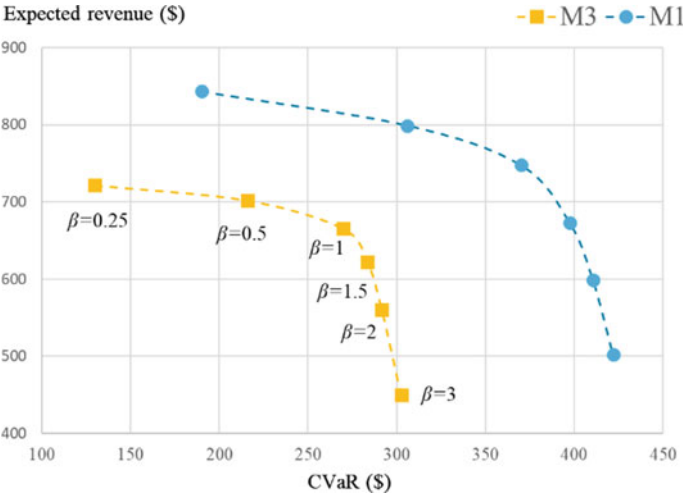


Fig. 7.20 CVaR efficient frontier in M1 and M3

7.4.2.4 Impact of the Balancing Market Prices

To evaluate the adaptability of the proposed bidding model in different market supply and demand situations, penalty factors based on day-ahead prices are introduced in this subsection to simulate different market conditions. Penalty factor are used to determine imbalance settlement prices, e.g. 1.2/0.8 indicate that the negative and positive imbalance prices are 120 and 80%, respectively, of the day-ahead market prices. The expected revenue in the day-ahead and the balancing market under different penalty factors in M1 and M3 are shown in Fig. 7.21.

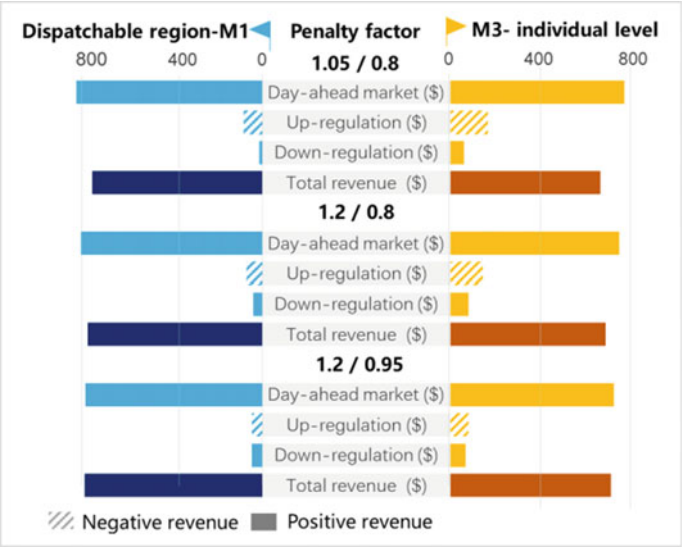


Fig. 7.21 Expected revenue of M1 and M3 under different penalty factors

As one can observe, in both M1 and M3, the expected market revenue with the penalty factor 1.2/0.95 outperforms the other two situations. This is because the PEVs required energy to fulfill the regular driving demand of PEV owners, and the battery degradation cost to some extent affects the economics of discharging, which results PEV aggregation in a lower potential for providing down-regulation. As a result, a high positive imbalance price can lead a higher expected market revenue in MG bidding. In addition, the expected revenue in the balancing market in M1 is generally higher than one in M3, which also demonstrates that the proposed dispatchable region method brings about high forecast accuracy.

7.5 Conclusion

This chapter introduces the aggregation of large-scale distributed flexible resources, which aggregates a large number of flexible loads into a small number of aggregated load models. By reducing the scale of the problem, the model is solved by traditional algorithms. In this chapter, the distributed flexible resources are divided into two parts: demand response (interruptible and shiftable load) and distributed energy storage, and some conclusions are obtained.

Firstly, focusing on the scheduling of the power system with large-scale flexible loads, this chapter proposes an equivalent aggregated model of flexible loads based on grouping equivalence, proves the equivalence between the models, estimates the

upper bound of equivalent deviations, and establishes a scheduling model applying the equivalent aggregated model.

In addition, for the aggregation of distributed energy storage, electric vehicles are selected as the research object, a dispatchable region formation method of the EV aggregation was proposed for MG bidding considering both day-ahead and balancing markets. A detailed model for EV fleets was formulated to dispatchable region in market bidding. The SOC constraint of each EV was replaced by the cumulative energy constraints of aggregation, and the dispatchable region was represented by an equivalent aggregation region, which was restricted by power and cumulative energy limits. The EV aggregation model can effectively reduce the prediction deviation of the dispatchable region.

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Chapter 8
Market Mechanism Design for Enhancing the Flexibility of Power Systems



This chapter takes balancing market mechanism design as an example, to studies the possible impact of balancing market mechanism design on improving power system flexibility. With the increasing proportion of renewable energy in the power system, it is necessary to improve it from the aspect of power market. Due to its short-term trading characteristics, power spot market (balancing market) has become a key link in designing market mechanism to improve flexibility. Taking the participation of wind power suppliers in the trading process of power spot market as an example, due to its inherent uncertainty and volatility, renewable energy such as wind power intensifies the real-time power imbalance in the market, which makes it more difficult to balance the operation of market links and related unbalancing settlement in the spot market. Considering the market environment, it is necessary to analyze and balance the market operation effect through effective methods, and simulate and evaluate the interaction between the behaviors of market members. Firstly, this chapter introduces the framework of balancing market and summarizes the key elements of unbalancing settlement in it, including assessment tolerance margin, programme time unit (PTU) duration and unbalancing power settlement price mechanism. Then two analysis models are proposed. The first is the optimization model for wind power suppliers who can offer strategically to participate in spot market transactions. The second model is based on the ABM method with multi criteria decision analysis (MCDA). The two models seek the optimal design with the minimum total unbalancing cost of the market from different perspectives, and then conduct numerical analysis based on the above models. Finally, the corresponding suggestions are provided for the design method of balancing market mechanism to improve the flexibility of power system.

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M. Zhou et al., *Power System Flexibility*, Power Systems,
https://doi.org/10.1007/978-981-19-9075-5_8

8.1 Introduction

With the increasing proportion of renewable energy connected to the power grid, it is necessary to improve the flexibility of power systems. We must pay attention to and timely deal with the impact of large scale of renewable energy power generation injected into the power grid on the power system. In addition, we can not ignore the important impact of renewable energy grid connection on the main bodies of all links in the power system. From the perspective of power market, the key is how to improve the flexibility of power system from the design of market mechanism. Improving the flexibility of power grid is reflected in updating the existing market mechanism for power system, including designing supporting market mechanism from many aspects, such as the impact of renewable energy participation in market transactions and the interaction behavior of market subjects. This is mainly reflected in the power spot market (or subdivided into balancing market). Due to the short-term trading characteristics of the power spot market, it has a close relationship and significant impact on the flexibility of power system.

Several existing studies have investigated balancing market design. Vandezande et al. [1] provided the first overview of balancing market design options, which divided balancing market design into three parts: balance planning, the provision of balancing services and balance settlement. Van der Veen and Hakvoort [2] present the relevant design space and performance criteria of balancing market design. Kristof et al. [3] presented an assessment of a specific support mechanism exempting balancing responsibilities for wind power producers in Belgium.

In practice, market participants can strategically adapt their market behaviours to respond to different imbalance settlement mechanisms. According to mechanism design theory, one of the basic principles of mechanism design is that a mechanism can incentivize participants to maximize individual interests while achieving the established goals of the designer, and this principle is known as the "incentive compatibility principle". Therefore, a well-designed imbalance settlement mechanism can incentivize market participants to act truthfully regarding energy and price; in this way, the actual balancing cost of the system can be revealed.

However, considering the differences of electricity spot markets in different regions and the construction stages, effective market design evaluation tools are urgently needed to investigate what kind of market design is appropriate, especially for real-time balancing mechanisms. Determining how to design an imbalance settlement mechanism that can consider the interaction between market participants and the market design and guarantee the fairness and effectiveness of balancing market operation remains an open question to be addressed.

Therefore, in the above background, it is necessary to optimize the existing market mechanism design to adapt to system changes. Moreover, it is also necessary to explore the strategy behavior of market members, such as offering strategy, market mechanism and the interaction of market members.

This chapter takes the balancing market design as the investigation object and aims to exploit an effective and applicable market design evaluation method for enhancing

the flexibility of power system. Firstly, the framework of balancing market and the key elements in the design process of unbalancing settlement mechanism are introduced. Then, two models are introduced, namely, the market clearing optimization model with wind power supplier offering strategy and the ABM method with MCDA evaluation. Then, two cases are analyzed respectively which purpose is to explore the strategic offering behavior of wind power suppliers in the balancing market and the interactive behavior among market subjects in the balancing market. The corresponding conclusions are given at the end of this chapter.

8.2 The Framework of Balancing Market

8.2.1 The Framework of Balancing Market

A balancing market consists of three main entities: (1) the system operator (SO): responsible for the operation of the entire power system, including organizing a balancing market, maintaining system power balance and implementing balance settlement; (2) balancing service providers (BSPs): in charge of providing various balancing services to cover any deviation between real-time power demand and that scheduled in advance; and (3) balancing responsible parties (BRPs): bearing the responsibility of incurring any imbalance in a generation and/or consumption portfolio [1]. The operation framework of a balancing market is summarized in and consists of the above three parts: the provision of balancing services, the division of balance responsibility and balance settlement.

Balancing service provision is an arrangement of bidding/offering balancing services by BSPs and the procurement and dispatch of these services by the SO. The division of balance responsibility refers to the financial accountability of BRPs according to their real output in each PTU. Accurate imbalance responsibility allocation is conducive to keeping BRPs in line with generation/consumption schedules as much as possible and reducing system balancing costs. Balance settlement comprises two forms of settlement: one is paying for the procured balancing services from BSPs at market-based prices, and the other is allocating the resulting balancing costs to BRPs. Theoretically, BSPs should deliver/consume the energy that they sell to/buy from the market in scheduling; otherwise, they are charged for imbalances, so settlement to BRPs is also called imbalance settlement.

Therefore, imbalance settlement can be considered as a set of ex-post settlement rules, including the division of balance responsibility, the setting of PTU length, and the imbalance pricing mechanism. Under this set of rules, BRPs are allowed to make strategic decisions to participate in electricity markets that are most beneficial to themselves. That is, these design elements of imbalance settlement have vital guidance for the behaviour of BRPs, as well as a large impact on balancing market performance. In a balancing market, strong interaction exists among imbalance settlement rules, the behaviour of BRPs and the whole balancing market performance, as

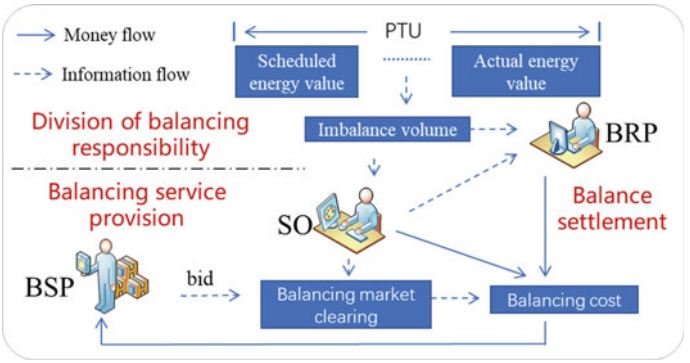


Fig. 8.1 Operation framework of a balancing market

shown in Fig. 8.1. Since BRPs generally have unsymmetrical and incomplete information, referring to the imbalance settlement rules, BRPs may continuously adjust their offering strategies to maximize their expected revenue. Additionally, they can create an 'intentional imbalance' by means of over- and under-contracting in the day-ahead market and then make an internal balancing in real time. Afterwards, the SO will evaluate market performance on a regular basis and may update the settlement rules to improve market efficiency according to the evaluation results.

8.2.2 Key Design Elements in Imbalance Settlement

In Fig. 8.2, the design of a balancing market will affect the decision making of market participants and the efficiency of the whole market. To achieve the incentive compatibility of market design, a market designer is supposed to properly consider the reactions of market participants to design elements when designing imbalance settlement, since these reactions are directly related to the economic benefits of market participants. In this character, the following key design elements related to imbalance settlement are considered.

(1) Imbalance pricing mechanism

An imbalance pricing mechanism is used to determine imbalance settlement prices and can be considered the process of balancing cost allocation. As an important part of a balancing market, the imbalance pricing mechanism directly affects the final income and market behaviour of BRPs and provides strong incentives for BRPs to balance their energy portfolios. In this chapter, the following three imbalance pricing mechanisms are investigated.

1) One-price

In a one-price pricing mechanism, real-time deviations of BRPs are settled at a unique regulation price regardless of the imbalance directions of both the BRPs and

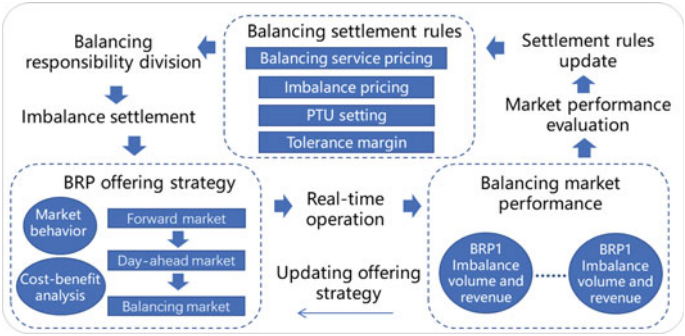


Fig. 8.2 The interrelation of balancing market components based on balance settlement rules

the system. In general, the regulation price is higher than the day-ahead price if the system needs up-regulation. In contrast, the regulation price is lower than the day-ahead price when the system has a surplus of power production.

2) Two-price

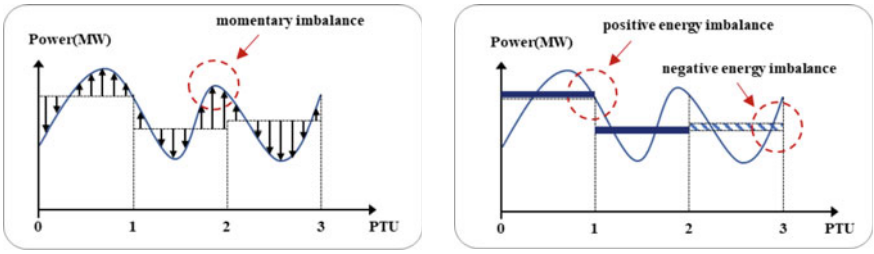
Under a two-price pricing mechanism, BRPs are settled at different prices depending on the imbalance direction. Deviations that are in the same direction as the system imbalance, which aggravate the imbalance of the system, are settled at the regulation price of the balancing market. Conversely, imbalances opposite to that of the system are settled at the day-ahead price.

3) Dual pricing

This pricing mechanism is different from the two-price pricing mechanism, in which imbalance prices depend on the imbalance sign between BRPs and the overall system. In general, under a dual pricing mechanism, if upward or downward regulation bids are activated, the short imbalance and long imbalance are settled at the upward and downward regulation prices, respectively. In this chapter, a new dual pricing scheme is proposed under which both sides of imbalances are settled at penalty prices based on day-ahead market prices and variable penalty factors, and this scheme aims to give stronger incentives for BRPs to minimize energy imbalances and is suitable for the initial operation of a spot market.

(2) Programme Time Unit (PTU)

Actually, the setting of a PTU can be considered a way of dividing balancing responsibilities between the SO and BRPs. Since the SO is responsible for the momentary imbalance intra-PTU, which is defined as a function of both the frequency deviation and the deviation from the agreed cross-border exchange, the BRPs are only concerned about the energy content of their trades in a PTU and are only responsible for the energy imbalance cross-PTU. For a BRP, positive and negative imbalances in the same PTU can be offset, and the offset part does not need to assume imbalance responsibility. The concepts of intra-PTU and cross-PTU are shown in Fig. 8.3.



(a) Intra-PTU (b) Cross-PTU

Fig. 8.3 Intra-PTU and Cross-PTU

Due to the uncertain nature of variable renewable power, it is particularly important to reduce partial imbalance responsibilities by this offset effect. Accordingly, the PTU length has a great impact on imbalance settlement results. In general, smaller PTUs will give stronger incentives for BRPs to balance their energy portfolios, as deviations from scheduled energy exchange in a PTU even out to a lesser degree. However, a smaller PTU also mitigates the effort of the SO and may aggravate the phenomenon of internal balancing, i.e., BRPs may keep balancing resources for individual balancing rather than providing balancing services to the SO, and this effect can reduce the utilization efficiency as well. Therefore, determining the appropriate PTU length is of great importance in imbalance settlement design.

(3) A tolerance margin

A tolerance margin refers to a specific support mechanism exempting certain kinds of BRPs from their imbalance responsibilities, and this mechanism has been enforced in some European countries and in China for long-term transactions. In European countries, only renewable generators may enjoy this support mechanism due to the nature of inherent variability and limited predictability. However, most BRPs can enjoy this support mechanism in long-term transactions in China, and for different kinds of BRPs, the tolerance margin ranges from 3–10%. This mechanism is used because a tolerance margin can give market participants a buffer in spot market construction.

An appropriate tolerance margin is of vital importance: no tolerance margin or a low tolerance margin may not be enough to support wind power producers and other BRPs participating in spot markets. However, an excessive tolerance margin may harm the incentives to invest in ways to reduce imbalance costs, such as improving forecasting accuracy and offering strategies and the incorporation of flexible portfolios, and other BRPs suffer higher balance costs, which can be considered cross-subsidization.

8.3 System Model

To comprehensively investigate the effect of imbalance settlement design in balancing market operation, this section proposes an integrated ABM method and MCDA evaluation, where ABM method is implemented to simulate balancing market clearing under different imbalance settlement mechanisms composed of the above key elements, while MCDA evaluation method is applied to comprehensively evaluate the impact of different imbalance settlement mechanisms on the balancing market.

To quantitatively assess the effect of each of the above design elements on the market clearing results, system balancing cost, market player income, and especially market behaviour and income of wind energy providers in the operation of the balancing market. Then, a balancing market clearing optimization model embedded with strategic offerings of wind power producers is established. We take the perspective of a wind power producer in a two-settlement electricity market. The above two models are introduced below respectively.

8.3.1 Balancing Market Clearing Optimization Model Embedded with the Offering Strategy of Wind Power Producers

The objective function of the balancing market clearing model is intended to minimize the balancing cost for the system, in which up- and down regulation offers of BSPs and the imbalance volume of wind power producers are considered. The clearing model in time slot t can be represented as follows:

$$\min \sum_{i \in \phi^{\mathrm{BSP}}} \left(C^{U}_{B,i,t} P^{U}_{B,i,t} - C^{D}_{B,i,t} P^{D}_{B,i,t} \right) \tag{8.1}$$

$$\sum_{i \in \Phi^{\mathrm{BSP}}} \left(P^{U}_{B,i,t} - P^{D}_{B,i,t} \right) = -P^{\mathrm{WPP}}_{im,t} \tag{8.2}$$

$$0 \le P^{U}_{B,i,t} \le P^{U}_{B,i,\max} \tag{8.3}$$

$$0 \le P^{D}_{B,i,t} \le P^{D}_{B,i,\max} \tag{8.4}$$

$$P^{\mathrm{WPP}}_{im,t} = P^{\mathrm{WPP}}_{t} - P^{\mathrm{WPP}}_{DA,t} \tag{8.5}$$

where $C^{U}_{B,i,t}$ and $C^{D}_{B,i,t}$ denote the up- and downregulation price offers by BSP i in time slot t and $P^{U}_{B,i,t}$ and $P^{D}_{B,i,t}$ denote the selected up- and downregulation energy

offers by BSP i in time slot t. $P^{U}_{B,i,max}$ and $P^{D}_{B,i,max}$ denote the maximum level of up- and downregulation for BSP i. P^{WPP}_{t} and $P^{\mathrm{WPP}}_{im,t}$ denote the actual output and energy deviation of wind power producer. $P^{\mathrm{WPP}}_{DA,t}$ is the energy offer of the wind power producer in time slot t, which is obtained from the offering strategy model in Sect. 8.3.1.1. Constraints (8.2)–(8.4) are the power balance constraints and limit the regulation offers of BSPs. In (8.5), the energy deviation of the wind power producer is calculated according to the actual output and energy offers in the day-ahead market.

8.3.2 Offering Strategy of the Wind Power Producer

The offering strategy of wind power producers can be formulated via the following two-stage risk-constrained stochastic optimization model.

8.3.3 Objective Function and Constraints

$$\max\ R_{DA} + R_{IM} + \beta\,\mathrm{CVaR} \tag{8.6}$$

The objective function to be maximized includes the following three parts: the profits in the day-ahead market, R_{DA}, the income from imbalance settlements, R_{IM}, and the conditional variance at risk (CVaR) of the profit multiplied by the risk preference coefficient $\beta \in [0,\infty]$. The risk preference coefficient enforces the tradeoff between the expected profit and risk in such a way that the higher value of β, the more risk averse is the wind power producer in offering. Therefore, if the wind power producer is risk-neutral, the value of β is 0.

The profits of wind power producers in the day-ahead market R_{DA} can be presented as follows:

$$R_{DA} = \sum_{t\in\Phi^{t}} \lambda_{DA,t} P^{\mathrm{WPP}}_{DA,t}, \forall t, \tag{8.7}$$

where $\lambda_{DA,t}$ denotes the market price of day-ahead markets in time slot t.

The income from imbalance settlement R_{IM} can be formulated as follows:

$$\begin{aligned}
R_{IM} &= \sum_{t\in\Phi^{t}} \sum_{s\in\Phi^{*}} A + B + C \\
A &= -\pi_s \lambda^{-}_{t,0}\left(\left(1-\psi^{-}_{s,t}\right) P^{\mathrm{WPP}-}_{s,t} + \gamma^{-} P_{s,t}\psi^{-}_{s,t}\right), \forall s, t, \\
B &= \pi_s \lambda^{+}_{t,0}\left(\left(1-\psi^{+}_{s,t}\right) P^{\mathrm{WPP}+}_{s,t}\right) + \gamma^{+} P^{\mathrm{WPP}}_{t}\psi^{+}_{s,t}\right), \forall s, t, \\
C &= \pi_s\left(\lambda^{-}_{t,1} P^{\mathrm{WPP}-}_{s,t,1} + \lambda^{+}_{t,1} P^{\mathrm{WPP}+}_{s,t,1}\right), \forall s, t,
\end{aligned} \tag{8.8}$$

where π_s is the probability of scenario s and $\lambda_{t,0}^{+}$ and $\lambda_{t,0}^{-}$ denote the positive imbalance price and negative imbalance price within the tolerance margin, respectively. $\lambda_{t,1}^{+}$ and $\lambda_{t,1}^{-}$ denote the positive imbalance price and negative imbalance price beyond the tolerance margin, respectively. $P_{s,t}^{\mathrm{WPP}+}$, $P_{s,t}^{\mathrm{WPP}-}$, $P_{s,t,1}^{\mathrm{WPP}+}$ and $P_{s,t,1}^{\mathrm{WPP}-}$ denote the total positive and negative imbalances of the wind power producer and the imbalances beyond the tolerance margin, respectively. $\Psi_{s,t}^{+}$ and $\Psi_{s,t}^{-}$ are binary variables active when positive and negative imbalances are higher than the tolerance margin, respectively. γ^{+} and γ^{-} denote the positive and negative tolerance margin, respectively.

The constraints are as follows:

(1) Imbalance equality constraints

$$P_{s,t}^{\mathrm{WPP}} - P_{DA,t}^{\mathrm{WPP}} = P_{s,t}^{\mathrm{WPP}-} + P_{s,t}^{\mathrm{WPP}+}, \forall s, t, \tag{8.9}$$

$$0 \le P_{DA,t}^{\mathrm{WPP}} \le P_{\max}^{\mathrm{WPP}}, \forall t, \tag{8.10}$$

$$-P_{\max}^{\mathrm{WPP}} \le P_{s,t}^{\mathrm{WPP}-} \le 0, \forall s, t, \tag{8.11}$$

$$0 \le P_{s,t}^{\mathrm{WPP}} + \le P_{s,t}^{\mathrm{WPP}}, \forall s, t, \tag{8.12}$$

$$P_{s,t,1}^{\mathrm{WPP}-} = P_{s,t}^{\mathrm{WPP}-}\psi_{s,t}^{-} + \gamma^{-}P_{s,t}^{\mathrm{WPP}}\psi_{s,t}^{-}, \forall s, t, \tag{8.13}$$

$$P_{s,t,1}^{\mathrm{WPP}+} = P_{s,t}^{\mathrm{WPP}+}\psi_{s,t}^{+} - \gamma^{+}P_{s,t}^{\mathrm{WPP}}\psi_{s,t}^{+}, \forall s, t, \tag{8.14}$$

$$\psi_{s,t}^{-} + \psi_{s,t}^{+} \le 1, \forall s, t \tag{8.15}$$

Constraint (8.9) calculates the positive and negative output deviation bias of the day-ahead offer. Constraint (8.10) limits the day-ahead offer power to its capacity $P_{i,max}^{\mathrm{WPP}}$. Constraints (8.11) and (8.12) present the imbalance limits of wind power producers. In (8.13)–(8.15), positive and negative energy deviations beyond the tolerance margin are calculated.

(2) CVaR constraints

For a given confidence level $\alpha \in (0, 1)$, the CVaR is defined as the expected value of the profit smaller than the (1-α)-quantile of the profit distribution. In this model, the CVaR is used to measure the risk of market revenue changes of wind power producers due to the uncertainty of market prices and wind power output. The CVaR constraints can be presented as follows:

$$\mathrm{CVaR} = \xi - \frac{1}{1-\alpha}\sum_{s\in\Phi^s}\pi_s\eta_s, \forall s \tag{8.16}$$

$$\xi - (R_{DA} + R_{IM}) \leq \eta_s, \forall s \tag{8.17}$$

$$\eta_s \geq 0, \forall s \tag{8.18}$$

where ξ is an auxiliary variable, which is also known as the value at risk (VaR), and η_s denotes the difference between the market revenue and VaR in scenario s.

8.3.4 Solution Method

The aforementioned balancing clearing model embedded with the offering strategy model needs to be solved sequentially. That is, the offering strategy model of wind power producers is solved first, and then the day-ahead offer $P_{DA,t}^{\mathrm{WPP}}$ can be obtained. Then, the balancing market clearing model can be solved, producing the final market clearing result. In addition, the previous formulation is non-linear due to being the product of two variables, $P_{s,t}^{\mathrm{WPP}-}\Psi_{s,t}^{-}$ and $P_{s,t}^{\mathrm{WPP}+}\Psi_{s,t}^{+}$ in constraints (8.13)–(8.14), and a linearization method has been proposed, involving replacing non-linear terms with two new continuous variables ($D_{s,t}$, $U_{s,t}$) and a set of constraints:

$$-P_{s,t}^{\mathrm{WPP}+} + D_{s,t} \leq 0, \forall s, t, \tag{8.19}$$

$$P_{s,t}^{\mathrm{WPP}-} - U_{s,t} \leq 0, \forall s, t, \tag{8.20}$$

$$-P_{s,t}^{\mathrm{WPP}+} - D_{s,t} + P_{s,t}^{\mathrm{WPP}}\psi_{s,t}^{+} \leq P_{s,t}^{\mathrm{WPP}}, \forall s, t, \tag{8.21}$$

$$-P_{s,t}^{\mathrm{WPP}-} + U_{s,t} + P_{\max}^{\mathrm{WPP}}\psi_{s,t}^{-} \leq P_{\max}^{\mathrm{WPP}}, \forall s, t, \tag{8.22}$$

$$-U_{s,t} \leq P_{\max}^{\mathrm{WPP}}\psi_{s,t}^{-}, \forall s, t, \tag{8.23}$$

$$D_{\omega t} \leq P_{s,t}^{\mathrm{WPP}}\psi_{s,t}^{+}, \forall s, t \tag{8.24}$$

Thus, the objective function (8.1) and the constraints (8.2)–(8.5) constitute a linear programming problem. The stochastic programming model consisting of the objective function and the constraints (8.9)–(8.24) is a mixed integer linear optimization problem.

8.3.5 ABM Method

8.3.5.1 Model Framework and Assumptions

An agent-based model is proposed to briefly represent the balancing market operation, with a focus on some vital design variables in imbalance settlement. The input data for these design variables and corresponding scenario settings are abstracted from a Chinese practical electricity market situation. The balancing market operation is simulated at two levels: the participant agent level and the system level. At the agent level, the BRPs are supposed to strategically respond to different balancing market rules, with the opportunity cost in imbalance settlement as the decision criterion. Then, the system imbalance is equal to the net sum of imbalance volumes of strategic behaviours of BRPs and is closely related to the activation of real-time balancing services. This activation then determines the imbalance prices, which can further incentivize BRPs to change their offering strategies; therefore, this process is reflected in the BRP imbalance at the agent level again.

For the sake of simplicity, the proposed model is based on the following assumptions:

(1) The strategic offerings of BRPs mean making a choice out of a fixed coefficient set of strategic imbalance options for each round. The strategic imbalance options represent different degrees of over- or under-contracting [5].
(2) The strategic imbalance options are selected according to the expected revenue loss of each offer option.
(3) The upward and downward regulation bid curves are fixed. The marginal pricing mechanism is applied, which means that the selected regulation power bid in real time is settled at the price of the last activated bid.

8.3.5.2 Model Description

A flow chart of the proposed agent-based model is shown in Fig. 8.4, and this model consists of the following steps:

(1) Strategic offerings of BRPs

The imbalance of a BRP consists of two parts: the strategic imbalance and the non-strategic imbalance. The strategic imbalance refers to the strategic offerings of BRPs and is the initial step; each BRP is supposed to select an intentional imbalance option from a fixed set of options. The offering strategies are specified in the form of percentages based on portfolio size. The intentional imbalance options for each BRP can be calculated through the intentional imbalance percentage multiplied by the portfolio size and the PTU length.

(2) Non-strategic imbalance calculation

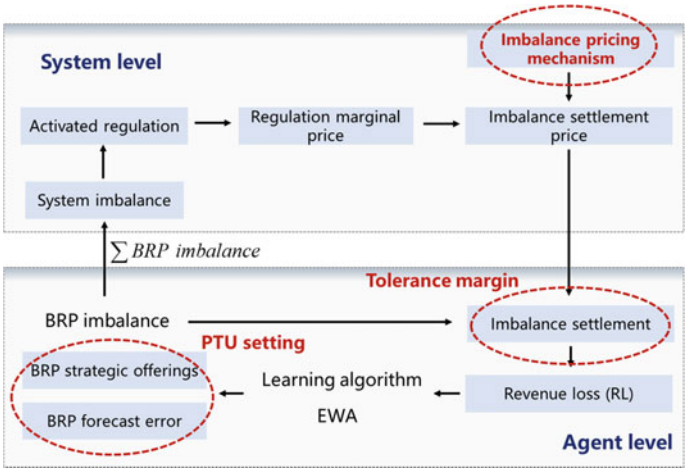


Fig. 8.4 Structure of the agent-based balancing market model

The non-strategic imbalance calculation refers to the prediction errors of BRPs, which are assumed to follow a normal distribution (other distributions are also applicative), and is calculated every minute. Then, the non-strategic imbalance within a PTU is the net sum of the prediction errors per minute.

(3) Determining the system imbalance quantity and imbalance price

The total imbalance quantity of each BRP is the sum of its strategic imbalance and non-strategic imbalance. In this way, the system imbalance quantity can be obtained and is equal to the net sum of all BRP imbalances. Based on the system imbalance, the corresponding upward (corresponding to negative system imbalance, also called market shortage) and downward regulations (positive system imbalance or market surplus) are activated. Once the regulation volume is determined, the marginal regulation price for upward or downward regulation is obtained.

(4) Revenue loss in imbalance settlement

Next, depending on the different imbalance pricing mechanisms, the long and short imbalance prices can be calculated based on the regulation price. The revenue loss of a BRP in imbalance settlement, which refers to the difference in revenues between the selected offering strategy and the optimal strategy without imbalance, can be calculated in each round by:

$$RI_{n,m} = \left(\lambda_m^{\mathrm{DA}} - \lambda_m^{\mathrm{up}}\right) \cdot \left(-IV_{n,m}\right)_+ + \left(\lambda_m^{\mathrm{DA}} - \lambda_m^{\mathrm{down}}\right) \cdot \left(IV_{n,m}\right)_+ \tag{8.25}$$

where $(\cdot) + = \max(0, \cdot)$; $RI_{n,m}$ denotes the revenue loss in imbalance settlement for BRP n in round m; λ_m^{DA}, λ_m^{up} and λ_m^{down} denote the price of the day-ahead market, the up-regulation price and the down-regulation price in round m, respectively; and $IV_{n,m}$ denotes the imbalance volume of BRP n in round m.

(5) Updating expected revenue loss in imbalance settlement

Finally, the expected revenue loss in imbalance settlement for each offering strategy is updated for every BRP according to the decision-making algorithm described in Sect. 8.3.1.3. These expected revenue losses are used as the input parameters in the decision-making process.

Decision-Making Algorithm

At the beginning of each round, each BRP is supposed to choose a specific intentional imbalance offering strategy based on a certain decision-making algorithm pursuing a lower revenue loss in imbalance settlement. In this section, a decision-making model based on the experience-weighted attraction (EWA) learning algorithm [6] is applied. EWA can mix appropriate elements of reinforcement learning and belief learning approaches in a way that makes sense and is more suitable for solving problems with multiple agents and multiple scenarios [7]. For each round and each BRP, the expected revenue losses in imbalance settlement under different offering strategies are updated by:

$$N_m = \rho\, N_{m-1} + 1 \tag{8.26}$$

$$A_{n,m}^{j} = N_m / \left(\varphi N_{m-1} A_{n,m-1}^{j} + \left(\delta + (1-\delta) I\left(s_{n,m}^{j}, s_{n,m} \right) \right) \pi_n \left(s_{n,m}^{j}, s_{-n,m} \right) \right) \tag{8.27}$$

$$I\left(s_{n,m}^{j}, s_{n,m}^{s} \right) = \begin{cases} 1 & S_{n,m}^{s} = S_{n,m}^{j} \\ 0 & \mathbf{S}_{n,m}^{s} \neq \mathcal{S}_{n,m}^{j} \end{cases} \tag{8.28}$$

$$P_{n,m}^{j} = e^{\gamma} A_{n,m-1}^{j} / \sum_{j} e^{\gamma} A_{n,m-1}^{j} \tag{8.29}$$

where $A_{n,m}^{j}$ and $P_{n,m}^{j}$ denote the attraction and selection probability of BRP n for strategy j in round m, respectively; Nm denotes the experience weight of round m; ρ and φ denote the learning ability from past experience and the discount rate of attraction, respectively; δ denotes the weight of an unselected strategy; $\pi_n(s_{n,m}^{j}, s_{-n,m})$ denotes the expected revenue loss for BRP n with strategy j when the strategy set of other BRPs is $s_{-n,m}$; $s_{n,m}^{s}$ denotes the selected strategy for BRP n in round m; and γ denotes the response sensitivity of attraction.

Based on the above decision-making algorithm, the final choice of a BRP for a specific strategy in each round is made with the probabilities being inversely proportional to the attraction of different offering strategies. The attractions of both selected and unselected strategies are updated in each round.

8.3.5.3 Model Input and Scenario Settings

The first inputs include the number, the characteristics and portfolio sizes of BRPs. The portfolio size represents the total capacity of production or consumption for those BRPs that have balancing responsibility, and BRPs with different characteristics are indicated by different prediction errors. The prediction error is used to calculate unintentional imbalance, while the intentional imbalance of a BRP is caused by strategic offerings. The second inputs concern market prices and the time scale. The day-ahead market prices and regulation prices and volume are derived from the simulative operation of the Guangdong provincial spot market, and the time scale of the proposed agent-based model is 24 h. The third inputs are all the strategic offerings of BRPs, which are represented by the decision-making algorithm in Sect. 8.3.1.3.

To study the impacts of the key imbalance settlement design elements mentioned in Sect. 8.2.2, several balancing market operation scenarios with different design elements are simulated by the proposed agent-based model.

(1) A-type scenarios: imbalance pricing mechanism

First, imbalance pricing mechanisms are designed for testing three popular rules: the one-price, dual pricing and two-price imbalance pricing mechanisms are denoted as three basic scenarios A1, A2, and A3, respectively. The other balancing market design variables are all tested under these three basic scenarios.

The penalty factors in the dual pricing mechanism are set according to related references and official codes in China as +0.2/–0.2, +0.25/–0.15, and +0.15/–0.25, which are designated as scenarios A21, A22 and A23, respectively.

(2) B-type scenarios: PTU

For the PTU setting in imbalance settlement, throughout Europe, PTUs of 15 min, 30 min and 60 min are in use, and according to the latest market rules in the Australian electricity market, the PTU is supposed to decrease to 5 min in 2021. Accordingly, we design 4 scenarios for the PTU: 5, 15, 30 and 60 min, denoted as scenarios B1–B4, respectively.

(3) C-type scenarios: tolerance margin

For the tolerance margin for renewable power producers, for example, a tolerance margin of 2.5% means that renewable generators are exempt from imbalance duties within 2.5% of deviation. In this model, the tolerance margins are set according to the short-term prediction accuracy of wind power as 0, 2.5%, 5%, and 7.5%, which are labelled as scenarios C1–C4, respectively.

In summary, the number of scenarios is 80, and the specific settings of the scenarios are listed in Table 8.1.

Table 8.1 Scenarios numbering mode

	A1(C1–C4)	A21(C1–C4)	A22(C1–C4)	A23(C1–C4)	A3(C1–C4)
B1	1–4	17–20	33–36	49–52	65–68
B2	5–8	21–24	37–40	53–56	69–72
B3	9–12	25–28	41–44	57–60	73–76
B4	13–16	29–32	45–48	61–64	77–80

8.3.6 The MCDA Evaluation

8.3.6.1 Criteria System Establishment

Apart from the seven output results from ABM, three critical qualitative criteria are introduced: BRP degree of satisfaction, utilization efficiency, and impacts on investment and construction of flexible generation, which are explained next in detail. To comprehensively evaluate various imbalance settlement design elements, a criteria system is established, as shown in Table 8.2. Based on the balancing market performance criteria set proposed by van der Veen R [8], the performance criteria for both the system level (including D2, D5-D7, and D10) and BRP level (including D1, D3-D4, and D8-D9) are included in the criteria system in this chapter. Overall, we formulate the criteria system from two aspects: the operation efficiency criteria and economic efficiency criteria, as follows.

(1) Operation efficiency criteria of the balancing market

- Balancing planning accuracy (D1)

Due to the necessity for the SO to maintain the system supply and demand balance in real time, the offset of positive or negative deviation in each PTU will have a

Table 8.2 Criteria system of balance market

Criteria classification	Criteria	Symbol
Quantitative criteria	Balance planning accuracy	D1
	Balance quality	D2
	Authenticity of offering	D3
	Responsibility efficiency for renewable generation	D4
	Cost allocation efficiency	D5
	Price efficiency	D6
	Balancing cost	D7
Qualitative criteria	BRP satisfaction degree	D8
	Utilization efficiency	D9
	Impact on investment and construction of flexible generation	D10

great impact on the real-time electricity market. The balance planning accuracy, as a measure quantifying the offset of unbalancing power in a PTU, can reflect actual balance accuracy from the viewpoint of energy injections and withdrawals. The value of criterion D1 for BRP n can be calculated as follows:

$$V_m^{D1} = \frac{1}{N} \sum_{n=1}^{N} \frac{IV_{n,m}^{\text{over}}}{IV_{n,m}^{\text{intra}}} \tag{8.30}$$

where V_m^{D1} denotes the value of criterion D1 in round m, N denotes the number of BRPs, and $IV_{n,m}^{over}$ and $IV_{n,m}^{intra}$ denote the imbalance volume for over- and intra-PTU, respectively.

- Balance quality (D2)

When the system is in large imbalance for a long time, the effectiveness of the balancing market will be greatly threatened. Thus, the balance quality is designed to measure the effectiveness of system operation from the viewpoint of imbalance status. The value of criterion D2 in each round is equal to the imbalance volume of the overall system.

- Authenticity of offerings (D3)

The authenticity of offerings represents the willingness of each participant to achieve optimal revenue by offering their true predictions. The value of criterion D3 in each round is equal to the average of the absolute value of the intentional imbalance volume for all BRPs.

- Responsibility efficiency for renewable generation (D4)

The responsibility efficiency for renewable generation is defined as the imbalance responsibilities exemption degree of renewable generators. The value of criterion D4 in each round is equal to the ratio of the imbalance volume within the tolerance margin to the total imbalance volume for renewable generators.

- BRP degree of satisfaction (D8)

For a BRP, the degree of satisfaction will greatly affect the smooth implementation of the imbalance settlement mechanism and is one of the most important criteria for measuring balancing market stability.

(2) Economic efficiency criteria of the balancing market

- Cost allocation efficiency (D5)

The cost allocation efficiency refers to whether the imbalance cost paid by BRPs can fully cover the balancing service costs. In this chapter, cost allocation is measured by the net settlement sum, and the value of criterion D5 in each round is equal to the

difference between the income that the SO receives from BRPs and its payments to BSPs.

- Price efficiency (D6)

D6 is calculated by the difference between the imbalance price and day-ahead electricity price and can effectively reflect the market liquidity and the competition situation of current balancing services. The value of criterion D6 in each round is equal to the differences between the day-ahead market prices and the balancing energy prices.

- Balancing cost (D7)

The balancing cost represents the total balancing cost of the system based on balancing service bids. The value of criterion D7 in each round is equal to the balancing cost of the overall system.

- Utilization efficiency (D9)

Generally, the system overall efficiency will be reduced when BRPs use balancing resources to minimize their own imbalance costs rather than providing them to the SO. Thus, the utilization efficiency refers to the utilization of the available balancing resources.

- Impacts on investment and construction of flexible generation (D10)

Imbalance settlement refers to a financial settlement mechanism for charging or paying BRPs for their imbalances, and this mechanism can incentivize market participants to help restore the system balance and have an impact on investment and construction of flexible generation. This criterion measures the effect of the imbalance settlement mechanism on the investment construction of flexible generation.

8.3.6.2 Multi-Criteria Decision Analysis Model

(1) Representation of qualitative information

As uncertain and imprecise information usually occurs in multiple criteria decision analysis [9], decision makers are not always certain of their given decision or preference information and often use a certain degree of uncertainty to express their subjective judgements [10]. Fortunately, fuzzy sets can effectively express various aspects of decision information from decision makers. As an extension of ordinary fuzzy sets, type-2 fuzzy sets introduced by Zadeh can process high computational complexity and are difficult to practically implement [11]. At present, the theory of IT2TrFN (interval type-2 trapezoidal fuzzy numbers) has been widely used with the advantages of modelling impressions and quantifying the ambiguous nature of subjective judgements.

(2) Weight-setting methods

The main two types of weight-setting methods are subjective and objective weight methods. For subjective weight methods, the analytic hierarchy process (AHP) is adopted due to its wide utilization and simple calculation. For objective weight methods, the entropy method is applied in this model [12].

(3) Ranking method

Under the IT2TrFN environment, various outranking methods such as AHP, VIKOR (Vlsekriterijumska Optimizacija I Kompromisno Resenje), ELECTRE (elimination and choice expressing reality) and PROMETHEE (preference ranking organization method for enrichment evaluations) are utilized to solve multi-attribute problems. Compared to other methods, PROMETHEE requires fewer parameters from decision makers and can provide more effective preference functions without being computationally expensive as well as satisfy decision makers' preferences.

PROMETHEE-II is an improved method of PROMETHEE-I and can better reflect the subjective wishes of decision makers by choosing the type of priority function and setting the corresponding parameters. In addition, PROMETHEE-II does not need to process the original data and avoids the errors caused by different methods in original data processing. Moreover, this method is easy to calculate and understand, with a clear concept; many schemes can be compared, and a complete ranking can be obtained. Hence, PROMETHEE-II is adopted in this model for ranking different scenarios.

8.3.6.3 Research Framework of Imbalance Settlement Design in the Balancing Market

To investigate an appropriate imbalance settlement design for balancing market, an effective assessment method is of great significance. The overall assessment process is presented in Fig. 3. The process is divided into three parts: critical criteria identification, imbalance settlement design assessment and result analysis. The criteria system is presented in Sect. 3.2.1, and the remaining information is shown as follows.

Step 1. Establishment of a primitive matrix of imbalance settlement design evaluation information.

The data of the first seven quantitative criteria are obtained by ABM simulation and can be directly utilized for MCDA. The remaining three qualitative criteria are represented through IT2TrFNs by an expert committee. The established criteria information matrix D_{ci} is then formulated as follows:

$$D_{ci} = \begin{bmatrix} d_{11} & d_{12} & \dots & d_{1,80} \\ d_{21} & d_{22} & \dots & d_{2,80} \\ \dots & \dots & \dots & \dots \\ d_{10,1} & d_{10,2} & \dots & d_{10,80} \end{bmatrix} \tag{8.31}$$

Table 8.3 Criteria weights

	Subjective weights	Objective weights	Integrated weights
D1	0.1189	0.1638	0.1414
D2	0.660	0.0346	0.0503
D3	0.1561	0.0865	0.1213
D4	0.1884	0.1826	0.1855
D5	0.0723	0.0561	0.0642
D6	0.130	0.1049	0.1220
D7	0.0435	0.1382	0.0909
D8	0.0352	0.0801	0.0576
D9	0.0593	0.0744	0.0669
D10	0.1213	0.0787	0.1000

where m is 10, the number of criteria, and n is 80, which represents the number of evaluation objects.

Step 2. Determine the weights of evaluation factors via combining the AHP and the entropy method.

To comprehensively consider decision makers' preferences and the differences among objective data, a weighting method combining subjective and objective weights is adopted, in which subjective and objective weights are calculated by the AHP and entropy weights, respectively. Considering the calculation requirements of the entropy method, a unified data type of the proposed criteria system is formulated by defuzzying IT2TrFNs into a crisp number via Eq. (B.1). Then, an integrated weight can be calculated with an α of 0.5. All types of criteria weights are shown in Table 8.3. (Fig. 8.5)

Step 3. Rank the potential settlement design by PROMETHEE-II.

According to the basic theory of PROMETHEE-II, the performance differences between every two alternatives with respect to each criterion are first calculated. In addition, the difference between two IT2TrFNs can be formulated by comparing the distance between IT2TrFN and $\tilde{0}$. Due to the diversity of criteria attributes, the setting of the preference function for each criterion has a certain particularity.

For the benefit criteria of D1 and D5, larger values are usually better. Thus, the distance between these criteria can effectively reflect the gap among different alternatives. The dominance reflected by such a gap is linear when it does not exceed a certain threshold. However, once the gap exceeds this threshold, the dominance can be determined to be completely superior. Thus, the preference functions of these two criteria are set as follows.

$$P_{D1,D5}\{d(A_i, A_j)\} = \begin{cases} 0 & d(A_i, A_j) \le 0 \\ \frac{d(A_i, A_j)}{p} & 0 < d(A_i, A_j) \le p \\ 1 & d(A_i, A_j) > p \end{cases} \tag{8.32}$$

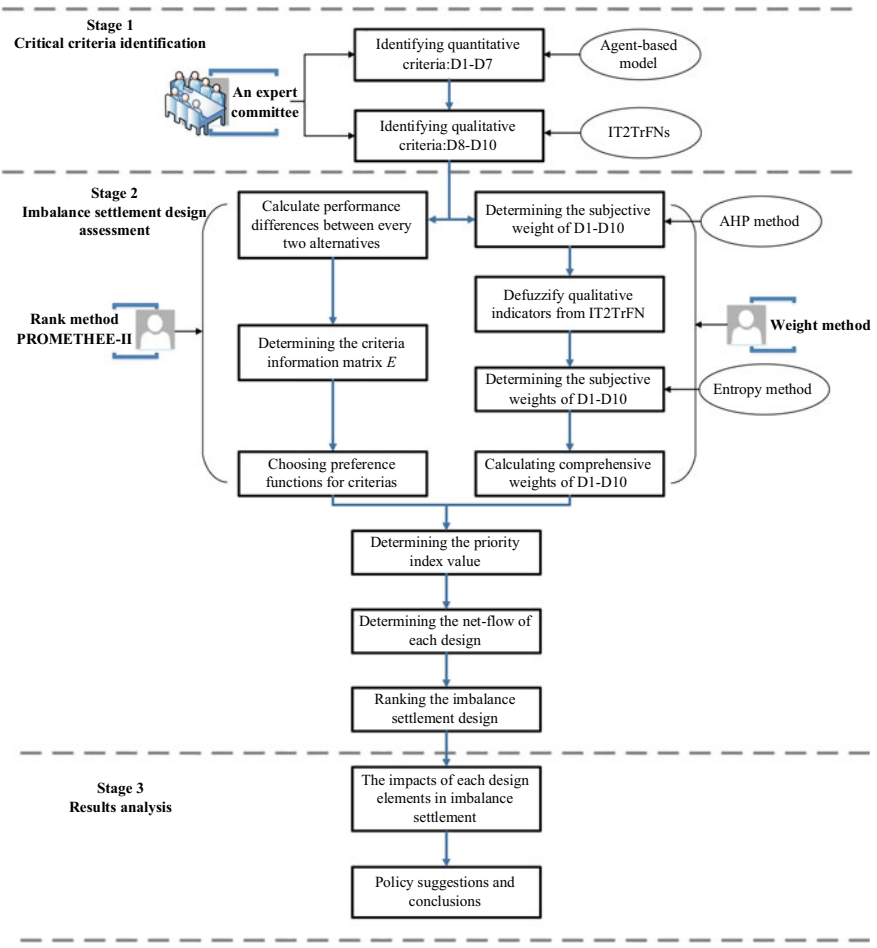


Fig. 8.5 The flow chart of the proposed assessment process

With regard to the cost criteria D2, D3, D6, and D7, the principle of the degree of dominance is roughly the same as that of the benefit criteria, but the expression form and threshold value change to some extent, as shown in Eq. (8.33).

$$P_{D2,D3,D6,D7}\left\{d\left(A_i, A_j\right)\right\} = \begin{cases} 0 & d\left(A_i, A_j\right) \geq p \\ 1 - d\left(A_i, A_j\right) & 0 \leq d\left(A_i, A_j\right) < p \\ 1 & d\left(A_i, A_j\right) < 0 \end{cases} \tag{8.33}$$

In contrast to the above criteria, criterion D4 is considered to be better when it approaches 0.7. Under such a setting, the imbalance settlement mechanism can support the development of renewable generation while avoiding excessive balancing

responsibilities for other BRPs. Thus, the distance from the best value of each alternative is calculated, and the gap of this distance value should be measured in a threshold.

$$P_{D4}\{d(A_i, A_j)\} = \begin{cases} 0 & d(A_i, A_j) \le 0 \\ \frac{d(A_i, A_j)}{p} & 0 < d(A_i, A_j) \le p \\ 1 & d(A_i, A_j) > p \end{cases} \tag{8.34}$$

For the qualitative criteria D8, D9 and D10, the criteria value is the benefit type. In general, when the distance of this scoring data is larger than 0.2, A_i is regarded to be completely dominant. As a result, 0.2 is the threshold of criteria D8–D10.

$$P_{D8,D9.D10}\{d(A_i, A_j)\} = \begin{cases} 0 & d(A_i, A_j) \le 0 \\ \frac{d(A_i, A_j)}{0.2} & 0 < d(A_i, A_j) \le 0.2 \\ 1 & d(A_i, A_j) > 0.2 \end{cases} \tag{8.35}$$

After determining the degree of preference of each alternative among all criteria, the ranking scores can be obtained by calculating the net flow of each scenario, where the net flow refers to the total priority of one scenario over the rest.

8.4 Case Studies

This section simulates and analyzes the two models.

8.4.1 Analysis of Wind Power Supplier's Strategic Offering

8.4.1.1 Impact of the Imbalance Pricing Mechanism on the Balancing Cost

Figure 8.6 shows the balancing cost of each duration under different imbalance pricing mechanisms based on scenarios 1 and 2, respectively. Additionally, the corresponding day-ahead prices of each period are also shown in Fig. 8.6 to facilitate comparison. As seen, the balancing cost during the peak load period is smaller due to the "anti-peak regulation" character of wind power. In addition, during the off-peak hour, the balancing costs in scenario 2 are higher than those in scenario 1. This is because scenario 2 uses a day-ahead weighted average price-based fixed penalty price in imbalance settlement, and the energy deviations of wind power producers are relatively high during off-peak load hours, which can lead to a higher balancing cost.

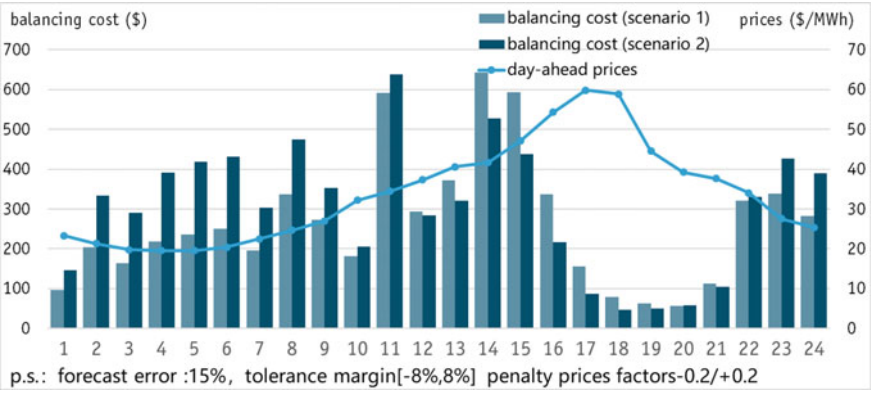


Fig. 8.6 Balancing costs under different penalty prices

8.4.1.2 Impact of Different PTU Settings on the Wind Power Revenue

To study the impact of the PTU on the clearing results of wind power producers under different imbalance pricing mechanisms, we investigate three specific periods (6, 12, and 18 h), and the revenue of wind power producers and imbalance cost proportions under different PTU settings in scenarios 1 and 3 are shown in Fig. 8.7.

Figure 8.7 clearly shows that when the PTU changes from 1 to 0.5 h, the clearing output of wind power producers decreases, while the proportion of the imbalance cost relative to the total revenue of the wind power producer significantly increases, with the highest proportion of 30%. This is mainly because as the PTU shortens, the settlement frequency increases, and the portion that can be offset through the positive and negative imbalances within the same PTU decreases, thereby increasing the imbalance costs. Additionally, for wind power producers, positive and negative

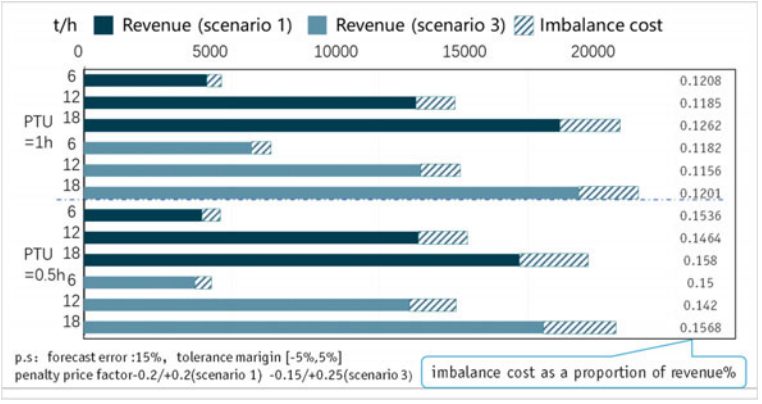


Fig. 8.7 The clearing results of the wind power producers under different PTUs

imbalance energy can be offset within the PTU; however, for the SO, the balance between the supply and demand of the overall system must be always maintained.

By comparing the clearing results under scenarios 1 and 3, we found that relative to the case in which the symmetric imbalance pricing mechanism (scenario 1) is applied, the penalty for upregulation is harsher than that for downregulation, so the over-contract characteristic is alleviated to some extend when adopting the asymmetric imbalance pricing mechanism in scenario 3. This outcome suggests that the wind power producer is more inclined to over-contract in the day-ahead market to maximize profits in a symmetric imbalance pricing mechanism, resulting in excessive day-ahead clearing and the need for more upregulation in real-time operation, thereby increasing the system balancing cost.

8.4.1.3 Impact of the Tolerance Margin on the Market Adequacy

(1) To study the impact of different tolerance margins on the operation of a balancing market, we obtained balancing market clearing results under different wind power prediction accuracies (15 and 25%) and tolerance margins, as shown in Fig. 8.8. The points on the line represent the proportions of imbalance cost relative to the total revenue of the wind power producer, and the tolerance margin refers to the upper and lower limits of the exemption interval of balance responsibility.

As can be seen from Fig. 8.8, with the increase of the tolerance margin, the proportion of the imbalance cost to wind power producers relative to the total revenue gradually decreases, and the system position gradually changes from over-supply (in which downregulation is needed) to a high level of under-supply (in which significant upregulation is needed). In particular, when the tolerance margin is 20%, the

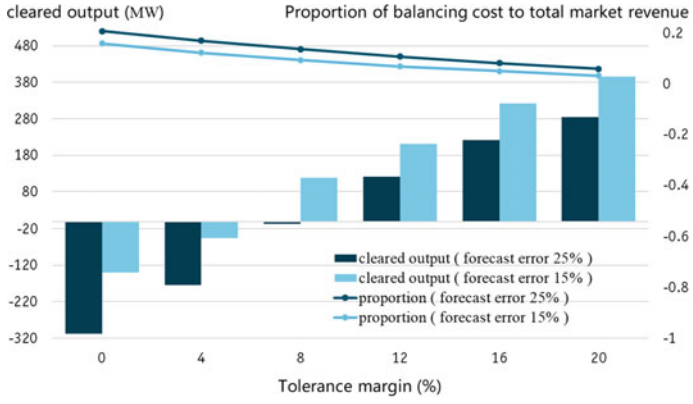


Fig. 8.8 Balancing market clearing results under different tolerance margins

required upregulation is approximately 10% of the predicted wind power output, indicating that the day-ahead offer of the wind power producer significantly exceeds the predicted output (110%), and the wind power producer is considered over-contract in the day-ahead market. This is because when the tolerance margin is zero or relatively low, the offering strategy of the wind power producer is conservative, and as the tolerance margin increases, the wind power producer chooses to submit a radical offer that exceeds the output prediction to maximize profits because the wind power producer needs to not assume the balance responsibilities for the energy deviation within the exemption interval.

In addition, by comparing the balancing market clearing results under different forecast errors, the system balancing cost is negatively correlated to the prediction error of wind power if the tolerance margin is larger than 8%. This is because the wind power producer deliberately chooses a more conservative offering strategy under a higher prediction error for risk aversion, and as the wind power prediction accuracy increases, the wind power producer is able to over-contract in the day-ahead market to earn higher profits by making better use of the tolerance margin, resulting in a higher system balancing cost. This phenomenon is even more profound when the tolerance margin is larger.

Since the wind power forecasting error is closely correlated with the time scale, the operation plan of the balancing market determines the forecasting error faced by wind power producers in day-ahead offerings to some extent. For different wind power forecasting errors, an appropriate tolerance margin helps reduce the total balancing cost of the system. As shown in Fig. 8.8, the system balancing costs are relatively low under a tolerance margin of 8% for a large prediction error scenario and a tolerance margin of 4% for a small prediction error scenario.

(2) Given that the tolerance margin is essentially intended to exempt the relevant BRP from the balance responsibility, it may have an impact on the balancing cost allocation. According to the mechanism design theory, a well-designed imbalance settlement mechanism should identify the different deviations of BRPs and allocate the appropriate balancing cost to them, including renewable generators. Here, we take the perspective of a wind power producer to explore whether the imbalance cost paid to the SO can cover the system balancing cost. The surplus and deficit of the balance account under different tolerance margins are shown in Fig. 8.8, and different market conditions are distinguished through risk preference factors.

Figure 8.9 shows that the balance account starts to have a deficit when the tolerance margin is increased to 15%, and the deficit increases with increasing tolerance margins. This result mainly occurs because when the tolerance margin is too large, the system balancing cost significantly increases due to the strategic offering of the wind power producer, as indicated in, while the tolerance margin reduces the balance responsibilities of the wind power producer to a greater extent, eventually leading to a deficit in the balance account.

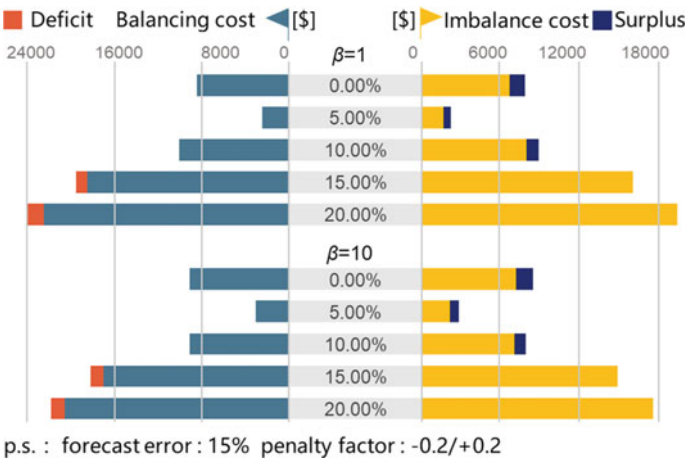


Fig. 8.9 Balance account surplus and deficit under different exemption coefficients

Additionally, as seen in Fig. 8.9, as the risk preference factor increases (that is, as the wind power producer becomes more risk-averse and may choose a more conservative offering strategy), both the imbalance cost to the wind power producer and the balancing cost of the system decrease slightly. In particular, the balance account deficit under a high tolerance margin is eased. In addition, the balance account surplus significantly increases when the tolerance margin is small, especially at 0. This is because the wind power producer is tending to choose an overly conservative offering strategy without the support of a tolerance margin, which in turn increases the imbalance, and the balance account surplus increases. This result is also consistent with the balancing market clearing results shown in Fig. 8.9.

8.4.2 Analysis of Strategic Interaction Behavior of Market Players

8.4.2.1 Imbalance Pricing Mechanism Exploration

Based on the above ABM simulation results and the MCDA method, the top 9 preferences among 80 scenarios are provided in Table 8.4.

where $\tilde{\phi}$(xi) is the standardized net-flow value from the eighty scenarios.

From the evaluation results shown in Table 8.4, we can see that the imbalance pricing mechanism used in the top 9 scenarios is the dual pricing mechanism. This result is because, under a dual pricing mechanism, both sides of imbalances are settled at a penalty price based on day-ahead market prices and penalty factors, and this approach leads to a higher price efficiency (D4). The dual pricing mechanism

Table 8.4 The final top nine net-flow information

Scenario	48	64	50	18	32	34	54	38	22
$\phi(x_i)$	15.75	13.90	13.58	12.98	12.76	11.66	10.27	9.81	9.77
$\tilde{\phi}(x_i)$	1	0.95	0.94	0.93	0.92	0.89	0.85	0.84	0.84
Rank	1	2	3	4	5	6	7	8	9

can give stronger incentives for BRPs to minimize energy imbalance and can incentivize BRPs to offer their true predictions (D3). In contrast, even though a one-price pricing mechanism results in lower imbalance costs for BRPs, the offering incentive compatibility of BRPs (D3) and the impacts on investment and construction of flexible generation (D10) are relatively poor, leading to a lower net flow of scenarios under a one-price pricing mechanism. Similarly, a two-price pricing mechanism can only accurately reflect the impact of a BRP's imbalance on the overall imbalance of the system, but the balancing responsibilities assumed by BRPs may not in accordance with the actual situation, which can lead to a lower value of balancing accuracy D1. Therefore, the comprehensive evaluation results of those scenarios under the dual pricing mechanism are best.

8.4.2.2 PTU Setting Exploration

To search for the appropriate setting of the PTU in balancing market design, the MCDA assessment results for those scenarios with different lengths of PTU, denoted as B1–B4, are shown in Fig. 8.10, and the dotted lines refer to the average levels under different lengths of PTUs.

As can be observed, with increasing PTU length from 5 to 30 min, the average net flow value decreases from 3.1 to –1.8, and the high net flow values are mostly

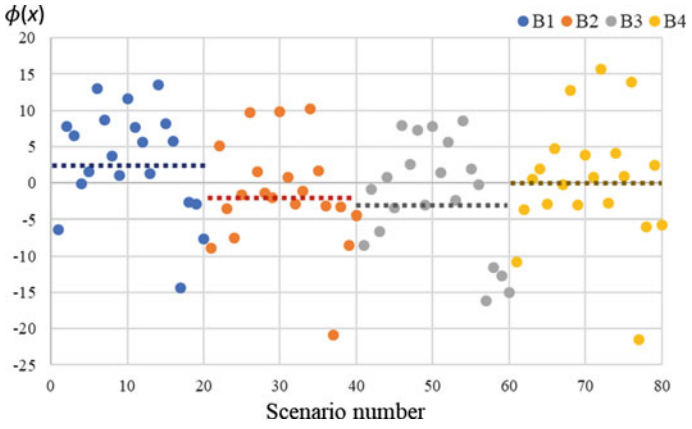


Fig. 8.10 The MCDA results for different lengths of PTU

concentrated in the scenarios with 5 min of PTU. This finding is because the SO is responsible for a momentary imbalance, while the BRPs are only concerned about the balancing energy in each PTU. A smaller PTU length will give stronger incentives to BRPs to balance their energy portfolios (D3), as deviations from a scheduled energy exchange during a PTU will even out to a lesser degree and lead to higher balancing planning accuracy (D1) and balancing quality (D2). However, when the PTU length is 60 min, the net flow value rebounds to 0.24, and the top 2 scenarios both have a PTU length of 60 min. This result is because a large part of momentary imbalance evens out within a PTU, leading to higher utilization efficiency (D9) and degree of satisfaction for BRPs (D8).

8.4.2.3 Tolerance Margin Exploration for Renewable Producers

The tolerance margin can partially exempt renewable producers from imbalance responsibilities, but such a mechanism may in general harm the interests of other BRPs. An appropriate tolerance margin selection for balancing market design is necessary. The MCDA evaluation results for those scenarios with different tolerance margins, labelled C1–C4, are shown in Fig. 8.11, and the dotted lines refer to the average levels under different tolerance margins.

As seen, with an increase in the tolerance margin, the average net flow value of scenarios rises first and then falls. When the tolerance margin is small, renewable producers have to take on much more balance responsibilities due to the inherent variability and unpredictability of the generation source, and this demand leads to a lower responsibility efficiency for renewable generation (D8) and utilization efficiency (D9). When the tolerance margin is relatively high, i.e., 7.5%, the renewable producers may make a strategic offering for a long position to obtain more profits (D3), and revenue adequacy may not be guaranteed in some extreme situations (D5).

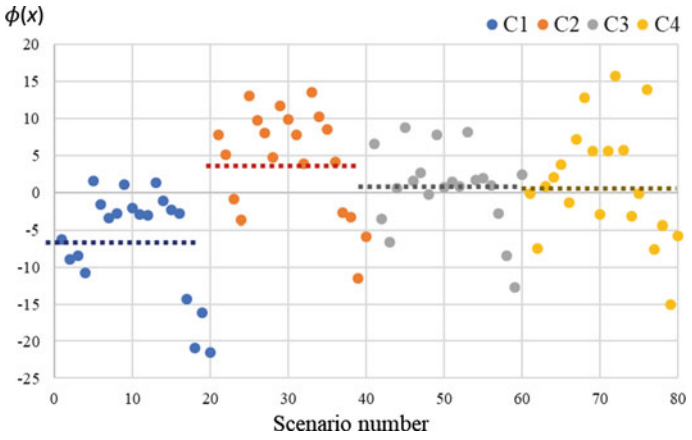


Fig. 8.11 The MCDA results with different tolerance margins

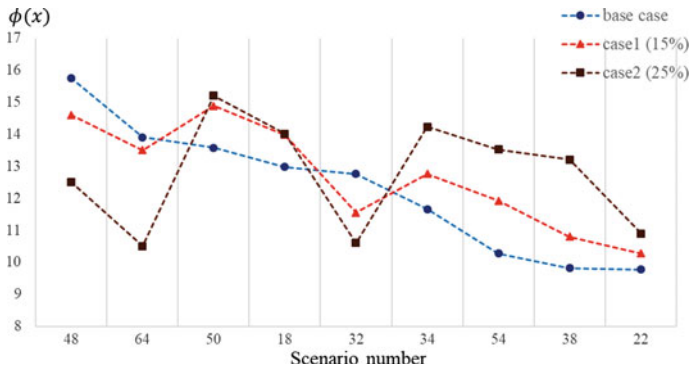


Fig. 8.12 The MCDA results with different penetration rate of variable renewables

In addition, an overly high tolerance margin may harm the incentive to invest to reduce imbalance costs at the system level (D10) and limit the transparency and efficiency of the balancing market.

8.4.2.4 Renewable Energy Penetration Rate Exploration

To investigate the effective imbalance settlement mechanism in different penetration levels of variable renewables, two additional cases with different penetration ratios of variable renewables are considered. Compared to the base case with a penetration ratio of 6.1% (which can be calculated based on the input data in Table D.1), the penetration of variable renewables in other two cases are set to 15%, and 25%, respectively. Figure 8.12 shows the MCDA results of the top nine market design scenarios under different penetration of variable renewables.

As seen in Fig. 8.12, the net flow value of original top two scenarios (48 and 64) significantly decreases in the two comparative cases. This is because the PTU and tolerance margin in both scenarios (48 and 64) are 60 min and 7.5%, respectively, which cannot accurately reflect the balancing cost caused by wind power producers, and leads to lower balance planning accuracy (D1), cost allocation efficiency (D5) and balancing cost (D7). Conversely, the net flow value under some scenarios (e.g. scenarios 34, 38 and 22) has risen in varying degrees in the two comparative cases because of the smaller PTU and lower tolerance margin of these scenarios. In addition, we find that the net flow value of the above top nine scenarios is still relatively high in the two cases. This phenomenon proves that the dual pricing mechanism is still effective under a high share of variable renewables scenario, because it can effectively regulate the market behaviours of BRPs compared to the other two pricing mechanisms.

Based on the above results and provided that the scope of the balancing market design task is vast and intricate, determining how to deal with the complexity and

uncertainties in a practical way, without ignoring vital balancing market design elements and their functions, is of vital importance for policy makers.

8.5 Conclusion

From the perspective of power market mechanism, this chapter focuses on how to update and design the existing market mechanism to improve the flexibility of power system. The core lies in the design of power spot market (balancing market), which is mainly reflected in the impact of renewable energy power generation suppliers' participation in the spot market and the interaction between balancing market design and the strategic behavior of market subjects. On the one hand, this chapter introduces the impact of single wind power supplier's participation in the power spot market, established the framework and balancing market clearing model with wind power supplier's strategic offering. On the other hand, through ABM method with MCDA evaluation is established to analysis strategic behaviors of market subjects. This chapter verifies the key elements of balancing market design from different perspectives through the above model, and puts forward the following suggestions. For the balancing market design elements proposed in this chapter, we propose the following recommendations:

(1) Imbalance pricing mechanism

Considering that China is currently in the early stage of spot market construction, the market operation may not be stable, under a dual pricing mechanism, both sides of imbalances are settled at a penalty price based on day-ahead market prices and corresponding penalty factors, and this approach can drive the imbalance prices more stable and closer to the day-ahead prices. In addition, we also find that under a high share of variable renewables cases, the dual pricing mechanism can effectively regulate the market behaviours of BRPs and reveal the real system balancing cost.

(2) Concerning PTU settings

The simulation results show that the PTU determines the distribution of balance responsibilities between BRPs and the SO. We also find that a longer PTU (e.g. 1 h) can ensure the smooth operation of the balancing market under a relatively low share of variable renewables, which is suitable at the early stage of spot market construction. With the increase share of variable renewables and further development of spot market construction, the PTU is suggested to be gradually shortened to 30, 15 min to divide the imbalance responsibilities more accurately and further promote justice and efficiency in the balancing market.

(3) Tolerance margin

For countries that are to set up effective spot markets, the enthusiasm of the players for participation in the market may be not high because of the expectation of guaranteed revenues stemming from long planned economy history. A tolerance margin

in imbalance settlement seems to be a good way to encourage the participation of generators and consumers. Besides, a tolerance margin for variable renewables can be considered as a compensation for the environmental benefits of renewable generation, which is advocated as a promising solution to promote the development of variable renewables at the initial stage of spot market construction. However, from another point of view, a tolerance margin limits the imbalance costs for corresponding BRPs and may reduce the incentives for planning flexible generation, and developing new forecasting tools and offering strategies. A tolerance margin for certain BRPs means that other BRPs have to undertake much more imbalance responsibilities, and this may affect the fairness and efficiency of the balancing market. It is worth mentioning that this phenomenon will intensify as the proportion of renewable energy rises, which suggests that the tolerance margin should be set carefully, approximately 5–10% only for renewables generators according to our simulation.

(4) Renewables generation suppliers

It is very important to improve the prediction accuracy of renewables generation for system balancing cost reduction and market operation efficiency improvement. Regardless of the balancing market design, to effectively reduce the imbalance cost and maximize profit, renewable generators need to strive to improve the prediction accuracy, thereby improving the operational efficiency of the entire market.

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